FINAL
TRIM AND STABILITY BOOKLET
OF
ARCADIA PARSHVA
(OFFICIAL NO. JMR-0025)

APPROVED
FOR COMPLIANCE WITH AS MENTIONED (SEE LETTER) REQUIREMENTS
ON BEHALF OF GOVT. OF INDIA
INDIAN REGISTER OF SHIPPING
MUMBAI
DATE: 04-Jun-2024

SEE LETTER E-219379-295955

AS AMENDED
SEE PAGE NO. 4, 9, 15, 20, 52-55, 77, 83, 101, 107, 233 & 250

Rev.	Description	Date	Prepared	Review
Issued for : Approval				

Owner : **ARVIND AND COMPANY SHIPPING AGENCIES LTD.** Builder: **CHISTI MARINE ENGINEERING WORKS**	Project: **PONTOON (YARD NO. CHISTI-037)**

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CONTENTS

General

FOR 250T CRAWLER CRANE

FOR 150T CRAWLER CRANE

Appendix

GENERAL PARTICULARS

Name of the Vessel	ARCADIA PARSHVA
Owner	ARVIND AND COMPANY SHIPPING AGENCIES LTD.
Builder	CHISTI MARINE ENGINEERING WORKS
Yard No.	CHISTI 037
Type of Vessel	PONTOON
Official No.	JMR-0025
Class Notation	卐SU,PONTOON,**INDIAN COASTAL SERVICE** "DECK STRENGTHENED FOR 10T/M^2" "FOR CRANE OPERATION"

Main Particulars

LOA	50.000 M
LBP	50.000 M
Breadth (Mld.)	18.000 M
Depth (Mld.)	03.000 M
Draft (Mld.)	2.200 M
Draft (Ext.)	2.212 M
Displacement(Ext.)	1893.06 T
Lightship	465.25 T
Deadweight	1427.81 T

Stability Criteria

REGULATION

Intact Stability Criteria in accordance **IS CODE 2008**

1) The area under a righting lever curve up to the angle of maximum righting lever should not be less than 0.080 meter-radian.

2) The static angle of heel due to a uniformly distributed wind load of 0.54 kPa (wind speed 30 m/s) should not exceed an angle corresponding to half freeboard for the relevant loading condition, where the lever of wind heeling moment is measured from the centroid of the windage area to half the draught.

3) The minimum range of stability should be:
   For L <= 100m: 20^0
   For L >= 150m: 15^0
   For intermediate length by interpolation

Intact stability criteria during cargo lifting

The following intact stability criteria are to be complied with:

- $\theta C \leq 15°$
- $GZC \leq 0,6\ GZMAX$
- $A1 \geq 0.4\ ATOT$

where:

θC : Heeling angle of equilibrium, corresponding to the first intersection between heeling and righting arms (see Fig 1)

GZC, GZMAX: Defined in Fig 1

A1 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θC to the heeling angle equal to the lesser of:

- heeling angle θR of loss of stability, corresponding to the second intersection between heeling and righting arms (see Fig 1)
- heeling angle θF, corresponding to flooding of unprotected openings

ATOT : Total area, in m.rad, below the righting lever curve.

In the above formula, the heeling arm, corresponding to the cargo lifting, is to be obtained, in m, from the following formula:

$$b=(Pd-Zz)/\Delta$$

where:

P : Cargo lifting mass, in t

d : Transversal distance, in m, of lifting cargo to the longitudinal plane (see Fig 1)

Z: Mass, in t, of ballast used for righting the pontoon, if applicable (see Fig 1)

z : Transversal distance, in m, of the centre of gravity of Z to the longitudinal plane (see Fig 1)

Δ : Displacement, in t, at the loading condition

The above check is to be carried out considering the most unfavourable situations of cargo lifting combined with the lesser initial metacentric height GM.The vertical position of the centre of gravity of cargo lifting is to be assumed in correspondence of the suspension point.

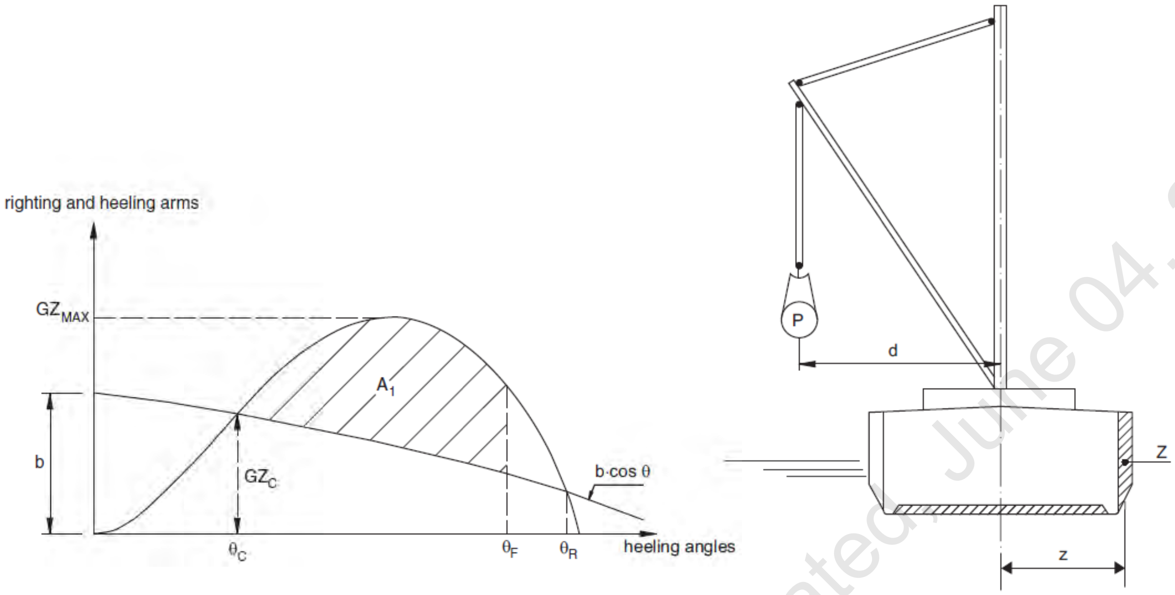


FIGURE 1: CARGO LIFTING

Intact stability criteria in the event of sudden loss of cargo during lifting

This additional requirement is compulsory when counterweights or ballasting of the ship are necessary or when deemed necessary by the Society taking into account the ship dimensions and the weights lifted.

The case of a hypothetical loss of cargo during lifting due to a break of the lifting cable is to be considered.

In this case, the following intact stability criteria are to be compiled with

- $A2 / A1 \geq 1$

- $\theta 2 - \theta 3 \geq 20°$

where:

A1 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle $\theta 1$ to the heeling angle θC (see Fig 2)

A2 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θC to the heeling angle $\theta 2$ (see Fig 2)

A3 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θC to the heeling angle θ3 (see Fig 2)

θ1 : Heeling angle of equilibrium during lifting (see Fig 2)

θ2 : Heeling angle corresponding to the lesser of θR and θF

θC : Heeling angle of equilibrium, corresponding to the first intersection between heeling and righting arms (see Fig 2)

θ3 : Maximum heeling angle due to roll, at which A3 = A1, to be taken not greater than 30° (angle in correspondence of which the loaded cargo on deck is assumed to shift (see Fig 2)

θR : Heeling angle of loss of stability, corresponding to the second intersection between heeling and righting arms (see Fig 2).

θF : Heeling angle at which progressive flooding may occur (see Fig 2)

In the above formulae, the heeling arm, induced on the ship by the cargo loss, is to be obtained, in m, from the following formula

$b = Zz \cos\theta/\Delta$

HEELING AND RIGHTING ARMS

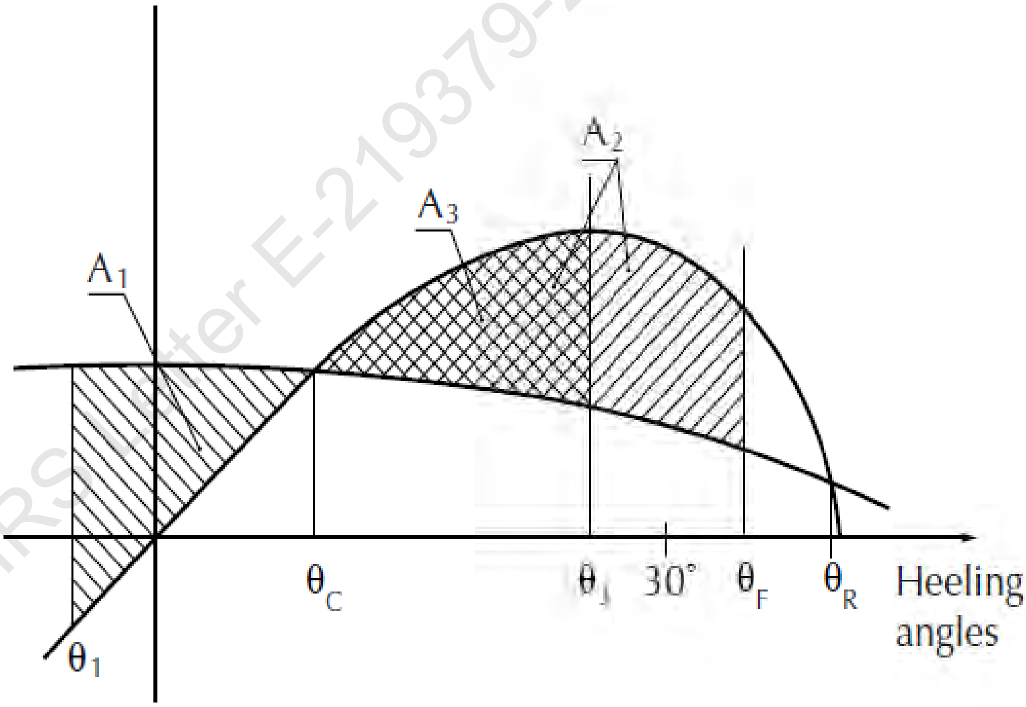


FIGURE 2: CARGO LOSS

Notes to the Master

General precautions against capsizing

1. Compliance with the required minimum stability criteria does not insure immunity against capsizing, regardless of the circumstances, or absolve the Master from his responsibilities. Masters should therefore exercise prudence and good seamanship, having regard to the season of the year, weather forecasts, the navigational zone, and should take appropriate action as to speed and course warranted by the prevailing circumstances.

2. Loading conditions shown in this booklet are typical of those which may be encountered in service and are worked out to be used for guidance only. Further the booklet contains sufficient data to enable the ship's staff to compute the ship’s stability in various loading conditions.

3. Excessive trim by forward is to be avoided in all cases as this may lead to difficulties of maneuverability and course keeping as also infringe the minimum Bow-height requirements.

4. All longitudinal levers, namely LCB, LCG, and LCF are computed with reference to the A.P of Ship.

5. In this booklet hydrostatic particulars and cross curves are given at following trims.

   -1.00m,-0.50m, 0.00m,0.50m, 1.00m

   These trims are the expected operational trims.

Notes on Use of Cross Curves of Stability

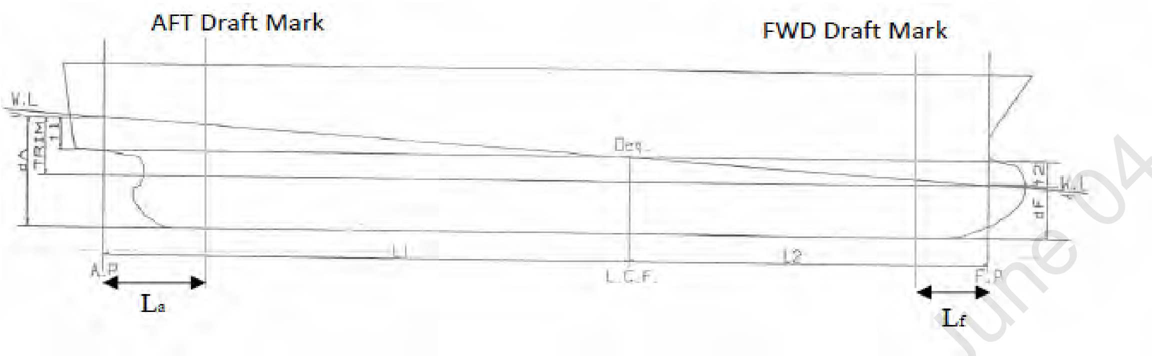
The purpose of the Cross-Curves of Stability is to enable Statical Stability curves to be drawn for the ship in any loading condition. Intact stability is then determined on the basis of this Curve. WORKED OUT EXAMPLE demonstrates the use of Cross Curves.

Note:

1. During unmanned tow minimum forward draft of 1.94m to be maintained at all times for compliance with slamming requirements or else it is recommended that unmanned tow of vessel should be carried out in fair weather and against favorable weather forecast.

2. Deck strength is examined considering Deck loading of 25 T/m2 in way of crane operating area & 10 T/m2 in way of other deck area as marked in Structural Plan. The selected crane's ground pressure as well as deck cargo shall not exceed above considered values for any orientation and its stated capacity.

Draft and Trim Calculation



The following data are required for ascertaining trim and draft:

a) Longitudinal Centre of gravity of the ship calculated as follows:

$$LCG = \frac{\text{Summation of longitudinal weight moments}}{\text{Displacement}}$$

b) Longitudinal centre of buoyancy, LCB

c) Displacement, Δ

d) Moment to change trim, MCT1cm

e) Longitudinal centre of floatation, LCF

LCB, LCF and MCT1cm are taken from the hydrostatic tables (trim = 0 m) corresponding to the actual displacement. The trim is calculated according to the following formula:

$$Trim = \frac{\Delta * (LCB-LCG)}{MCT1cm * 100}$$

The Extreme drafts at AP (TAP) and at FP (TFP) are calculated according to the following formulae:

$$T_{AP} = \frac{LCF * Trim}{LBP.} + T\ (extreme)$$

$$T_{FP} = T_{AP} - Trim$$

Where T is the Extreme draft taken from the hydrostatic tables (at trim = 0) corresponding to the actual displacement.

The draft at the Aft and Forward reading marks are as follows:

$$T_a = T_{AP} - \frac{Trim * L_a}{LBP}$$

$$T_f = T_{FP} + \frac{Trim * L_f}{LBP}$$

Where,

T_a = Draft at Aft Draft Mark

T_f = Draft at Fwd Draft Mark

L_a = Distance of Aft Draft Mark from AP and it is consider positive when Aft Draft Mark is Fwd of AP

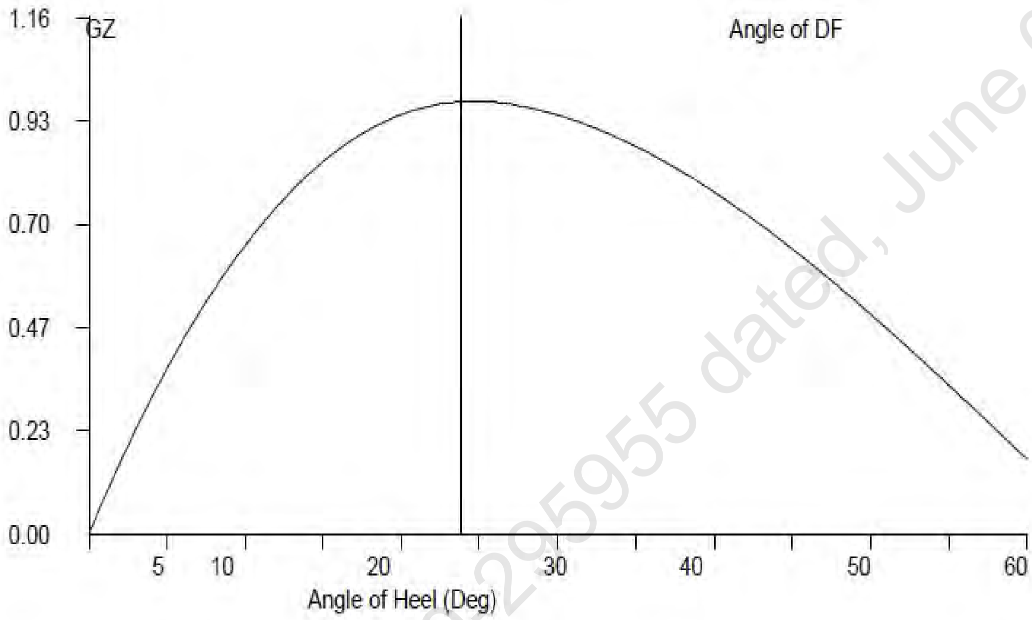
L_f = Distance of Fwd Draft Mark from FP and it is consider positive when Fwd Draft Mark is Aft of FP

GZ Calculation:

1) GZ = KN - KGo Sin (theta)
2) Area under 0-10 = 10*0.01745*[GZ0 + GZ10] / 2
3) Area under 10-20 = 10*0.01745*[GZ10 + GZ20] / 2
4) Area under 20-30 = 10*0.01745*[GZ20 + GZ30] / 2
5) Area under 30-40 = 10*0.01745*[GZ30 + GZ40] / 2
6) Area under 40-50 = 10*0.01745*[GZ40 + GZ50] / 2
7) Area under 50-60 = 10*0.01745*[GZ50 + GZ60] / 2
8) Area upto 40 deg / Angle of DF = sum of 2+3+4+5
9) Max Gz = Read from Gz Curve

Plot GZ values against the angles of heel as above. At 57.3 degree erect an ordinate equal to GM fluid on the same scale as GZ ordinates. Join the point to the origin. This is the Tangent to the GZ Curve at the Origin. Find the

Maxima of GZ curve, ordinates under GZ curve upto 30 degrees, 40 degrees/angle of flooding & between 30 degrees to 40 degrees/angle of flooding. If maximum GZ occurs at an angle less than 30 degrees then area upto this angle must be calculated.



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Worked Out Examples

1 Loading the Vessel

List out the weights, LCG, VCG, TCG and work out longitudinal moment and vertical moment for each load and finally sum up the entire loads to get Displacement, LCG and VCG of the vessel.

For this worked out Example,

Condition 1: Lightship + Crane + Ballast

Displacement	1100 12	Tonnes
LCG	24.288	M Fwd of AP
VCG (KG)	3.923	M above Base Line
TCG	0.000	M Off centerline

2 Free Surface Correction

A tank which is partially filled, adversely affects the stability of the vessel. This is known as Free Surface Effect and is expressed as 'Loss of GM' or 'Virtual Rise in KG' and is calculated as follows:

$$\text{Free Surface Correction (M)} = \frac{\text{Sum of Free Surface Moments of all Tanks}}{\text{Displacement of Vessel}}$$

Free Surface moment for a Tank = Moment of Inertia (M^4) for a Tank x Density of Liquid in Tank

KGo (Fluid) = KG (Solid) + Free Surface Correction.

Hydrostatic Particulars

The trim of a particular loading condition is obtained by

Trim = (LCB-LCG) X displacement / (MCT x 100) + Trim*

Where,

Trim* = Trim at which hydrostatic particulars have been computed.

Condition 1 : Lightship + Crane + Ballast

TANK NAME	VOL(Cu.M.)	SpGr	Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT	FSM
				METER	T-M	METER	T-M	METER	T-M	T-M
06_WBTK.C	132.36	1.025	135.67	42.251	5732.19	0.000	0.00	1.499	203.37	135.61
07_WBTK.P	48.39	1.025	49.60	47.660	2363.94	-5.971	-296.16	1.880	93.25	72.32
07_WBTK.S	48.39	1.025	49.60	47.660	2363.94	5.971	296.16	1.880	93.25	72.32
TOTAL TANK			234.87	44.536	10460.07	0.000	0.00	1.660	389.87	280.25
FIXED WEIGHT			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT	
				METER	T-M	METER	T-M	METER	T-M	
LIGHT SHIP			465.25	25.062	11,660.10	0.000	0.00	3.000	1,395.75	
CRANE			400.00	11.500	4,600.00	0.000	0.00	6.325	2,530.00	
TOTAL FIXED WEIGHT			865.25	18.792	16,260.10	0.000	0.00	4.537	3,925.75	
ITEM			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT	
				METER	T-M	METER	T-M	METER	T-M	
TOTAL TANK			234.870	44.536	10460.065	0.000	0.000	1.660	389.865	
TOTAL FIXED WEIGHT			865.250	18.792	16260.096	0.000	0.000	4.537	3925.750	
TOTAL WEIGHT			1100.120	24.288	26720.161	0.000	0.000	3.923	4315.615	

Condition 1 : Lightship + Crane + Ballast

Free Surface Correction for VCG = FSM/Displacement

	0.255	m
VCGc	4.178	m

From the MaxVCG table,

Displacement	1100.12	T
Allowable Max VCG	14.476	m

 # MAX VCG DOES NOT COVER CRANE OPERATIONS/LIFTING OF WEIGHTS AND IS VALID ONLY FOR UNMANNED TOW, WITH CRANE IN STOWED POSITION

VCGc < Maxvcg PASS

Note : The Master should always check the obtained VCG value against Maximum limiting VCG curve for intact
Refer to the Hydrostatic Table for even trim & Even Heel Condition of Stability Booklet.

Displacement =	1100.120	T
Draft ,Tm @ 25.0f =	1.337	m
LCB =	25.000	m
LCF =	25.000	m
MCT =	33.282	T-m/cm
KMT =	22.272	m

The vessels trim is to be determined as follow,

Trim=(LCB-LCG)xDisplacement /(MCTX100)
0.235 m,(+) is aft trim,(-) is fwd trim

Determining the Forward perendicular,Midship and aft Perpendicular Draft

LBP = 50m

Draft @ 25.00 (MS),Tmid 1.337 m

Draft @ 50.0f (FP),Tfwd =Tm - Trim/2
1.219 m

Draft@0(AP),Taft =Tm + Trim/2
1.454 m

Condition 1 : Lightship + Crane + Ballast

From HYDROSTATICS

Hydrostatic Particulars			**Drafts & Trims**	
MCT	33.282	T-m	Draft (Mean)	1.337 m
LCB	25.000	m	Draft AP (USK)	1.454 m
			Draft FP (USK)	1.219 m
Metacentric Height				
KMt	22.272	m		
KG	3.923	m		
KG0	4.178	m		
GMt	18.094	m	**Trim**	0.235 m

The condition shown above is first worked
using Even Keel hydrostatic perticulars
Program then interpolates the hydrostatic perticulars
from the values available for the displacement

From HYDROSTATICS

Hydrostatic Particulars			**Drafts & Trims**	
MCT	32.280	T-m	Draft (Mean)	1.315 m
LCB	24.233	m	Draft AP (USK)	1.441 m
			Draft FP (USK)	1.190 m
Metacentric Height				
KMt	22.574	m		
KG	3.938	m		
KG0	4.197	m		
GMt	18.377	m	**Trim**	0.252 m

Trim is calculaed as follows:
Trim = (LCB - LCG) X displacement / (MCT X 100) + Trim*
Where Trim* is trim at which hydrostatics were calculated.

Condition 1 : Lightship + Crane + Ballast

Intact Stability (Primary Criterion)

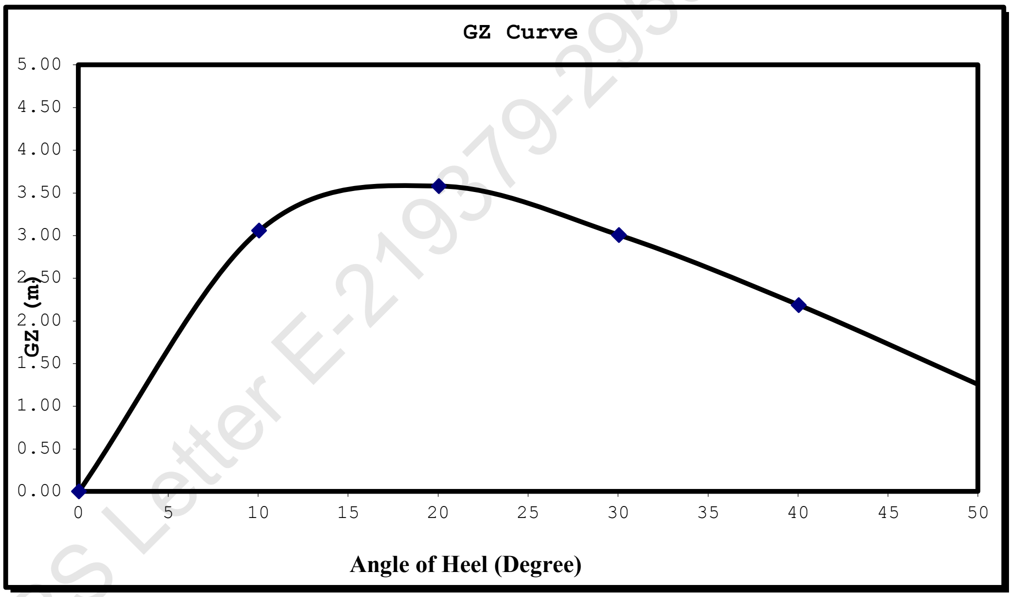
			Minimum Required
Displacement	1100.12	T	
KG0	4.197	M	
GMt	18.377	M	0.150 M

Residual Arm Curve

Angle (Deg)	0	10	20	30	40	50
KN (M)	0.000	3.788	5.018	5.107	4.885	4.468
GZ (M)	0.000	3.056	3.579	3.005	2.184	1.251

	Computed values		Minimum Required	
Residual Area from abs 0 deg to MAxRA	0.6908	m-rad	0.080	m-rad
Angle from Equilibrium to RAZero or flood	LARGE	m-rad	20.00	Deg.
Angle from Equilibrium to 50% DK Imm Angle	5.05	m-rad	0.00	Deg.

Plot GZ values against the angles of heel as above.
At 57.3 degree erect an ordinate equal to GM fluid on the same scale as GZ ordinates.Join the point to the origin.
This is the Tangent to the GZ Curve at the Origin.Find the Maxima of GZ curve, ordinates under GZ curve upto 30 degrees, 40 degrees/angle of flooding
If maximum GZ occurs at an angle less than 30 degrees then area upto this angle must be calculated.



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Worked Out Example Contd.

Mean Draft	1.315	M
LCF from AP	24.766	M
LCB from AP	24.233	M
MCT	32.280	T-M/Cm
Trim	0.252	M
Draft (AP)	1.441	M
Draft (FP)	1.190	M
KG	3.938	M
KGo	4.197	M
GoMt	18.377	M

LOADLINE DRAFT: Check that Mean Draft (Midship) does not exceed the Summer Draft (Ext) for the corresponding season, as per the assigned loadline.

BLANK CALCULATION SHEET

TANK NAME	VOL(Cu.M.)	SpGr	Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT	FSM
				METER	T-M	METER	T-M	METER	T-M	T-M
TOTAL TANK										

FIXED WEIGHT			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
LIGHT SHIP			465.250	25.062	11,660.096	0.000	0.000	3.000	1,395.750
CRANE									
DECK CARGO									
LOAD LIFTED									
TOTAL FIXED WEIGHT									

ITEM			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
TOTAL TANK									
TOTAL FIXED WEIGHT									
TOTAL WEIGHT									

Free Surface Correction for VCG = FSM/Displacement

m

VCGc m

From the MaxVCG table,

Displacement T

Allowable Max VCG m

 # MAX VCG DOES NOT COVER CRANE OPERATIONS/LIFTING OF WEIGHTS AND IS VALID ONLY FOR UNMANNED TOW, WITH CRANE IN STOWED POSITION

$\boxed{\text{VCGc < Maxvcg}}$

Note : The Master should always check the obtained VCG value against Maximum limiting VCG curve for intact
Refer to the Hydrostatic Table for even trim & Even Heel Condition of Stability Booklet.

Displacement = T

Draft ,Tm @ 25.0f = m

LCB = m

LCF = m

MCT = T-m/cm

KMT = m

The vessels trim is to be determined as follow,

Trim=(LCB-LCG)xDisplacement /(MCTX100)

m,(+) is aft trim,(-) is fwd trim

Determining the Forward perendicular,Midship and aft Perpendicular Draft

LBP = 50m

Draft @ 25.00 (MS),Tmid m

Draft @ 50.0f (FP),Tfwd =Tm - Trim/2

m

Draft@0(AP),Taft =Tm + Trim/2

m

Definitions, Symbols and Sign Conventions:

AP	Fr.0
FP	Fr.100
Midship	Fr.50
Frame Spacing	500mm throughout

List of Symbols

Symbol	Unit	Description
LOA	M	Length Overall
LBP	M	Length between perpendiculars
B (MLD)	M	Moulded Breadth
D (MLD)	M	Moulded Depth
TPCm	T/Cm	Tonnes per 1 Cm immersion
MCTCm	T-M/Cm	Moment to change Trim by 1 Cm
Draft AP (USK)	M	Draft at AP wrt Keel line
Draft FP (USK)	M	Draft at FP wrt Keel line
Draft (Mean)	M	Draft wrt Keel line at Midship
Draft (Mean)	M	[Draft AP(USK) + Draft FP(USK)] x 0.5
VCG, KG	M	Vertical Center of Gravity above Base Line before correcting for free surface
KGo	M	Vertical Center of Gravity above Base Line after correcting for free surface
KB	M	Vertical Center of Buoyancy from Base Line
KMt	M	Transverse Metacenter above Base Line
GoMt	M	Transverse Metacenter above KGo
GZ	M	Righting Arm
FSM	T-M	Free Surface Moment
sp.gr.		Specific Gravity
LCG	M	Longitudinal Center of Gravity
LCB	M	Longitudinal Center of Buoyancy
LCF	M	Longitudinal Center of Floatation

Sign Convention

LCB, LCG, LCF	Fwd of AP is denoted by f and aft of AP is denoted by a
TCG	port is negative and stbd is positive
Trim by Aft	Positive
Trim by Ford	Negative

Metric Conversion Table

MULTIPLY BY	TO CONVERT FROM	TO OBTAIN	
0.03937	millimeters	Inches	25.400
0.39370	Centimeters	Inches	2.5400
3.28080	Meters	Feet	0.3050
2.20460	kilograms	Pounds	0.4540
0.00098	kilogram’s	tons(2240 lb.)	1016.1
0.98420	Metric tonnes (i.e. tonnes of 1000 kg.)	tons(2240 lb.)	1.0160
2.49980	Metric tonnes per centimeter of immersion	tons per inch (immersion)	0.4000
8.20140	moment to change trim one centimeter one (tonne meter units)	moment to change trim inch(foot ton units)	0.1220
187.976	Meter Radians	Feet degrees	0.0053
	TO OBTAIN	**TO CONVERT FROM**	**MULTIPLY**

Relation between Weight and volume

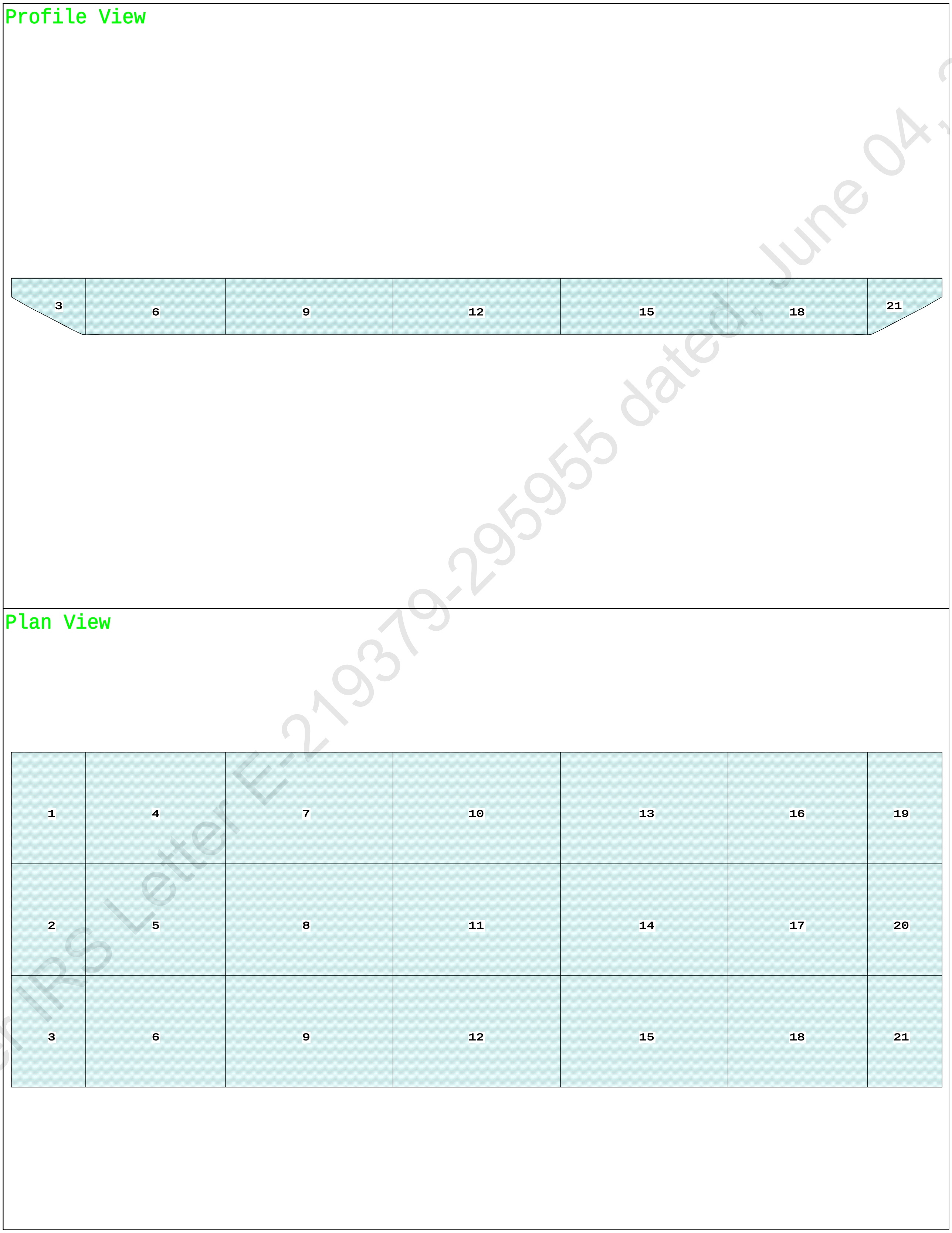
1000mm cubed	1 Cubic Centimeters
1 Cubic Centimeter of Fresh Water (S.G.1.0)	1 Gram
1000 cubic centimeter of fresh Water (S.G.1.0)	1 Kilogram (1000 Gm.)
1 Cubic Meter of fresh water (S.G.1.0)	1 Tonne (1000 Kg.)
1 Cubic Meter of fresh water (S.G.1.0)	1 tonne
1 Cubic Meter of Salt water (S.G. 1.025)	1.025 Tonnes
1 Tonne of Salt Water (S.G.1.025)	0.975 Cubic Meters

Angle of Downflooding

1. The angle of flooding is the angle of immersion of the sill or coaming of the opening on or above the freeboard deck through which progressive downflooding can occur as the vessel heels.
2. If this opening has a weathertight closing device which cannot be kept closed and secured at all times while the vessel is underway (e.g. engine room air supply trunk), or has no weathertight closing device, progressive downflooding will occur when the sill of the opening immerses. This angle is to be marked on the GZ curve if it comes within the apparent positive range of the GZ curve, and the GZ curve terminates at this angle: i.e., the actual angle of flooding becomes the range of the GZ curve.
3. If an opening is fitted with a weathertight closing device which can be kept closed and secured at all times while the vessel is underway, progressive downflooding can only occur through the opening if its closing device is not closed and secured. The angle of immersion of the lower sill off this opening is a \`potential' angle of flooding. If the closing appliance is not closed and secured, downflooding will occur when the lower sill of this opening reaches the water level.
   If the lower sill of the opening through which downflooding can occur immerses at 40 degree or smaller angles of heel, its closing appliances must be kept closed and secured at all times when the vessel is underway or at sea where this opening is used for access in the normal working of the vessel, an alternate access should be used in heavy weather. In moderate weather, the opening may be used for access, but the closing appliances must be closed and secured again immediately after use.

   As there is no any unprotected opening exists above freeboard deck, hence the downflooding angle is considered at 50 deg.

Condition Graphic



Tanks

1 01_WBTK.P:100% SALT WATER Intact	8 03_WBTK.C:100% SALT WATER Intact	16 06_WBTK.P:100% SALT WATER Intact
2 01_WBTK.C:100% SALT WATER Intact	9 03_WBTK.S:100% SALT WATER Intact	17 06_WBTK.C:100% SALT WATER Intact
3 01_WBTK.S:100% SALT WATER Intact	10 04_WBTK.P:100% SALT WATER Intact	18 06_WBTK.S:100% SALT WATER Intact
4 02_WBTK.P:100% SALT WATER Intact	11 04_WBTK.C:100% SALT WATER Intact	19 07_WBTK.P:100% SALT WATER Intact
5 02_WBTK.C:100% SALT WATER Intact	12 04_WBTK.S:100% SALT WATER Intact	20 07_WBTK.C:100% SALT WATER Intact
6 02_WBTK.S:100% SALT WATER Intact	13 05_WBTK.P:100% SALT WATER Intact	21 07_WBTK.S:100% SALT WATER Intact
7 03_WBTK.P:100% SALT WATER Intact	14 05_WBTK.C:100% SALT WATER Intact	
	15 05_WBTK.S:100% SALT WATER Intact	

20-03-24 10:44:31 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

TANK STATUS
Trim: zero, Heel: zero

Part	Cu.M.	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
01_WBTK.P	48.4	1.025	49.60	2.340f	5.971p	1.880	72.32*
01_WBTK.C	48.9	1.025	50.11	2.336f	0.000	1.872	72.33*
01_WBTK.S	48.4	1.025	49.60	2.340f	5.971s	1.880	72.32*
02_WBTK.P	131.4	1.025	134.72	7.749f	5.980p	1.509	135.61*
02_WBTK.C	132.4	1.025	135.67	7.749f	0.000	1.499	135.61*
02_WBTK.S	131.4	1.025	134.72	7.749f	5.980s	1.509	135.61*
03_WBTK.P	157.7	1.025	161.60	16.000f	5.980p	1.509	162.73*
03_WBTK.C	158.8	1.025	162.73	16.000f	0.000	1.500	162.73*
03_WBTK.S	157.7	1.025	161.60	16.000f	5.980s	1.509	162.73*
04_WBTK.P	157.7	1.025	161.60	25.000f	5.980p	1.509	162.73*
04_WBTK.C	158.8	1.025	162.73	25.000f	0.000	1.500	162.73*
04_WBTK.S	157.7	1.025	161.60	25.000f	5.980s	1.509	162.73*
05_WBTK.P	157.7	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.C	158.8	1.025	162.73	34.000f	0.000	1.500	162.73*
05_WBTK.S	157.7	1.025	161.60	34.000f	5.980s	1.509	162.73*
06_WBTK.P	131.4	1.025	134.72	42.251f	5.980p	1.509	135.61*
06_WBTK.C	132.4	1.025	135.67	42.251f	0.000	1.499	135.61*
06_WBTK.S	131.4	1.025	134.72	42.251f	5.980s	1.509	135.61*
07_WBTK.P	48.4	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	48.9	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	48.4	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks--------->			2,566.63	25.000f	0.000	1.549	2712.14

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

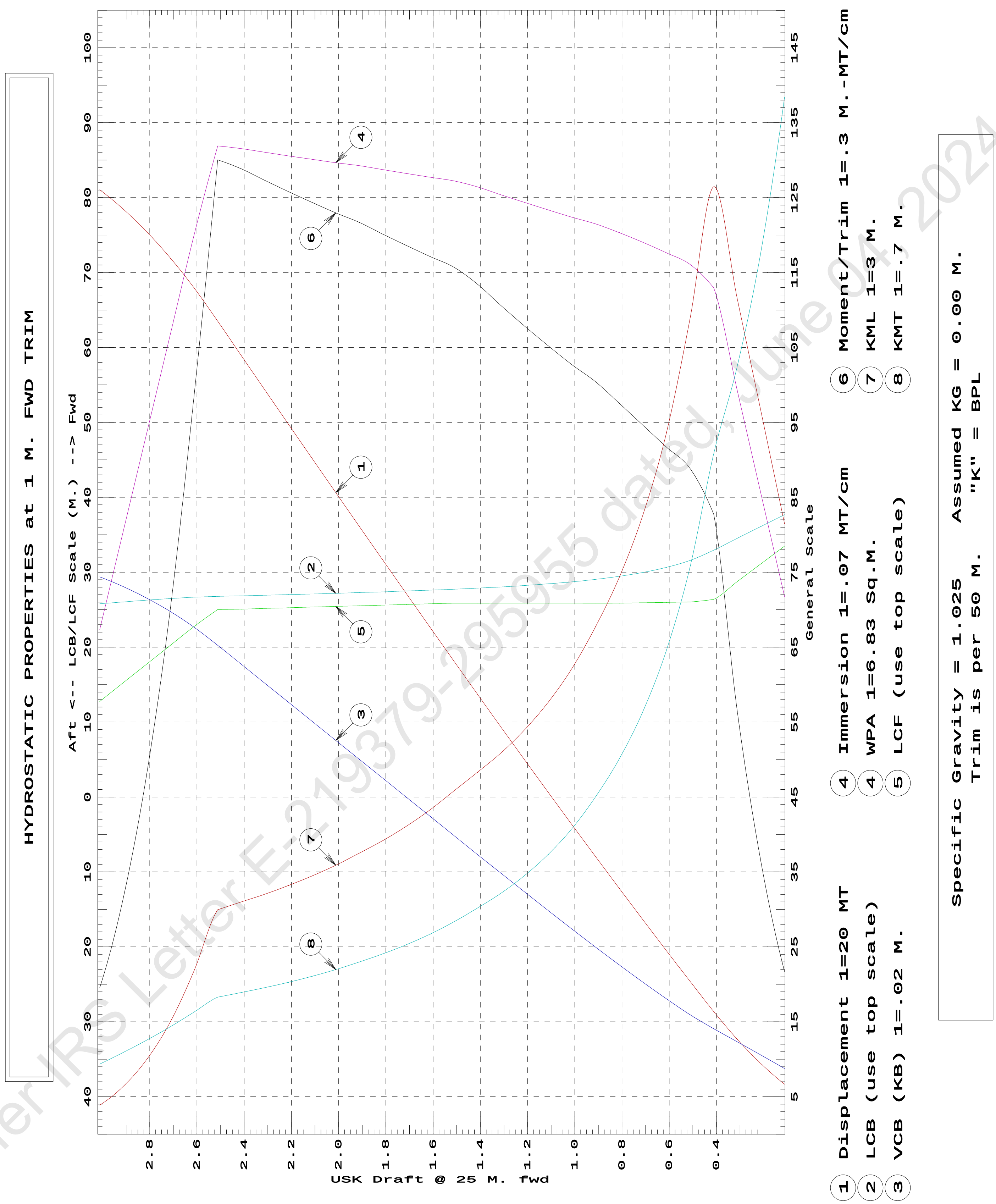
HYDROSTATIC PARTICULARS

HYDROSTATIC PROPERTIES
Trim: Fwd 1.000/50.000, No Heel, Fixed VCG = 0.000

Draft@ 25.000f	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	KML	KMT
0.112	133.29	37.689f	0.175	5.02	33.490f	6.52	244.36	96.881
0.212	188.30	36.107f	0.211	5.98	31.124f	10.90	289.46	82.456
0.312	252.92	34.523f	0.246	6.94	28.751f	16.92	334.36	71.772
0.412	327.16	32.938f	0.281	7.91	26.375f	24.82	379.30	63.564
0.512	407.76	31.583f	0.319	8.13	26.034f	26.65	326.70	53.012
0.612	489.54	30.651f	0.361	8.23	25.975f	27.51	280.91	45.226
0.712	572.35	29.971f	0.406	8.33	25.923f	28.38	247.92	39.491
0.812	656.10	29.452f	0.452	8.42	25.886f	29.26	222.97	35.068
0.912	740.70	29.044f	0.499	8.50	25.870f	30.12	203.29	31.535
1.012	826.02	28.717f	0.548	8.57	25.880f	30.81	186.46	28.540
1.112	912.01	28.450f	0.597	8.63	25.878f	31.57	173.02	26.147
1.212	998.69	28.226f	0.646	8.70	25.875f	32.35	161.94	24.171
1.312	1,086.07	28.037f	0.696	8.78	25.872f	33.17	152.69	22.519
1.412	1,174.15	27.875f	0.746	8.85	25.869f	34.03	144.90	21.124
1.512	1,262.91	27.735f	0.797	8.91	25.870f	34.71	137.41	19.834
1.612	1,352.12	27.609f	0.848	8.94	25.783f	35.14	129.91	18.700
1.712	1,441.67	27.493f	0.899	8.97	25.694f	35.58	123.36	17.712
1.812	1,531.58	27.384f	0.950	9.01	25.603f	36.03	117.61	16.848
1.912	1,621.83	27.283f	1.000	9.05	25.508f	36.51	112.52	16.088
2.012	1,712.42	27.187f	1.051	9.08	25.436f	36.88	107.68	15.374
2.112	1,803.32	27.097f	1.102	9.11	25.351f	37.30	103.41	14.750
2.212	1,894.54	27.011f	1.153	9.14	25.263f	37.74	99.58	14.190
2.312	1,986.08	26.928f	1.203	9.17	25.174f	38.19	96.11	13.689
2.412	2,077.95	26.849f	1.254	9.21	25.081f	38.65	92.98	13.238
2.512	2,170.15	26.772f	1.305	9.23	25.013f	39.01	89.86	12.803
2.612	2,258.72	26.664f	1.353	8.40	22.760f	29.59	65.48	11.408
2.712	2,338.10	26.490f	1.395	7.48	20.257f	21.09	45.09	10.043
2.812	2,408.27	26.272f	1.431	6.56	17.755f	14.46	30.01	8.792
2.912	2,469.20	26.032f	1.462	5.63	15.252f	9.46	19.15	7.632
3.012	2,520.92	25.786f	1.488	4.71	12.750f	5.86	11.62	6.543

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

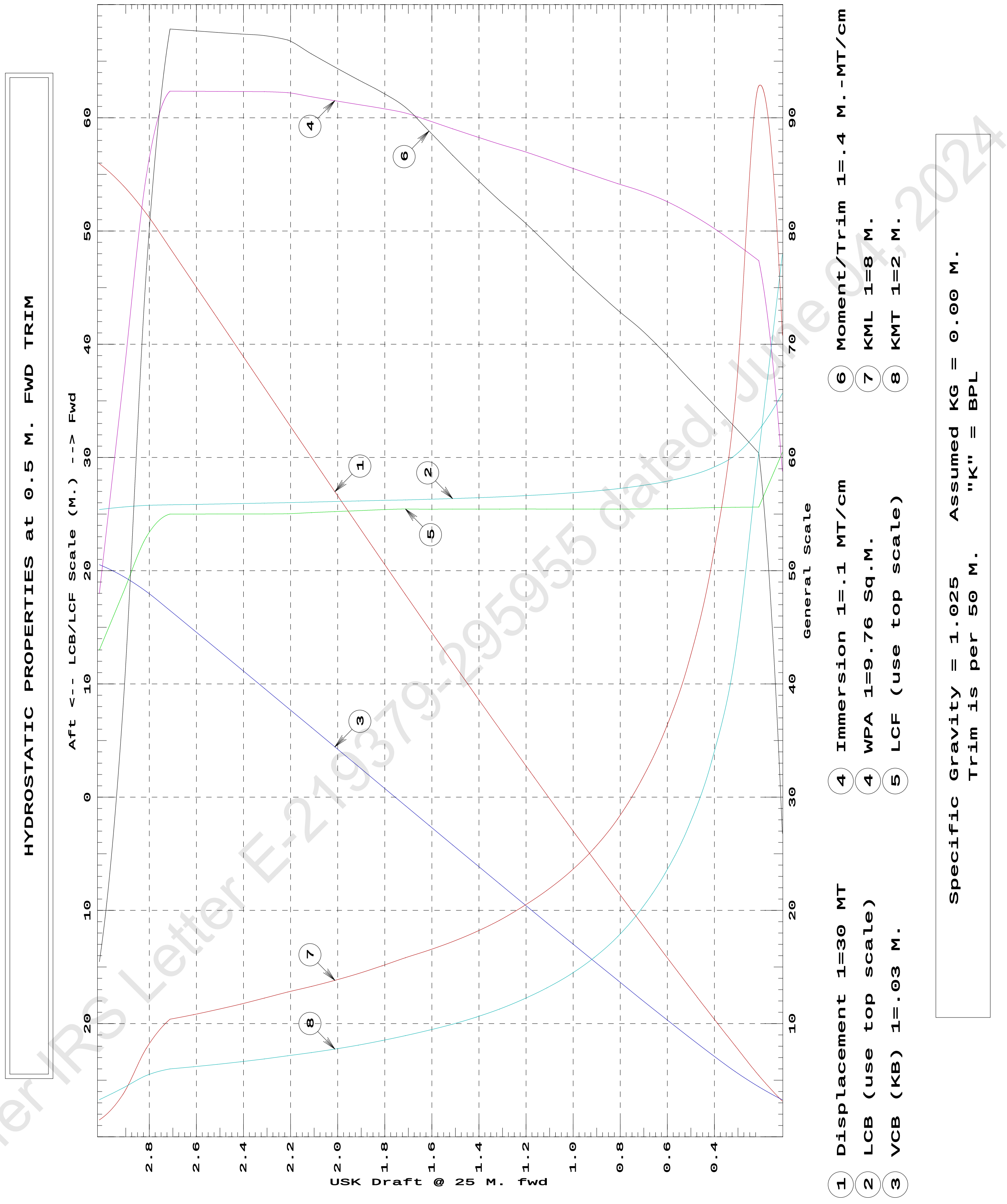

HYDROSTATIC PROPERTIES at 1 M. FWD TRIM
Aft <-- LCB/LCF Scale (M.) --> Fwd
General Scale
USK Draft @ 25 M. fwd
1 Displacement 1=20 MT
2 LCB (use top scale)
3 VCB (KB) 1=.02 M.
4 Immersion 1=.07 MT/cm
4 WPA 1=6.83 Sq.M.
5 LCF (use top scale)
6 Moment/Trim 1=.3 M.-MT/cm
7 KML 1=3 M.
8 KMT 1=.7 M.
Specific Gravity = 1.025 Assumed KG = 0.00 M.
Trim is per 50 M. "K" = BPL

HYDROSTATIC PROPERTIES
Trim: Fwd 0.500/50.000, No Heel, Fixed VCG = 0.000

Draft@ 25.000f	Displacement Weight(MT)	Buoyancy-Ctr. LCB	Buoyancy-Ctr. VCB	Weight/ cm	LCF	Moment/ cm trim	KML	KMT
0.112	94.61	35.657f	0.098	5.88	30.439f	10.72	566.32	155.843
0.212	162.65	32.417f	0.132	7.74	25.617f	24.15	742.47	120.816
0.312	240.79	30.208f	0.173	7.89	25.602f	25.06	520.34	85.097
0.412	320.44	29.059f	0.218	8.04	25.566f	25.94	404.70	66.389
0.512	401.45	28.349f	0.266	8.16	25.509f	26.81	333.87	54.610
0.612	483.61	27.862f	0.315	8.27	25.457f	27.70	286.38	46.395
0.712	566.72	27.508f	0.365	8.35	25.449f	28.52	251.60	40.187
0.812	650.57	27.242f	0.415	8.42	25.440f	29.21	224.49	35.343
0.912	735.11	27.034f	0.466	8.49	25.439f	29.96	203.76	31.632
1.012	820.35	26.868f	0.517	8.56	25.438f	30.74	187.34	28.689
1.112	906.31	26.733f	0.567	8.63	25.436f	31.56	174.08	26.309
1.212	992.99	26.620f	0.619	8.71	25.451f	32.35	162.89	24.309
1.312	1,080.33	26.524f	0.670	8.77	25.439f	33.07	153.05	22.576
1.412	1,168.34	26.443f	0.722	8.83	25.438f	33.86	144.89	21.132
1.512	1,257.02	26.372f	0.773	8.90	25.437f	34.68	137.92	19.898
1.612	1,346.39	26.310f	0.825	8.97	25.435f	35.53	131.95	18.836
1.712	1,436.46	26.255f	0.877	9.04	25.448f	36.37	126.58	17.889
1.812	1,527.11	26.206f	0.929	9.08	25.389f	36.90	120.81	16.991
1.912	1,618.11	26.157f	0.981	9.12	25.304f	37.34	115.37	16.198
2.012	1,709.44	26.109f	1.033	9.15	25.217f	37.79	110.53	15.496
2.112	1,801.11	26.062f	1.085	9.19	25.127f	38.26	106.21	14.872
2.212	1,893.13	26.014f	1.137	9.22	25.034f	38.75	102.33	14.317
2.312	1,985.42	25.968f	1.189	9.23	25.012f	38.91	97.97	13.759
2.412	2,077.73	25.925f	1.240	9.23	25.009f	38.96	93.75	13.255
2.512	2,170.06	25.886f	1.292	9.23	25.006f	39.02	89.90	12.799
2.612	2,262.39	25.850f	1.343	9.23	25.004f	39.08	86.36	12.384
2.712	2,354.73	25.817f	1.394	9.24	25.001f	39.14	83.10	12.006
2.812	2,445.61	25.772f	1.445	8.50	23.000f	30.67	62.71	10.846
2.912	2,521.33	25.617f	1.485	6.65	18.000f	15.11	29.97	8.622
3.012	2,578.59	25.395f	1.516	4.80	13.000f	6.19	12.01	6.555

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

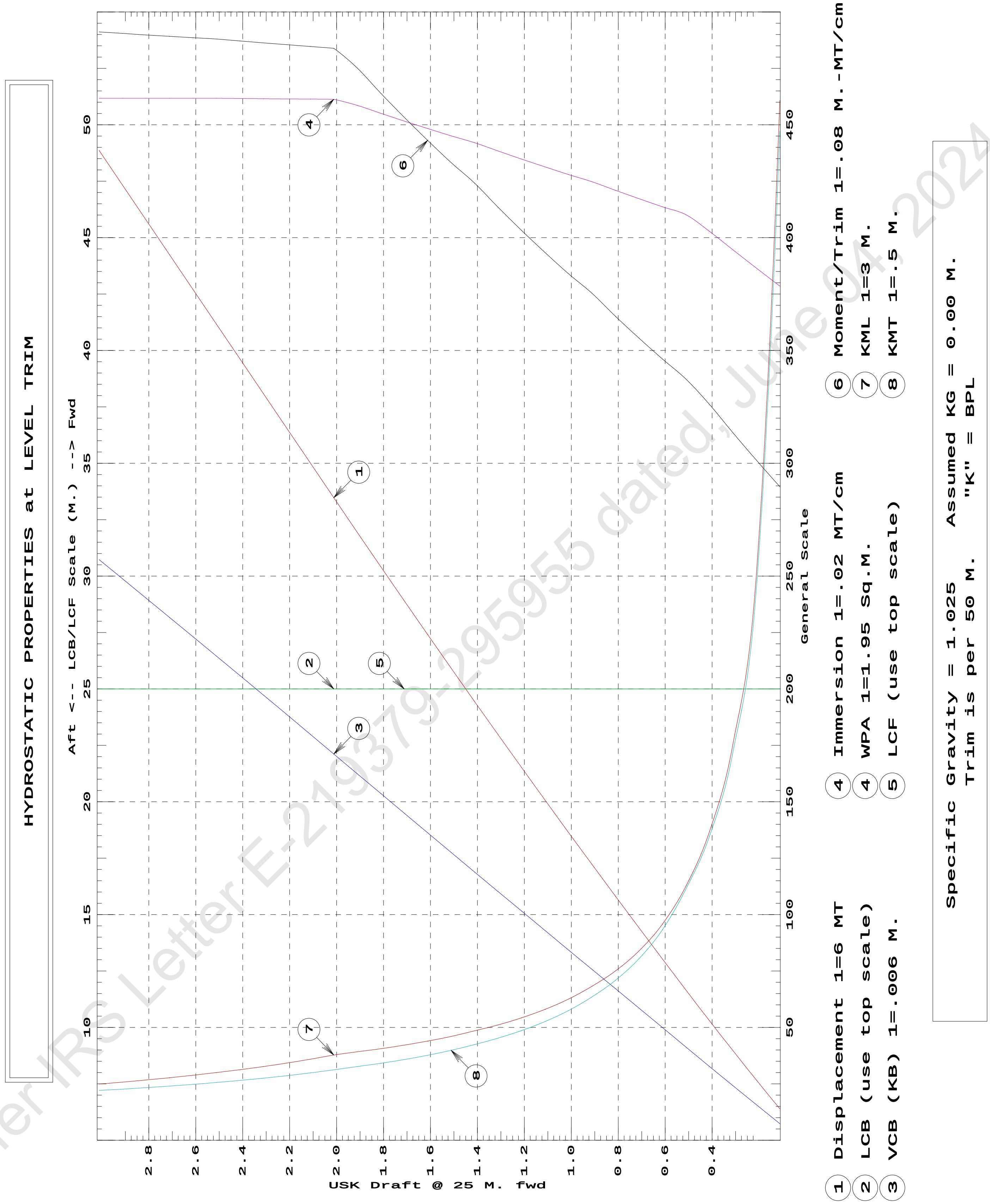

HYDROSTATIC PROPERTIES at 0.5 M. FWD TRIM
Aft <-- LCB/LCF Scale (M.) --> Fwd
General Scale
USK Draft @ 25 M. fwd
1 Displacement 1=30 MT
2 LCB (use top scale)
3 VCB (KB) 1=.03 M.
4 Immersion 1=.1 MT/cm
4 WPA 1=9.76 Sq.M.
5 LCF (use top scale)
6 Moment/Trim 1=.4 M.-MT/cm
7 KML 1=8 M.
8 KMT 1=2 M.
Specific Gravity = 1.025 Assumed KG = 0.00 M.
Trim is per 50 M. "K" = BPL

HYDROSTATIC PROPERTIES
No Trim, No Heel, Fixed VCG = 0.000

Draft@ 25.000f	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	KML	KMT
0.112	83.79	25.000f	0.044	7.57	25.000f	23.17	1382.58	223.643
0.212	160.29	25.000f	0.095	7.73	25.000f	24.11	752.15	122.246
0.312	238.38	25.000f	0.145	7.89	25.000f	25.08	526.13	85.934
0.412	318.09	25.000f	0.196	8.06	25.000f	26.09	410.04	67.317
0.512	399.40	25.000f	0.248	8.20	25.000f	26.98	337.80	55.685
0.612	481.78	25.000f	0.300	8.27	25.000f	27.70	287.43	46.641
0.712	564.87	25.000f	0.351	8.35	25.000f	28.44	251.70	40.223
0.812	648.69	25.000f	0.403	8.42	25.000f	29.21	225.14	35.449
0.912	733.24	25.000f	0.454	8.50	25.000f	30.02	204.72	31.774
1.012	818.51	25.000f	0.506	8.56	25.000f	30.72	187.65	28.736
1.112	904.44	25.000f	0.558	8.63	25.000f	31.47	173.99	26.304
1.212	991.06	25.000f	0.610	8.70	25.000f	32.26	162.74	24.297
1.312	1,078.37	25.000f	0.661	8.77	25.000f	33.07	153.35	22.621
1.412	1,166.40	25.000f	0.713	8.84	25.000f	33.93	145.46	21.208
1.512	1,255.11	25.000f	0.765	8.90	25.000f	34.66	138.07	19.915
1.612	1,344.46	25.000f	0.818	8.97	25.000f	35.45	131.83	18.822
1.712	1,434.46	25.000f	0.870	9.03	25.000f	36.27	126.41	17.872
1.812	1,525.13	25.000f	0.922	9.10	25.000f	37.12	121.69	17.042
1.912	1,616.48	25.000f	0.974	9.17	25.000f	38.01	117.57	16.315
2.012	1,708.51	25.000f	1.027	9.23	25.000f	38.71	113.30	15.610
2.112	1,800.77	25.000f	1.079	9.23	25.000f	38.78	107.66	14.924
2.212	1,893.06	25.000f	1.132	9.23	25.000f	38.84	102.58	14.309
2.312	1,985.36	25.000f	1.184	9.23	25.000f	38.90	97.98	13.756
2.412	2,077.68	25.000f	1.236	9.23	25.000f	38.97	93.78	13.257
2.512	2,170.03	25.000f	1.287	9.24	25.000f	39.04	89.95	12.804
2.612	2,262.38	25.000f	1.339	9.24	25.000f	39.09	86.38	12.385
2.712	2,354.73	25.000f	1.390	9.24	25.000f	39.14	83.10	12.003
2.812	2,447.09	25.000f	1.442	9.24	25.000f	39.19	80.07	11.654
2.912	2,539.44	25.000f	1.493	9.24	25.000f	39.24	77.26	11.334
3.012	2,631.79	25.000f	1.544	9.24	25.000f	39.29	74.65	11.040

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

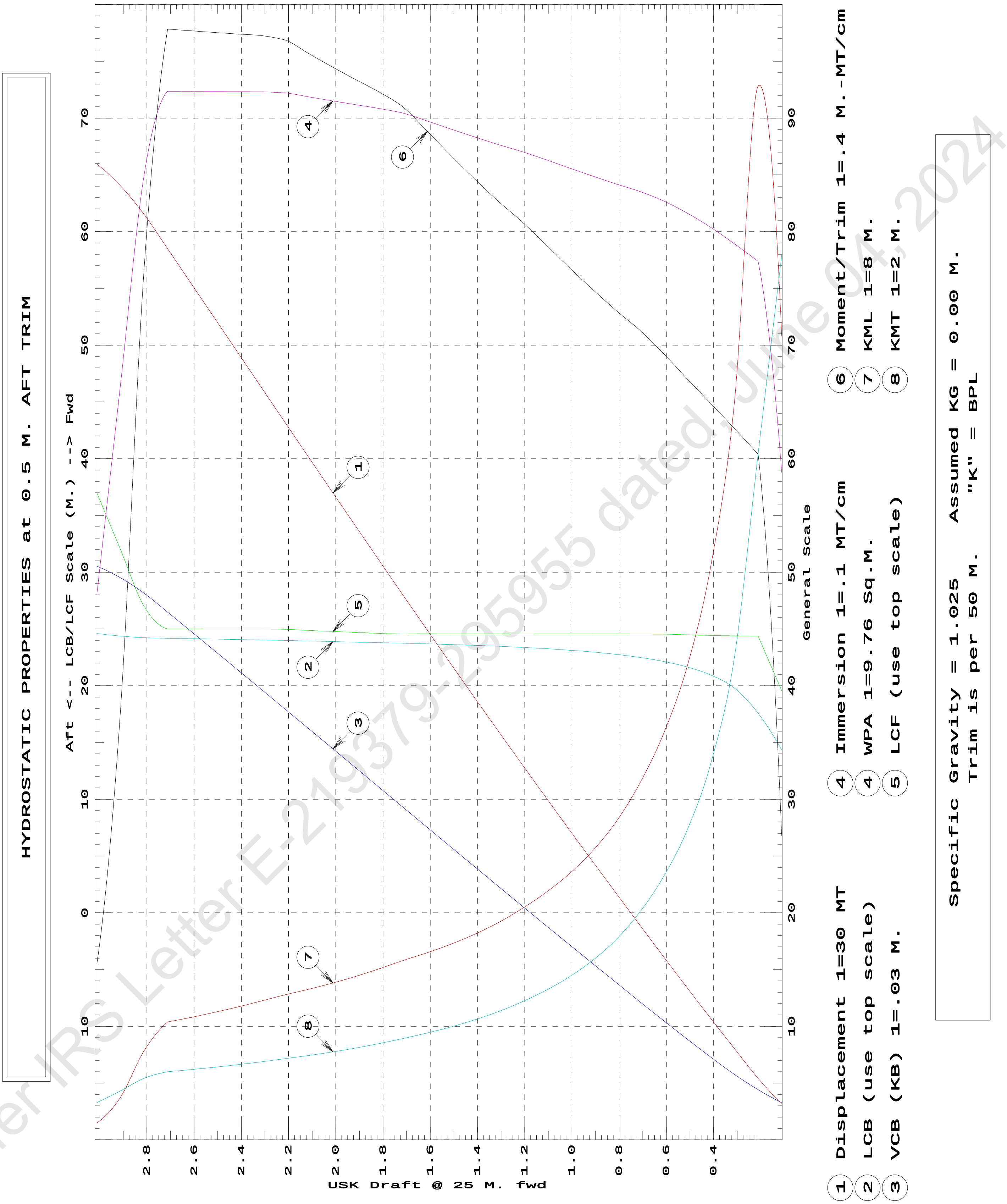

HYDROSTATIC PROPERTIES at LEVEL TRIM
Aft <-- LCB/LCF Scale (M.) --> Fwd
General Scale
USK Draft @ 25 M. fwd
1 Displacement 1=6 MT
2 LCB (use top scale)
3 VCB (KB) 1=.006 M.
4 Immersion 1=.02 MT/cm
4 WPA 1=1.95 Sq.M.
5 LCF (use top scale)
6 Moment/Trim 1=.08 M.-MT/cm
7 KML 1=3 M.
8 KMT 1=.5 M.
Specific Gravity = 1.025 Assumed KG = 0.00 M.
Trim is per 50 M. "K" = BPL

HYDROSTATIC PROPERTIES
Trim: Aft 0.500/50.000, No Heel, Fixed VCG = 0.000

Draft@ 25.000f	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	KML	KMT
0.112	94.61	14.343f	0.098	5.88	19.561f	10.72	566.32	155.836
0.212	162.65	17.583f	0.132	7.74	24.383f	24.15	742.47	120.811
0.312	240.79	19.792f	0.173	7.89	24.398f	25.06	520.34	85.098
0.412	320.44	20.941f	0.218	8.04	24.434f	25.94	404.70	66.390
0.512	401.45	21.651f	0.266	8.16	24.491f	26.81	333.87	54.611
0.612	483.61	22.138f	0.315	8.27	24.543f	27.70	286.38	46.395
0.712	566.72	22.492f	0.365	8.35	24.551f	28.52	251.60	40.186
0.812	650.57	22.758f	0.415	8.42	24.560f	29.21	224.49	35.343
0.912	735.11	22.966f	0.466	8.49	24.561f	29.96	203.76	31.632
1.012	820.35	23.132f	0.517	8.56	24.562f	30.74	187.34	28.689
1.112	906.31	23.267f	0.567	8.63	24.564f	31.56	174.08	26.309
1.212	992.99	23.380f	0.619	8.71	24.549f	32.35	162.89	24.308
1.312	1,080.33	23.476f	0.670	8.77	24.561f	33.07	153.05	22.576
1.412	1,168.34	23.557f	0.722	8.83	24.562f	33.86	144.89	21.132
1.512	1,257.02	23.628f	0.773	8.90	24.563f	34.68	137.92	19.898
1.612	1,346.39	23.690f	0.825	8.97	24.565f	35.53	131.95	18.836
1.712	1,436.46	23.745f	0.877	9.04	24.552f	36.37	126.58	17.889
1.812	1,527.11	23.794f	0.929	9.08	24.611f	36.90	120.81	16.991
1.912	1,618.10	23.843f	0.981	9.12	24.696f	37.34	115.37	16.198
2.012	1,709.44	23.891f	1.033	9.15	24.783f	37.79	110.53	15.496
2.112	1,801.11	23.938f	1.085	9.19	24.873f	38.26	106.21	14.872
2.212	1,893.13	23.986f	1.137	9.22	24.966f	38.75	102.33	14.317
2.312	1,985.42	24.032f	1.189	9.23	24.988f	38.91	97.97	13.758
2.412	2,077.73	24.075f	1.240	9.23	24.991f	38.96	93.75	13.255
2.512	2,170.06	24.114f	1.292	9.23	24.994f	39.02	89.90	12.799
2.612	2,262.39	24.150f	1.343	9.23	24.996f	39.08	86.36	12.384
2.712	2,354.73	24.183f	1.394	9.24	24.999f	39.14	83.10	12.006
2.812	2,445.61	24.228f	1.445	8.50	27.000f	30.67	62.71	10.846
2.912	2,521.33	24.383f	1.485	6.65	32.000f	15.11	29.97	8.622
3.012	2,578.59	24.605f	1.516	4.80	37.000f	6.19	12.01	6.555

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

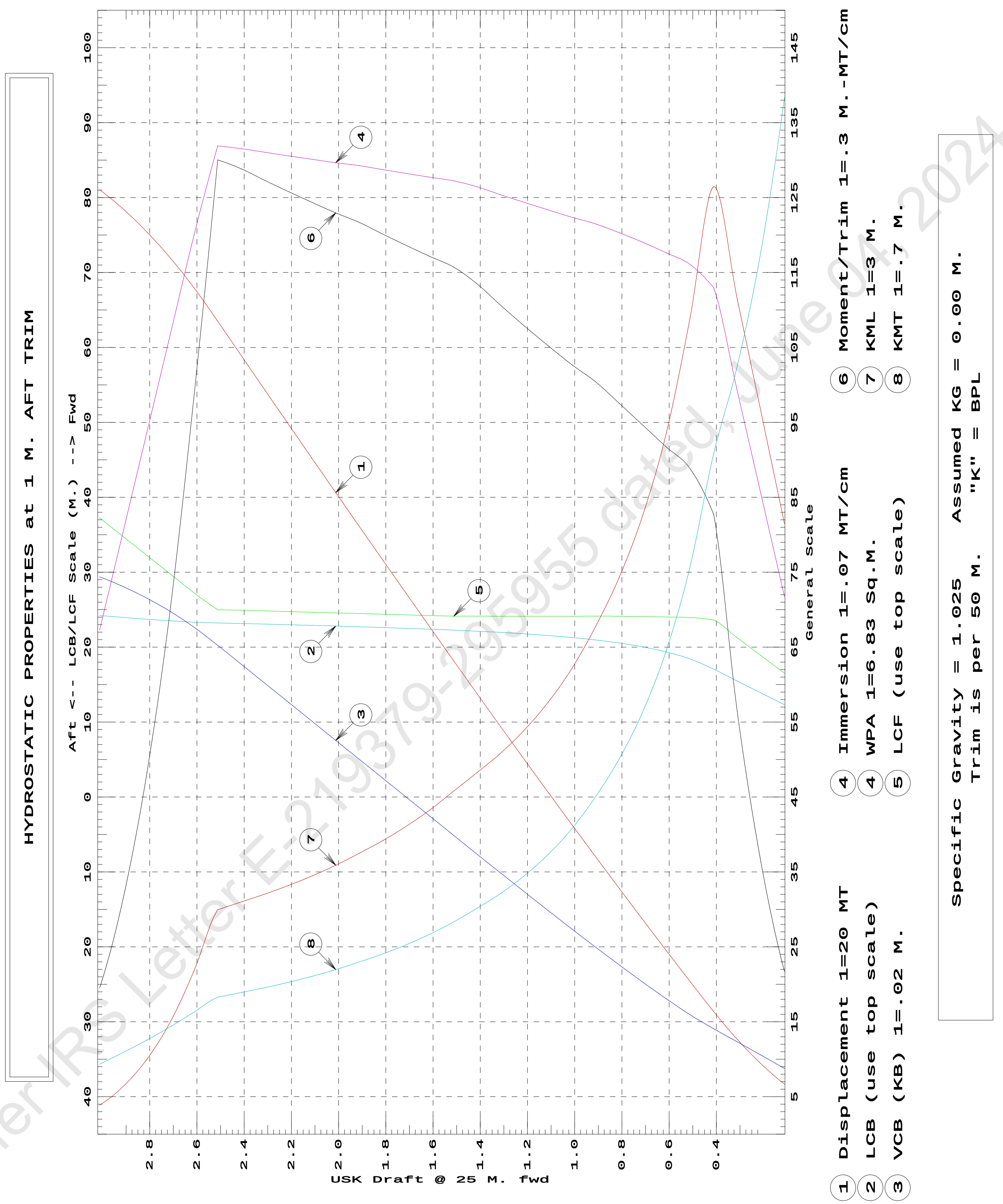

HYDROSTATIC PROPERTIES at 0.5 M. AFT TRIM
Aft <-- LCB/LCF Scale (M.) --> Fwd
General Scale
USK Draft @ 25 M. fwd
1 Displacement 1=30 MT
2 LCB (use top scale)
3 VCB (KB) 1=.03 M.
4 Immersion 1=.1 MT/cm
4 WPA 1=9.76 Sq.M.
5 LCF (use top scale)
6 Moment/Trim 1=.4 M.-MT/cm
7 KML 1=8 M.
8 KMT 1=2 M.
Specific Gravity = 1.025 Assumed KG = 0.00 M.
Trim is per 50 M. "K" = BPL

HYDROSTATIC PROPERTIES
Trim: Aft 1.000/50.000, No Heel, Fixed VCG = 0.000

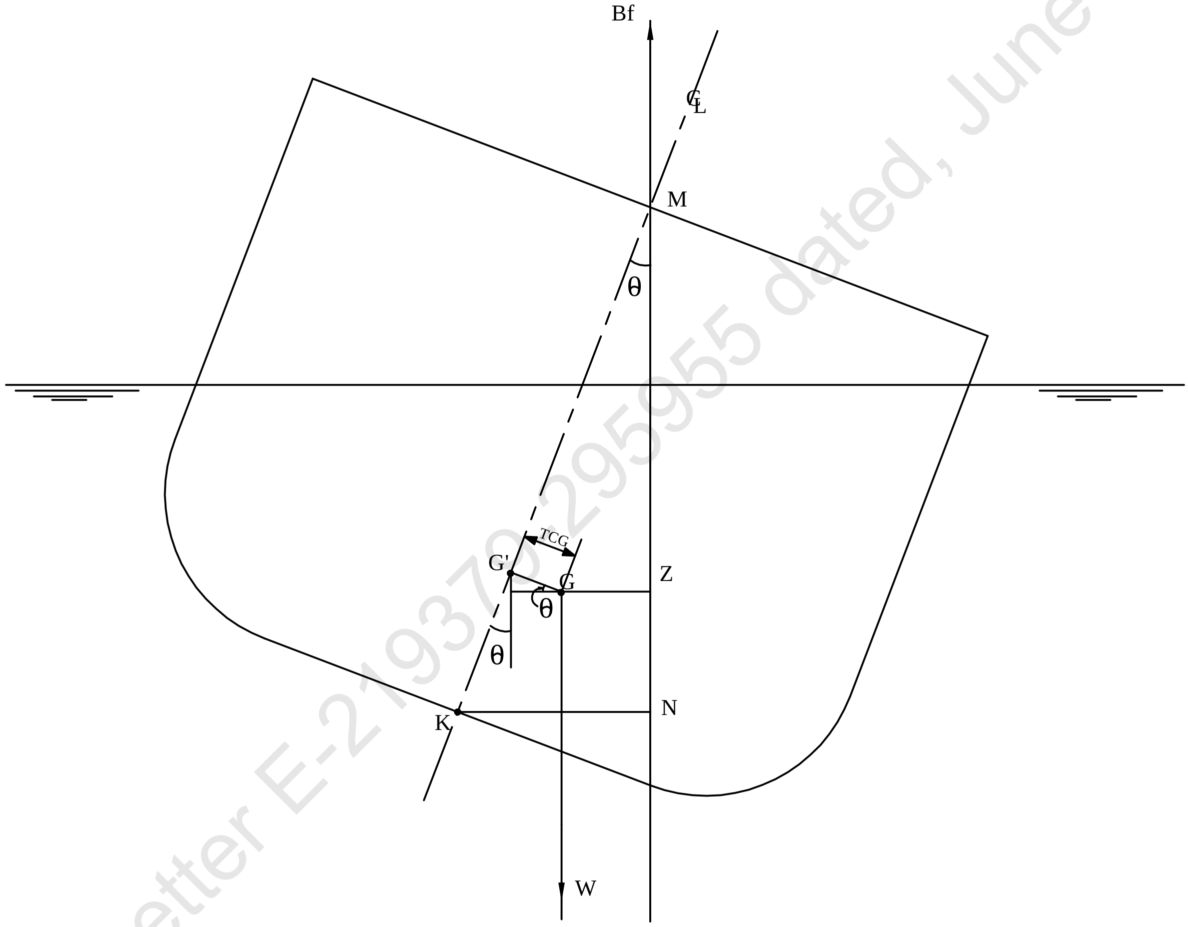
Draft@ 25.000f	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	KML	KMT
0.112	133.29	12.311f	0.175	5.02	16.510f	6.52	244.36	96.880
0.212	188.30	13.893f	0.211	5.98	18.876f	10.90	289.46	82.454
0.312	252.92	15.477f	0.246	6.94	21.249f	16.92	334.36	71.771
0.412	327.16	17.062f	0.281	7.91	23.625f	24.82	379.31	63.562
0.512	407.76	18.417f	0.319	8.13	23.966f	26.65	326.70	53.012
0.612	489.54	19.349f	0.361	8.23	24.026f	27.51	280.91	45.226
0.712	572.35	20.029f	0.406	8.33	24.077f	28.38	247.92	39.491
0.812	656.10	20.548f	0.452	8.42	24.114f	29.26	222.97	35.069
0.912	740.70	20.956f	0.499	8.50	24.130f	30.12	203.29	31.535
1.012	826.02	21.283f	0.548	8.57	24.120f	30.81	186.46	28.540
1.112	912.01	21.550f	0.597	8.63	24.122f	31.57	173.02	26.147
1.212	998.69	21.774f	0.646	8.70	24.125f	32.35	161.94	24.171
1.312	1,086.07	21.963f	0.696	8.78	24.128f	33.17	152.69	22.519
1.412	1,174.15	22.125f	0.746	8.85	24.131f	34.03	144.90	21.125
1.512	1,262.91	22.265f	0.797	8.91	24.130f	34.71	137.41	19.834
1.612	1,352.12	22.391f	0.848	8.94	24.217f	35.14	129.91	18.700
1.712	1,441.67	22.507f	0.899	8.97	24.306f	35.58	123.36	17.712
1.812	1,531.58	22.616f	0.950	9.01	24.397f	36.03	117.61	16.848
1.912	1,621.83	22.717f	1.000	9.05	24.492f	36.51	112.52	16.088
2.012	1,712.42	22.813f	1.051	9.08	24.564f	36.88	107.68	15.374
2.112	1,803.32	22.903f	1.102	9.11	24.649f	37.30	103.41	14.749
2.212	1,894.54	22.989f	1.153	9.14	24.737f	37.74	99.58	14.190
2.312	1,986.08	23.072f	1.203	9.17	24.826f	38.19	96.11	13.689
2.412	2,077.95	23.151f	1.254	9.21	24.919f	38.65	92.98	13.238
2.512	2,170.15	23.228f	1.305	9.23	24.987f	39.01	89.86	12.803
2.612	2,258.72	23.336f	1.353	8.40	27.240f	29.59	65.48	11.408
2.712	2,338.10	23.510f	1.395	7.48	29.743f	21.09	45.09	10.042
2.812	2,408.27	23.728f	1.431	6.56	32.245f	14.46	30.01	8.792
2.912	2,469.21	23.968f	1.462	5.63	34.748f	9.46	19.15	7.632
3.012	2,520.92	24.214f	1.488	4.71	37.250f	5.86	11.62	6.543

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.


HYDROSTATIC PROPERTIES at 1 M. AFT TRIM
Aft <-- LCB/LCF Scale (M.) --> Fwd
General Scale
USK Draft @ 25 M. fwd
1 Displacement 1=20 MT
2 LCB (use top scale)
3 VCB (KB) 1=.02 M.
4 Immersion 1=.07 MT/cm
4 WPA 1=6.83 Sq.M.
5 LCF (use top scale)
6 Moment/Trim 1=.3 M.-MT/cm
7 KML 1=3 M.
8 KMT 1=.7 M.
Specific Gravity = 1.025 Assumed KG = 0.00 M.
Trim is per 50 M. "K" = BPL

KN CONCEPT



GZ = KN - KG' SIN θ - TCG COS θ

VOLUME USED FOR KN COMPUTATION

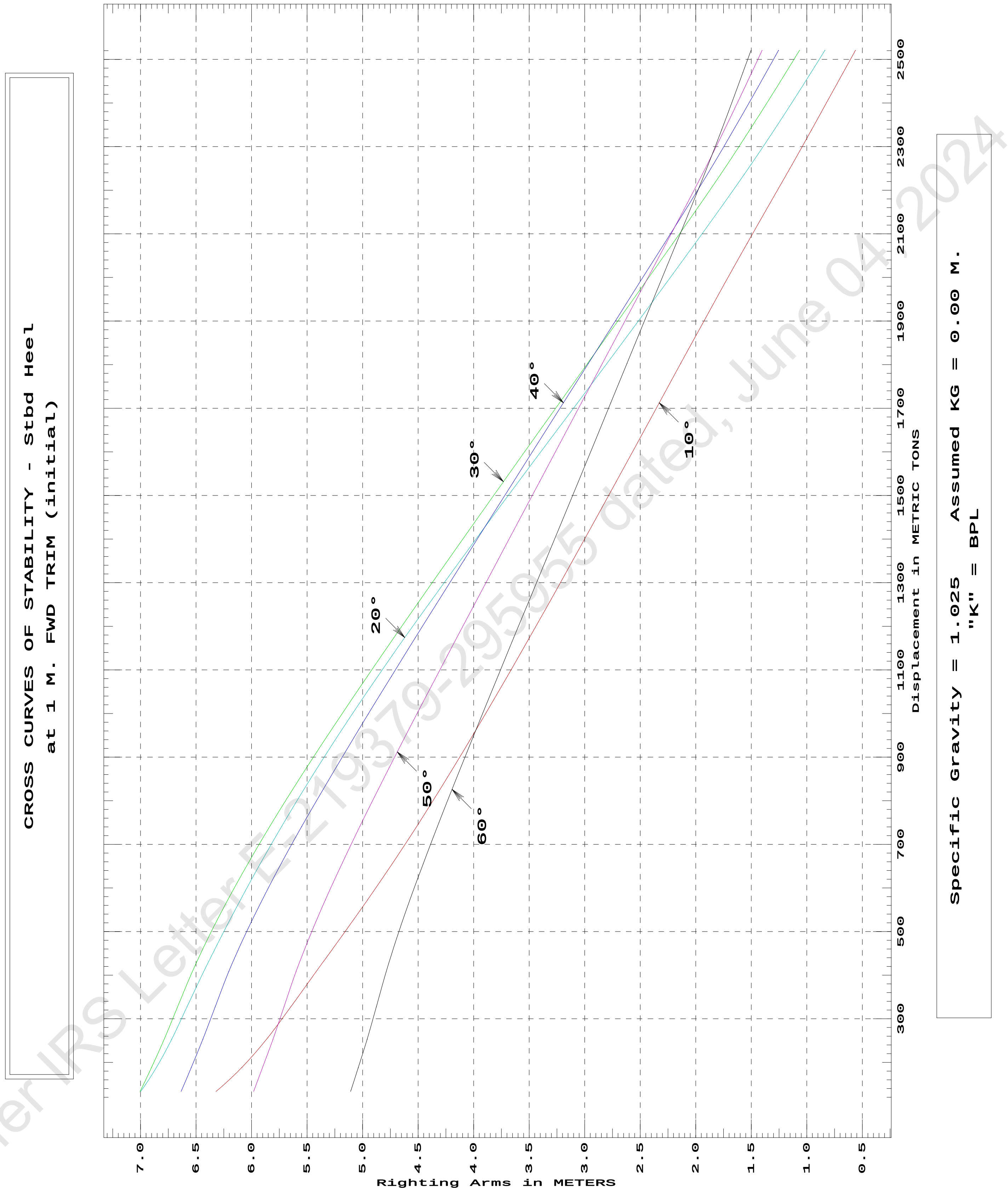
Condition Graphic

CROSS CURVE TABLES

CROSS CURVES OF STABILITY
Showing righting arms in heel at VCG = 0.00
Trim: Fwd 1.000/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees 10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
133.29	6.320s	6.999s	7.003s	6.634s	5.981s	5.107s
188.30	6.087s	6.857s	6.900s	6.542s	5.901s	5.038s
252.92	5.865s	6.719s	6.786s	6.438s	5.806s	4.959s
327.16	5.646s	6.581s	6.666s	6.327s	5.707s	4.876s
407.76	5.419s	6.431s	6.534s	6.205s	5.600s	4.788s
489.54	5.186s	6.270s	6.382s	6.063s	5.474s	4.686s
572.35	4.956s	6.100s	6.214s	5.904s	5.335s	4.574s
656.10	4.731s	5.920s	6.031s	5.732s	5.184s	4.453s
740.70	4.513s	5.729s	5.836s	5.548s	5.025s	4.326s
826.02	4.303s	5.528s	5.629s	5.355s	4.859s	4.194s
912.01	4.097s	5.316s	5.412s	5.155s	4.687s	4.059s
998.69	3.895s	5.094s	5.186s	4.949s	4.512s	3.921s
1,086.07	3.695s	4.861s	4.955s	4.739s	4.333s	3.781s
1,174.15	3.497s	4.620s	4.717s	4.524s	4.151s	3.638s
1,262.91	3.300s	4.371s	4.475s	4.306s	3.966s	3.494s
1,352.12	3.105s	4.116s	4.229s	4.086s	3.781s	3.349s
1,441.67	2.910s	3.858s	3.982s	3.864s	3.594s	3.203s
1,531.58	2.716s	3.596s	3.732s	3.641s	3.406s	3.057s
1,621.83	2.523s	3.331s	3.480s	3.417s	3.218s	2.911s
1,712.42	2.329s	3.065s	3.227s	3.192s	3.029s	2.765s
1,803.32	2.134s	2.798s	2.972s	2.965s	2.840s	2.618s
1,894.54	1.938s	2.532s	2.717s	2.738s	2.649s	2.471s
1,986.08	1.740s	2.268s	2.462s	2.511s	2.459s	2.323s
2,077.95	1.538s	2.006s	2.207s	2.283s	2.268s	2.175s
2,170.15	1.332s	1.746s	1.954s	2.056s	2.077s	2.027s
2,258.72	1.132s	1.503s	1.718s	1.843s	1.897s	1.888s
2,338.10	0.958s	1.294s	1.514s	1.660s	1.743s	1.767s
2,408.27	0.806s	1.115s	1.338s	1.502s	1.610s	1.664s
2,469.20	0.674s	0.963s	1.190s	1.368s	1.497s	1.576s
2,520.92	0.561s	0.836s	1.066s	1.255s	1.402s	1.502s

Distances in METERS.---Specific Gravity = 1.025.-------------

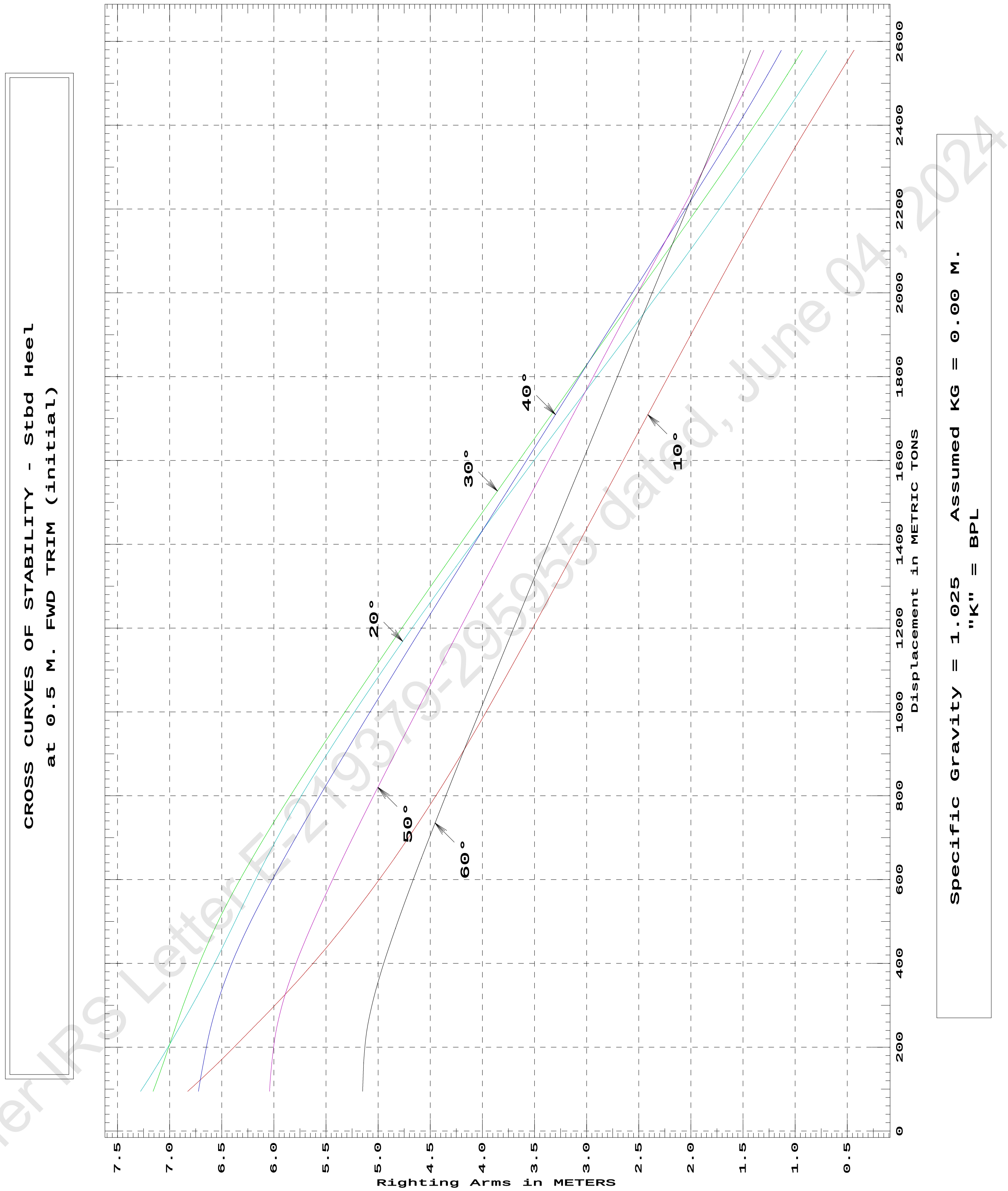

CROSS CURVES OF STABILITY - Stbd Heel
at 1 M. FWD TRIM (initial)
Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL
Displacement in METRIC TONS
Righting Arms in METERS
10°
20°
30°
40°
50°
60°

CROSS CURVES OF STABILITY
Showing righting arms in heel at VCG = 0.00
Trim: Fwd 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
94.61	6.827s	7.278s	7.158s	6.723s	6.041s	5.149s
162.65	6.533s	7.105s	7.059s	6.676s	6.020s	5.139s
240.79	6.221s	6.920s	6.953s	6.612s	5.977s	5.107s
320.44	5.912s	6.738s	6.839s	6.522s	5.897s	5.039s
401.45	5.619s	6.567s	6.712s	6.401s	5.785s	4.944s
483.61	5.346s	6.404s	6.566s	6.254s	5.649s	4.833s
566.72	5.088s	6.242s	6.397s	6.085s	5.499s	4.712s
650.57	4.845s	6.071s	6.209s	5.902s	5.340s	4.585s
735.11	4.615s	5.887s	6.005s	5.710s	5.173s	4.454s
820.35	4.395s	5.688s	5.790s	5.510s	5.002s	4.318s
906.31	4.184s	5.473s	5.567s	5.305s	4.826s	4.180s
992.99	3.980s	5.246s	5.336s	5.094s	4.647s	4.039s
1,080.33	3.782s	5.009s	5.100s	4.880s	4.465s	3.896s
1,168.34	3.587s	4.764s	4.859s	4.662s	4.280s	3.752s
1,257.02	3.393s	4.512s	4.614s	4.441s	4.093s	3.605s
1,346.39	3.198s	4.253s	4.364s	4.217s	3.903s	3.457s
1,436.46	3.003s	3.990s	4.111s	3.990s	3.712s	3.308s
1,527.11	2.807s	3.722s	3.856s	3.761s	3.519s	3.157s
1,618.11	2.610s	3.451s	3.598s	3.530s	3.325s	3.006s
1,709.44	2.413s	3.177s	3.338s	3.299s	3.130s	2.854s
1,801.11	2.215s	2.901s	3.077s	3.066s	2.934s	2.702s
1,893.13	2.016s	2.624s	2.815s	2.833s	2.738s	2.549s
1,985.42	1.816s	2.349s	2.551s	2.598s	2.540s	2.396s
2,077.73	1.613s	2.077s	2.287s	2.363s	2.343s	2.242s
2,170.06	1.408s	1.811s	2.025s	2.128s	2.145s	2.088s
2,262.39	1.199s	1.551s	1.766s	1.894s	1.947s	1.934s
2,354.73	0.983s	1.295s	1.512s	1.662s	1.751s	1.781s
2,445.61	0.763s	1.048s	1.268s	1.440s	1.562s	1.633s
2,521.33	0.578s	0.848s	1.074s	1.263s	1.411s	1.515s
2,578.59	0.436s	0.699s	0.931s	1.133s	1.301s	1.428s

Distances in METERS.---Specific Gravity = 1.025.-------------

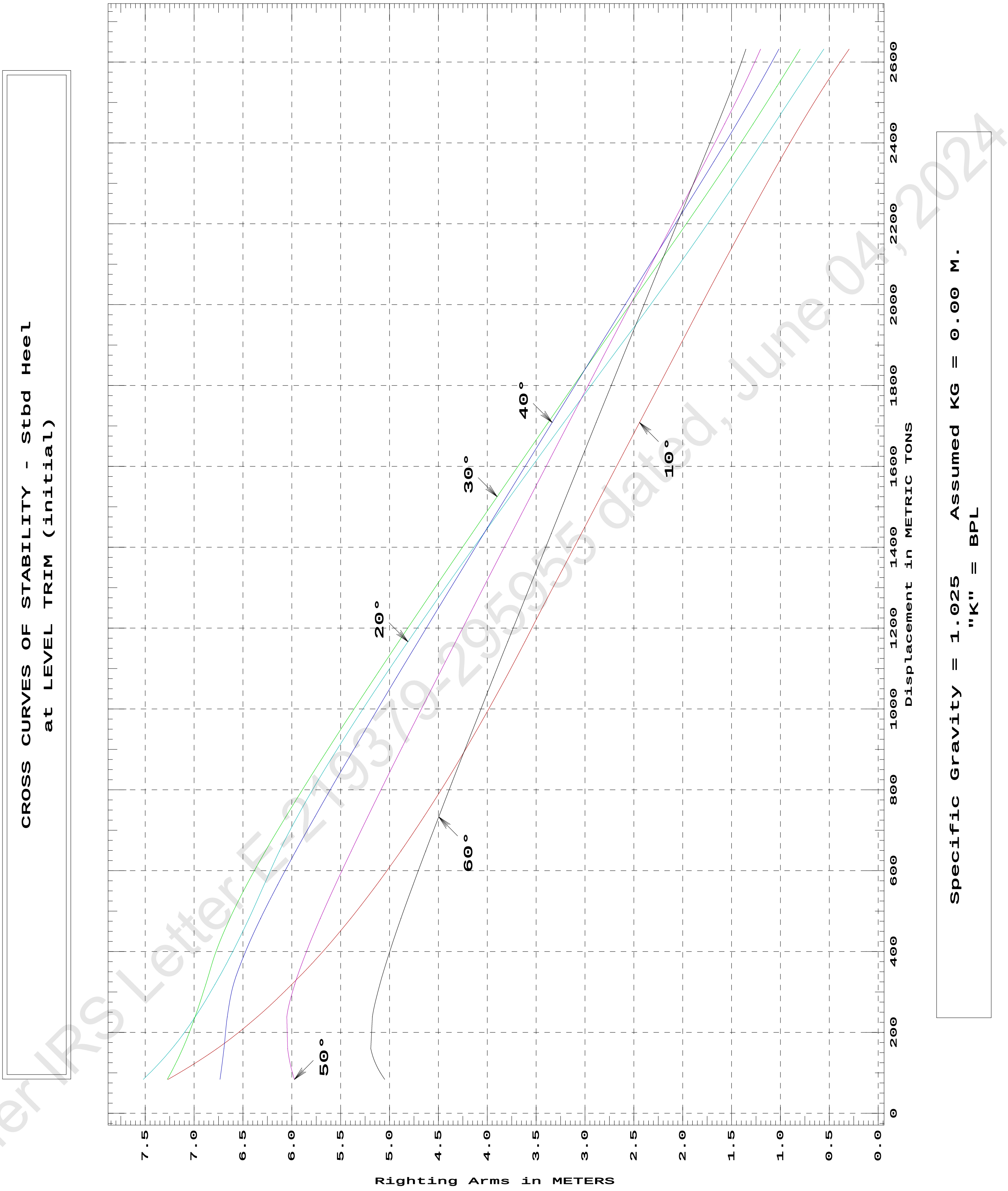
31-07-20 12:20:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 **PONTOON**



CROSS CURVES OF STABILITY
Showing righting arms in heel at VCG = 0.00
Trim: zero at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees 10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
83.79	7.268s	7.519s	7.274s	6.735s	5.973s	5.049s
160.29	6.763s	7.226s	7.116s	6.694s	6.043s	5.192s
238.38	6.354s	6.990s	6.990s	6.660s	6.051s	5.176s
318.09	6.003s	6.788s	6.881s	6.597s	5.971s	5.100s
399.40	5.685s	6.606s	6.776s	6.474s	5.851s	4.999s
481.78	5.397s	6.438s	6.634s	6.319s	5.709s	4.883s
564.87	5.132s	6.283s	6.462s	6.146s	5.555s	4.760s
648.69	4.884s	6.127s	6.269s	5.960s	5.392s	4.630s
733.24	4.648s	5.945s	6.061s	5.764s	5.224s	4.497s
818.51	4.425s	5.743s	5.844s	5.562s	5.050s	4.360s
904.44	4.211s	5.525s	5.619s	5.355s	4.873s	4.221s
991.06	4.006s	5.297s	5.387s	5.143s	4.693s	4.080s
1,078.37	3.807s	5.058s	5.149s	4.927s	4.509s	3.936s
1,166.40	3.615s	4.812s	4.907s	4.708s	4.324s	3.791s
1,255.11	3.427s	4.559s	4.660s	4.486s	4.135s	3.643s
1,344.46	3.231s	4.300s	4.410s	4.261s	3.945s	3.495s
1,434.46	3.036s	4.035s	4.156s	4.033s	3.753s	3.345s
1,525.13	2.840s	3.767s	3.899s	3.803s	3.559s	3.193s
1,616.48	2.642s	3.493s	3.640s	3.571s	3.363s	3.040s
1,708.51	2.442s	3.216s	3.377s	3.336s	3.165s	2.886s
1,800.77	2.241s	2.937s	3.113s	3.100s	2.966s	2.731s
1,893.06	2.041s	2.656s	2.848s	2.864s	2.767s	2.576s
1,985.36	1.840s	2.374s	2.582s	2.627s	2.568s	2.421s
2,077.68	1.638s	2.100s	2.316s	2.390s	2.369s	2.265s
2,170.03	1.433s	1.832s	2.049s	2.153s	2.169s	2.110s
2,262.38	1.225s	1.570s	1.786s	1.916s	1.969s	1.954s
2,354.73	1.010s	1.314s	1.529s	1.679s	1.769s	1.798s
2,447.09	0.789s	1.063s	1.280s	1.449s	1.571s	1.642s
2,539.44	0.554s	0.813s	1.038s	1.228s	1.380s	1.491s
2,631.79	0.298s	0.558s	0.799s	1.016s	1.204s	1.354s

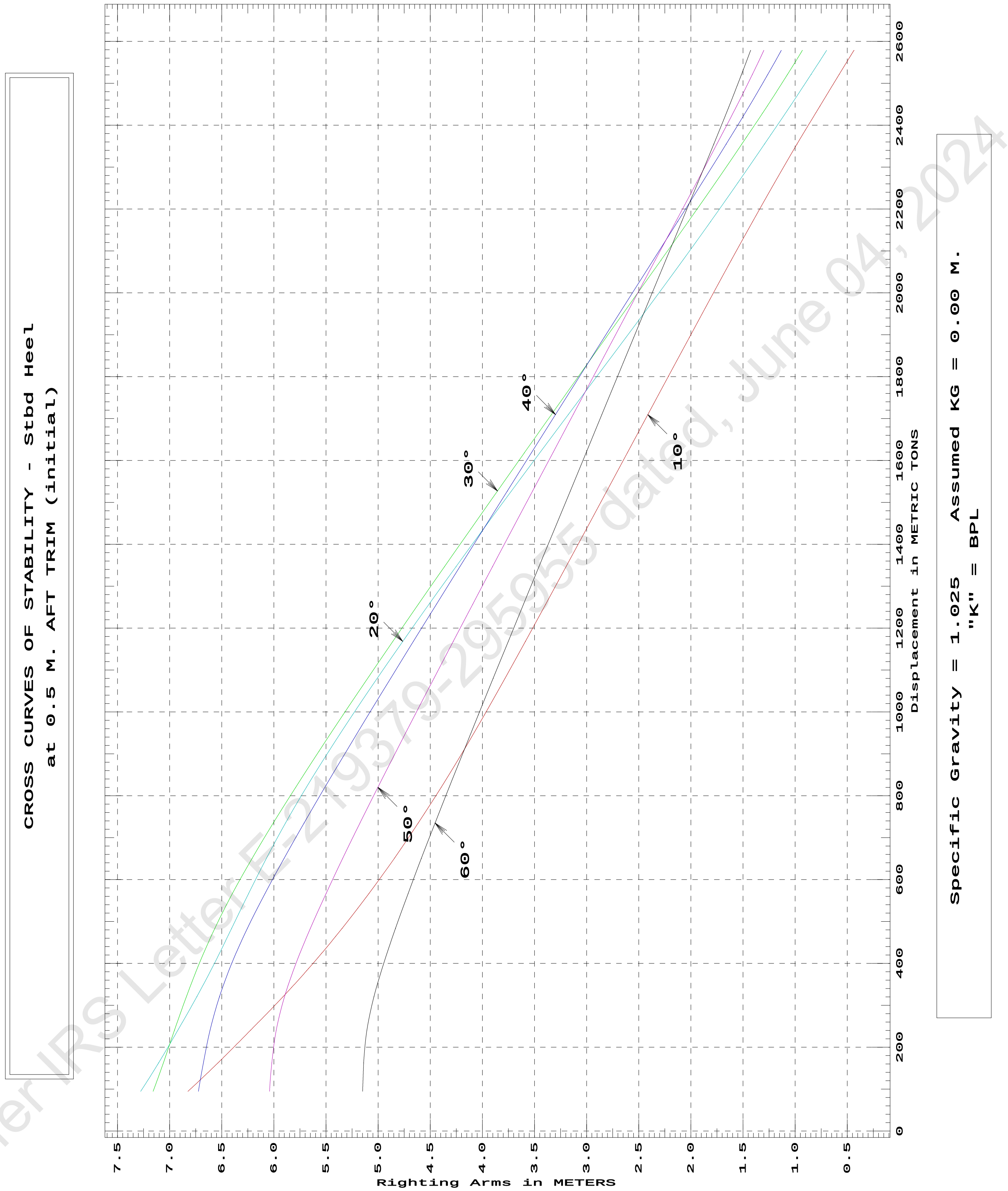
Distances in METERS.---Specific Gravity = 1.025.-------------


CROSS CURVES OF STABILITY - Stbd Heel
at LEVEL TRIM (initial)
Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL
Displacement in METRIC TONS
Righting Arms in METERS
10°
20°
30°
40°
50°
60°
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200
2400
2600
7.5
7.0
6.5
6.0
5.5
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0

CROSS CURVES OF STABILITY
Showing righting arms in heel at VCG = 0.00
Trim: Aft 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement	Heel Angles in Degrees					
METRIC TONS	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
94.61	6.824s	7.277s	7.158s	6.723s	6.041s	5.149s
162.65	6.535s	7.105s	7.059s	6.676s	6.020s	5.139s
240.79	6.222s	6.920s	6.953s	6.612s	5.977s	5.107s
320.44	5.912s	6.738s	6.839s	6.522s	5.897s	5.040s
401.45	5.619s	6.567s	6.712s	6.401s	5.785s	4.944s
483.61	5.346s	6.405s	6.566s	6.254s	5.649s	4.833s
566.72	5.088s	6.242s	6.397s	6.085s	5.499s	4.712s
650.57	4.845s	6.071s	6.209s	5.902s	5.340s	4.585s
735.11	4.615s	5.887s	6.005s	5.710s	5.173s	4.454s
820.35	4.395s	5.688s	5.790s	5.510s	5.002s	4.318s
906.31	4.184s	5.473s	5.567s	5.305s	4.826s	4.180s
992.99	3.980s	5.246s	5.336s	5.094s	4.647s	4.039s
1,080.33	3.782s	5.009s	5.100s	4.880s	4.465s	3.896s
1,168.34	3.587s	4.764s	4.859s	4.662s	4.280s	3.752s
1,257.02	3.393s	4.511s	4.614s	4.441s	4.093s	3.605s
1,346.39	3.198s	4.253s	4.364s	4.216s	3.903s	3.457s
1,436.46	3.003s	3.990s	4.111s	3.990s	3.712s	3.308s
1,527.11	2.807s	3.722s	3.855s	3.761s	3.519s	3.157s
1,618.10	2.610s	3.451s	3.598s	3.530s	3.325s	3.006s
1,709.44	2.413s	3.177s	3.338s	3.299s	3.130s	2.854s
1,801.11	2.215s	2.901s	3.077s	3.066s	2.934s	2.702s
1,893.13	2.016s	2.624s	2.815s	2.833s	2.738s	2.549s
1,985.42	1.816s	2.349s	2.551s	2.598s	2.540s	2.396s
2,077.73	1.613s	2.077s	2.287s	2.363s	2.343s	2.242s
2,170.06	1.408s	1.811s	2.025s	2.128s	2.145s	2.088s
2,262.39	1.198s	1.551s	1.766s	1.894s	1.947s	1.934s
2,354.73	0.983s	1.295s	1.512s	1.662s	1.751s	1.781s
2,445.61	0.763s	1.048s	1.268s	1.440s	1.562s	1.633s
2,521.33	0.578s	0.848s	1.074s	1.263s	1.411s	1.515s
2,578.59	0.436s	0.698s	0.931s	1.133s	1.301s	1.428s

Distances in METERS.---Specific Gravity = 1.025.-------------

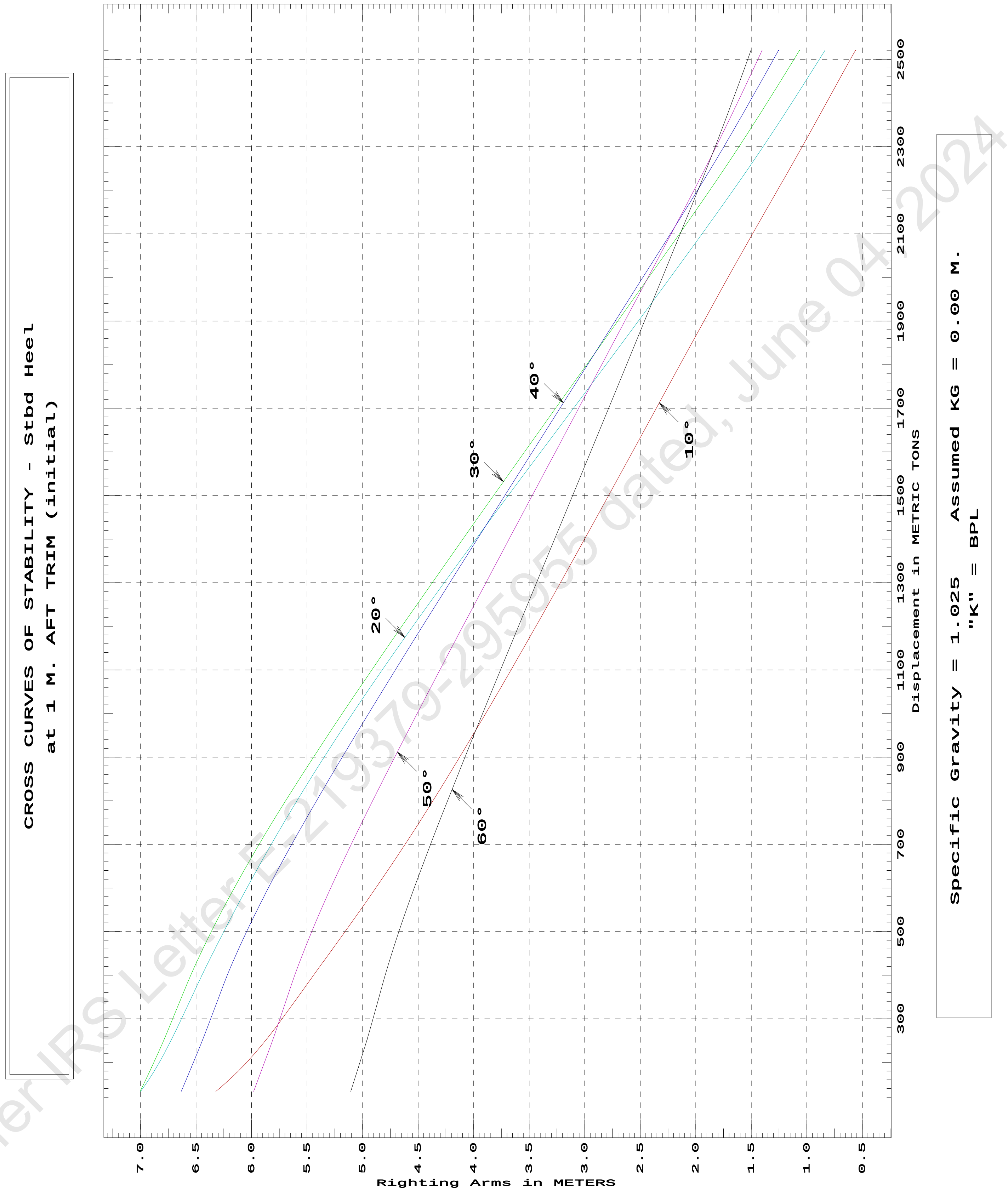

CROSS CURVES OF STABILITY - Stbd Heel
at 0.5 M. AFT TRIM (initial)
Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL
Displacement in METRIC TONS
Righting Arms in METERS
10°
20°
30°
40°
50°
60°
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200
2400
2600
7.5
7.0
6.5
6.0
5.5
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5

CROSS CURVES OF STABILITY
Showing righting arms in heel at VCG = 0.00
Trim: Aft 1.000/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees 10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
133.29	6.322s	6.999s	7.004s	6.633s	5.981s	5.107s
188.30	6.087s	6.857s	6.900s	6.542s	5.901s	5.038s
252.92	5.864s	6.719s	6.786s	6.438s	5.807s	4.959s
327.16	5.646s	6.580s	6.666s	6.327s	5.708s	4.876s
407.76	5.419s	6.431s	6.533s	6.205s	5.600s	4.787s
489.54	5.186s	6.270s	6.382s	6.063s	5.474s	4.686s
572.35	4.956s	6.100s	6.214s	5.904s	5.335s	4.574s
656.10	4.731s	5.920s	6.031s	5.732s	5.184s	4.453s
740.70	4.513s	5.729s	5.836s	5.548s	5.025s	4.326s
826.02	4.303s	5.528s	5.629s	5.355s	4.858s	4.194s
912.01	4.097s	5.316s	5.411s	5.155s	4.687s	4.059s
998.69	3.895s	5.093s	5.186s	4.949s	4.511s	3.921s
1,086.07	3.695s	4.861s	4.954s	4.738s	4.333s	3.780s
1,174.15	3.497s	4.619s	4.717s	4.524s	4.151s	3.638s
1,262.91	3.300s	4.370s	4.474s	4.306s	3.966s	3.494s
1,352.12	3.105s	4.116s	4.229s	4.085s	3.780s	3.348s
1,441.67	2.910s	3.858s	3.981s	3.863s	3.593s	3.203s
1,531.58	2.716s	3.596s	3.732s	3.640s	3.406s	3.057s
1,621.83	2.523s	3.330s	3.480s	3.416s	3.218s	2.911s
1,712.42	2.329s	3.065s	3.227s	3.192s	3.029s	2.764s
1,803.32	2.134s	2.798s	2.973s	2.966s	2.840s	2.618s
1,894.54	1.938s	2.532s	2.717s	2.738s	2.650s	2.471s
1,986.08	1.740s	2.268s	2.462s	2.511s	2.458s	2.323s
2,077.95	1.538s	2.006s	2.207s	2.283s	2.268s	2.175s
2,170.15	1.332s	1.746s	1.954s	2.056s	2.077s	2.027s
2,258.72	1.132s	1.503s	1.718s	1.843s	1.898s	1.888s
2,338.10	0.958s	1.294s	1.513s	1.660s	1.743s	1.768s
2,408.27	0.806s	1.115s	1.339s	1.502s	1.610s	1.664s
2,469.21	0.674s	0.963s	1.189s	1.367s	1.497s	1.576s
2,520.92	0.561s	0.836s	1.065s	1.254s	1.402s	1.503s

Distances in METERS.---Specific Gravity = 1.025.-------------

31-07-20 12:20:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 **PONTOON**



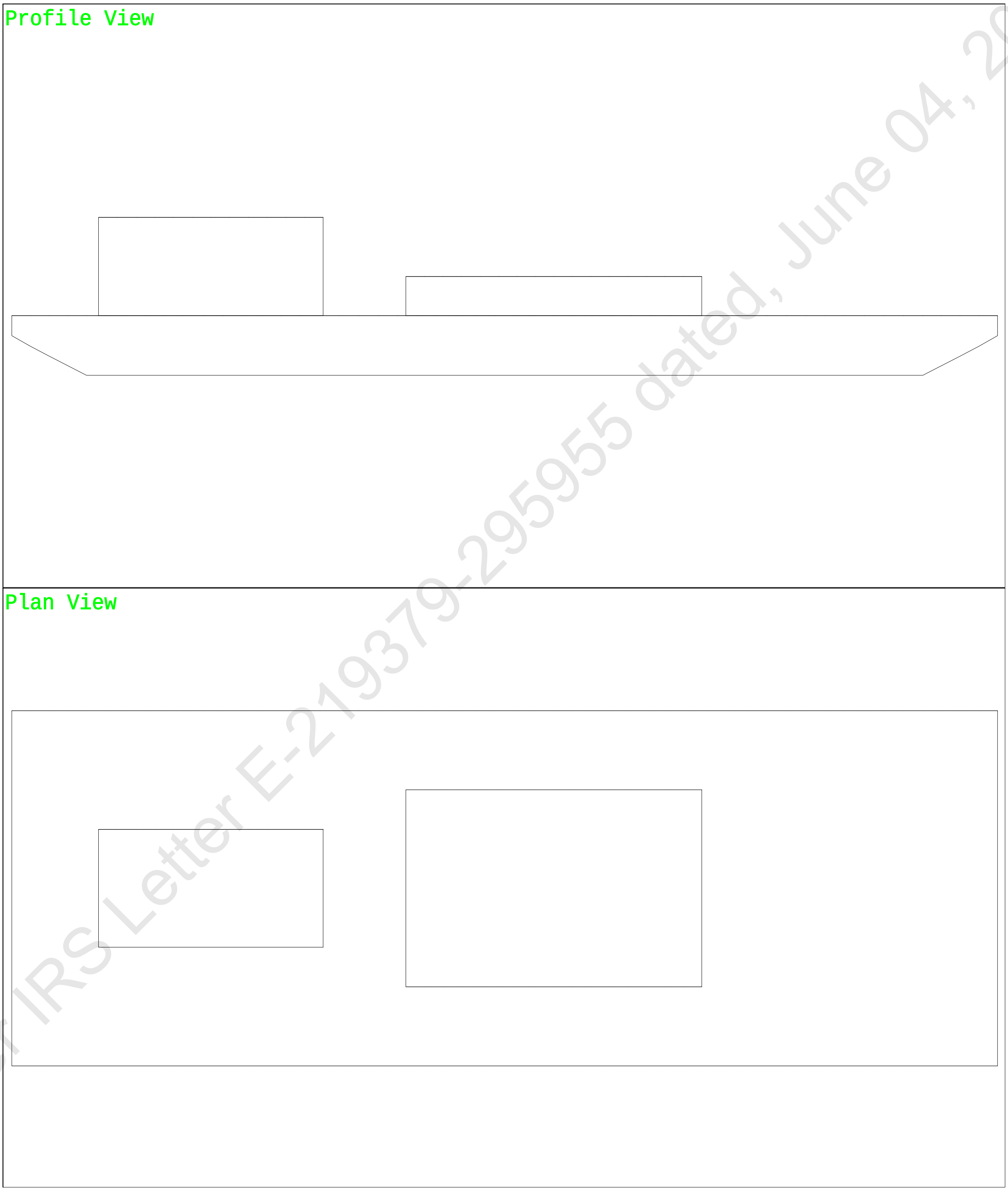
DEADWEIGHT TABLE

Draft M	Disp T	Dwt T	
0.612	481.78	16.53	
0.712	564.87	99.62	
0.812	648.69	183.44	
0.912	733.24	267.99	
1.012	818.51	353.26	
1.112	904.44	439.19	
1.212	991.06	525.81	
1.312	1078.37	613.12	
1.412	1166.40	701.15	
1.512	1255.11	789.86	
1.612	1344.46	879.21	
1.712	1434.46	969.21	
1.812	1525.13	1059.88	
1.912	1616.48	1151.23	
2.012	1708.51	1243.26	
2.112	1800.77	1335.52	
2.212	**1893.06**	**1427.81**	**Full Load Displacement**

01-08-20 22:26:06   Cybermarine Technologies PTE. Ltd.
GHS 17.22   **PONTOON**

USK draft refers to the line:
0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000

Condition Graphic



MAXIMUM VCG vs. DISPLACEMENT

Heeling moment is present from: wind
Trim = Fwd 1.000/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Max VCG	--- Margins --- LIM1	LIM2	LIM3
400.00	~~18.489~~ 18.466	122%	0d	7d
600.00	17.349	115%	0d	6d
800.00	16.087	112%	0d	5d
1,000.00	14.631	73%	0d	4d
1,200.00	13.079	51%	0d	3d
1,400.00	~~11.429~~ 11.407	30%	0d	3d
1,600.00	~~9.770~~ 9.697	12%	0d	2d
1,800.00	~~7.778~~ 7.619	0%	1d	1d
2,000.00	~~5.433~~ 5.293	0%	5d	1d

Heeling moment is present from: wind
Trim = Fwd 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Max VCG	--- Margins --- LIM1	LIM2	LIM3
400.00	18.792	138%	0d	10d
600.00	17.834	103%	0d	7d
800.00	16.573	83%	0d	6d
1,000.00	15.128	70%	0d	5d
1,200.00	13.510	57%	0d	4d
1,400.00	11.824	35%	0d	3d
1,600.00	10.137	7%	0d	3d
1,800.00	7.959	0%	2d	2d
2,000.00	~~5.618~~ 5.373	3%	6d	1d

Heeling moment is present from: wind
Trim = zero at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Max VCG	--- Margins --- LIM1	LIM2	LIM3
400.00	18.899	137%	0d	14d
600.00	17.990	117%	0d	10d
800.00	16.745	80%	0d	7d
1,000.00	15.270	69%	0d	6d
1,200.00	13.683	59%	0d	5d
1,400.00	11.943	37%	0d	4d
1,600.00	10.176	8%	0d	3d
1,800.00	8.007	0%	2d	3d
2,000.00	~~5.806~~ 5.351	0%	5d	2d

Heeling moment is present from: wind
Trim = Aft 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Max VCG	--- Margins --- LIM1	LIM2	LIM3
400.00	18.793	138%	0d	10d
600.00	17.835	103%	0d	7d
800.00	16.573	83%	0d	6d
1,000.00	15.129	70%	0d	5d
1,200.00	13.510	57%	0d	4d

01-08-20 22:26:06 Cybermarine Technologies PTE. Ltd.
GHS 17.22 **PONTOON**

1,400.00	11.825	35%	0d	3d
1,600.00	10.136	7%	0d	3d
1,800.00	7.958	0%	2d	2d
2,000.00	~~5.599~~ 5.363	4%	6d	1d

Heeling moment is present from: wind
Trim = Aft 1.000/50.000 at zero heel (trim righting arm held at zero)

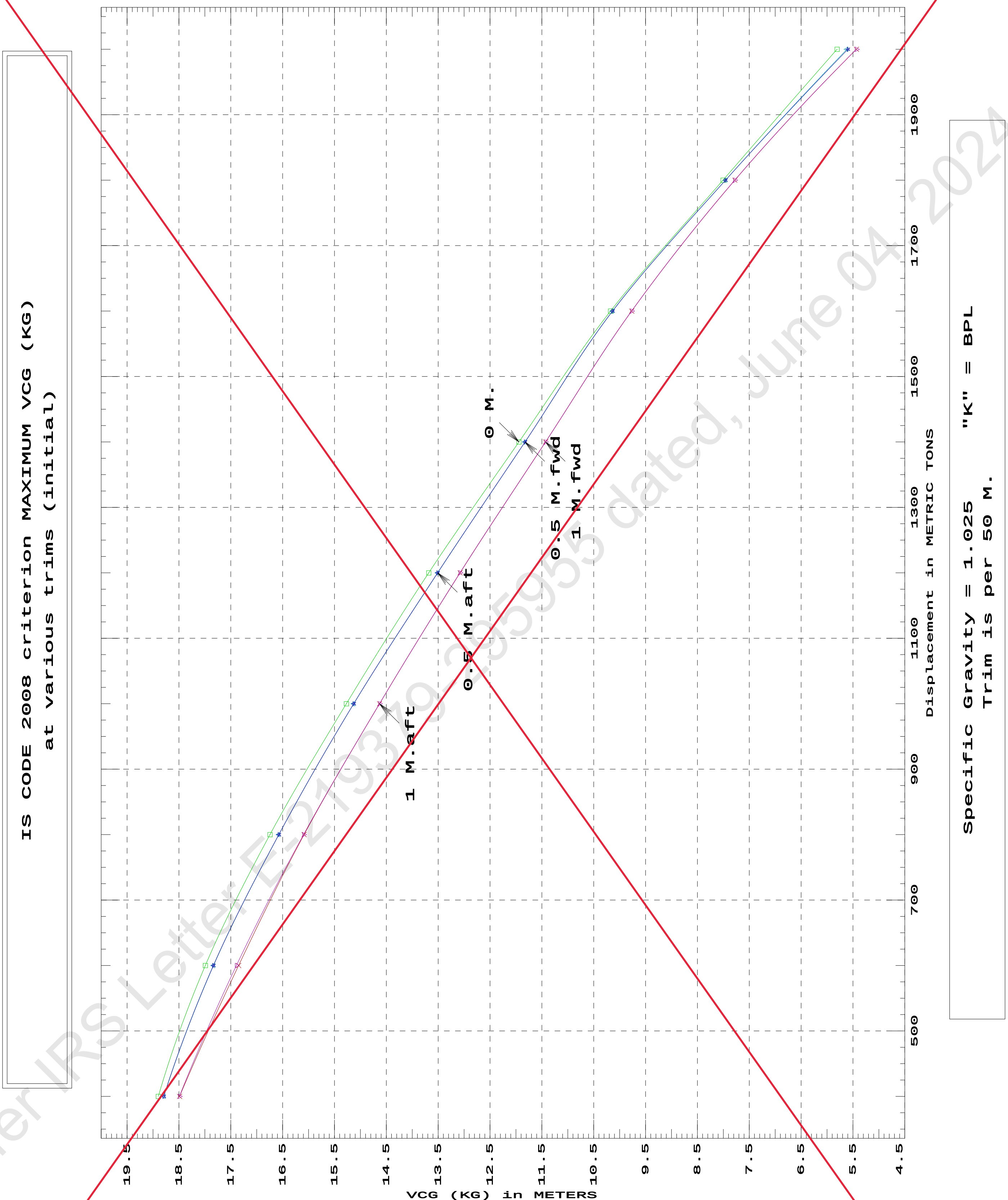
Displacement		--- Margins ---		
METRIC TONS	Max VCG	LIM1	LIM2	LIM3
400.00	~~18.491~~ 18.452	119%	0d	7d
600.00	17.392	114%	0d	6d
800.00	16.095	112%	0d	5d
1,000.00	14.632	73%	0d	4d
1,200.00	13.079	51%	0d	3d
1,400.00	~~11.430~~ 11.408	31%	0d	3d
1,600.00	~~9.769~~ 9.699	12%	0d	2d
1,800.00	~~7.781~~ 7.661	0%	1d	1d
2,000.00	~~5.434~~ 5.312	0%	5d	1d

Distances in METERS.---Specific Gravity = 1.025.---d = degrees.

LIM	IS CODE 2008 CRITERION		Min/Max
(1)	Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad
(2)	Angle from Equilibrium to RAzero or Flood	>	20.00 deg
(3)	Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg

1. MAX VCG DOES NOT COVER CRANE OPERATIONS/LIFTING OF WEIGHTS AND IS VALID ONLY FOR UNMANNED TOW, WITH CRANE IN STOWED POSITION.

2. MAX VCG VALUES ARE GENERATED BASED ON WINDAGE AREA AS SHOWN IN PAGE 51 OF THE DOCUMENT.


IS CODE 2008 criterion MAXIMUM VCG (KG)
at various trims (initial)
Specific Gravity = 1.025 "K" = BPL
Trim is per 50 M.
Displacement in METRIC TONS
VCG (KG) in METERS
0 M.
0.5 M.fwd
1 M.fwd
0.5 M.aft
1 M.aft
500
700
900
1100
1300
1500
1700
1900
19.5
18.5
17.5
16.5
15.5
14.5
13.5
12.5
11.5
10.5
9.5
8.5
7.5
6.5
5.5
4.5

SUMMARY OF CONDITIONS				
CONDITION NO.	1	2	3	4
CONDITION	Condition No. 0 - Lightship	Condition No. 1 - Lightship + Crane + Ballast	Condition No. 2 - Lightship + Crane + Deck Cargo + Ballast - Maximum Draft Condition	Condition No. 3 - Lightship + Deck Cargo - Maximum Deck Cargo Condition
TOTAL TANK LOAD (T)	0	234.87	234.87	0
DWT CONST.(T)	0	~~0~~ 400.00	~~0~~ 1192.50	~~0~~ 1428.00
DEADWEIGHT (T)	0	~~0~~ 634.87	~~0~~ 1427.37	~~0~~ 1428.00
LIGHT WEIGHT (T)	465.25	465.25	465.25	465.25
DISPLACEMENT (T)	465.25	1 100.12	1 892.62	1 893.25
FWD DRAFT (M)	0.597	1.215	2.206	2.216
AFT DRAFT (M)	0.587	1.457	2.217	2.208
MEAN DRAFT (M)	0.592	1.336	2.212	2.212
TRIM (M)	0.011f	0.242a	0.010a	0.008f
HEEL (DEG.)	0	0	0	0
L.C.G. (FROM AP) (M)	25.062f	24.288f	24.980f	25.015f
L.C.B. (FROM AP) (M)	25.063f	24.273f	24.979f	25.016f
M.C.T. (T-CM)	27.27	32.44	37.34	37.42
KM_T (M)	48.197	22.263	14.311	14.307
KG (M)	3	3.923	3.955	3.754
GG0	0	0.255	0.148	0
KG0 (M)	3	4.178	4.103	3.754
GM_T (M)	45.197	18.085	10.208	10.553
MAX. GZ (M)	5.424	3.598	1.365	1.453

SUMMARY OF CONDITIONS					
5	6	7	8	9	10
Condition No. 4 - Operating Condition of 400T Crane, Crane Lifting 40T @ 30m Radius Aft	Condition No. 5 - Operating Condition of 400T Crane, Crane Lifting 40T @ 30m Radius Stbd	Condition No. 6 - Operating Condition of 400T Crane, Crane Lifting 40T @ 30m Radius Fwd	Condition No. 7 - Operating Condition of 350T Crane, Crane Lifting 40T @ 35m Radius Aft	Condition No. 8 - Operating Condition of 350T Crane, Crane Lifting 40T @ 35m Radius Stbd	Condition No. 9 - Operating Condition of 350T Crane, Crane Lifting 40T @ 35m Radius Fwd
201.09	201.09	201.09	422.4	422.4	422.4
440	440	440	390	390	390
641.09	641.09	641.09	812.4	812.4	812.4
465.25	465.25	465.25	465.25	465.25	465.25
1 106.35	1 106.35	1 106.35	1 277.65	1 277.65	1 277.56
0.868	1.066	1.26	1.317	1.526	1.731
1.801	1.615	1.424	1.754	1.549	1.341
1.335	1.341	1.342	1.536	1.537	1.536
0.932a	0.548a	0.164a	0.437a	0.023a	0.390f
0.93s	0.93s	0.93s	0	0	0
22.282f	23.405f	24.527f	23.842f	24.938f	26.034f
22.201f	23.357f	24.513f	23.814f	24.936f	26.059f
31.6	32.58	31.61	33.84	33.83	33.85
22.046	22.159	22.018	19.638	19.624	19.64
5.058	5.058	5.058	4.025	4.025	4.025
0.319	0.319	0.319	0.368	0.368	0.368
5.377	5.377	5.377	4.393	4.393	4.393
16.669	16.782	16.641	15.245	15.231	15.247
2.836	2.939	2.989	3.031	3.064	3.037

SUMMARY OF CONDITIONS

11	12	13	14	15	16	17
Condition No. 10 - Operating Condition of 250T Crane, Crane Lifting 36.6T @ 18m Radius Aft	Condition No. 11 - Operating Condition of 250T Crane, Crane Lifting 36.6T @ 18m Radius Stbd	Condition No. 12 - Operating Condition of 250T Crane, Crane Lifting 36.6T @ 18m Radius Fwd	Condition No. 13 - Operating Condition of 150T Crane, Crane Lifting 51.3T @ 12m Radius Aft	Condition No. 14 - Operating Condition of 150T Crane, Crane Lifting 51.3T @ 12m Radius Stbd	Condition No. 15 - Operating Condition of 150T Crane, Crane Lifting 51.3T @ 12m Radius Fwd	Condition No. 16 - Lightship + Crane + Ballast (Towing Condition)
422.4	422.4	422.4	323.2	323.2	323.2	924.21
286.6	286.6	286.6	201.3	201.3	201.3	0
709	709	709	524.5	524.5	524.5	0
465.25	465.25	465.25	465.25	465.25	465.25	465.25
1 174.23	1 174.21	1 174.20	989.65	989.71	989.74	1 789.46
1.346	1.447	1.547	1.018	1.117	1.215	2.064
1.495	1.395	1.294	1.4	1.303	1.206	2.136
1.421	1.421	1.42	1.209	1.21	1.21	2.1
0.149a	0.052f	0.253f	0.383a	0.186a	0.010f	0.072a
0	0	0	0	0	0	0
24.584f	25.145f	25.706f	23.787f	24.409f	25.031f	24.848f
24.570f	25.151f	25.729f	23.758f	24.395f	25.031f	24.846f
32.71	32.77	32.69	31.41	31.38	31.37	37.57
21.058	21.095	21.05	24.358	24.333	24.325	15.004
5.269	5.269	5.269	4.413	4.413	4.413	2.936
0.4	0.4	0.4	0.329	0.329	0.329	0.611
5.669	5.669	5.669	4.742	4.742	4.742	3.547
15.389	15.426	15.381	19.616	19.591	19.583	11.457
3.001	3.005	2.993	3.705	3.726	3.732	1.801

LOADING CONDITIONS

Condition 00 : Lightship

Lightship condition

Condition Graphic - Draft: 0.597 @ 50.000f, 0.587 @ 0.000 Heel: zero

```
18-03-24 11:29:48    Cybermarine Technologies PTE. Ltd.
GHS 18.66                            PONTOON
                              Condition 00 : Lightship
                                Lightship condition

                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
          USK draft: 0.597 @ 50.00f, 0.592 @ 25.00f, 0.587 @ 0.00
                    Trim: Fwd 0.011/50.000,  Heel: zero
Part---------------------------------Weight(MT)----LCG-----TCG-----VCG-------FSM
WEIGHT                                  465.25  25.062f  0.000   3.000
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG
  Total Tanks--------->                   0.00                                0.00
                                   Displ(MT)----LCB-----TCB-----VCB
HULL                  1.025         465.25  25.063f  0.000   0.289
------------------------------------------------------------------------------
               Righting Arms:                    0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025          805.7  25.009f  0.000    295.80  47.907
                                       MT/cm--------m.-MT/cm-------GML-----GMT
                                        8.26           27.27    293.09  45.197
Distances in METERS.----------------------------------------Moments in m.-MT.

                            SUMMARY OF LOADING
         0.0 Cu.M.   (0%) SALT WATER

DRAFT AT FP (M) =    0.597
DRAFT AT MS (M) =    0.592
DRAFT AT AP (M) =    0.587
DRAFT AT FWD MARK (M) =    0.596
DRAFT AT AFT MARK (M) =    0.588
                          HYDROSTATIC PROPERTIES
                Trim: Fwd 0.011/50.000, No Heel, VCG = 3.000

   LCF   Displacement   Buoyancy-Ctr.  Weight/         Moment/
  Draft----Weight(MT)----LCB-----VCB-------cm-----LCF---cm trim----GML-----GMT
  0.592       465.25  25.063f  0.289     8.26  25.009f   27.27  293.09  45.197
 Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
                            Trim is per 50.00m.
 Draft is from USK.

                         HEELING MOMENT specification
                            Lateral Plane Method
                        Wind pressure toward starboard
          USK draft: 0.597 @ 50.00f, 0.592 @ 25.00f, 0.587 @ 0.00
                    Trim: Fwd 0.011/50.000,  Heel: zero

    Part---------------LPA*SF------HCP------Arm----Pressure----------Moment
    HULL                117.8    1.239    1.531     0.05500            9.92
    Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT
```

18-03-24 11:29:48 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 00 : Lightship

Lightship condition

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
LCG = 25.062f TCG = 0.000 VCG = 3.000

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area	50% DeckImm Angle
0.575	0.01f	0.00	465.27	0.000	-0.021	0.0000	12.220
0.575	0.01f	0.03s	465.24	0.000	0.000	-0.0000	12.194
0.539	0.01f	5.03s	465.25	0.000	3.618	0.1579	7.194
0.284	0.02f	10.03s	465.25	0.000	4.914	0.5417	2.194
0.124	0.02f	12.22s	465.25	0.000	5.151	0.7350	0.000
-0.106	0.02f	15.03s	465.41	0.000	5.324	0.9925	-2.806
-0.567	0.02f	20.03s	465.37	0.000	5.423	1.4651	-7.806
-0.575	0.02f	20.10s	465.24	0.000	5.424	1.4725	-7.885
-1.071	0.03f	25.03s	465.25	0.000	5.357	1.9367	-12.806
-1.584	0.03f	30.03s	465.25	0.000	5.143	2.3959	-17.806
-2.086	0.04f	35.03s	465.25	0.000	4.811	2.8311	-22.806
-2.572	0.04f	40.03s	465.25	0.000	4.400	3.2336	-27.806
-3.038	0.05f	45.03s	465.25	0.000	3.931	3.5976	-32.806
-3.481	0.05f	50.03s	465.25	0.000	3.416	3.9185	-37.806
-3.898	0.05f	55.03s	465.25	0.000	2.865	4.1928	-42.806
-4.285	0.06f	60.03s	465.25	0.000	2.285	4.4177	-47.806

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: No tank loads are present.

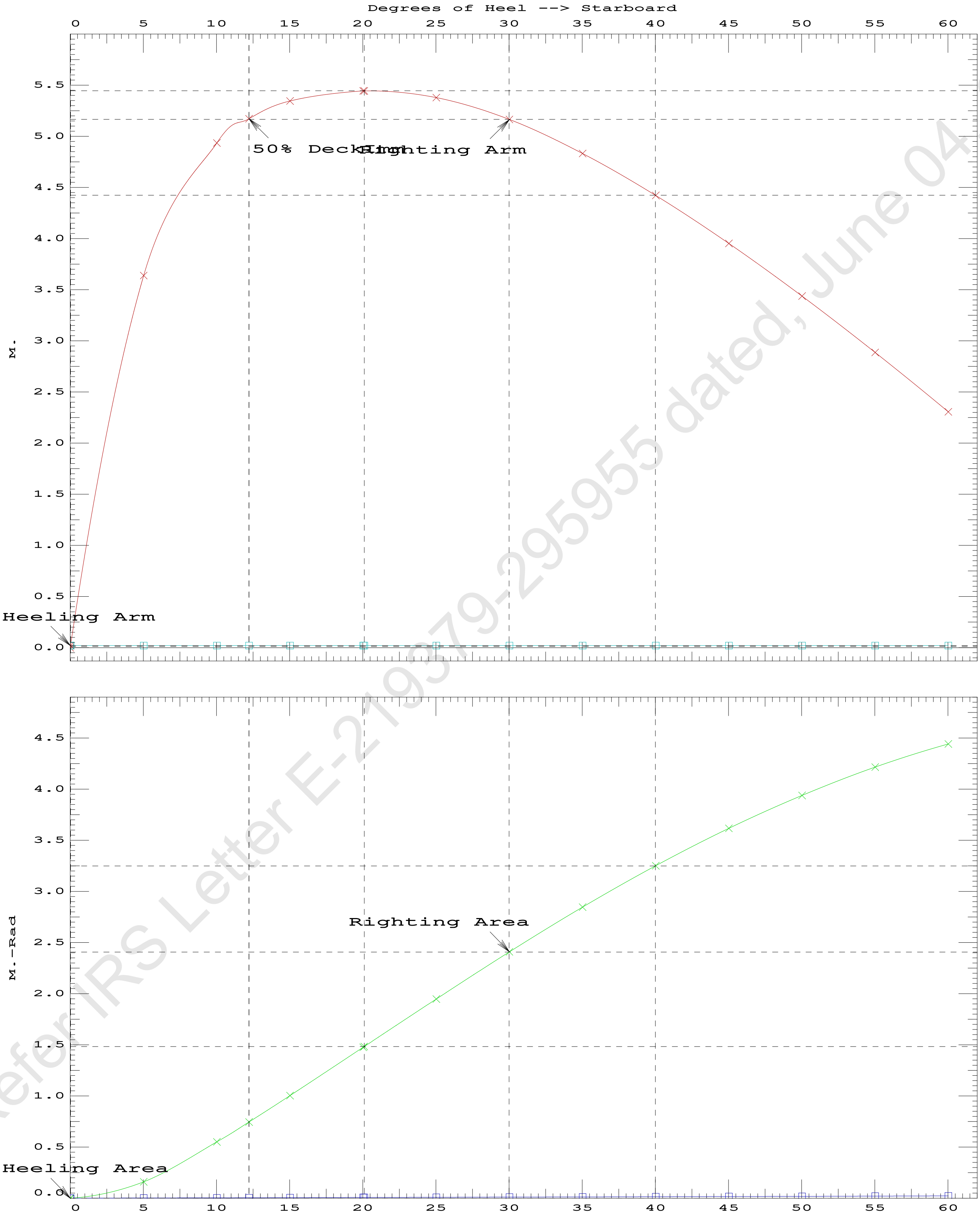
Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 9.92 (constant)

LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2		Min/Max		Attained
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	1.4725 P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00	deg	LARGE
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	12.19 P

18-03-24 11:29:48 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 00 : Lightship

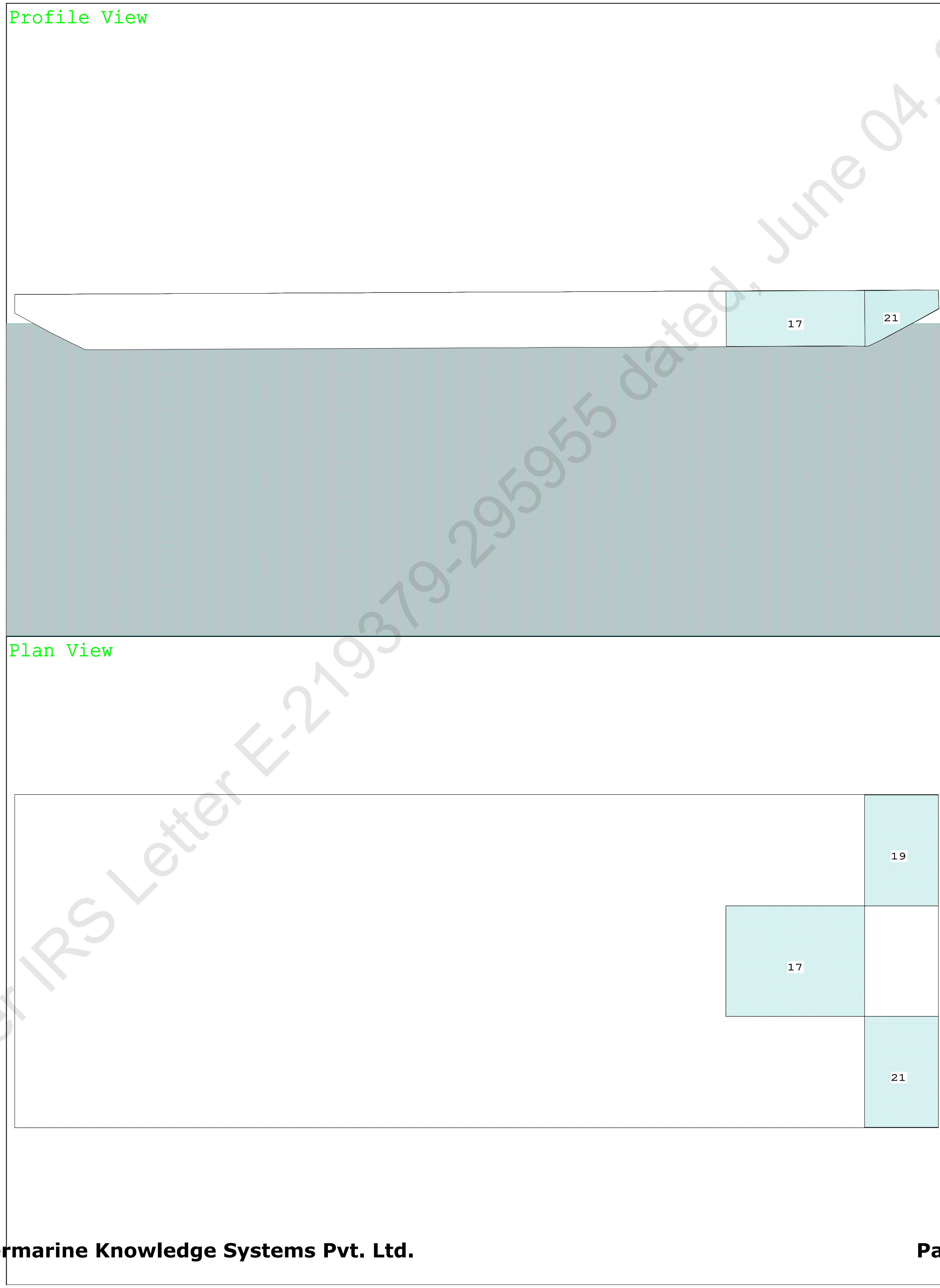
Lightship condition



24-04-10 18:24:44 Cybermarine Technologies PTE. Ltd.
GHS 18.90 **PONTOON**

Condition 01 : Lightship + Crane + Ballast

Condition Graphic - Draft: 1.215 @ 50.000f, 1.457 @ 0.000 Heel: zero



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24-04-10 18:24:44    Cybermarine Technologies PTE. Ltd.
GHS 18.90                        PONTOON
```

Condition 01 : Lightship + Crane + Ballast

```
                 WEIGHT and DISPLACEMENT and WATERPLANE STATUS
            USK draft: 1.215 @ 50.00f, 1.336 @ 25.00f, 1.457 @ 0.00
                      Trim: Aft 0.242/50.000,  Heel: zero
Part------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                            465.25  25.062f  0.000   3.000
Crane                                 400.00  11.500f  0.000   6.325
  Total Fixed--------->               865.25  18.792f  0.000   4.537
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
06_WBTK.C      1.000    1.025         135.67  42.251f  0.000   1.499    135.61*
07_WBTK.P      1.000    1.025          49.60  47.660f  5.971p  1.880     72.32*
07_WBTK.S      1.000    1.025          49.60  47.660f  5.971s  1.880     72.32*
  Total Tanks--------->               234.87  44.536f  0.000   1.660    280.25
  Total Weight-------->             1,100.12  24.288f  0.000   3.923
                                    Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025       1,100.12  24.273f  0.000   0.676
------------------------------------------------------------------------------
               Righting Arms:                  0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025          857.3  24.773f  0.000    150.69  21.332*
                                       MT/cm--------m.-MT/cm-------GML-----GMT
                                        8.79            32.44    147.45  18.085
Distances in METERS.----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


                          SUMMARY OF LOADING
       229.1 Cu.M.   (9%) SALT WATER

      400.00 MT  Crane

DRAFT AT FP (M) =    1.215
DRAFT AT MS (M) =    1.336
DRAFT AT AP (M) =    1.457
DRAFT AT FWD MARK (M) =    1.235
DRAFT AT AFT MARK (M) =    1.438
                         HYDROSTATIC PROPERTIES
                Trim: Aft 0.242/50.000, No Heel, VCG = 3.923

    LCF   Displacement   Buoyancy-Ctr.  Weight/         Moment/
   Draft----Weight(MT)----LCB-----VCB-------cm-----LCF---cm trim----GML-----GMT
   1.337     1,100.12  24.273f  0.676     8.79  24.773f   32.44  147.45  18.085
  Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
                           Trim is per 50.00m.
  Draft is from USK.                             Formal Free Surface included.

      Note:  GMT includes the formal free surface moment 280.2 m.-MT
```

24-04-10 18:24:44 Cybermarine Technologies PTE. Ltd.
GHS 18.90 **PONTOON**

Condition 01 : Lightship + Crane + Ballast

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 1.215 @ 50.00f, 1.336 @ 25.00f, 1.457 @ 0.00
Trim: Aft 0.242/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	83.4	0.851	1.507	0.05500	6.91
CRANE	57.0	4.104	4.760	0.05500	14.92
Total wind heeling moment to starboard------------->					21.83

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 24.288f TCG = 0.000 VCG = 3.923
Free Surface Adjustment: 0.255
Adjusted CG: LCG = 24.290f TCG = 0.000 VCG = 4.178

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area	50% DeckImm Angle
1.445	0.28a	0.00	1,100.1	0.000	-0.020	0.0000	5.047
1.445	0.28a	0.05s	1,100.1	0.000	0.000	-0.0000	4.999
1.430	0.28a	5.05s	1,100.1	0.000	1.583	0.0691	-0.001
1.385	0.30a	10.05s	1,100.1	0.000	3.020	0.2710	-5.001
1.290	0.43a	15.05s	1,100.1	0.000	3.575	0.5652	-10.001
1.254	0.49a	16.87s	1,100.1	0.000	3.598	0.6797	-11.826
1.190	0.58a	20.05s	1,100.1	0.000	3.537	0.8783	-15.001
1.084	0.74a	25.05s	1,100.1	0.000	3.299	1.1790	-20.001
0.973	0.89a	30.05s	1,100.1	0.000	2.967	1.4531	-25.001
0.857	1.04a	35.05s	1,100.1	0.000	2.579	1.6954	-30.001
0.738	1.19a	40.05s	1,100.1	0.000	2.153	1.9022	-35.001
0.616	1.34a	45.05s	1,100.1	0.000	1.701	2.0706	-40.001
0.490	1.48a	50.05s	1,100.1	0.000	1.229	2.1986	-45.001
0.361	1.61a	55.05s	1,100.1	0.000	0.744	2.2848	-50.001
0.229	1.73a	60.05s	1,100.1	0.000	0.250	2.3282	-55.001

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 280.2 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.83 (constant)

LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2		Min/Max		Attained
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	0.6797 P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00	deg	LARGE
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	5.00 P

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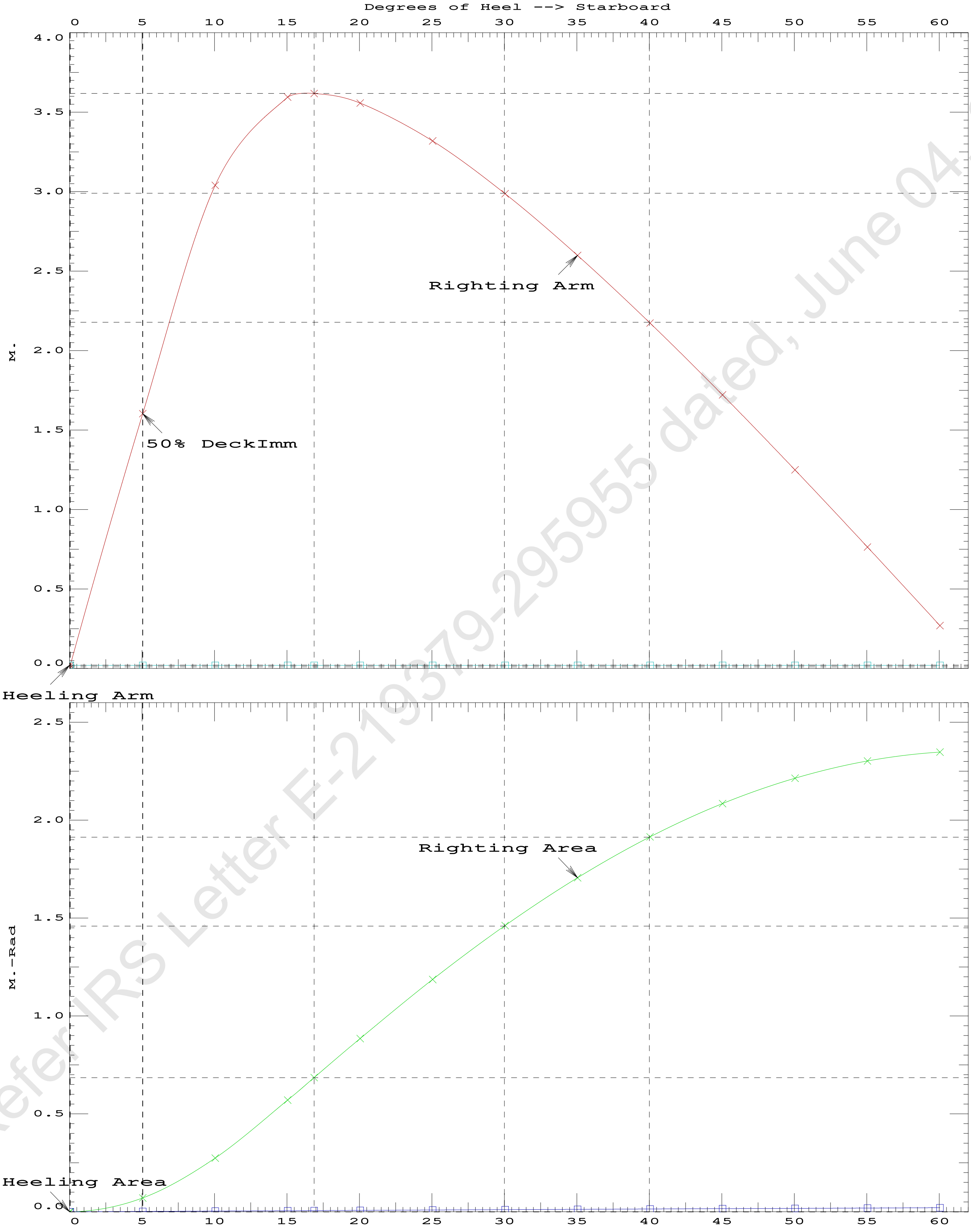
GHS 18.90 **PONTOON**

Condition 01 : Lightship + Crane + Ballast

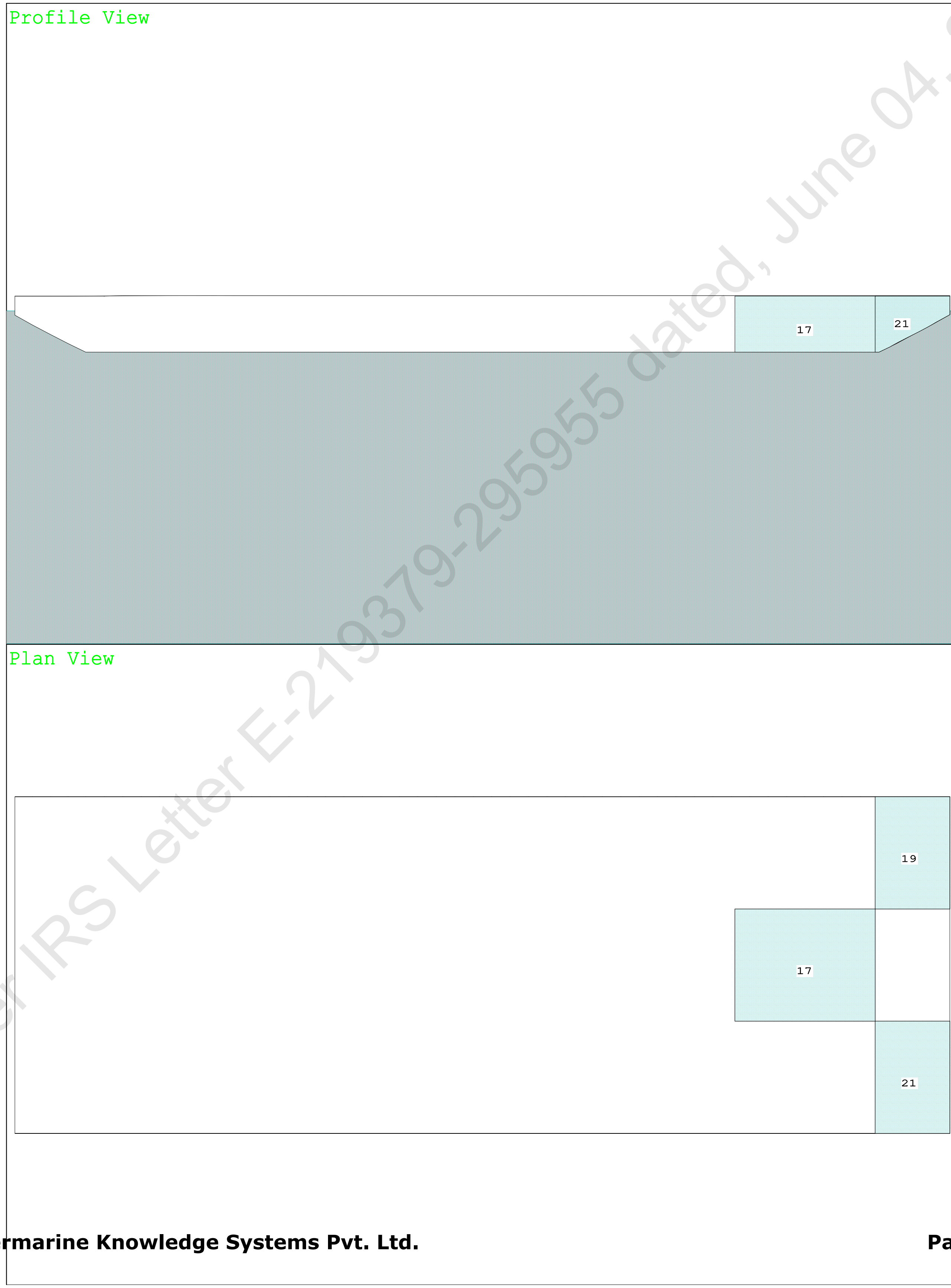


24-04-10 18:28:57 Cybermarine Technologies PTE. Ltd.
GHS 18.90 **PONTOON**

Condition 02 : Lightship + Crane + Deck Cargo + Ballast

Maximum Draft Condition

Condition Graphic - Draft: 2.206 @ 50.000f, 2.217 @ 0.000 Heel: zero



24-04-10 18:28:57 Cybermarine Technologies PTE. Ltd.
GHS 18.90 **PONTOON**

Condition 02 : Lightship + Crane + Deck Cargo + Ballast
Maximum Draft Condition

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 2.206 @ 50.00f, 2.212 @ 25.00f, 2.217 @ 0.00
Trim: Aft 0.010/50.000, Heel: zero

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			465.25	25.062f	0.000	3.000	
Crane			400.00	11.500f	0.000	6.325	
Deck Cargo			792.50	25.940f	0.000	4.000	
Total Fixed--------->			1,657.75	22.209f	0.000	4.280	
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000	1.499	135.61*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks--------->			234.87	44.536f	0.000	1.660	280.25
Total Weight-------->			1,892.62	24.980f	0.000	3.955	

	SpGr	Displ(MT)	LCB	TCB	VCB
HULL	1.025	1,892.62	24.979f	0.000	1.132

Righting Arms: 0.000 0.000

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	900.4	25.000f	0.000	101.47	13.032*

MT/cm	m.-MT/cm	GML	GMT
9.23	37.34	98.65	10.208

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

229.1 Cu.M. (9%) SALT WATER

400.00 MT Crane 792.50 MT Deck Cargo

DRAFT AT FP (M) = 2.206
DRAFT AT MS (M) = 2.212
DRAFT AT AP (M) = 2.217
DRAFT AT FWD MARK (M) = 2.207
DRAFT AT AFT MARK (M) = 2.216

HYDROSTATIC PROPERTIES
Trim: Aft 0.010/50.000, No Heel, VCG = 3.955

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
2.212	1,892.62	24.979f	1.132	9.23	25.000f	37.34	98.65	10.208

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 280.2 m.-MT

24-04-10 18:28:57 Cybermarine Technologies PTE. Ltd.
GHS 18.90 **PONTOON**

Condition 02 : Lightship + Crane + Deck Cargo + Ballast
Maximum Draft Condition

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 2.206 @ 50.00f, 2.212 @ 25.00f, 2.217 @ 0.00
Trim: Aft 0.010/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	40.5	0.405	1.478	0.05500	3.29
CRANE	57.0	3.297	4.370	0.05500	13.70
DECKCARGO.C	29.4	1.801	2.874	0.05500	4.65
Total wind heeling moment to starboard------------->					21.64

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 24.980f TCG = 0.000 VCG = 3.955
Free Surface Adjustment: 0.148
Adjusted CG: LCG = 24.980f TCG = 0.000 VCG = 4.103

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area	50% DeckImm Angle
2.205	0.01a	0.00	1,892.6	0.000	-0.011	0.0000	2.561
2.205	0.01a	0.06s	1,892.6	0.000	0.000	-0.0000	2.499
2.202	0.01a	2.56s	1,892.6	0.000	0.450	0.0098	-0.000
2.194	0.01a	5.06s	1,892.6	0.000	0.893	0.0391	-2.501
2.292	0.02a	10.06s	1,892.6	0.000	1.320	0.1408	-7.501
2.437	0.02a	13.41s	1,892.6	0.000	1.365	0.2201	-10.845
2.520	0.02a	15.06s	1,892.6	0.000	1.354	0.2594	-12.501
2.793	0.03a	20.06s	1,892.6	0.000	1.240	0.3742	-17.501
3.058	0.04a	25.06s	1,892.6	0.000	1.035	0.4742	-22.501
3.299	0.04a	30.06s	1,892.6	0.000	0.783	0.5538	-27.501
3.516	0.05a	35.06s	1,892.6	0.000	0.505	0.6102	-32.501
3.706	0.06a	40.06s	1,892.6	0.000	0.213	0.6416	-37.501
3.824	0.06a	43.62s	1,892.6	0.000	0.000	0.6482	-41.057
3.868	0.07a	45.06s	1,892.6	0.000	-0.087	0.6471	-42.501
4.000	0.07a	50.06s	1,892.6	0.000	-0.390	0.6263	-47.501
4.102	0.08a	55.06s	1,892.6	0.000	-0.693	0.5791	-52.501
4.172	0.08a	60.06s	1,892.6	0.000	-0.992	0.5055	-57.501

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 280.2 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.64 (constant)

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24-04-10 18:28:57    Cybermarine Technologies PTE. Ltd.
GHS 18.90                           PONTOON
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Condition 02 : Lightship + Crane + Deck Cargo + Ballast

Maximum Draft Condition

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LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max--------Attained
(1) Residual Area from abs 0 deg to MaxRA         >   0.0800 m.-Rad 0.2201 P
(2) Angle from Equilibrium to RAzero or Flood     >    20.00  deg    43.56 P
(3) Angle from Equilibrium to 50% Dk Imm Angle    >     0.00  deg     2.50 P
-----------------------------------------------------------------------------
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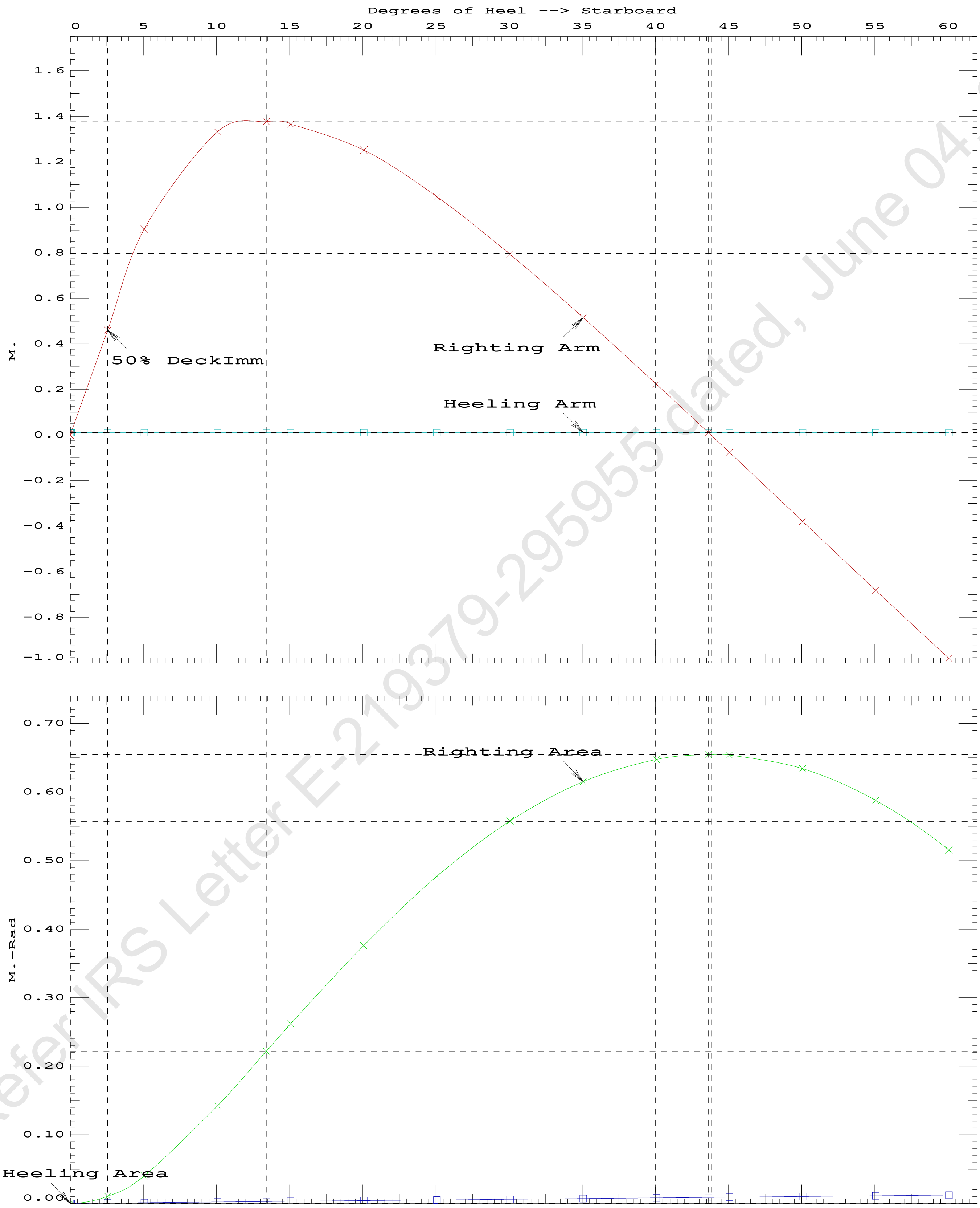
PONTOON

Condition 02 : Lightship + Crane + Deck Cargo + Ballast

Maximum Draft Condition

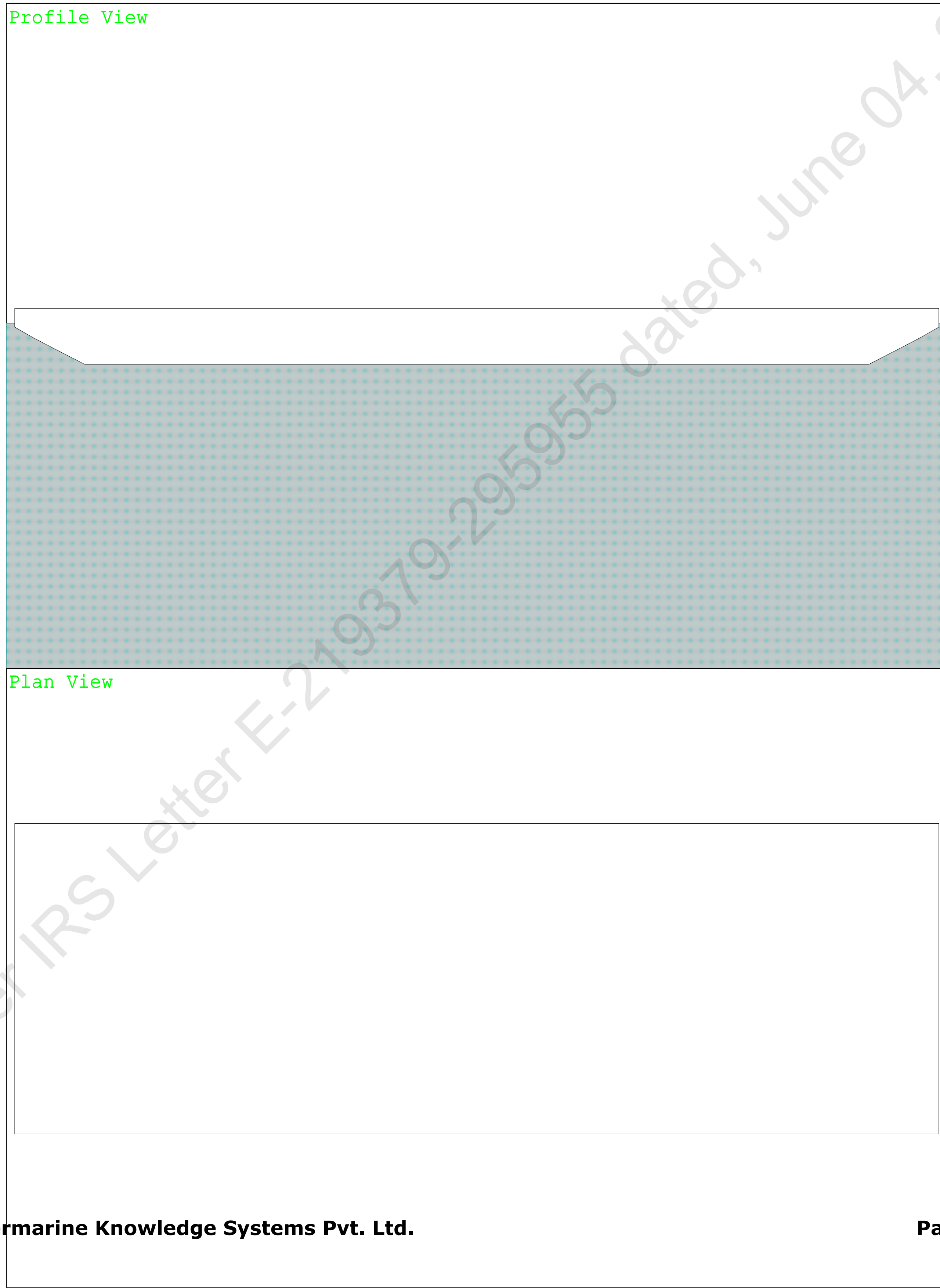


24-04-10 18:32:26 Cybermarine Technologies PTE. Ltd.
GHS 18.90 **PONTOON**

Condition 03 : Lightship + Deck Cargo

Maximum Deck Cargo Condition

Condition Graphic - Draft: 2.216 @ 50.000f, 2.208 @ 0.000 Heel: zero



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24-04-10 18:32:26    Cybermarine Technologies PTE. Ltd.
GHS 18.90                      PONTOON
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Condition 03 : Lightship + Deck Cargo

Maximum Deck Cargo Condition

```
                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 2.216 @ 50.00f, 2.212 @ 25.00f, 2.208 @ 0.00
                     Trim: Fwd 0.008/50.000,  Heel: zero
Part----------------------------------Weight(MT)----LCG-----TCG-----VCG-------FSM
LIGHT SHIP                               465.25  25.062f  0.000   3.000
Deck Cargo                             1,428.00  25.000f  0.000   4.000
  Total Weight-------->                1,893.25  25.015f  0.000   3.754
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG
  Total Tanks--------->                    0.00                               0.00
                                       Displ(MT)----LCB-----TCB-----VCB
HULL                        1.025      1,893.25  25.016f  0.000   1.132
-------------------------------------------------------------------------------
             Righting Arms:                        0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025             900.4  25.000f  0.000   101.44  13.176
                                          MT/cm--------m.-MT/cm-------GML-----GMT
                                           9.23             37.42     98.82  10.553
Distances in METERS.-----------------------------------------Moments in m.-MT.

                            SUMMARY OF LOADING
         0.0 Cu.M.   (0%) SALT WATER

    1,428.00 MT  Deck Cargo

DRAFT AT FP (M) =    2.216
DRAFT AT MS (M) =    2.212
DRAFT AT AP (M) =    2.208
DRAFT AT FWD MARK (M) =    2.215
DRAFT AT AFT MARK (M) =    2.209
                              HYDROSTATIC PROPERTIES
                 Trim: Fwd 0.008/50.000, No Heel, VCG = 3.754

   LCF   Displacement   Buoyancy-Ctr.  Weight/          Moment/
  Draft----Weight(MT)----LCB-----VCB-------cm-----LCF---cm trim----GML-----GMT
  2.212     1,893.25  25.016f  1.132     9.23  25.000f   37.42   98.82  10.553
 Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
                            Trim is per 50.00m.
 Draft is from USK.

                          HEELING MOMENT specification
                              Lateral Plane Method
                          Wind pressure toward starboard
           USK draft: 2.216 @ 50.00f, 2.212 @ 25.00f, 2.208 @ 0.00
                     Trim: Fwd 0.008/50.000,  Heel: zero

    Part---------------LPA*SF------HCP------Arm----Pressure----------Moment
    HULL                   40.5    0.405    1.478     0.05500             3.29
    DECKCARGO.C            29.4    1.799    2.873     0.05500             4.65
      Total wind heeling moment to starboard------------->               7.94
    Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT
```

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GHS 18.90

PONTOON

Condition 03 : Lightship + Deck Cargo

Maximum Deck Cargo Condition

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
LCG = 25.015f TCG = 0.000 VCG = 3.754

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel--->	Res. Area	50% DeckImm Angle
2.196	0.01f	0.00	1,893.1	0.000	-0.004	0.0000	2.563
2.196	0.01f	0.02s	1,893.1	0.000	0.000	-0.0000	2.541
2.193	0.01f	2.56s	1,893.3	0.000	0.472	0.0105	-0.000
2.186	0.01f	5.02s	1,893.3	0.000	0.923	0.0404	-2.459
2.278	0.01f	10.02s	1,893.3	0.000	1.386	0.1463	-7.459
2.459	0.02f	14.17s	1,893.3	0.000	1.453	0.2509	-11.609
2.502	0.02f	15.02s	1,893.2	0.000	1.450	0.2724	-12.459
2.770	0.02f	20.02s	1,893.2	0.000	1.366	0.3970	-17.459
3.030	0.03f	25.02s	1,893.3	0.000	1.190	0.5092	-22.459
3.266	0.03f	30.02s	1,893.3	0.000	0.965	0.6035	-27.459
3.477	0.04f	35.02s	1,893.2	0.000	0.713	0.6769	-32.459
3.662	0.04f	40.02s	1,893.2	0.000	0.445	0.7275	-37.459
3.819	0.05f	45.02s	1,893.2	0.000	0.168	0.7543	-42.459
3.898	0.05f	48.00s	1,893.2	0.000	0.000	0.7587	-45.438
3.946	0.05f	50.02s	1,893.2	0.000	-0.115	0.7567	-47.459
4.044	0.06f	55.02s	1,893.3	0.000	-0.399	0.7343	-52.459
4.111	0.06f	60.02s	1,893.2	0.000	-0.681	0.6872	-57.459

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: No tank loads are present.

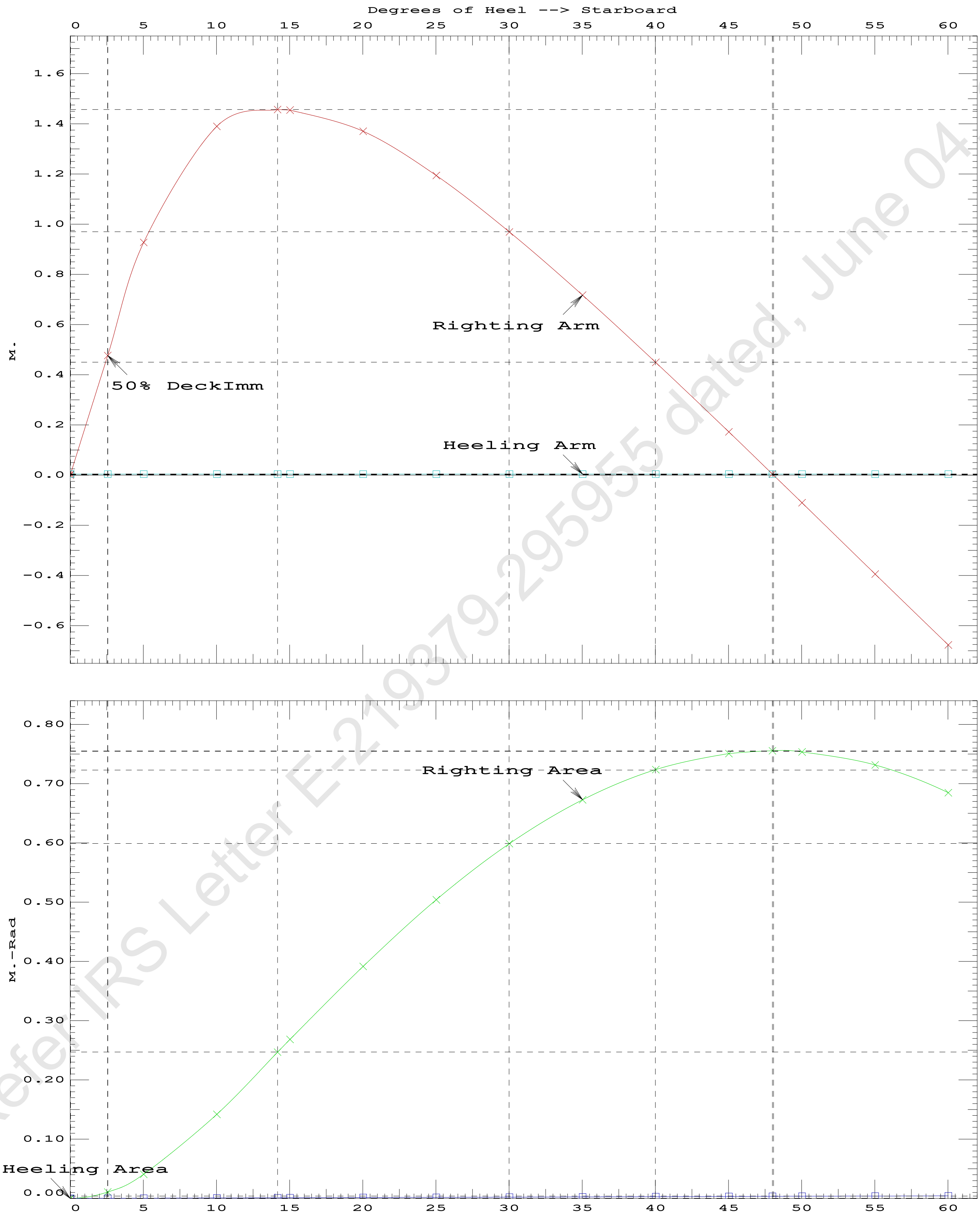
Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 7.94 (constant)

LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2		Min/Max		Attained	
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	0.2509	P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00	deg	47.98	P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	2.54	P

PONTOON

Condition 03 : Lightship + Deck Cargo

Maximum Deck Cargo Condition

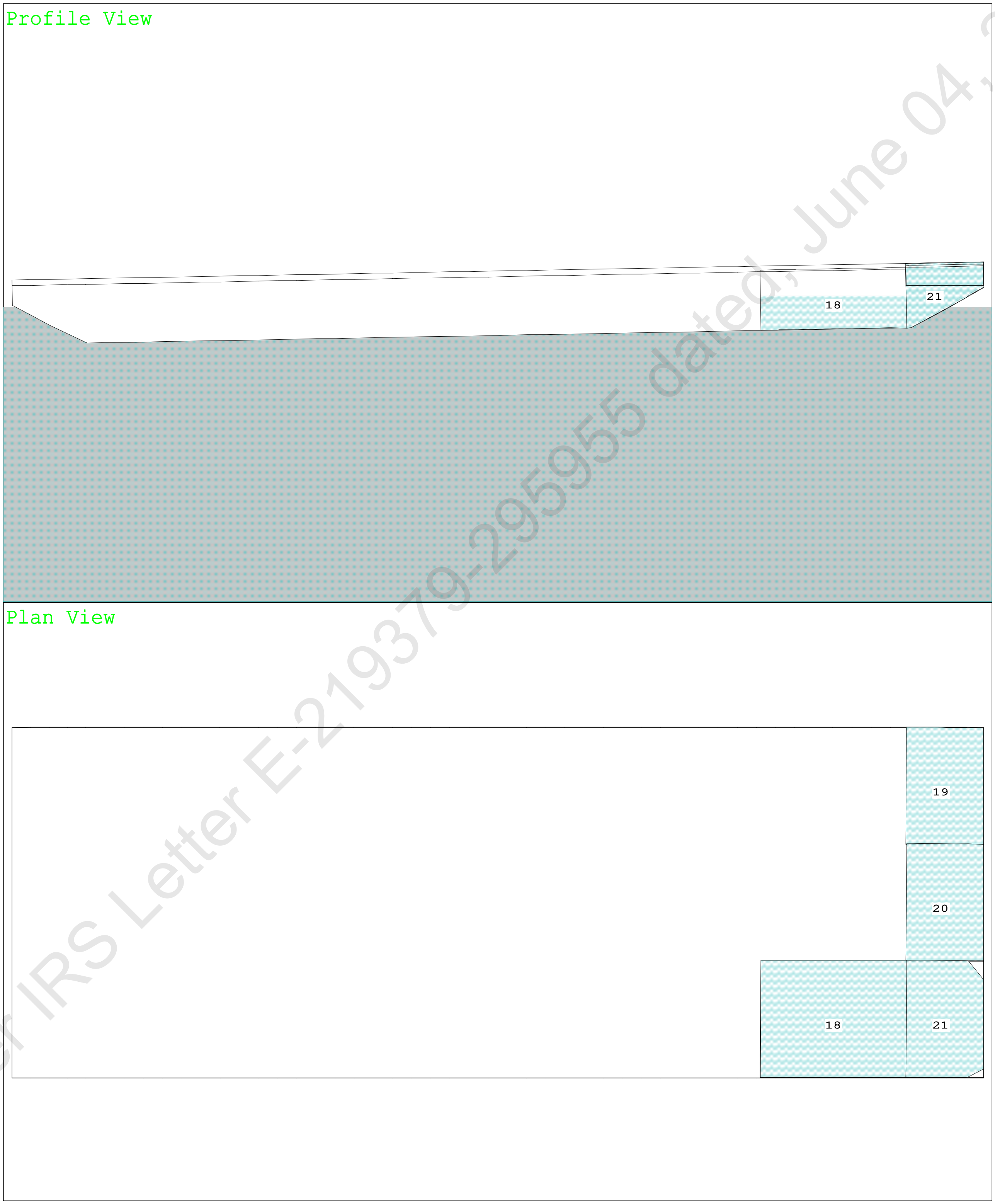


29-04-24 11:48:51 Cybermarine Technologies PTE. Ltd.
GHS 19.00 **PONTOON**

Condition 04 :Operating Condition of 400T Crane

Crane Lifting 40.0T @ 30m Radius Aft

CG - Draft: 0.868 @ 50.000f, 1.801 @ 0.000 Heel: stbd 0.93 deg.



Tanks
18 06_WBTK.S..55% SALT WATER Intact
19 07_WBTK.P.100% SALT WATER Intact
20 07_WBTK.C.100% SALT WATER Intact
21 07_WBTK.S..55% SALT WATER Intact

29-04-24 11:48:51 Cybermarine Technologies PTE. Ltd.
GHS 19.00 **PONTOON**

Condition 04 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Aft

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 0.868 @ 50.00f, 1.335 @ 25.00f, 1.801 @ 0.00
Trim: Aft 0.932/50.000, Heel: Stbd 0.93 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			465.25	25.062f	0.000	3.000	
Crane			400.00	11.500f	0.000	6.325	
Crane Load			40.00	~~19.500a~~ 18.500a	0.000	34.602	
Total Fixed--------->			905.25	17.100f	0.000	5.866	
06_WBTK.S	0.550	1.025	74.10	42.199f	5.994s	0.838	135.61
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	0.550	1.025	27.28	47.362f	6.001s	1.342	72.32*
Total Tanks--------->			201.09	45.608f	1.550s	1.421	352.57
Total Weight-------->			1,106.34	22.282f	0.282s	5.058	
			Displ(MT)	LCB	TCB	VCB	
HULL		1.025	1,106.35	22.201f	0.352s	0.707	

Righting Arms:			0.000	0.000s		
Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	852.6	24.134f	0.084s	147.17	21.022*
		MT/cm		m.-MT/cm	GML	GMT
		8.74		31.60	142.82	16.669

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

196.2 Cu.M. (8%) SALT WATER

400.00 MT Crane 40.00 MT Crane Load

DRAFT AT FP (M) = 0.868
DRAFT AT MS (M) = 1.335
DRAFT AT AP (M) = 1.801
DRAFT AT FWD MARK (M) = 0.943
DRAFT AT AFT MARK (M) = 1.726

HYDROSTATIC PROPERTIES
Trim: Aft 0.932/50.000, Heel: Stbd 0.93 deg., VCG = 5.058

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
1.351	1,106.35	22.201f	0.707	8.74	24.134f	31.60	142.82	16.669

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 352.6 m.-MT

29-04-24 11:48:51 Cybermarine Technologies PTE. Ltd.
GHS 19.00 **PONTOON**

Condition 04 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Aft

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 0.870 @ 50.00f, 1.336 @ 25.00f, 1.802 @ 0.00
Trim: Aft 0.932/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	83.1	0.872	1.548	0.05500	7.07
CRANE	57.0	3.897	4.573	0.05500	14.34
Total wind heeling moment to starboard------------->					21.41

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 22.282f TCG = 0.282s VCG = 5.058
Free Surface Adjustment: 0.319
Adjusted CG: LCG = 22.288f TCG = 0.276s VCG = 5.376

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area	50% DeckImm Angle
1.791	1.07a	0.00	1,106.3	0.000	-0.296	0.0000	3.868
1.788	1.07a	1.00s	1,106.3	0.000	0.000	-0.0026	2.872
1.783	1.07a	3.87s	1,106.3	0.000	0.845	0.0187	0.000
1.776	1.08a	6.00s	1,106.3	0.000	1.457	0.0614	-2.128
1.782	1.27a	11.00s	1,106.3	0.000	2.589	0.2402	-7.128
1.854	1.74a	15.49s	1,106.3	0.000	2.836	0.4577	-11.621
1.864	1.80a	16.00s	1,106.3	0.000	2.834	0.4828	-12.128
1.958	2.38a	21.00s	1,106.4	0.000	2.647	0.7261	-17.128
2.048	2.98a	26.00s	1,106.4	0.000	2.298	0.9431	-22.128
2.135	3.59a	31.00s	1,106.4	0.000	1.873	1.1256	-27.128
2.216	4.18a	36.00s	1,106.4	0.000	1.405	1.2689	-32.128
2.289	4.77a	41.00s	1,106.4	0.000	0.912	1.3702	-37.128
2.349	5.33a	46.00s	1,106.4	0.000	0.405	1.4278	-42.128
2.385	5.75a	49.94s	1,106.3	0.000	0.000	1.4418	-46.075
2.394	5.86a	51.00s	1,106.3	0.000	-0.108	1.4408	-47.128
2.418	6.34a	56.00s	1,106.4	0.000	-0.623	1.4089	-52.128
2.418	6.75a	61.00s	1,106.4	0.000	-1.133	1.3322	-57.128

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 352.6 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.41 (constant)

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GHS 19.00 **PONTOON**

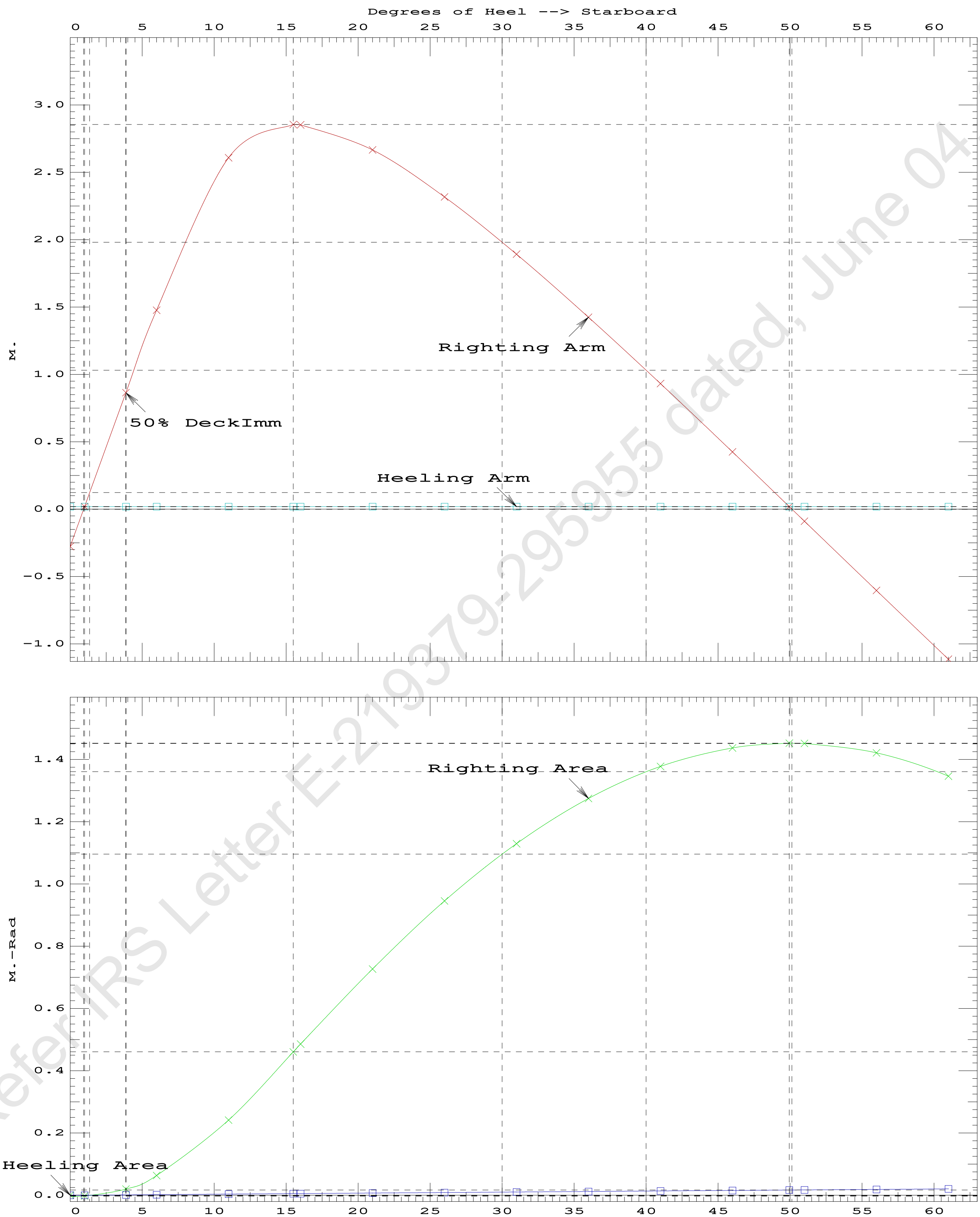
Condition 04 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Aft

LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2		Min/Max		Attained	
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	0.4577	P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00	deg	48.95	P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	2.87	P

PONTOON

Condition 04 :Operating Condition of 400T Crane

Crane Lifting 40.0T @ 30m Radius Aft



29-04-24 11:48:51 Cybermarine Technologies PTE. Ltd.
GHS 19.00 **PONTOON**

Condition 04 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 22.282f TCG = 0.282s VCG = 5.058
Free Surface Adjustment: 0.319
Adjusted CG: LCG = 22.288f TCG = 0.277s VCG = 5.376

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel --->	Res. Area
1.788	1.07a	0.95s	1,106.3	0.000	0.000	0.0000
1.776	1.08a	5.95s	1,106.3	0.000	1.458	0.0636
1.782	1.27a	10.95s	1,106.3	0.000	2.597	0.2429
1.854	1.74a	15.49s	1,106.3	0.000	2.851	0.4639
1.863	1.79a	15.95s	1,106.3	0.000	2.849	0.4865
1.957	2.38a	20.95s	1,106.4	0.000	2.664	0.7313
2.048	2.98a	25.95s	1,106.4	0.000	2.316	0.9498
2.134	3.58a	30.95s	1,106.4	0.000	1.892	1.1340
2.215	4.18a	35.95s	1,106.4	0.000	1.425	1.2790
2.288	4.77a	40.95s	1,106.4	0.000	0.932	1.3821
2.349	5.33a	45.95s	1,106.4	0.000	0.425	1.4414
2.387	5.77a	50.09s	1,106.3	0.000	0.000	1.4568
2.393	5.85a	50.95s	1,106.3	0.000	-0.089	1.4561
2.418	6.33a	55.95s	1,106.4	0.000	-0.603	1.4260
2.418	6.75a	60.95s	1,106.4	0.000	-1.113	1.3510

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 352.6 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM------------STABILITY CRITERIA FOR CRANE OPERAT		----Min/Max		--------Attained
(1) Absolute Angle at Equilibrium	<	15.00	deg	0.95 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	63078.4 P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		376.849 P

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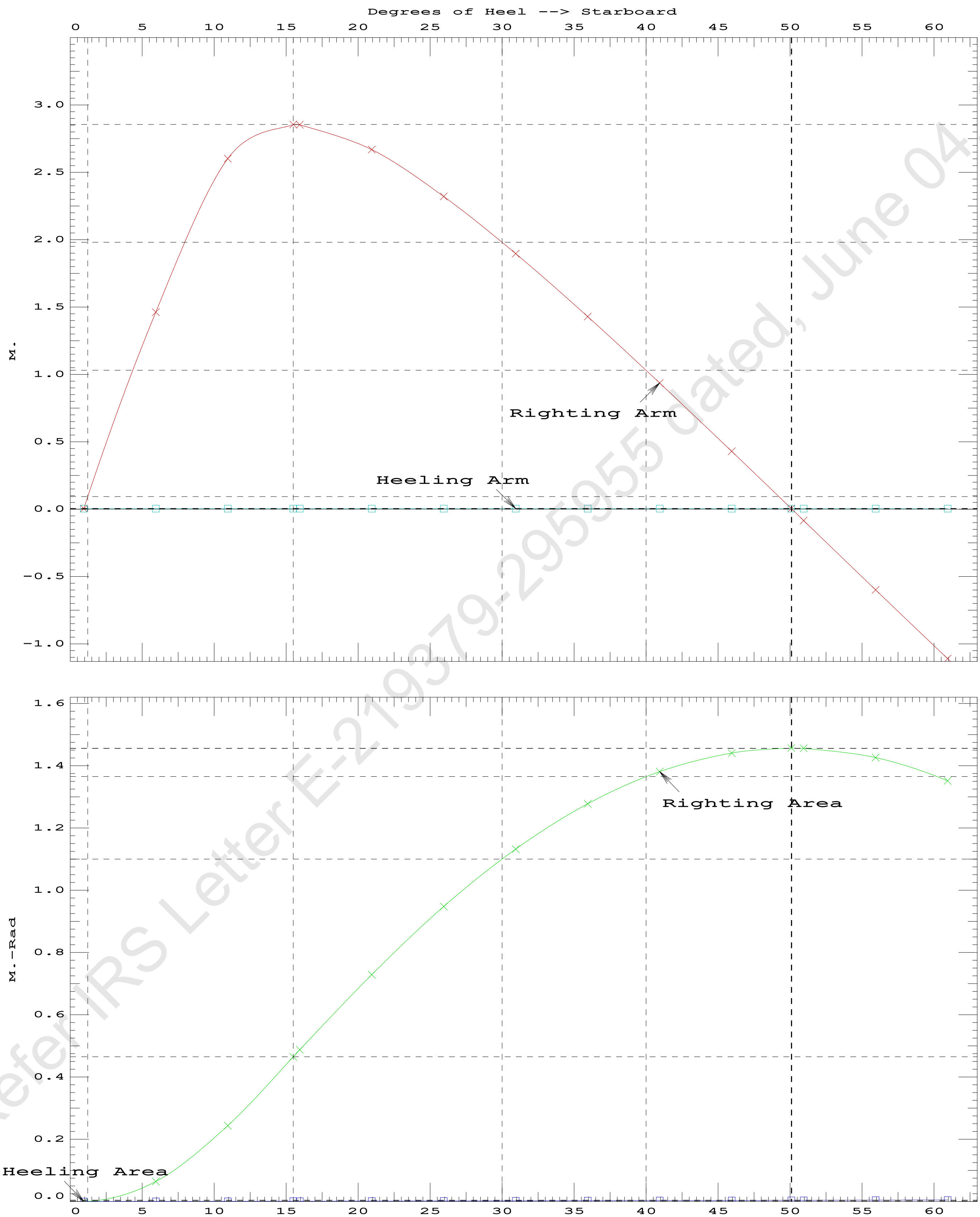
GHS 19.00 **PONTOON**

Condition 04 :Operating Condition of 400T Crane

Crane Lifting 40.0T @ 30m Radius Aft



29-04-24 16:41:33 Cybermarine Technologies PTE. Ltd.
GHS 19.00 **PONTOON**

Condition 04 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Aft

Crane in operation with load in place
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 0.868 @ 50.00f, 1.335 @ 25.00f, 1.801 @ 0.00
Trim: Aft 0.932/50.000, Heel: Stbd 0.93 deg.

Part	Load	SpGr	Weight (MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			465.25	25.062f	0.000	3.000	
Crane			400.00	11.500f	0.000	6.325	
Crane Load			40.00	~~19.500a~~ 18.500a	0.000	34.602	
Total Fixed--------->			905.25	17.100f	0.000	5.866	
06_WBTK.S	0.550	1.025	74.10	42.199f	5.994s	0.838	135.61
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	0.550	1.025	27.28	47.362f	6.001s	1.342	72.32*
Total Tanks--------->			201.09	45.608f	1.550s	1.421	352.57
Total Weight-------->			1,106.34	22.282f	0.282s	5.058	

Part	SpGr	Displ (MT)	LCB	TCB	VCB
HULL	1.025	1,106.35	22.201f	0.354s	0.707

Righting Arms:	0.000	0.001s
External Arms:	0.000	0.001
Residual Righting Arms:	0.000	0.000s

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	852.6	24.134f	0.084s	147.18	21.022*
		MT/cm		m.-MT/cm	GML	GMT
		8.74		31.60	142.83	16.670

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.93

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29-04-24 16:41:33    Cybermarine Technologies PTE. Ltd.
GHS 19.00                        PONTOON

                        After hook load drop
               WEIGHT and DISPLACEMENT and WATERPLANE STATUS
          USK draft: 1.105 @ 50.00f, 1.296 @ 25.00f, 1.487 @ 0.00
               Trim: Aft 0.381/50.000,  Heel: Stbd 0.88 deg.
Part------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                           465.25  25.062f  0.000   3.000
Crane                                400.00  11.500f  0.000   6.325
  Total Fixed--------->              865.25  18.792f  0.000   4.537
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
06_WBTK.S      0.550    1.025         74.10  42.231f  5.992s  0.837    135.61
07_WBTK.P      1.000    1.025         49.60  47.660f  5.971p  1.880     72.32*
07_WBTK.C      1.000    1.025         50.11  47.663f  0.000   1.872     72.33*
07_WBTK.S      0.550    1.025         27.28  47.375f  5.996s  1.342     72.32*
  Total Tanks--------->              201.09  45.622f  1.549s  1.421    352.57
  Total Weight-------->            1,066.34  23.852f  0.292s  3.949
                                  Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025      1,066.33  23.827f  0.348s  0.661
------------------------------------------------------------------------------
               Righting Arms:                  0.000   0.005s
               External Arms:                  0.000   0.005s
      Residual Righting Arms:                  0.000   0.000s
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025          853.9  24.762f  0.063s   153.44  21.753*
                                       MT/cm
                                        8.75
Distances in METERS.----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


limmarg: 23398374.00; t3: 30.0; t: 15.00
limmarg: 9369666.00; t3: 15.0; t: 7.50
limmarg: 2853896.00; t3: 7.5; t: 3.75
limmarg: 876691.88; t3: 3.8; t: 1.88
limmarg: 322254.84; t3: 1.9; t: 0.94
limmarg: 139829.09; t3: 1.0; t: 0.47
limmarg: 82388.41; t3: 0.5; t: 0.23
limmarg: 54987.58; t3: 0.3; t: 0.12
limmarg: 46773.21; t3: 0.2; t: 0.06
limmarg: 43324.39; t3: 0.1; t: 0.03
limmarg: 37544.19; t3: 0.1; t: 0.01

                  Theta3 is 0.07 degrees, as demonstrated below.
```

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 18.792f TCG = 0.000 VCG = 4.537

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.475	0.44a	0.93s	1,066.3	0.000	-0.016	0.0000
1.475	0.44a	0.88s	1,066.3	0.000	0.000	-0.0000
1.476	0.44a	0.07p	1,066.3	0.000	0.318	0.0026
1.465	0.44a	4.12p	1,066.3	0.000	1.646	0.0720
1.431	0.46a	9.12p	1,066.3	0.000	3.182	0.2834
1.351	0.63a	14.12p	1,066.3	0.000	3.919	0.5991
1.305	0.77a	17.02p	1,066.3	0.000	3.981	0.8002
1.270	0.87a	19.12p	1,066.3	0.000	3.949	0.9454
1.184	1.10a	24.12p	1,066.3	0.000	3.730	1.2834
1.092	1.34a	29.12p	1,066.3	0.000	3.404	1.5955

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 865.25 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

LIM	STABILITY CRITERION		Min/Max	Attained
(1)	Res. Area Ratio from abs -0.9 deg to abs 0.1	>	1.000	376.442 P

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.792f TCG = 0.000 VCG = 4.537

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.475	0.44a	0.93s	1,066.3	0.000	-0.016	0.0000
1.475	0.44a	0.88s	1,066.3	0.000	0.000	-0.0000
1.476	0.44a	0.07p	1,066.3	0.000	0.318	0.0026
1.465	0.44a	4.12p	1,066.3	0.000	1.646	0.0720
1.431	0.46a	9.12p	1,066.3	0.000	3.182	0.2834
1.351	0.63a	14.12p	1,066.3	0.000	3.919	0.5991
1.305	0.77a	17.02p	1,066.3	0.000	3.981	0.8002
1.270	0.87a	19.12p	1,066.3	0.000	3.949	0.9454
1.184	1.10a	24.12p	1,066.3	0.000	3.730	1.2834
1.092	1.34a	29.12p	1,066.3	0.000	3.404	1.5955
0.998	1.58a	34.12p	1,066.3	0.000	3.016	1.8761
0.900	1.82a	39.12p	1,066.3	0.000	2.589	2.1209
0.799	2.05a	44.12p	1,066.3	0.000	2.132	2.3271
0.695	2.27a	49.12p	1,066.3	0.000	1.653	2.4924
0.588	2.48a	54.12p	1,066.3	0.000	1.158	2.6151
0.475	2.67a	59.12p	1,066.3	0.000	0.652	2.6942
0.358	2.83a	64.12p	1,066.3	0.000	0.140	2.7288
0.326	2.88a	65.48p	1,066.3	0.000	0.000	2.7305
0.235	2.97a	69.12p	1,066.3	0.000	-0.374	2.7186

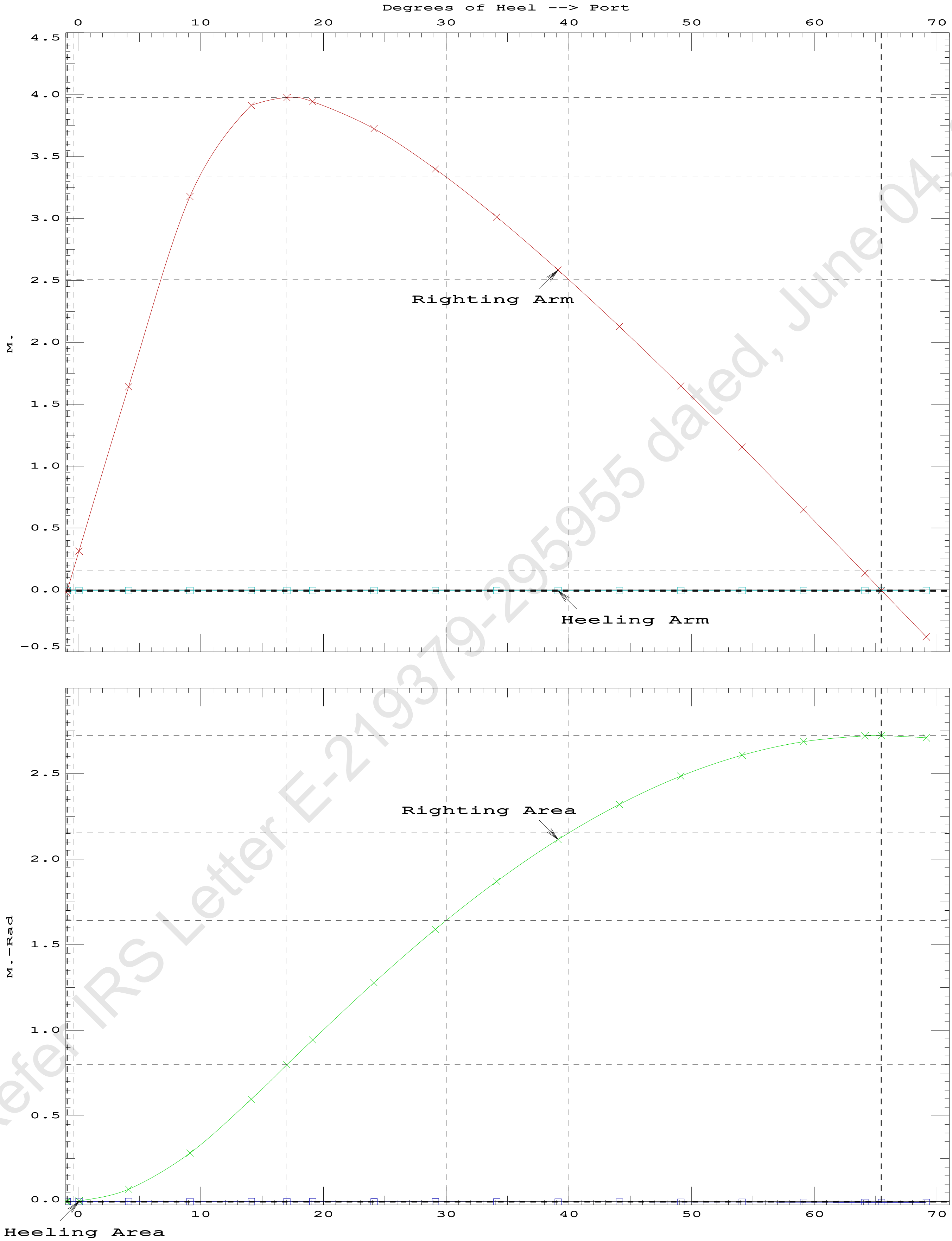
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 865.25 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):

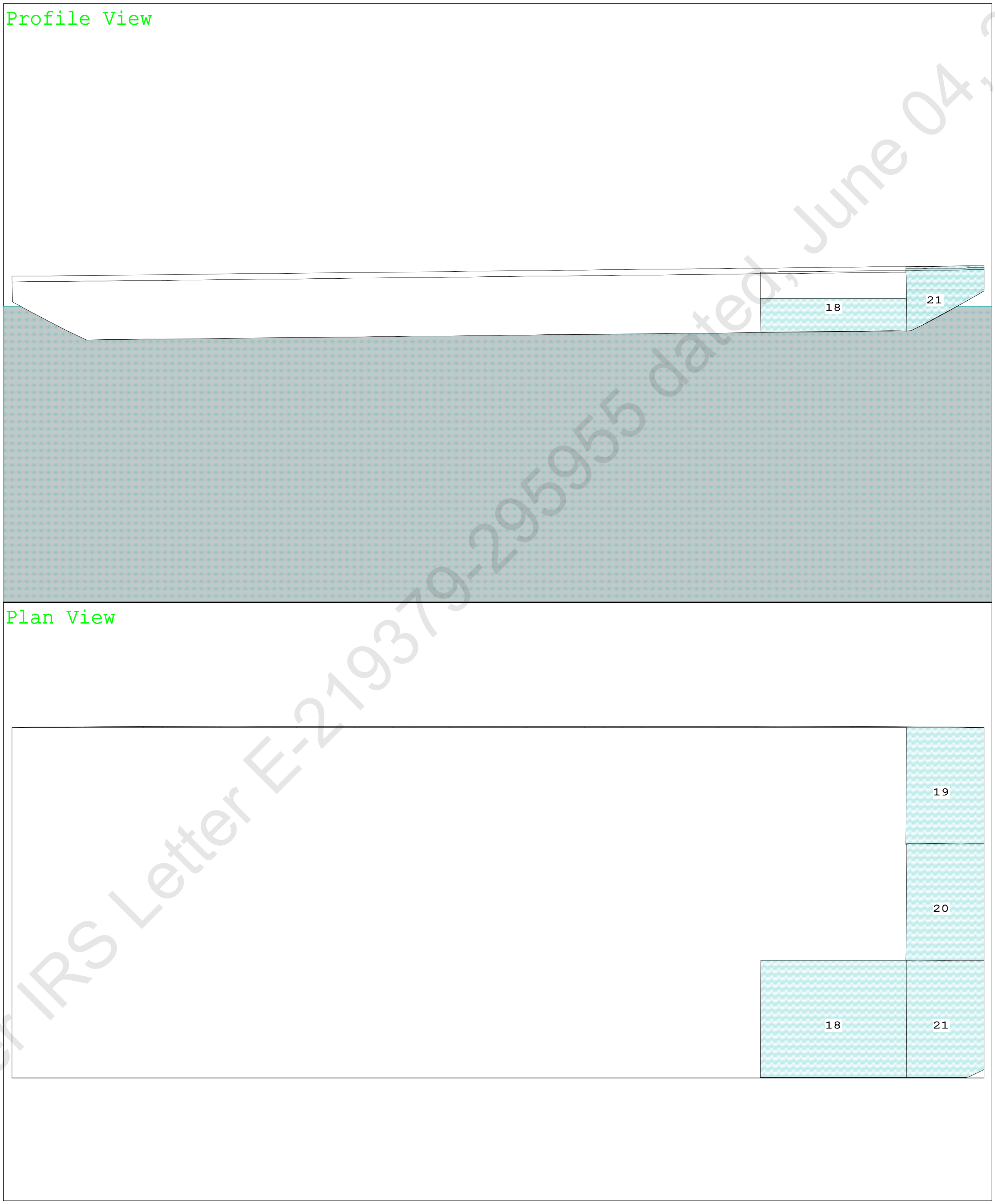
Stbd heeling moment = 5.00

LIM--------------RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Res. Area Ratio from abs -0.9 deg to RAzero	>	1.000		387827 P
(2) Angle from abs 0.1 deg to RAzero	>	20.00	deg	65.41 P


Degrees of Heel --> Port
M.
Righting Arm
Heeling Arm
M.-Rad
Righting Area
Heeling Area

Condition 05 : Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Stbd

CG - Draft: 1.066 @ 50.000f, 1.615 @ 0.000 Heel: stbd 0.93 deg.



Tanks
18 06_WBTK.S..55% SALT WATER Intact
19 07_WBTK.P.100% SALT WATER Intact
20 07_WBTK.C.100% SALT WATER Intact
21 07_WBTK.S..55% SALT WATER Intact

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GHS 19.00 **PONTOON**

Condition 05 : Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Stbd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.066 @ 50.00f, 1.341 @ 25.00f, 1.615 @ 0.00
Trim: Aft 0.548/50.000, Heel: Stbd 0.93 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			465.25	25.062f	0.000	3.000	
Crane			400.00	11.500f	0.000	6.325	
Crane Load			40.00	11.500f	0.000	34.602	
Total Fixed--------->			905.25	18.470f	0.000	5.866	
06_WBTK.S	0.550	1.025	74.10	42.221f	5.994s	0.838	135.61
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	0.550	1.025	27.28	47.371f	5.999s	1.342	72.32*
Total Tanks--------->			201.09	45.618f	1.550s	1.421	352.57
Total Weight-------->			1,106.34	23.405f	0.282s	5.058	
			Displ(MT)	LCB	TCB	VCB	
HULL		1.025	1,106.35	23.357f	0.352s	0.690	

Righting Arms:			0.000	0.000s			
Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	861.1	24.513f	0.041s	151.59	21.151*
		MT/cm		m.-MT/cm	GML	GMT
		8.83		32.58	147.22	16.782

Distances in METERS.---Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

196.2 Cu.M. (8%) SALT WATER

400.00 MT Crane 40.00 MT Crane Load

DRAFT AT FP (M) = 1.066
DRAFT AT MS (M) = 1.341
DRAFT AT AP (M) = 1.615
DRAFT AT FWD MARK (M) = 1.110
DRAFT AT AFT MARK (M) = 1.571

HYDROSTATIC PROPERTIES

Trim: Aft 0.548/50.000, Heel: Stbd 0.93 deg., VCG = 5.058

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
1.346	1,106.35	23.357f	0.690	8.83	24.513f	32.58	147.22	16.782

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 352.6 m.-MT

29-04-24 11:57:24 Cybermarine Technologies PTE. Ltd.
GHS 19.00 **PONTOON**

Condition 05 : Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Stbd

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 1.067 @ 50.00f, 1.341 @ 25.00f, 1.615 @ 0.00
Trim: Aft 0.548/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	83.0	0.855	1.519	0.05500	6.94
CRANE	57.0	4.007	4.671	0.05500	14.64
Total wind heeling moment to starboard------------->					21.58

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 23.405f TCG = 0.282s VCG = 5.058
Free Surface Adjustment: 0.319
Adjusted CG: LCG = 23.408f TCG = 0.276s VCG = 5.376

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area	50% DeckImm Angle
1.603	0.63a	0.00	1,106.3	0.000	-0.296	0.0000	4.492
1.602	0.63a	0.99s	1,106.3	0.000	0.000	-0.0026	3.502
1.591	0.63a	4.49s	1,106.3	0.000	1.029	0.0289	0.000
1.585	0.63a	5.99s	1,106.3	0.000	1.463	0.0615	-1.498
1.553	0.73a	10.99s	1,106.3	0.000	2.663	0.2432	-6.498
1.542	1.02a	15.53s	1,106.3	0.000	2.939	0.4696	-11.038
1.541	1.05a	15.99s	1,106.3	0.000	2.937	0.4931	-11.498
1.529	1.40a	20.99s	1,106.4	0.000	2.753	0.7459	-16.498
1.513	1.75a	25.99s	1,106.4	0.000	2.405	0.9721	-21.498
1.493	2.11a	30.99s	1,106.4	0.000	1.980	1.1641	-26.498
1.467	2.46a	35.99s	1,106.4	0.000	1.512	1.3167	-31.498
1.436	2.81a	40.99s	1,106.4	0.000	1.016	1.4272	-36.498
1.398	3.15a	45.99s	1,106.4	0.000	0.504	1.4937	-41.498
1.352	3.45a	50.82s	1,106.3	0.000	0.000	1.5149	-46.327
1.350	3.46a	50.99s	1,106.3	0.000	-0.018	1.5149	-46.498
1.292	3.75a	55.99s	1,106.4	0.000	-0.542	1.4905	-51.498
1.220	4.00a	60.99s	1,106.3	0.000	-1.065	1.4203	-56.498

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 352.6 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.58 (constant)

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29-04-24 11:57:24    Cybermarine Technologies PTE. Ltd.
GHS 19.00                          PONTOON
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Condition 05 : Operating Condition of 400T Crane

Crane Lifting 40.0T @ 30m Radius Stbd

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LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max--------Attained
(1) Residual Area from abs 0 deg to MaxRA         >   0.0800 m.-Rad 0.4696 P
(2) Angle from Equilibrium to RAzero or Flood     >    20.00  deg    49.83 P
(3) Angle from Equilibrium to 50% Dk Imm Angle    >     0.00  deg     3.50 P
-----------------------------------------------------------------------------
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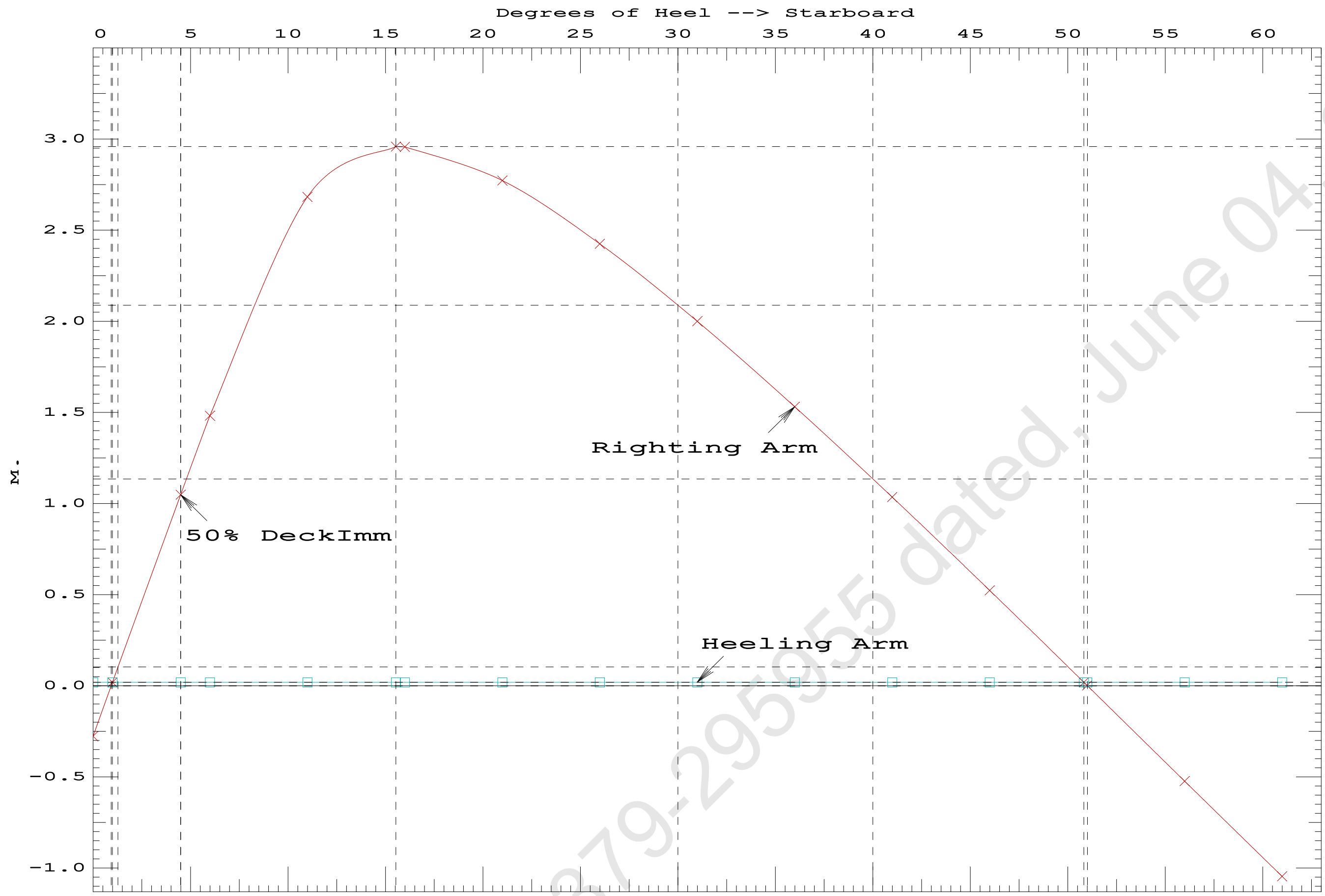
29-04-24 11:57:24 Cybermarine Technologies PTE. Ltd.

GHS 19.00 **PONTOON**

Condition 05 : Operating Condition of 400T Crane

Crane Lifting 40.0T @ 30m Radius Stbd



29-04-24 11:57:24 Cybermarine Technologies PTE. Ltd.
GHS 19.00 **PONTOON**

Condition 05 : Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 23.405f TCG = 0.294s VCG = 5.058
Free Surface Adjustment: 0.319
Adjusted CG: LCG = 23.408f TCG = 0.268s VCG = 5.376

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.591	0.63a	4.59s	1,106.3	0.000	0.000	0.0000
1.561	0.68a	9.59s	1,106.3	0.000	1.352	0.0590
1.544	0.96a	14.59s	1,106.3	0.000	1.874	0.2070
1.542	1.03a	15.58s	1,106.3	0.000	1.882	0.2395
1.533	1.30a	19.59s	1,106.3	0.000	1.770	0.3695
1.518	1.65a	24.59s	1,106.3	0.000	1.455	0.5127
1.499	2.01a	29.59s	1,106.4	0.000	1.047	0.6226
1.475	2.36a	34.59s	1,106.4	0.000	0.588	0.6943
1.446	2.72a	39.59s	1,106.4	0.000	0.098	0.7245
1.439	2.78a	40.57s	1,106.3	0.000	0.000	0.7254
1.410	3.06a	44.59s	1,106.4	0.000	-0.411	0.7110
1.365	3.38a	49.59s	1,106.4	0.000	-0.931	0.6526
1.310	3.67a	54.59s	1,106.4	0.000	-1.456	0.5485
1.242	3.93a	59.59s	1,106.4	0.000	-1.980	0.3986
1.159	4.15a	64.59s	1,106.3	0.000	-2.498	0.2032

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 352.6 m.-MT was applied to artificially modify the CG.

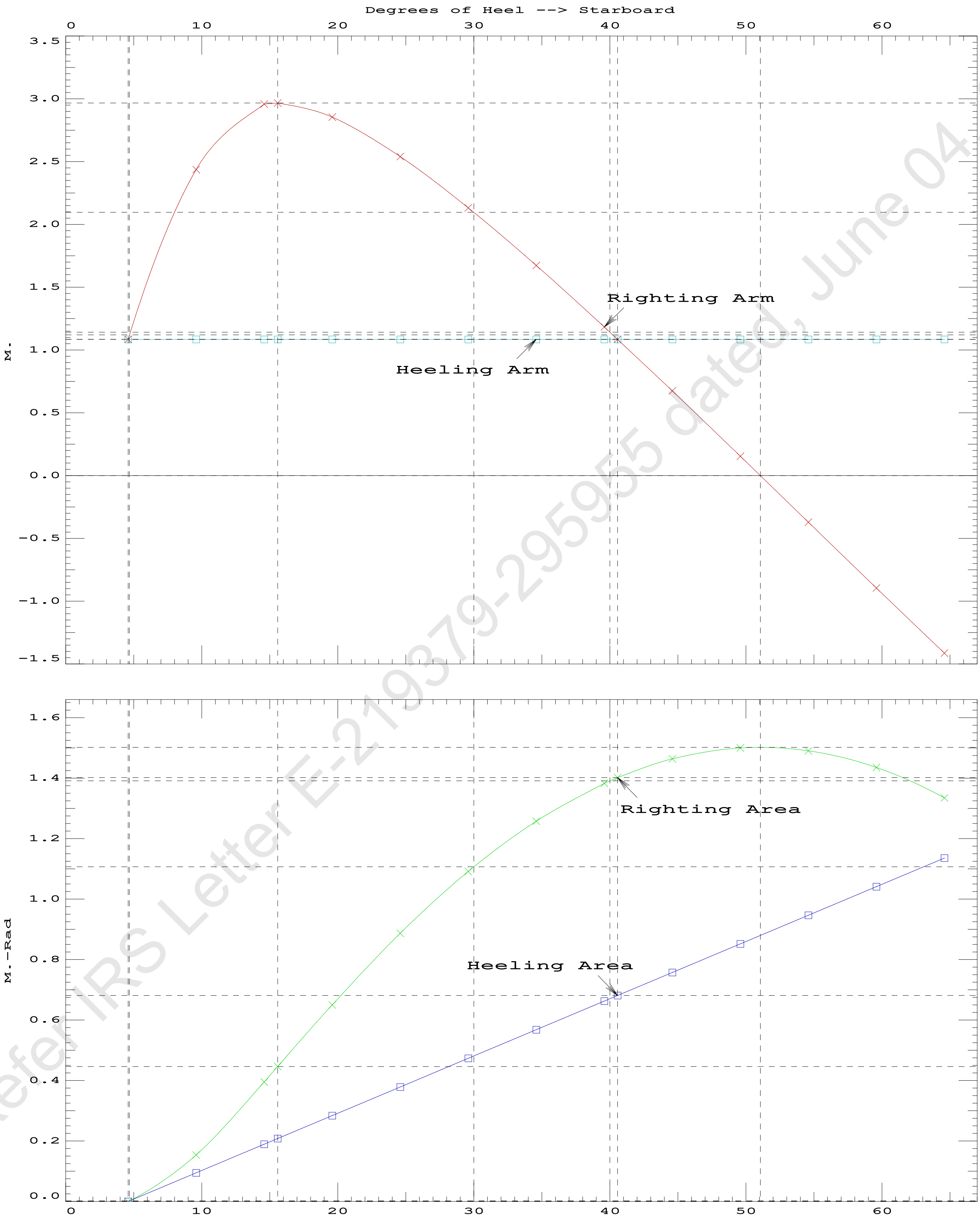
Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1200.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM------------STABILITY CRITERIA FOR CRANE OPERAT		----Min/Max		--------Attained	
(1) Absolute Angle at Equilibrium	<	15.00	deg	4.59	P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	173.5	P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		2.065	P

Condition 05 : Operating Condition of 400T Crane

Crane Lifting 40.0T @ 30m Radius Stbd



Condition 05 :Operating Condition of 400T Crane

Crane Lifting 40.0T @ 30m Radius Stbd

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.058 @ 50.00f, 1.333 @ 25.00f, 1.608 @ 0.00

Trim: Aft 0.548/50.000, Heel: Stbd 4.59 deg.

Part			Weight (MT)	LCG	TCG	VCG	
LIGHT SHIP			465.25	25.062f	0.000	3.000	
Crane			400.00	11.500f	0.000	6.325	
Crane Load			40.00	11.500f	0.000	34.602	
Total Fixed--------->			905.25	18.470f	0.000	5.866	
	Load	SpGr	Weight (MT)	LCG	TCG	VCG	FSM
06_WBTK.S	0.550	1.025	74.10	42.221f	6.111s	0.843	135.61*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	0.550	1.025	27.28	47.375f	6.165s	1.350	72.32*
Total Tanks--------->			201.09	45.618f	1.615s	1.424	352.57
Total Weight-------->			1,106.34	23.405f	0.294s	5.058	
			Displ (MT)	LCB	TCB	VCB	
HULL		1.025	1,106.35	23.356f	1.727s	0.756	

Righting Arms:	0.000	1.085s
External Arms:	0.000	1.085s
Residual Righting Arms:	0.000	0.000s

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	856.8	24.533f	0.180s	149.01	21.142*
		MT/cm				
		8.78				

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 4.59

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                         After hook load drop
                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 1.105 @ 50.00f, 1.296 @ 25.00f, 1.487 @ 0.00
                Trim: Aft 0.381/50.000,  Heel: Stbd 0.87 deg.
Part------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                           465.25  25.062f  0.000   3.000
Crane                                400.00  11.500f  0.000   6.325
  Total Fixed--------->              865.25  18.792f  0.000   4.537
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
06_WBTK.S      0.550    1.025          74.10  42.231f  5.992s  0.837    135.61
07_WBTK.P      1.000    1.025          49.60  47.660f  5.971p  1.880     72.32*
07_WBTK.C      1.000    1.025          50.11  47.663f  0.000   1.872     72.33*
07_WBTK.S      0.550    1.025          27.28  47.375f  5.995s  1.342     72.32*
  Total Tanks--------->               201.09  45.622f  1.548s  1.421    352.57
  Total Weight-------->             1,066.34  23.852f  0.292s  3.949
                                    Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025       1,066.34  23.827f  0.343s  0.661
------------------------------------------------------------------------------
               Righting Arms:                  0.000   0.001s
               External Arms:                  0.000   0.001s
      Residual Righting Arms:                  0.000   0.000s
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025         853.9  24.761f  0.063s   153.42  21.751*
                                     MT/cm--------m.-MT/cm-------GML-----GMT
                                      8.75            32.02    150.13  18.462
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


limmarg: 4063.92; t3: 30.0; t: 15.0
limmarg: 1566.79; t3: 15.0; t: 6.0
limmarg: 599.33; t3: 9.0; t: 2.4
limmarg: 304.87; t3: 6.6; t: 1.0
limmarg: 204.22; t3: 5.6; t: 0.4
limmarg: 167.87; t3: 5.2; t: 0.2
limmarg: 150.54; t3: 5.0; t: 0.1
limmarg: 142.09; t3: 4.9; t: 0.0
limmarg: 142.09; t3: 4.9; t: 0.0

                  Theta3 is 4.90 degrees, as demonstrated below.

                        RESIDUAL RIGHTING ARMS vs HEEL ANGLE
                 Fixed CG:  LCG = 18.792f  TCG = 0.000   VCG =  4.537

        Origin    Degrees of   Displacement    Residual Arms       Res.
         Depth---Trim----Heel----Weight(MT)---in Trim--in Heel---> Area
         1.463   0.44a   4.59s     1,066.3      0.000   -1.219    0.0000
         1.475   0.44a   0.87s     1,066.3      0.000    0.000   -0.0395
         1.465   0.44a   4.13p     1,066.3      0.000    1.646    0.0322
         1.461   0.44a   4.90p     1,066.3      0.000    1.898    0.0561
         1.431   0.46a   9.13p     1,066.3      0.000    3.181    0.2440
         1.351   0.63a  14.13p     1,066.3      0.000    3.916    0.5578
         1.305   0.77a  17.02p     1,066.3      0.000    3.978    0.7580
```

1.270	0.87a	19.13p	1,066.3	0.000	3.944	0.9039
1.183	1.10a	24.13p	1,066.3	0.000	3.726	1.2415
1.092	1.34a	29.13p	1,066.3	0.000	3.399	1.5532

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 865.25 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM---STABILITY CRITERION		Min/Max	Attained
(1) Res. Area Ratio from abs -4.6 deg to abs 4.9	>	1.000	2.421 P

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 18.792f TCG = 0.000 VCG = 4.537

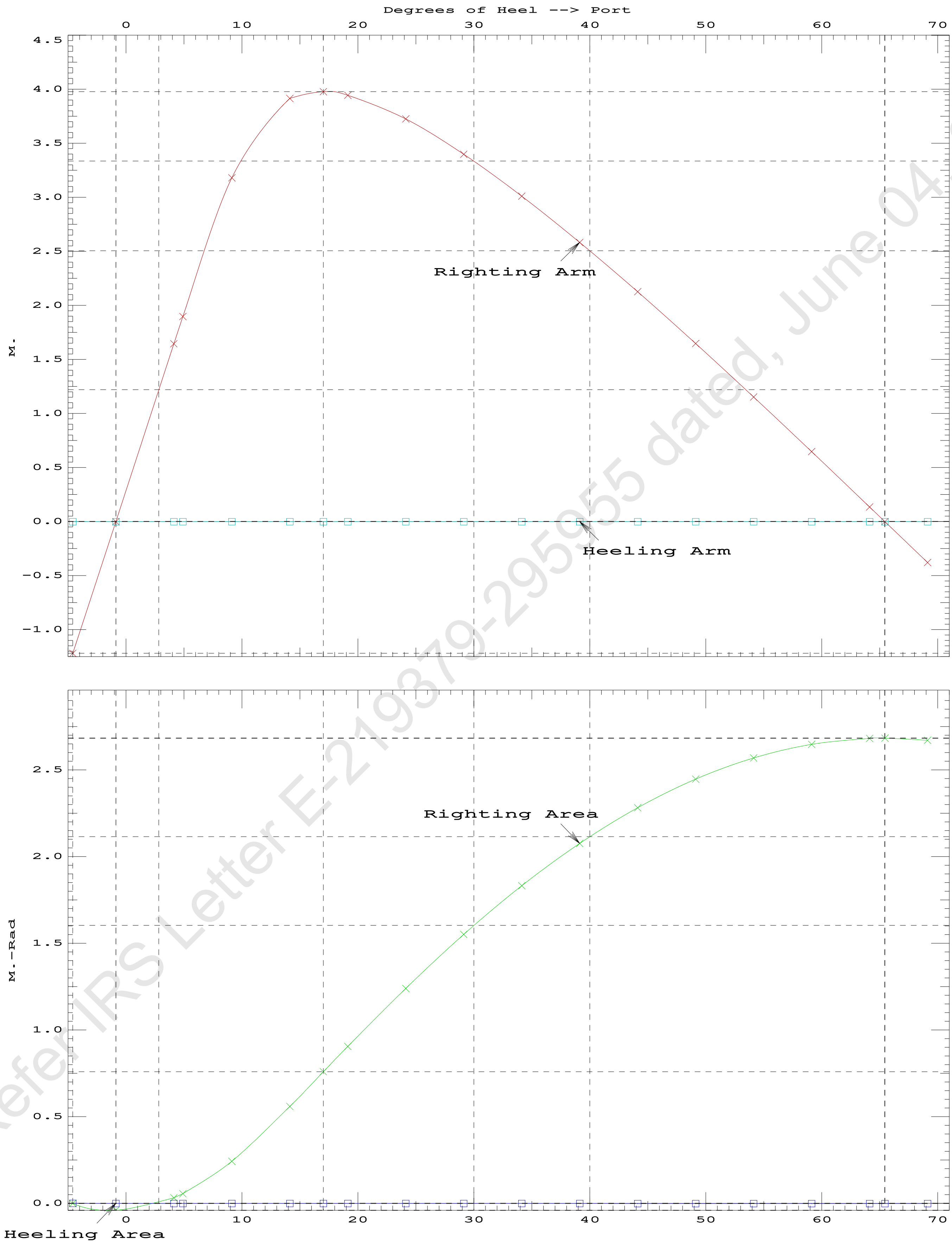
Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight (MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.463	0.44a	4.59s	1,066.3	0.000	-1.219	0.0000
1.475	0.44a	0.87s	1,066.3	0.000	0.000	-0.0395
1.465	0.44a	4.13p	1,066.3	0.000	1.646	0.0322
1.461	0.44a	4.90p	1,066.3	0.000	1.898	0.0561
1.431	0.46a	9.13p	1,066.3	0.000	3.181	0.2440
1.351	0.63a	14.13p	1,066.3	0.000	3.916	0.5578
1.305	0.77a	17.02p	1,066.3	0.000	3.978	0.7580
1.270	0.87a	19.13p	1,066.3	0.000	3.944	0.9039
1.183	1.10a	24.13p	1,066.3	0.000	3.726	1.2415
1.092	1.34a	29.13p	1,066.3	0.000	3.399	1.5532
0.998	1.58a	34.13p	1,066.3	0.000	3.012	1.8334
0.900	1.82a	39.13p	1,066.3	0.000	2.584	2.0778
0.799	2.05a	44.13p	1,066.3	0.000	2.127	2.2835
0.695	2.27a	49.13p	1,066.3	0.000	1.648	2.4484
0.587	2.48a	54.13p	1,066.3	0.000	1.153	2.5707
0.475	2.67a	59.13p	1,066.3	0.000	0.647	2.6494
0.358	2.84a	64.13p	1,066.3	0.000	0.135	2.6836
0.327	2.88a	65.44p	1,066.3	0.000	0.000	2.6851
0.235	2.97a	69.13p	1,066.3	0.000	-0.379	2.6730

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 865.25 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM--------------RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Res. Area Ratio from abs -4.6 deg to RAzero	>	1.000		68.955 P
(2) Angle from abs 4.9 deg to RAzero	>	20.00	deg	60.54 P


Degrees of Heel --> Port
0
10
20
30
40
50
60
70
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
-0.5
-1.0
M.
Righting Arm
Heeling Arm
2.5
2.0
1.5
1.0
0.5
0.0
M.-Rad
Righting Area
Heeling Area

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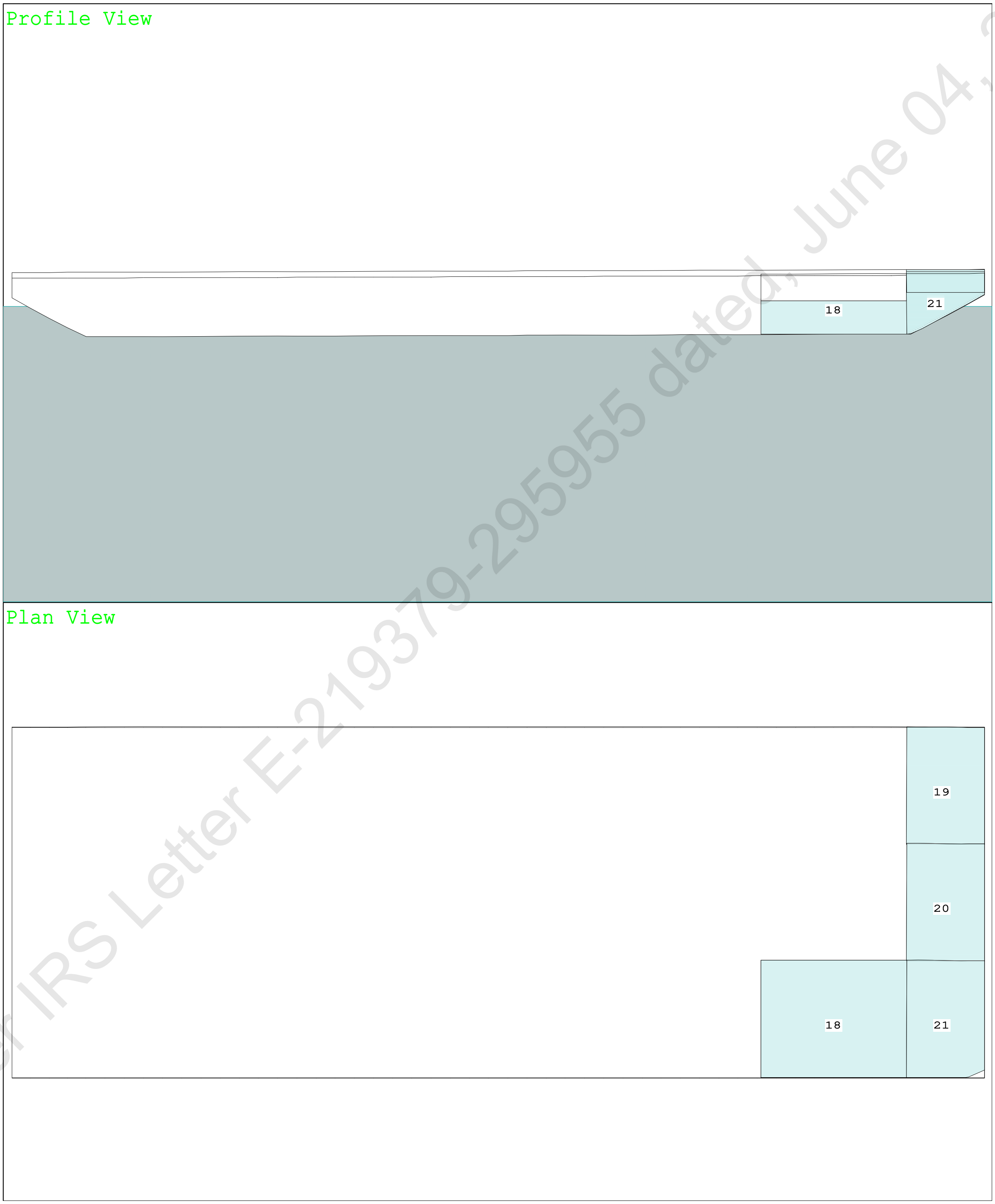
GHS 19.00 **PONTOON**

Condition 06 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Fwd

CG - Draft: 1.260 @ 50.000f, 1.424 @ 0.000 Heel: stbd 0.93 deg.



Tanks
18 06_WBTK.S..55% SALT WATER Intact
19 07_WBTK.P.100% SALT WATER Intact
20 07_WBTK.C.100% SALT WATER Intact
21 07_WBTK.S..55% SALT WATER Intact

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29-04-24 11:48:13    Cybermarine Technologies PTE. Ltd.
GHS 19.00                           PONTOON
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Condition 06 :Operating Condition of 400T Crane

Crane Lifting 40.0T @ 30m Radius Fwd

```
                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 1.260 @ 50.00f, 1.342 @ 25.00f, 1.424 @ 0.00
                Trim: Aft 0.164/50.000,  Heel: Stbd 0.93 deg.
Part---------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                              465.25  25.062f  0.000   3.000
Crane                                   400.00  11.500f  0.000   6.325
                                                41.500f
Crane Load                               40.00  42.500f  0.000  34.602
  Total Fixed--------->                 905.25  19.840f  0.000   5.866
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
06_WBTK.S      0.550    1.025            74.10  42.243f  5.994s  0.837    135.61
07_WBTK.P      1.000    1.025            49.60  47.660f  5.971p  1.880     72.32*
07_WBTK.C      1.000    1.025            50.11  47.663f  0.000   1.872     72.33*
07_WBTK.S      0.550    1.025            27.28  47.380f  5.998s  1.342     72.32*
  Total Tanks--------->                 201.09  45.627f  1.549s  1.421    352.57
  Total Weight-------->               1,106.34  24.527f  0.282s  5.058
                                      Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025         1,106.35  24.513f  0.353s  0.681
------------------------------------------------------------------------------
               Righting Arms:                    0.000   0.000s
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025             852.7  24.905f  0.083s   147.24  21.018*
                                          MT/cm--------m.-MT/cm-------GML-----GMT
                                           8.74            31.61    142.86  16.641
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


                         SUMMARY OF LOADING
       196.2 Cu.M.   (8%) SALT WATER

      400.00 MT  Crane                      40.00 MT  Crane Load

DRAFT AT FP (M) =     1.260
DRAFT AT MS (M) =     1.342
DRAFT AT AP (M) =     1.424
DRAFT AT FWD MARK (M) =    1.273
DRAFT AT AFT MARK (M) =    1.411
                       HYDROSTATIC PROPERTIES
          Trim: Aft 0.164/50.000, Heel: Stbd 0.93 deg., VCG = 5.058

   LCF   Displacement   Buoyancy-Ctr.  Weight/         Moment/
  Draft----Weight(MT)----LCB-----VCB-------cm-----LCF---cm trim----GML-----GMT
  1.343     1,106.35  24.513f  0.681     8.74  24.905f   31.61  142.86  16.641
 Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
                         Trim is per 50.00m.
 Draft is from USK.                             Formal Free Surface included.

     Note:  GMT includes the formal free surface moment 352.6 m.-MT
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GHS 19.00 **PONTOON**

Condition 06 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Fwd

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 1.262 @ 50.00f, 1.344 @ 25.00f, 1.426 @ 0.00
Trim: Aft 0.164/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	83.1	0.847	1.505	0.05500	6.88
CRANE	57.0	4.120	4.778	0.05500	14.98
Total wind heeling moment to starboard------------->					21.86

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 24.527f TCG = 0.282s VCG = 5.058
Free Surface Adjustment: 0.319
Adjusted CG: LCG = 24.528f TCG = 0.276s VCG = 5.376

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area	50% DeckImm Angle
1.413	0.19a	0.00	1,106.3	0.000	-0.296	0.0000	5.171
1.412	0.19a	1.00s	1,106.3	0.000	0.000	-0.0026	4.172
1.397	0.19a	5.17s	1,106.3	0.000	1.229	0.0422	0.000
1.393	0.19a	6.00s	1,106.3	0.000	1.469	0.0616	-0.828
1.329	0.21a	11.00s	1,106.3	0.000	2.699	0.2451	-5.828
1.228	0.30a	15.54s	1,106.3	0.000	2.989	0.4755	-10.374
1.218	0.31a	16.00s	1,106.3	0.000	2.987	0.4991	-10.828
1.100	0.42a	21.00s	1,106.3	0.000	2.803	0.7563	-15.828
0.975	0.52a	26.00s	1,106.3	0.000	2.456	0.9870	-20.828
0.846	0.63a	31.00s	1,106.3	0.000	2.031	1.1834	-25.828
0.712	0.73a	36.00s	1,106.3	0.000	1.562	1.3405	-30.828
0.574	0.84a	41.00s	1,106.3	0.000	1.065	1.4553	-35.828
0.433	0.94a	46.00s	1,106.3	0.000	0.550	1.5259	-40.828
0.289	1.03a	51.00s	1,106.3	0.000	0.025	1.5510	-45.828
0.283	1.03a	51.23s	1,106.3	0.000	0.000	1.5511	-46.064
0.143	1.11a	56.00s	1,106.3	0.000	-0.504	1.5301	-50.828
-0.005	1.19a	61.00s	1,106.3	0.000	-1.033	1.4631	-55.828

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 352.6 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.86 (constant)

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GHS 19.00 **PONTOON**

Condition 06 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Fwd

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LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max--------Attained
(1) Residual Area from abs 0 deg to MaxRA         >   0.0800 m.-Rad 0.4755 P
(2) Angle from Equilibrium to RAzero or Flood     >    20.00  deg    50.24 P
(3) Angle from Equilibrium to 50% Dk Imm Angle    >     0.00  deg     4.17 P
-----------------------------------------------------------------------------
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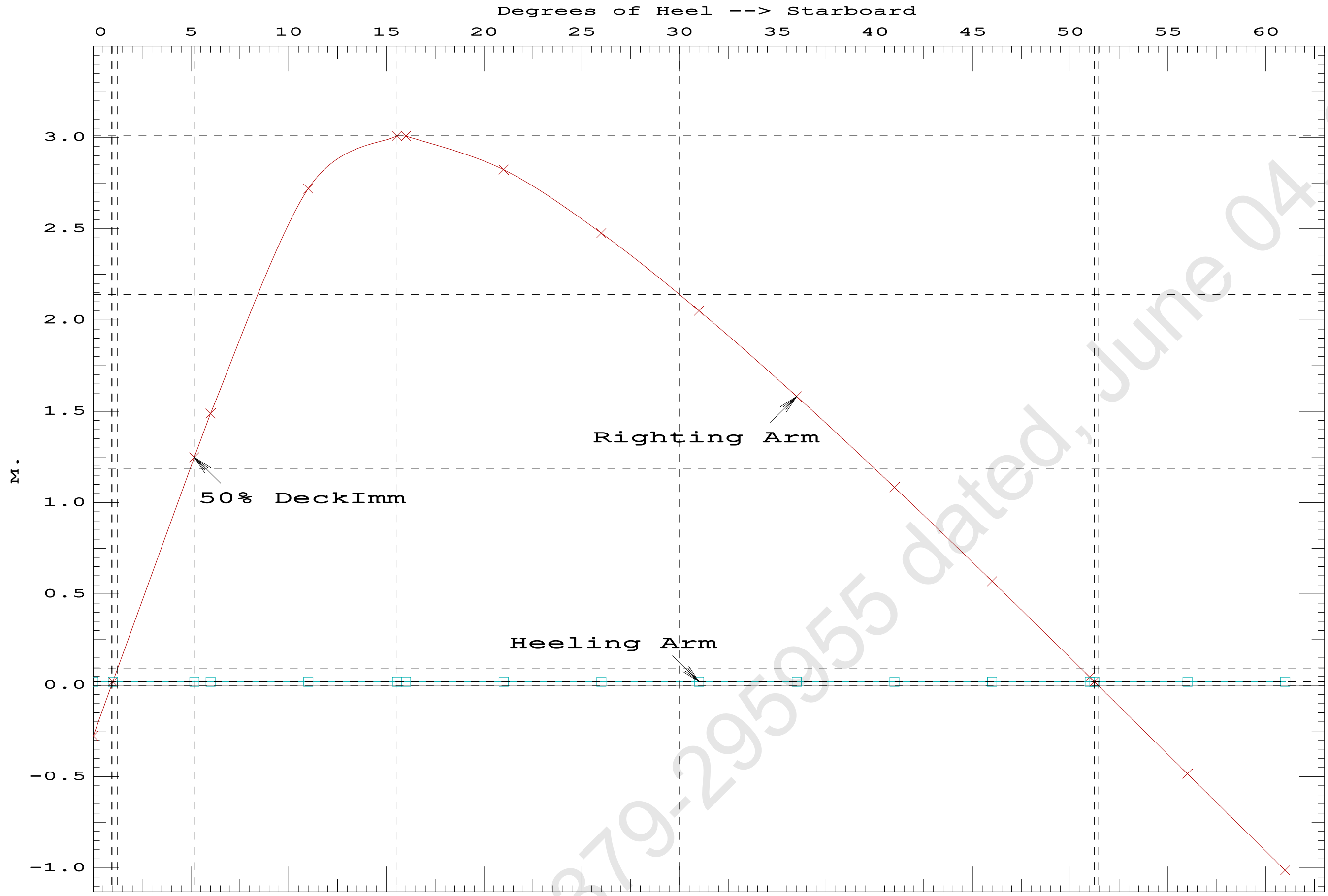
GHS 19.00 **PONTOON**

Condition 06 :Operating Condition of 400T Crane

Crane Lifting 40.0T @ 30m Radius Fwd



29-04-24 11:48:13 Cybermarine Technologies PTE. Ltd.
GHS 19.00 **PONTOON**

Condition 06 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.527f TCG = 0.282s VCG = 5.058
Free Surface Adjustment: 0.319
Adjusted CG: LCG = 24.528f TCG = 0.276s VCG = 5.376

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.412	0.19a	0.93s	1,106.3	0.000	0.000	0.0000
1.393	0.19a	5.93s	1,106.3	0.000	1.470	0.0641
1.330	0.21a	10.93s	1,106.3	0.000	2.708	0.2481
1.228	0.30a	15.54s	1,106.3	0.000	3.007	0.4827
1.219	0.31a	15.93s	1,106.3	0.000	3.006	0.5034
1.101	0.41a	20.93s	1,106.3	0.000	2.825	0.7613
0.977	0.52a	25.93s	1,106.3	0.000	2.480	0.9940
0.848	0.62a	30.93s	1,106.3	0.000	2.056	1.1925
0.714	0.73a	35.93s	1,106.3	0.000	1.587	1.3517
0.576	0.83a	40.93s	1,106.3	0.000	1.090	1.4687
0.435	0.93a	45.93s	1,106.3	0.000	0.575	1.5415
0.291	1.03a	50.93s	1,106.3	0.000	0.050	1.5689
0.277	1.04a	51.41s	1,106.3	0.000	0.000	1.5691
0.145	1.11a	55.93s	1,106.3	0.000	-0.479	1.5502
-0.003	1.19a	60.93s	1,106.3	0.000	-1.007	1.4854

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 352.6 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

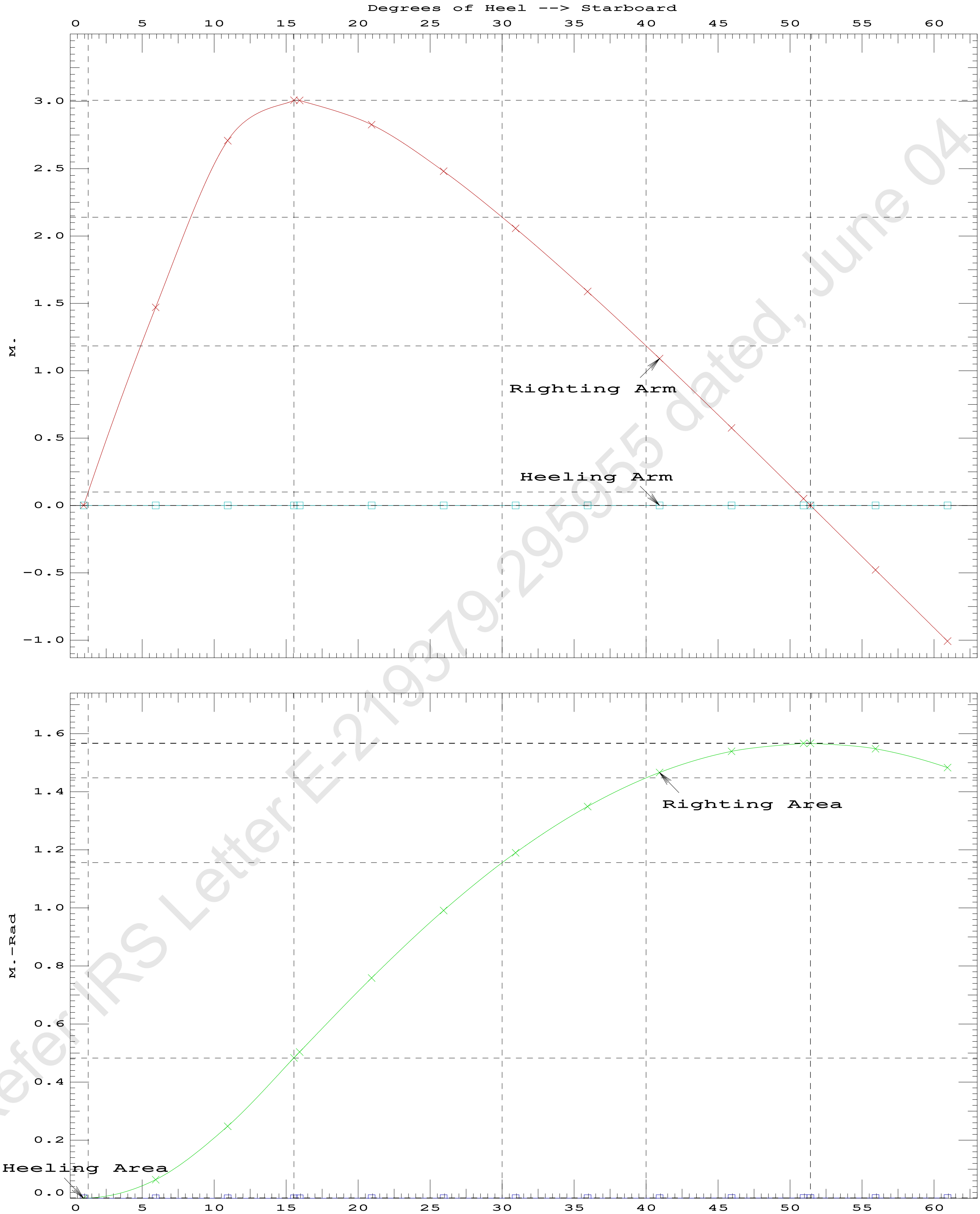
LIM------------STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained	
(1) Absolute Angle at Equilibrium	<	15.00	deg	0.93	P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	332730	P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		1971.51	P

29-04-24 11:48:13 Cybermarine Technologies PTE. Ltd.

GHS 19.00 **PONTOON**

Condition 06 :Operating Condition of 400T Crane

Crane Lifting 40.0T @ 30m Radius Fwd



Condition 06 :Operating Condition of 400T Crane

Crane Lifting 40.0T @ 30m Radius Fwd

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.260 @ 50.00f, 1.342 @ 25.00f, 1.424 @ 0.00

Trim: Aft 0.164/50.000, Heel: Stbd 0.95 deg.

Part	Load	SpGr	Weight (MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			465.25	25.062f	0.000	3.000	
Crane			400.00	11.500f	0.000	6.325	
Crane Load			40.00	~~42.500f~~ 41.500f	0.000	34.602	
Total Fixed--------->			905.25	19.840f	0.000	5.866	
06_WBTK.S	0.550	1.025	74.10	42.243f	5.994s	0.837	135.61
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	0.550	1.025	27.28	47.380f	5.998s	1.342	72.32*
Total Tanks--------->			201.09	45.627f	1.550s	1.421	352.57
Total Weight-------->			1,106.34	24.527f	0.282s	5.058	
			Displ (MT)	**LCB**	**TCB**	**VCB**	
HULL		1.025	1,106.35	24.513f	0.359s	0.681	

Righting Arms:	0.000	0.005s
External Arms:	0.000	0.005s
Residual Righting Arms:	0.000	0.000s

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	852.8 MT/cm 8.74	24.905f	0.084s	147.27	21.020*

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.95

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29-04-24 11:47:48    Cybermarine Technologies PTE. Ltd.
GHS 19.00                          PONTOON

                     After hook load drop
           WEIGHT and DISPLACEMENT and WATERPLANE STATUS
      USK draft: 1.105 @ 50.00f, 1.296 @ 25.00f, 1.487 @ 0.00
           Trim: Aft 0.382/50.000,  Heel: Stbd 0.88 deg.
Part------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                           465.25  25.062f  0.000   3.000
Crane                                400.00  11.500f  0.000   6.325
  Total Fixed--------->              865.25  18.792f  0.000   4.537
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
06_WBTK.S      0.550    1.025         74.10  42.231f  5.992s  0.837    135.61
07_WBTK.P      1.000    1.025         49.60  47.660f  5.971p  1.880     72.32*
07_WBTK.C      1.000    1.025         50.11  47.663f  0.000   1.872     72.33*
07_WBTK.S      0.550    1.025         27.28  47.375f  5.996s  1.342     72.32*
  Total Tanks--------->              201.09  45.622f  1.549s  1.421    352.57
  Total Weight-------->            1,066.34  23.852f  0.292s  3.949
                                   Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025      1,066.31  23.826f  0.347s  0.661
------------------------------------------------------------------------------
               Righting Arms:                  0.001   0.005s
               External Arms:                  0.000   0.005s
      Residual Righting Arms:                  0.001   0.000s
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025          854.0  24.762f  0.063s   153.45  21.753*
                                       MT/cm
                                        8.75
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.
```

```
limmarg: 14672817.00; t3: 30; t: 15.00
limmarg: 5875573.00; t3: 15; t: 13.50
limmarg: 146255.66; t3: 2; t: 12.15
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limmarg: 85761.54; t3: 1; t: 9.84
limmarg: ; t3: -9; t: 8.86
limmarg: 14093.49; t3: 0; t: 7.97
limmarg: ; t3: -8; t: 7.17
limmarg: -23.84; t3: -1; t: 6.46
limmarg: 1029044.94; t3: 5; t: 5.81
limmarg: 42.27; t3: -1; t: 5.23
limmarg: ; t3: -6; t: 4.71
limmarg: ; t3: -1; t: 4.24
limmarg: 435709.31; t3: 3; t: 3.81
limmarg: 42.27; t3: -1; t: 3.43
limmarg: ; t3: -4; t: 3.09
limmarg: -83.38; t3: -1; t: 2.78
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limmarg: 3683.62; t3: -1; t: 2.25
limmarg: ; t3: -3; t: 2.03
limmarg: ; t3: -1; t: 1.82
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limmarg: ; t3: -2; t: 1.08
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limmarg: ; t3: -1; t: 0.52
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limmarg: 4511.32; t3: 0; t: 0.42
limmarg: 5420.81; t3: 0; t: 0.38
limmarg: 6412.14; t3: 0; t: 0.34
limmarg: 7485.39; t3: 0; t: 0.30
limmarg: 8640.70; t3: 0; t: 0.27
limmarg: 9561.21; t3: 0; t: 0.25
limmarg: 10200.71; t3: 0; t: 0.22
limmarg: 11198.87; t3: 0; t: 0.20
limmarg: 11890.46; t3: 0; t: 0.18
limmarg: 12603.18; t3: 0; t: 0.16
limmarg: 13337.36; t3: 0; t: 0.15
limmarg: 13712.65; t3: 0; t: 0.13
limmarg: 14480.02; t3: 0; t: 0.12
limmarg: 14872.41; t3: 0; t: 0.11
limmarg: 15270.84; t3: 0; t: 0.10
limmarg: 15675.65; t3: 0; t: 0.09
limmarg: 16087.26; t3: 0; t: 0.08
limmarg: 16506.37; t3: 0; t: 0.07
limmarg: 16933.41; t3: 0; t: 0.06
limmarg: 17366.28; t3: 0; t: 0.06
limmarg: 17366.28; t3: 0; t: 0.05
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limmarg: 18248.20; t3: 0; t: 0.03
limmarg: 18697.29; t3: 0; t: 0.03
limmarg: 18697.29; t3: 0; t: 0.03
limmarg: 18697.29; t3: 0; t: 0.02
limmarg: 19151.80; t3: 0; t: 0.02
limmarg: 19151.80; t3: 0; t: 0.02

Theta3 is -0.02 degrees, as demonstrated below.

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 18.792f TCG = 0.000 VCG = 4.537

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight (MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.475	0.44a	0.95s	1,066.3	0.000	-0.021	0.0000
1.475	0.44a	0.88s	1,066.3	0.000	0.000	-0.0000
1.476	0.44a	0.02s	1,066.3	0.000	0.286	0.0022
1.465	0.44a	4.12p	1,066.3	0.000	1.646	0.0719
1.431	0.46a	9.12p	1,066.3	0.000	3.182	0.2833
1.351	0.63a	14.12p	1,066.3	0.000	3.919	0.5990
1.305	0.77a	17.02p	1,066.3	0.000	3.981	0.8002
1.270	0.87a	19.12p	1,066.3	0.000	3.949	0.9454
1.184	1.10a	24.12p	1,066.3	0.000	3.730	1.2833
1.092	1.34a	29.12p	1,066.3	0.000	3.404	1.5954

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 865.25 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

LIM	STABILITY CRITERION		Min/Max	Attained
(1)	Res. Area Ratio from abs -0.9 deg to abs 0	>	1.000	192.518 P

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 18.792f TCG = 0.000 VCG = 4.537

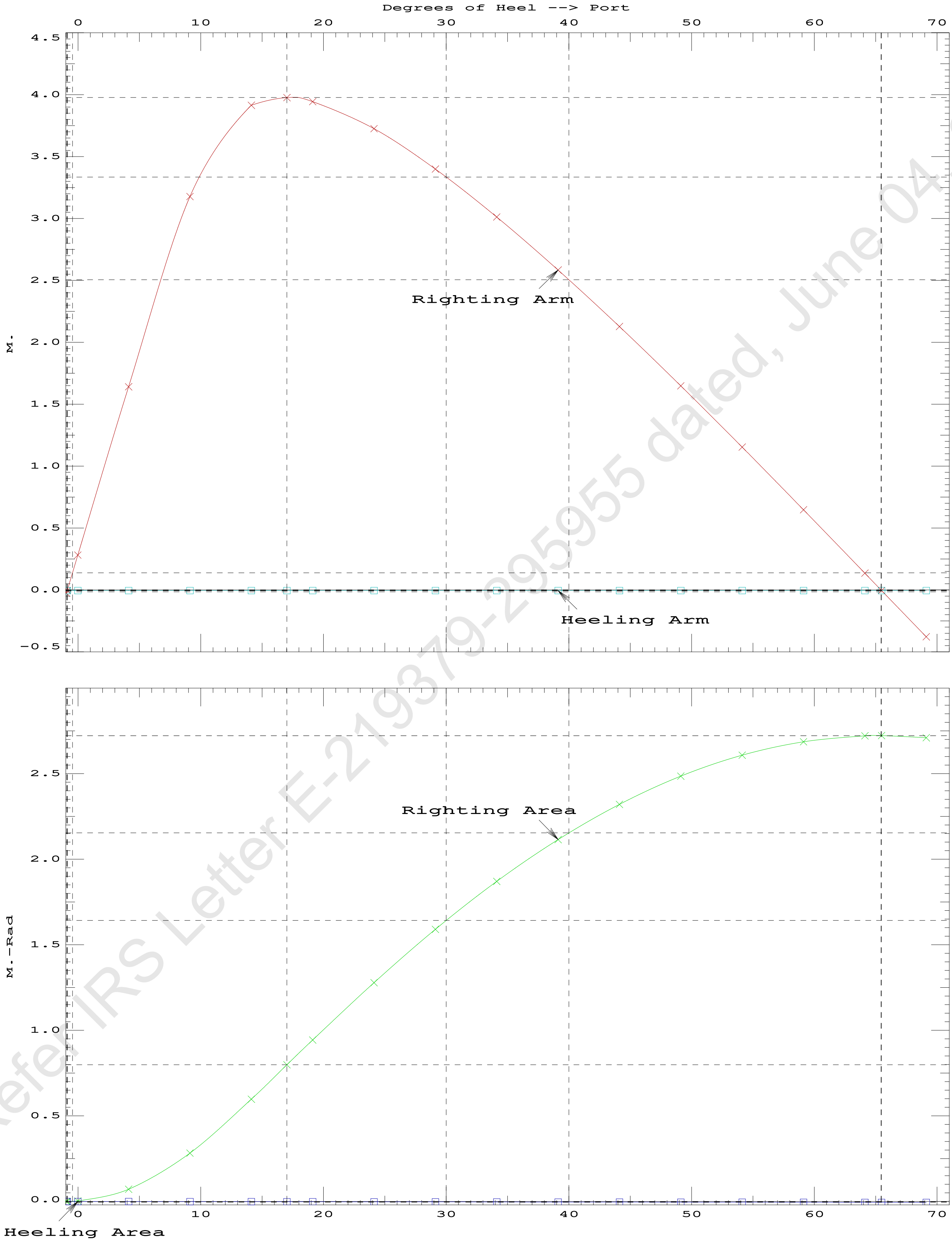
Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.475	0.44a	0.95s	1,066.3	0.000	-0.021	0.0000
1.475	0.44a	0.88s	1,066.3	0.000	0.000	-0.0000
1.476	0.44a	0.02s	1,066.3	0.000	0.286	0.0022
1.465	0.44a	4.12p	1,066.3	0.000	1.646	0.0719
1.431	0.46a	9.12p	1,066.3	0.000	3.182	0.2833
1.351	0.63a	14.12p	1,066.3	0.000	3.919	0.5990
1.305	0.77a	17.02p	1,066.3	0.000	3.981	0.8002
1.270	0.87a	19.12p	1,066.3	0.000	3.949	0.9454
1.184	1.10a	24.12p	1,066.3	0.000	3.730	1.2833
1.092	1.34a	29.12p	1,066.3	0.000	3.404	1.5954
0.998	1.58a	34.12p	1,066.3	0.000	3.016	1.8760
0.900	1.82a	39.12p	1,066.3	0.000	2.589	2.1208
0.799	2.05a	44.12p	1,066.3	0.000	2.132	2.3270
0.695	2.27a	49.12p	1,066.3	0.000	1.653	2.4923
0.588	2.48a	54.12p	1,066.3	0.000	1.158	2.6150
0.475	2.67a	59.12p	1,066.3	0.000	0.652	2.6941
0.358	2.83a	64.12p	1,066.3	0.000	0.140	2.7287
0.326	2.88a	65.48p	1,066.3	0.000	0.000	2.7304
0.235	2.97a	69.12p	1,066.3	0.000	-0.374	2.7185

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 865.25 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

LIM--------------RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Res. Area Ratio from abs -0.9 deg to RAzero	>	1.000		243194 P
(2) Angle from abs 0 deg to RAzero	>	20.00	deg	65.50 P


Degrees of Heel --> Port
0
10
20
30
40
50
60
70
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
-0.5
M.
Righting Arm
Heeling Arm
2.5
2.0
1.5
1.0
0.5
0.0
M.-Rad
Righting Area
Heeling Area

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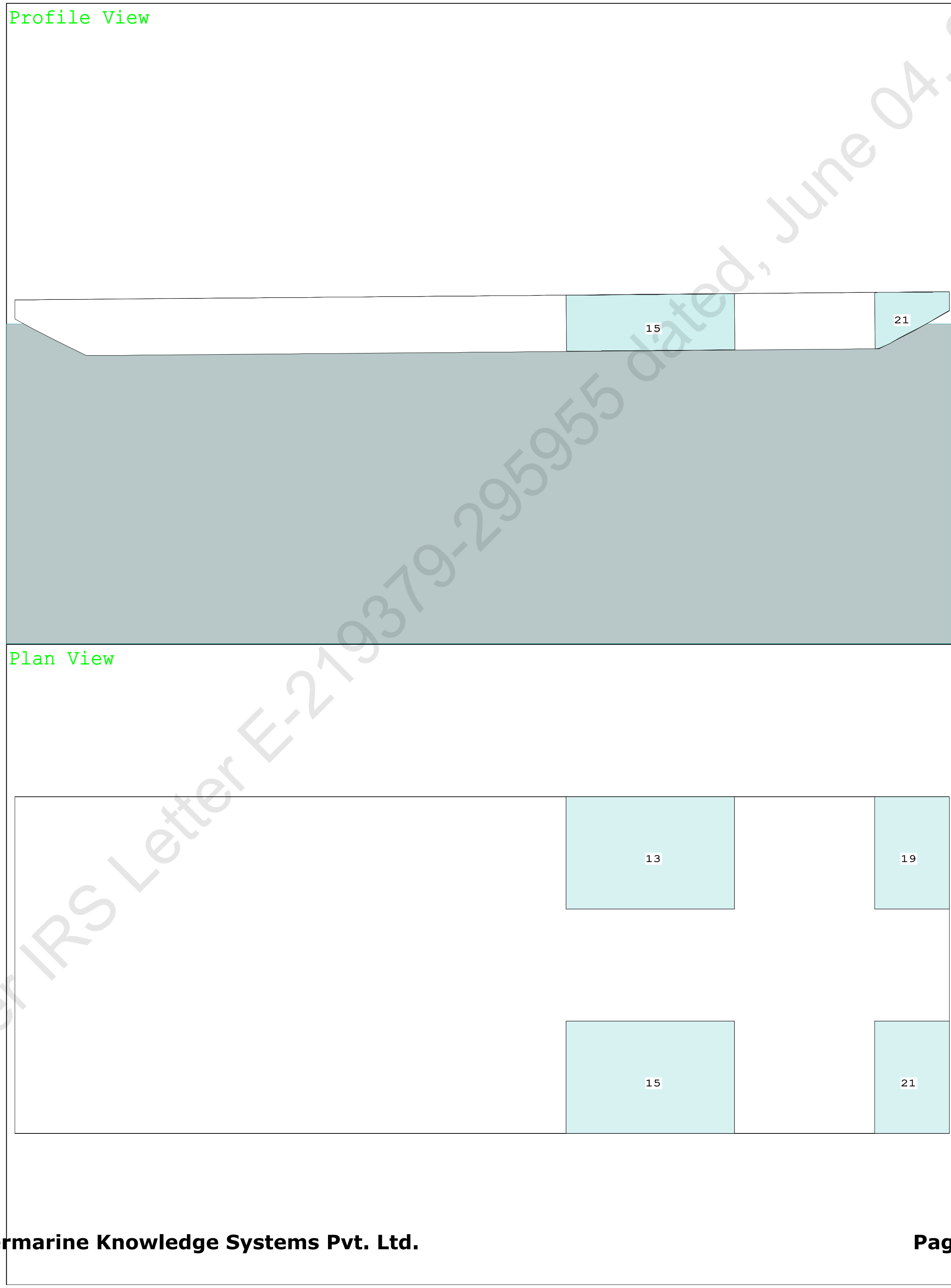
GHS 18.90 **PONTOON**

Condition 07 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Aft

Condition Graphic - Draft: 1.317 @ 50.000f, 1.754 @ 0.000 Heel: zero



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GHS 18.90 **PONTOON**

Condition 07 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Aft

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.317 @ 50.00f, 1.536 @ 25.00f, 1.754 @ 0.00
Trim: Aft 0.437/50.000, Heel: zero

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			465.25	25.062f	0.000	3.000	
Crane			350.00	11.500f	0.000	6.325	
Crane Load			40.00	23.500a	0.000	21.478	
Total Fixed--------->			855.25	17.241f	0.000	5.225	
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks--------->			422.40	37.208f	0.000	1.596	470.09
Total Weight-------->			1,277.65	23.842f	0.000	4.025	

	SpGr	Displ(MT)	LCB	TCB	VCB
HULL	1.025	1,277.65	23.814f	0.000	0.784
Righting Arms:				0.000	0.000

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	870.2	24.621f	0.000	135.69	18.487*
		MT/cm		m.-MT/cm	GML	GMT
		8.92		33.84	132.45	15.245

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

412.1 Cu.M. (16%) SALT WATER

350.00 MT Crane 40.00 MT Crane Load

DRAFT AT FP (M) = 1.317
DRAFT AT MS (M) = 1.536
DRAFT AT AP (M) = 1.754
DRAFT AT FWD MARK (M) = 1.352
DRAFT AT AFT MARK (M) = 1.719

HYDROSTATIC PROPERTIES
Trim: Aft 0.437/50.000, No Heel, VCG = 4.025

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	Buoyancy-Ctr. VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
1.539	1,277.65	23.814f	0.784	8.92	24.621f	33.84	132.45	15.245

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 470.1 m.-MT

24-04-10 18:52:55 Cybermarine Technologies PTE. Ltd.
GHS 18.90 **PONTOON**

Condition 07 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Aft

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 1.317 @ 50.00f, 1.536 @ 25.00f, 1.754 @ 0.00
Trim: Aft 0.437/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	73.7	0.753	1.507	0.05500	6.11
CRANE	57.0	3.846	4.600	0.05500	14.42
Total wind heeling moment to starboard------------->					20.53

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 23.842f TCG = 0.000 VCG = 4.025
Free Surface Adjustment: 0.368
Adjusted CG: LCG = 23.845f TCG = 0.000 VCG = 4.393

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area	50% DeckImm Angle
1.742	0.50a	0.00	1,277.6	0.000	-0.016	0.0000	4.041
1.742	0.50a	0.05s	1,277.7	0.000	0.000	-0.0000	3.989
1.733	0.50a	4.04s	1,277.6	0.000	1.067	0.0371	0.000
1.729	0.51a	5.05s	1,277.7	0.000	1.336	0.0583	-1.011
1.708	0.55a	10.05s	1,277.6	0.000	2.586	0.2301	-6.011
1.773	0.83a	15.05s	1,277.7	0.000	3.025	0.4815	-11.011
1.785	0.88a	15.99s	1,277.7	0.000	3.031	0.5310	-11.946
1.834	1.11a	20.05s	1,277.7	0.000	2.940	0.7447	-16.011
1.886	1.40a	25.05s	1,277.7	0.000	2.683	0.9921	-21.011
1.930	1.70a	30.05s	1,277.7	0.000	2.347	1.2122	-26.011
1.966	2.00a	35.05s	1,277.7	0.000	1.964	1.4006	-31.011
1.992	2.29a	40.05s	1,277.7	0.000	1.551	1.5542	-36.011
2.007	2.58a	45.05s	1,277.7	0.000	1.118	1.6708	-41.011
2.009	2.85a	50.05s	1,277.7	0.000	0.671	1.7490	-46.011
1.997	3.10a	55.05s	1,277.7	0.000	0.216	1.7877	-51.011
1.986	3.21a	57.40s	1,277.7	0.000	0.000	1.7922	-53.360
1.968	3.33a	60.05s	1,277.7	0.000	-0.244	1.7865	-56.011

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 20.53 (constant)

24-04-10 18:52:55 Cybermarine Technologies PTE. Ltd.
GHS 18.90 **PONTOON**

Condition 07 : Operating Condition of 350T Crane

Crane Lifting 40.0T @ 35m Radius Aft

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LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max--------Attained
(1) Residual Area from abs 0 deg to MaxRA         >   0.0800 m.-Rad 0.5310 P
(2) Angle from Equilibrium to RAzero or Flood     >    20.00  deg    57.35 P
(3) Angle from Equilibrium to 50% Dk Imm Angle    >     0.00  deg     3.99 P
-----------------------------------------------------------------------------
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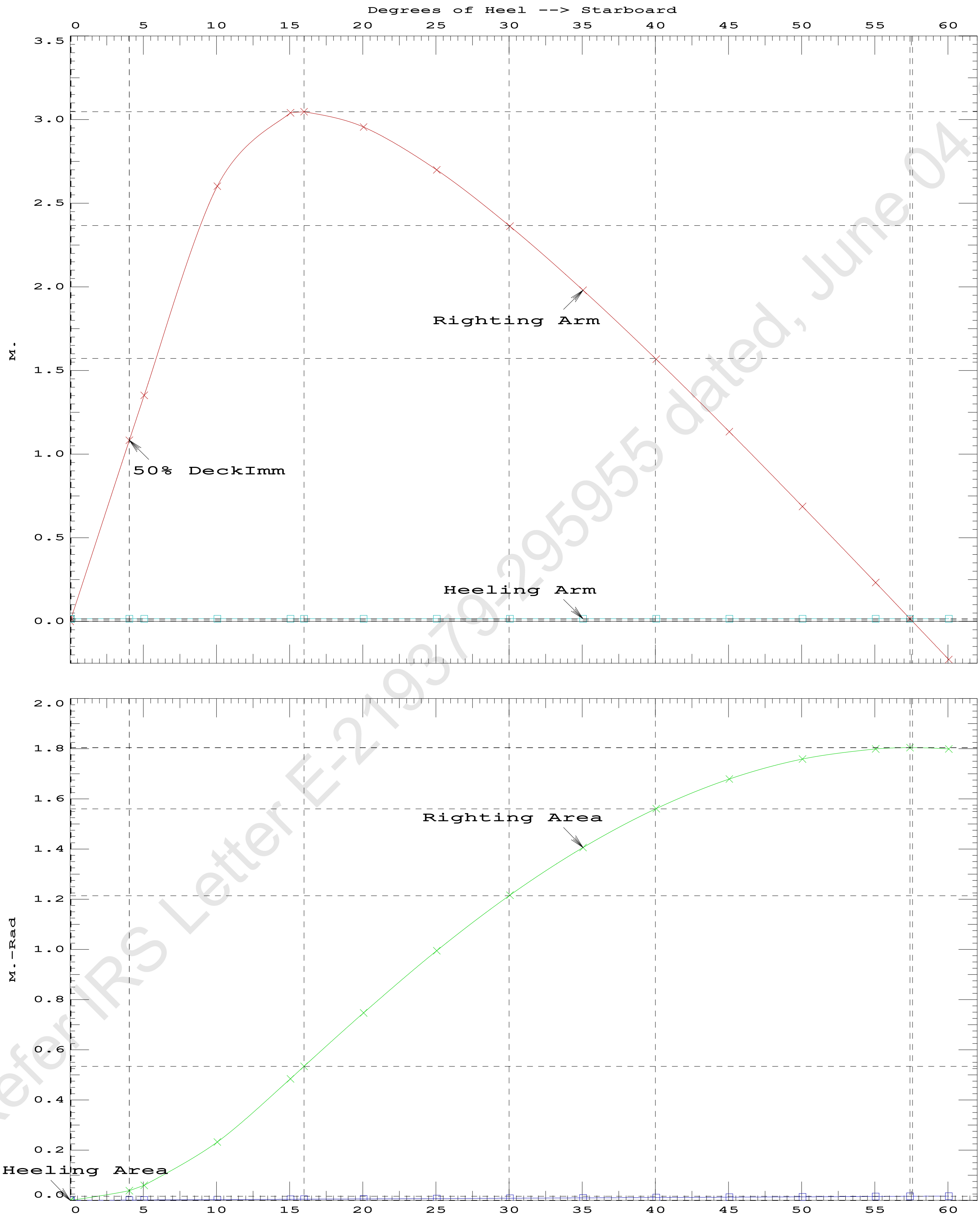
24-04-10 18:52:55 Cybermarine Technologies PTE. Ltd.
GHS 18.90 **PONTOON**

Condition 07 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Aft



24-04-10 18:52:55 Cybermarine Technologies PTE. Ltd.
GHS 18.90 **PONTOON**

Condition 07 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 23.842f TCG = 0.000 VCG = 4.025
Free Surface Adjustment: 0.368
Adjusted CG: LCG = 23.845f TCG = 0.000 VCG = 4.393

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.742	0.50a	0.00	1,277.6	0.000	-0.001	0.0000
1.730	0.51a	5.00s	1,277.7	0.000	1.337	0.0583
1.708	0.55a	10.00s	1,277.6	0.000	2.591	0.2303
1.773	0.82a	15.00s	1,277.7	0.000	3.039	0.4827
1.785	0.88a	16.00s	1,277.7	0.000	3.046	0.5355
1.833	1.11a	20.00s	1,277.7	0.000	2.957	0.7471
1.886	1.40a	25.00s	1,277.7	0.000	2.701	0.9961
1.930	1.70a	30.00s	1,277.7	0.000	2.366	1.2178
1.966	2.00a	35.00s	1,277.7	0.000	1.983	1.4079
1.992	2.29a	40.00s	1,277.7	0.000	1.571	1.5632
2.007	2.58a	45.00s	1,277.7	0.000	1.138	1.6815
2.010	2.85a	50.00s	1,277.7	0.000	0.691	1.7614
1.997	3.10a	55.00s	1,277.7	0.000	0.235	1.8019
1.985	3.22a	57.57s	1,277.7	0.000	0.000	1.8071
1.968	3.33a	60.00s	1,277.7	0.000	-0.224	1.8024

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM---STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained	
(1) Absolute Angle at Equilibrium	<	15.00	deg	0.00	P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	389167	P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		2299.08	P

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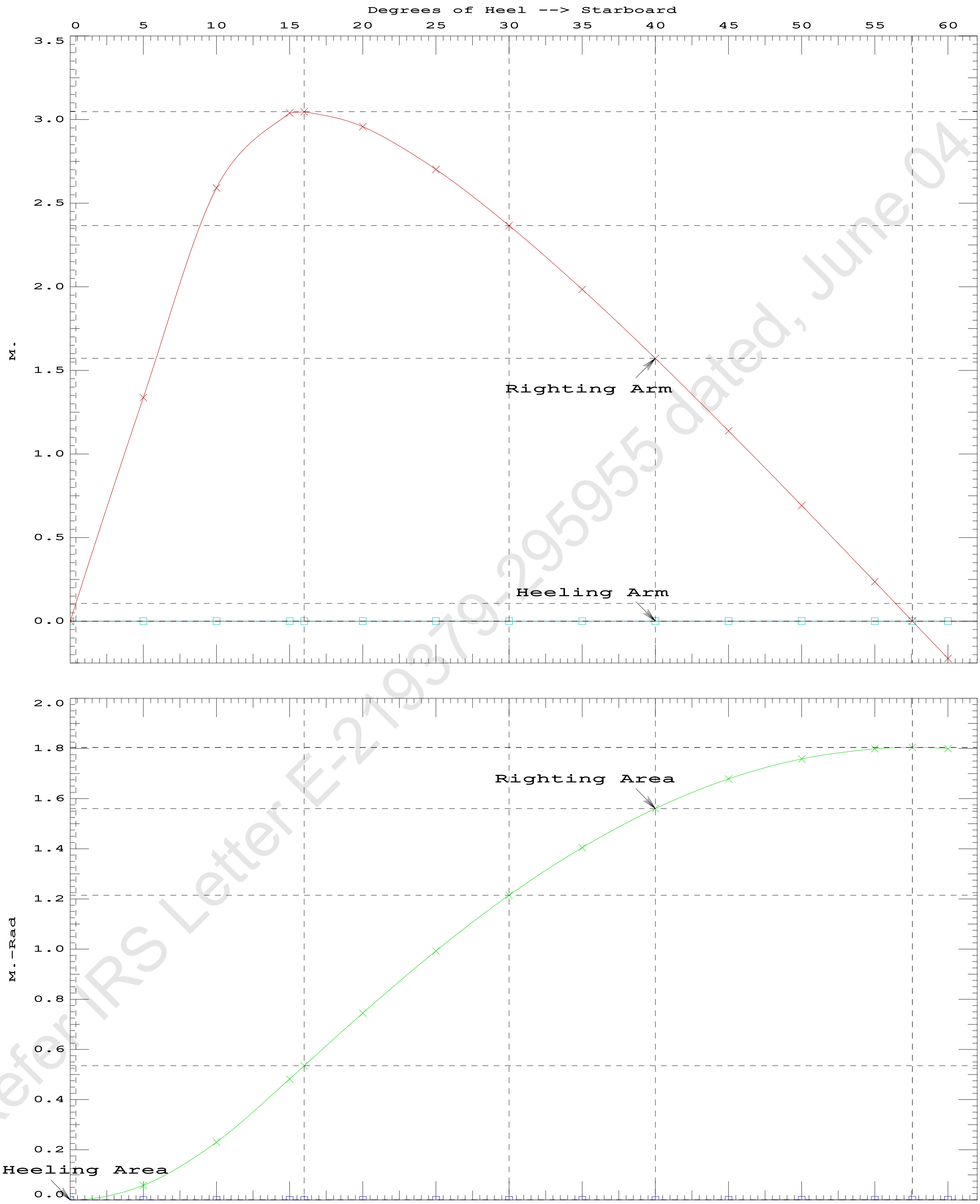
GHS 18.90 **PONTOON**

Condition 07 : Operating Condition of 350T Crane

Crane Lifting 40.0T @ 35m Radius Aft



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GHS 18.90 **PONTOON**

Condition 07 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Aft

Crane in operation with load in place
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.316 @ 50.00f, 1.536 @ 25.00f, 1.755 @ 0.00
Trim: Aft 0.439/50.000, Heel: 0.00 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			465.25	25.062f	0.000	3.000	
Crane			350.00	11.500f	0.000	6.325	
Crane Load			40.00	23.500a	0.000	21.478	
Total Fixed--------->			855.25	17.241f	0.000	5.225	
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks--------->			422.40	37.208f	0.000	1.596	470.09
Total Weight-------->			1,277.65	23.842f	0.000	4.025	
			Displ(MT)	LCB	TCB	VCB	
HULL		1.025	1,277.66	23.810f	0.003p	0.784	

Righting Arms:	0.003a	-0.003s
External Arms:	0.000	0.001
Residual Righting Arms:	0.003a	-0.004s

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	863.8 MT/cm 8.85	24.566f	0.045p	132.72	18.314*

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

```
24-04-10 18:55:39    Cybermarine Technologies PTE. Ltd.
GHS 18.90                       PONTOON

                          After hook load drop
                 WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 1.561 @ 50.00f, 1.492 @ 25.00f, 1.424 @ 0.00
                     Trim: Fwd 0.137/50.000,  Heel: zero
Part----------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                              465.25  25.062f  0.000   3.000
Crane                                   350.00  11.500f  0.000   6.325
  Total Fixed--------->                 815.25  19.240f  0.000   4.427
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025           161.60  34.000f  5.980p  1.509    162.73*
05_WBTK.S      1.000    1.025           161.60  34.000f  5.980s  1.509    162.73*
07_WBTK.P      1.000    1.025            49.60  47.660f  5.971p  1.880     72.32*
07_WBTK.S      1.000    1.025            49.60  47.660f  5.971s  1.880     72.32*
  Total Tanks--------->                 422.40  37.208f  0.000   1.596    470.09
  Total Weight-------->               1,237.65  25.372f  0.000   3.461
                                      Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025         1,237.64  25.380f  0.000   0.756
-------------------------------------------------------------------------------
               Righting Arms:                    0.000   0.000
               External Arms:                    0.000   0.001
      Residual Righting Arms:                    0.000  -0.001s
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025          867.5  25.108f  0.000    138.79  19.034*
                                       MT/cm--------m.-MT/cm-------GML-----GMT
                                        8.89            33.68    136.08  16.328
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


limmarg: INFINITY; t3: 30.0; t: 15.00
limmarg: INFINITY; t3: 15.0; t: 7.50
limmarg: INFINITY; t3: 7.5; t: 3.75
limmarg: INFINITY; t3: 3.8; t: 1.88
limmarg: INFINITY; t3: 1.9; t: 0.94
limmarg: INFINITY; t3: 0.9; t: 0.47
limmarg: INFINITY; t3: 0.5; t: 0.23
limmarg: INFINITY; t3: 0.2; t: 0.12
limmarg: INFINITY; t3: 0.1; t: 0.06
limmarg: INFINITY; t3: 0.1; t: 0.03
limmarg: INFINITY; t3: 0.0; t: 0.01

               Theta3 is 0.03 degrees, as demonstrated below.
```

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 19.240f TCG = 0.000 VCG = 4.427

Origin Depth	Degrees of Trim	Heel	Displacement Weight (MT)	Residual Arms in Trim	in Heel	Res. Area
1.412	0.16f	0.00	1,237.6	0.000	-0.001	0.0000
1.412	0.16f	0.03s	1,237.7	0.000	0.011	0.0000
1.396	0.16f	5.00s	1,237.7	0.000	1.465	0.0640
1.347	0.17f	10.00s	1,237.7	0.000	2.860	0.2532
1.233	0.25f	15.00s	1,237.7	0.000	3.420	0.5333
1.171	0.30f	17.52s	1,237.6	0.000	3.454	0.6852
1.108	0.34f	20.00s	1,237.7	0.000	3.421	0.8339
0.972	0.43f	25.00s	1,237.7	0.000	3.241	1.1268
0.828	0.52f	30.00s	1,237.7	0.000	2.974	1.3986

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 815.25 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):

Stbd heeling moment = 1.00

LIM	STABILITY CRITERION		Min/Max	Attained
(1)	Residual Area Ratio from abs 0 deg to abs 0	>	1.000	LARGE

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 19.240f TCG = 0.000 VCG = 4.427

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.412	0.16f	0.00	1,237.6	0.000	-0.001	0.0000
1.412	0.16f	0.03s	1,237.7	0.000	0.011	0.0000
1.396	0.16f	5.00s	1,237.7	0.000	1.465	0.0640
1.347	0.17f	10.00s	1,237.7	0.000	2.860	0.2532
1.233	0.25f	15.00s	1,237.7	0.000	3.420	0.5333
1.171	0.30f	17.52s	1,237.6	0.000	3.454	0.6852
1.108	0.34f	20.00s	1,237.7	0.000	3.421	0.8339
0.972	0.43f	25.00s	1,237.7	0.000	3.241	1.1268
0.828	0.52f	30.00s	1,237.7	0.000	2.974	1.3986
0.675	0.61f	35.00s	1,237.7	0.000	2.656	1.6446
0.515	0.70f	40.00s	1,237.7	0.000	2.301	1.8611
0.350	0.78f	45.00s	1,237.7	0.000	1.919	2.0455
0.182	0.87f	50.00s	1,237.7	0.000	1.517	2.1955
0.011	0.94f	55.00s	1,237.7	0.000	1.100	2.3099
-0.159	1.02f	60.00s	1,237.7	0.000	0.672	2.3872
-0.328	1.08f	65.00s	1,237.7	0.000	0.236	2.4269
-0.418	1.11f	67.70s	1,237.6	0.000	0.000	2.4325
-0.493	1.13f	70.00s	1,237.7	0.000	-0.202	2.4284

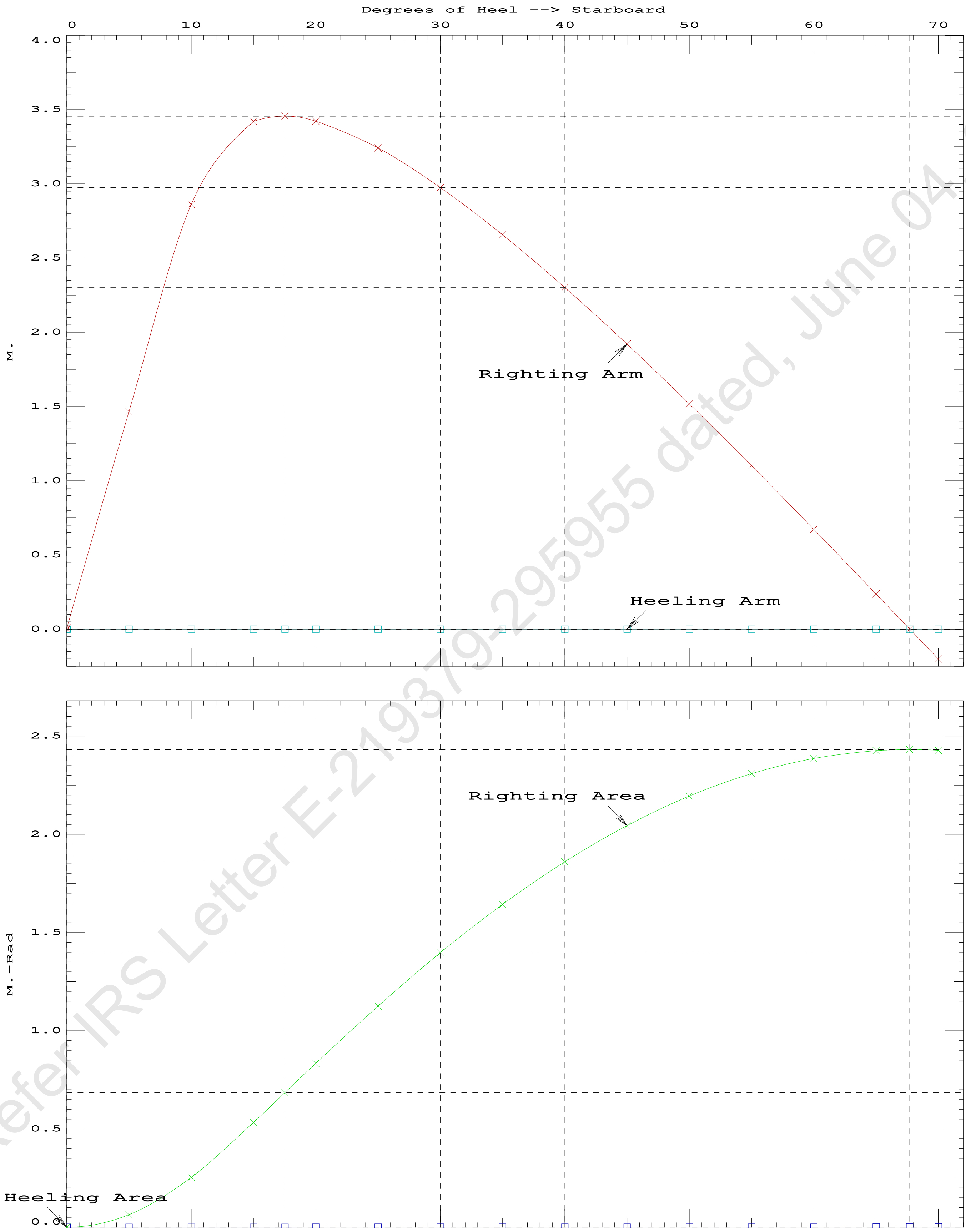
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 815.25 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):

Stbd heeling moment = 1.00

LIM--------------RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Residual Area Ratio from abs 0 deg to RAzero	>	1.000		LARGE
(2) Angle from abs 0 deg to RAzero	>	20.00	deg	67.67 P


Degrees of Heel --> Starboard
0
10
20
30
40
50
60
70
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
M.
Righting Arm
Heeling Arm
2.5
2.0
1.5
1.0
0.5
0.0
M.-Rad
Righting Area
Heeling Area

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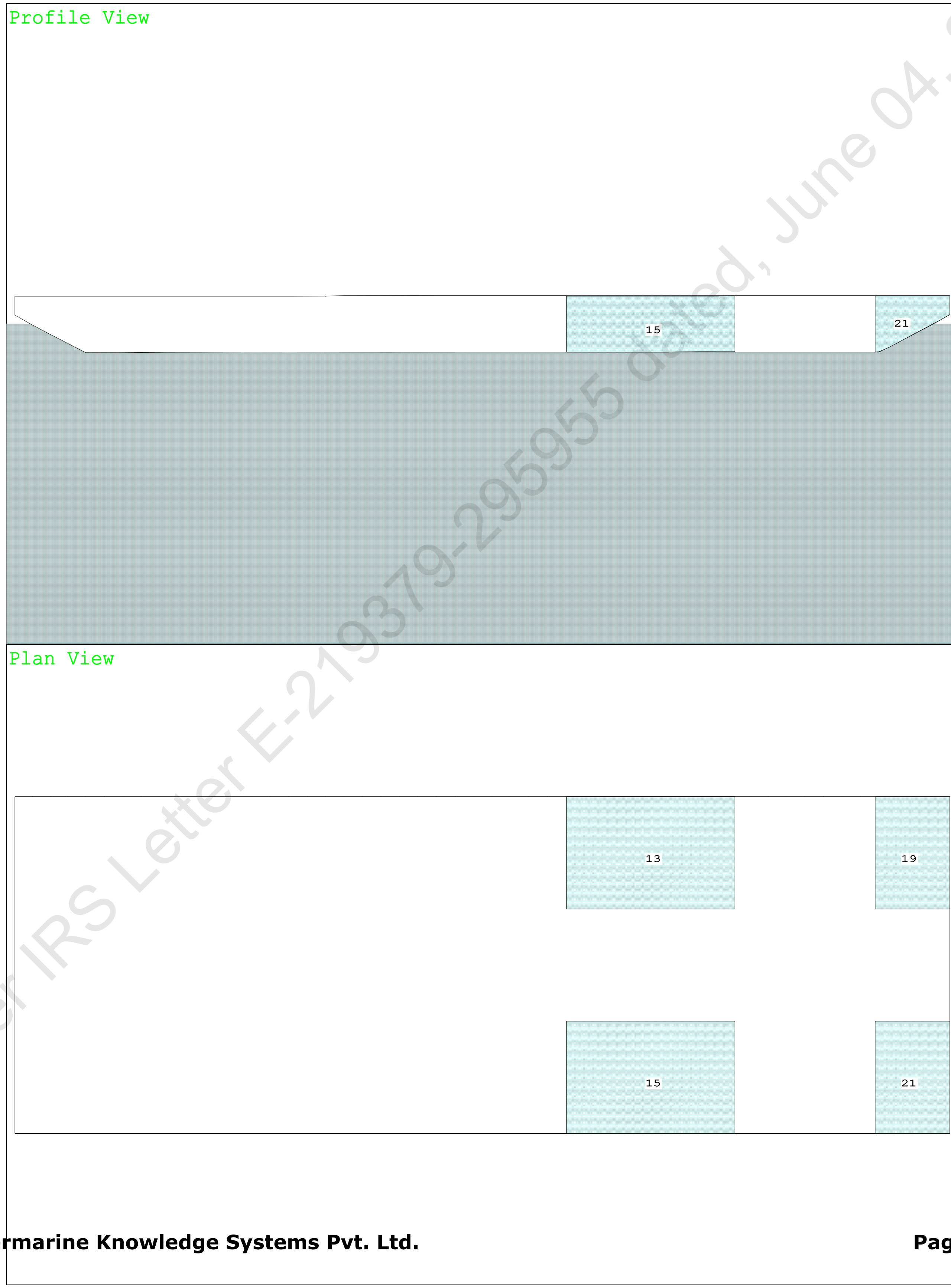
GHS 18.90 **PONTOON**

Condition 08 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Stbd

Condition Graphic - Draft: 1.526 @ 50.000f, 1.549 @ 0.000 Heel: zero



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GHS 18.90 **PONTOON**

Condition 08 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Stbd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.526 @ 50.00f, 1.537 @ 25.00f, 1.549 @ 0.00
Trim: Aft 0.023/50.000, Heel: zero

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			465.25	25.062f	0.000	3.000	
Crane			350.00	11.500f	0.000	6.325	
Crane Load			40.00	11.500f	0.000	21.478	
Total Fixed--------->			855.25	18.878f	0.000	5.225	
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks--------->			422.40	37.208f	0.000	1.596	470.09
Total Weight-------->			1,277.65	24.938f	0.000	4.025	
			Displ(MT)	LCB	TCB	VCB	
HULL		1.025	1,277.65	24.936f	0.000	0.779	

Part	SpGr	WPA	LCF	TCF	BML	BMT
Righting Arms:			0.000	0.000		
Total Waterplane---->	1.025	870.1	24.980f	0.000	135.63	18.477*
		MT/cm		m.-MT/cm	GML	GMT
		8.92		33.83	132.38	15.231

Distances in METERS.---Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

412.1 Cu.M. (16%) SALT WATER

350.00 MT Crane 40.00 MT Crane Load

DRAFT AT FP (M) = 1.526
DRAFT AT MS (M) = 1.537
DRAFT AT AP (M) = 1.549
DRAFT AT FWD MARK (M) = 1.527
DRAFT AT AFT MARK (M) = 1.547

HYDROSTATIC PROPERTIES
Trim: Aft 0.023/50.000, No Heel, VCG = 4.025

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	Buoyancy-Ctr. VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
1.537	1,277.65	24.936f	0.779	8.92	24.980f	33.83	132.38	15.231

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 470.1 m.-MT

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GHS 18.90

PONTOON

Condition 08 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Stbd

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 1.526 @ 50.00f, 1.537 @ 25.00f, 1.549 @ 0.00
Trim: Aft 0.023/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	73.8	0.746	1.497	0.05500	6.07
CRANE	57.0	3.968	4.718	0.05500	14.79
Total wind heeling moment to starboard------------->					20.87

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 24.938f TCG = 0.000 VCG = 4.025
Free Surface Adjustment: 0.368
Adjusted CG: LCG = 24.938f TCG = 0.000 VCG = 4.393

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. 50% Area	DeckImm Angle
1.537	0.03a	0.00	1,277.6	0.000	-0.016	0.0000	4.736
1.537	0.03a	0.06s	1,277.6	0.000	0.000	-0.0000	4.678
1.524	0.03a	4.74s	1,277.7	0.000	1.253	0.0511	0.000
1.522	0.03a	5.06s	1,277.7	0.000	1.338	0.0584	-0.322
1.480	0.03a	10.06s	1,277.7	0.000	2.611	0.2312	-5.322
1.433	0.04a	15.06s	1,277.7	0.000	3.058	0.4855	-10.322
1.423	0.05a	16.00s	1,277.7	0.000	3.064	0.5356	-11.261
1.375	0.06a	20.06s	1,277.7	0.000	2.973	0.7515	-15.322
1.306	0.08a	25.06s	1,277.7	0.000	2.717	1.0018	-20.322
1.229	0.09a	30.06s	1,277.7	0.000	2.380	1.2248	-25.322
1.142	0.11a	35.06s	1,277.7	0.000	1.998	1.4162	-30.322
1.047	0.12a	40.06s	1,277.7	0.000	1.584	1.5727	-35.322
0.944	0.14a	45.06s	1,277.7	0.000	1.150	1.6921	-40.322
0.834	0.15a	50.06s	1,277.7	0.000	0.701	1.7729	-45.322
0.717	0.17a	55.06s	1,277.7	0.000	0.242	1.8142	-50.322
0.654	0.17a	57.68s	1,277.6	0.000	0.000	1.8197	-52.944
0.596	0.18a	60.06s	1,277.7	0.000	-0.220	1.8151	-55.322

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 20.87 (constant)

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24-04-10 18:57:19    Cybermarine Technologies PTE. Ltd.
GHS 18.90                         PONTOON
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Condition 08 : Operating Condition of 350T Crane

Crane Lifting 40.0T @ 35m Radius Stbd

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LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max--------Attained
(1) Residual Area from abs 0 deg to MaxRA         >   0.0800 m.-Rad 0.5356 P
(2) Angle from Equilibrium to RAzero or Flood     >    20.00  deg    57.62 P
(3) Angle from Equilibrium to 50% Dk Imm Angle    >     0.00  deg     4.68 P
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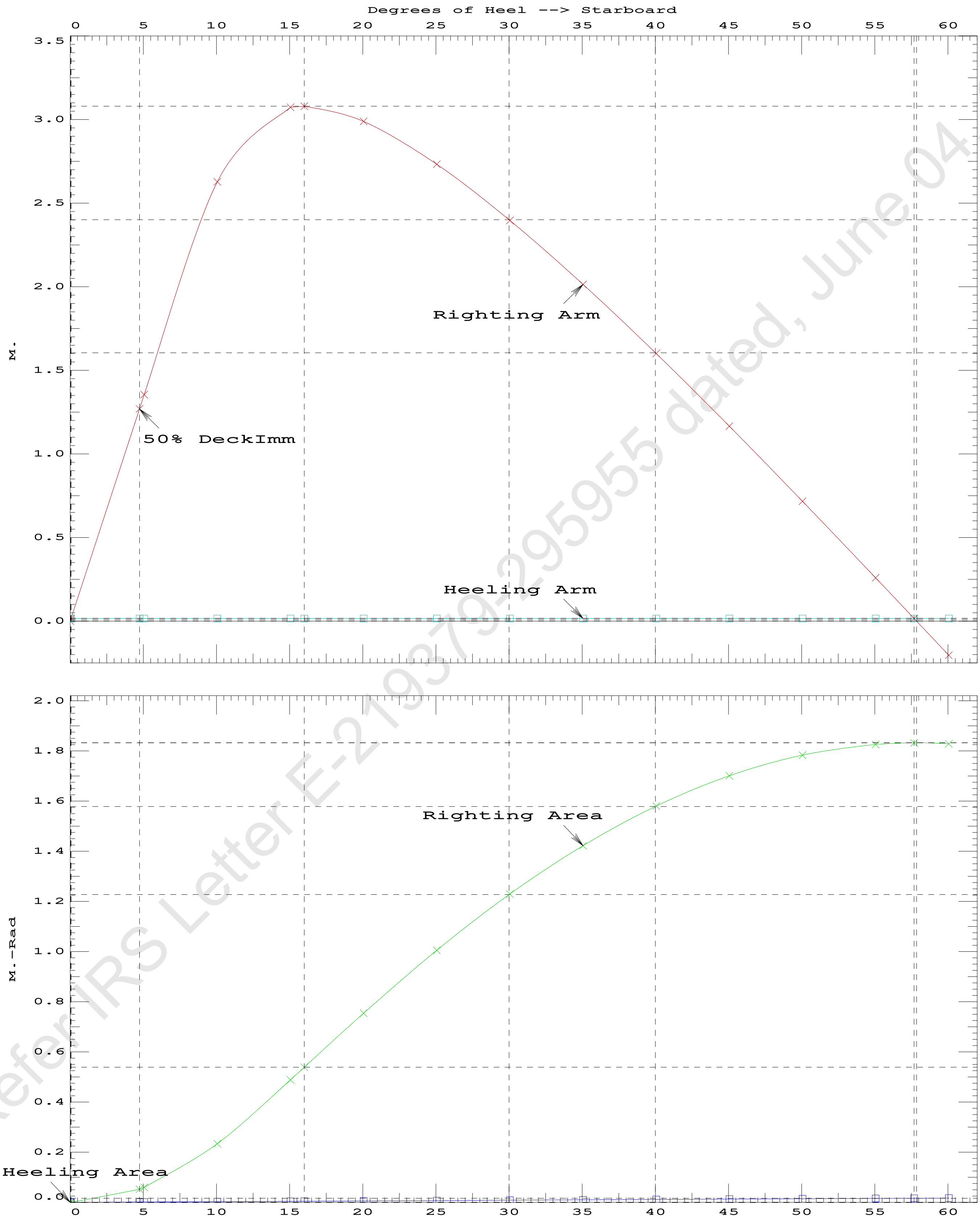
GHS 18.90 **PONTOON**

Condition 08 : Operating Condition of 350T Crane

Crane Lifting 40.0T @ 35m Radius Stbd



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GHS 18.90 **PONTOON**

Condition 08 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.938f TCG = 0.000 VCG = 4.025
Free Surface Adjustment: 0.368
Adjusted CG: LCG = 24.938f TCG = 0.026p VCG = 4.392

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.527	0.03a	3.99s	1,277.6	0.000	0.000	0.0000
1.492	0.03a	8.99s	1,277.7	0.000	1.303	0.0568
1.444	0.04a	13.99s	1,277.7	0.000	1.977	0.2088
1.421	0.05a	16.13s	1,277.7	0.000	2.008	0.2839
1.388	0.06a	18.99s	1,277.7	0.000	1.955	0.3835
1.322	0.07a	23.99s	1,277.6	0.000	1.724	0.5467
1.246	0.09a	28.99s	1,277.7	0.000	1.400	0.6837
1.161	0.10a	33.99s	1,277.7	0.000	1.024	0.7898
1.068	0.12a	38.99s	1,277.7	0.000	0.615	0.8616
0.966	0.14a	43.99s	1,277.7	0.000	0.183	0.8966
0.922	0.14a	46.05s	1,277.6	0.000	0.000	0.9000
0.858	0.15a	48.99s	1,277.6	0.000	-0.265	0.8932
0.743	0.16a	53.99s	1,277.7	0.000	-0.723	0.8502
0.622	0.18a	58.99s	1,277.6	0.000	-1.187	0.7669
0.497	0.19a	63.99s	1,277.7	0.000	-1.652	0.6431

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1400.00

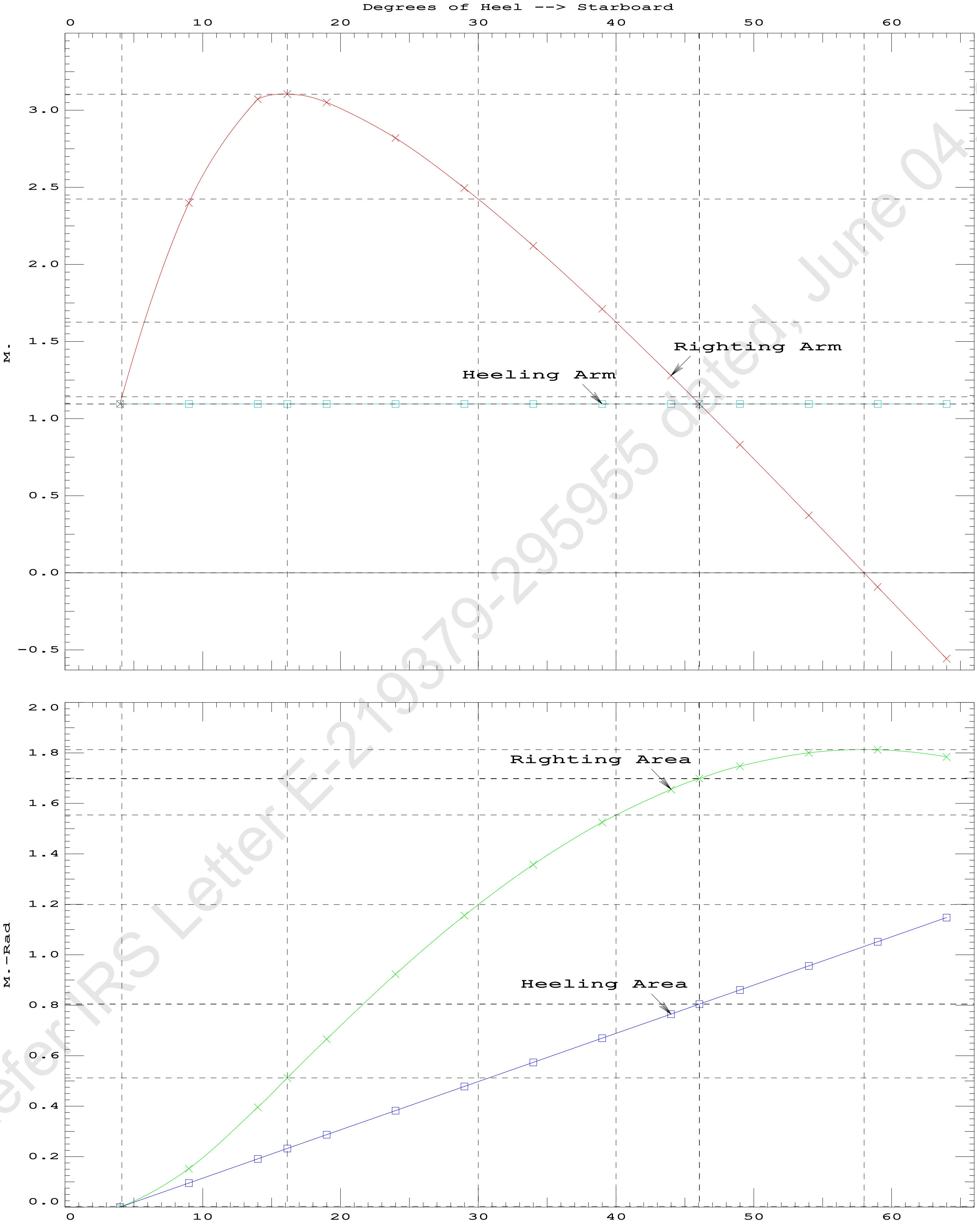
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM---STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained	
(1) Absolute Angle at Equilibrium	<	15.00	deg	3.99	P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	183.3	P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		2.119	P

PONTOON

Condition 08 : Operating Condition of 350T Crane

Crane Lifting 40.0T @ 35m Radius Stbd



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24-04-10 18:58:33    Cybermarine Technologies PTE. Ltd.
GHS 18.90                           PONTOON

              Condition 08 : Operating Condition of 350T Crane
                    Crane Lifting 40.0T @ 35m Radius Stbd
                     Crane in operation with load in place
                  WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 1.519 @ 50.00f, 1.531 @ 25.00f, 1.543 @ 0.00
                  Trim: Aft 0.023/50.000,  Heel: Stbd 3.99 deg.
Part------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                           465.25  25.062f  0.000   3.000
Crane                                350.00  11.500f  0.000   6.325
Crane Load                            40.00  11.500f  0.000  21.478
  Total Fixed--------->              855.25  18.878f  0.000   5.225
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025        161.60  34.000f  5.980p  1.509    162.73*
05_WBTK.S      1.000    1.025        161.60  34.000f  5.980s  1.509    162.73*
07_WBTK.P      1.000    1.025         49.60  47.660f  5.971p  1.880     72.32*
07_WBTK.S      1.000    1.025         49.60  47.660f  5.971s  1.880     72.32*
  Total Tanks--------->              422.40  37.208f  0.000   1.596    470.09
  Total Weight-------->            1,277.65  24.938f  0.000   4.025
                                   Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025      1,277.65  24.936f  1.322s  0.825
------------------------------------------------------------------------------
               Righting Arms:                 0.000    1.096s
               External Arms:                 0.000    1.096s
      Residual Righting Arms:                 0.000    0.000s
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025         870.7  24.978f  0.151s   135.55  18.563*
                                      MT/cm
                                       8.92
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


                            Angle Theta1 = 3.99
```

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24-04-10 18:58:33    Cybermarine Technologies PTE. Ltd.
GHS 18.90                        PONTOON

                          After hook load drop
                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
          USK draft: 1.561 @ 50.00f, 1.492 @ 25.00f, 1.424 @ 0.00
                    Trim: Fwd 0.137/50.000,  Heel: zero
Part------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                           465.25  25.062f  0.000   3.000
Crane                                350.00  11.500f  0.000   6.325
  Total Fixed--------->              815.25  19.240f  0.000   4.427
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025        161.60  34.000f  5.980p  1.509    162.73*
05_WBTK.S      1.000    1.025        161.60  34.000f  5.980s  1.509    162.73*
07_WBTK.P      1.000    1.025         49.60  47.660f  5.971p  1.880     72.32*
07_WBTK.S      1.000    1.025         49.60  47.660f  5.971s  1.880     72.32*
  Total Tanks--------->              422.40  37.208f  0.000   1.596   470.09
  Total Weight-------->            1,237.65  25.372f  0.000   3.461
                                   Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025      1,237.64  25.380f  0.000   0.756
------------------------------------------------------------------------------
               Righting Arms:                    0.000   0.000
               External Arms:                    0.000   0.001
      Residual Righting Arms:                    0.000  -0.001s
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025          867.5  25.108f  0.000    138.79  19.034*
                                       MT/cm---------m.-MT/cm-------GML-----GMT
                                        8.89            33.68    136.08  16.328
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


limmarg: 3334.53; t3: 30.0; t: 15.0
limmarg: 1209.83; t3: 15.0; t: 7.5
limmarg: 253.01; t3: 7.5; t: 3.8
limmarg: -13.70; t3: 3.7; t: 1.9
limmarg: 97.20; t3: 5.6; t: 0.9
limmarg: 39.02; t3: 4.7; t: 0.5
limmarg: 11.12; t3: 4.2; t: 0.2
limmarg: 0.83; t3: 4.0; t: 0.1
limmarg: -4.13; t3: 3.9; t: 0.1
limmarg: 0.83; t3: 4.0; t: 0.0
limmarg: 0.83; t3: 4.0; t: 0.0

                 Theta3 is 4.00 degrees, as demonstrated below.
```

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 19.240f TCG = 0.000 VCG = 4.427

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.401	0.16f	3.99s	1,237.6	0.000	-1.171	0.0000
1.412	0.16f	0.00	1,237.6	0.000	0.001	-0.0407
1.401	0.16f	4.00p	1,237.7	0.000	1.176	0.0003
1.396	0.16f	5.00p	1,237.7	0.000	1.467	0.0234
1.347	0.17f	10.00p	1,237.7	0.000	2.861	0.2126
1.233	0.25f	15.00p	1,237.7	0.000	3.421	0.4928
1.171	0.30f	17.52p	1,237.6	0.000	3.456	0.6448
1.108	0.34f	20.00p	1,237.7	0.000	3.423	0.7935
0.972	0.43f	25.00p	1,237.7	0.000	3.242	1.0866
0.828	0.52f	30.00p	1,237.7	0.000	2.976	1.3585

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 815.25 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM	STABILITY CRITERION		Min/Max	Attained
(1)	Residual Area Ratio from abs -4 deg to abs 4	>	1.000	1.008 P

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 19.240f TCG = 0.000 VCG = 4.427

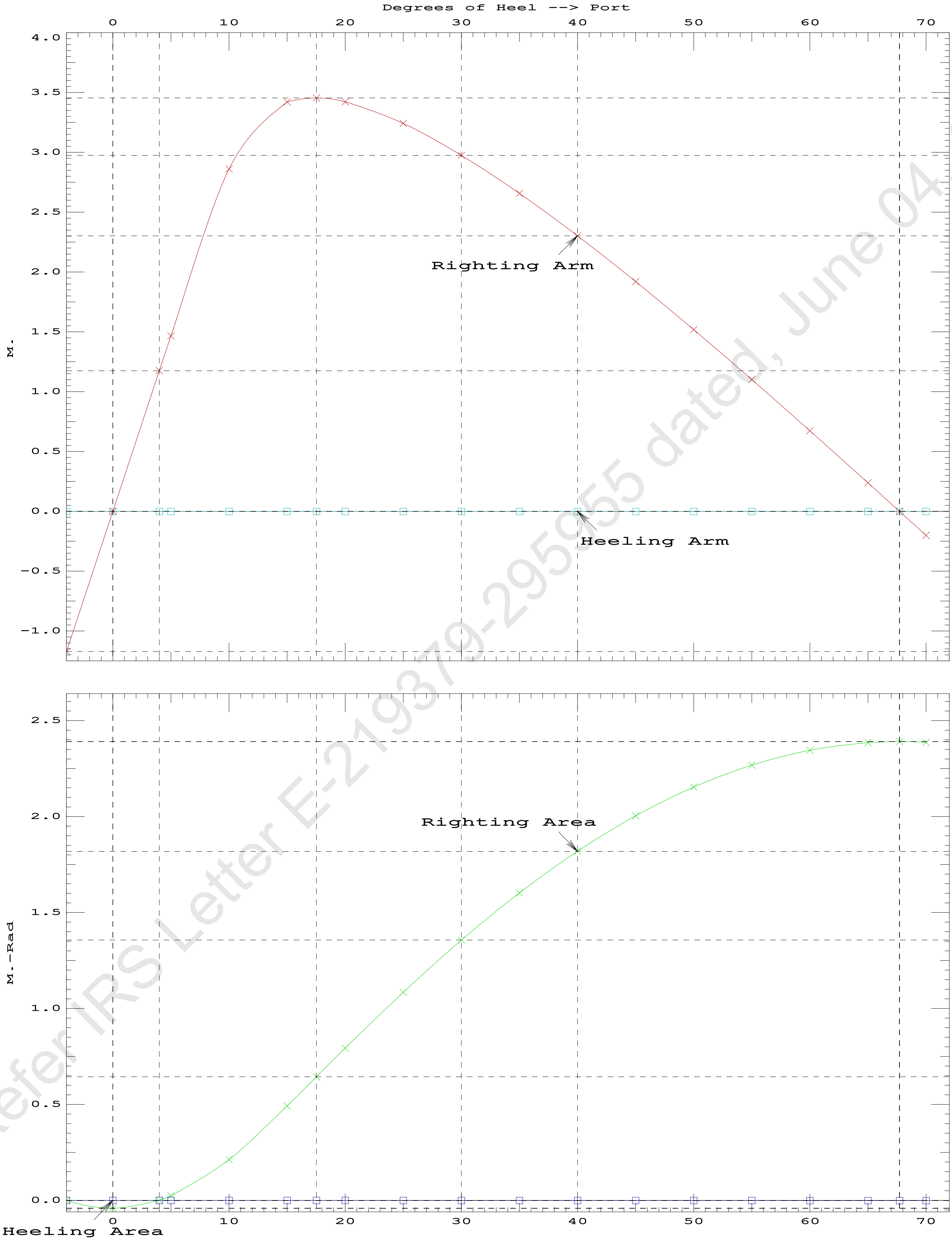
Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.401	0.16f	3.99s	1,237.6	0.000	-1.171	0.0000
1.412	0.16f	0.00	1,237.6	0.000	0.001	-0.0407
1.401	0.16f	4.00p	1,237.7	0.000	1.176	0.0003
1.396	0.16f	5.00p	1,237.7	0.000	1.467	0.0234
1.347	0.17f	10.00p	1,237.7	0.000	2.861	0.2126
1.233	0.25f	15.00p	1,237.7	0.000	3.421	0.4928
1.171	0.30f	17.52p	1,237.6	0.000	3.456	0.6448
1.108	0.34f	20.00p	1,237.7	0.000	3.423	0.7935
0.972	0.43f	25.00p	1,237.7	0.000	3.242	1.0866
0.828	0.52f	30.00p	1,237.7	0.000	2.976	1.3585
0.675	0.61f	35.00p	1,237.7	0.000	2.657	1.6047
0.515	0.70f	40.00p	1,237.7	0.000	2.302	1.8214
0.350	0.78f	45.00p	1,237.7	0.000	1.921	2.0058
0.182	0.87f	50.00p	1,237.7	0.000	1.519	2.1561
0.011	0.94f	55.00p	1,237.7	0.000	1.102	2.2705
-0.159	1.02f	60.00p	1,237.7	0.000	0.673	2.3480
-0.328	1.08f	65.00p	1,237.7	0.000	0.238	2.3878
-0.418	1.11f	67.72p	1,237.7	0.000	0.000	2.3935
-0.493	1.13f	70.00p	1,237.7	0.000	-0.200	2.3895

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 815.25 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

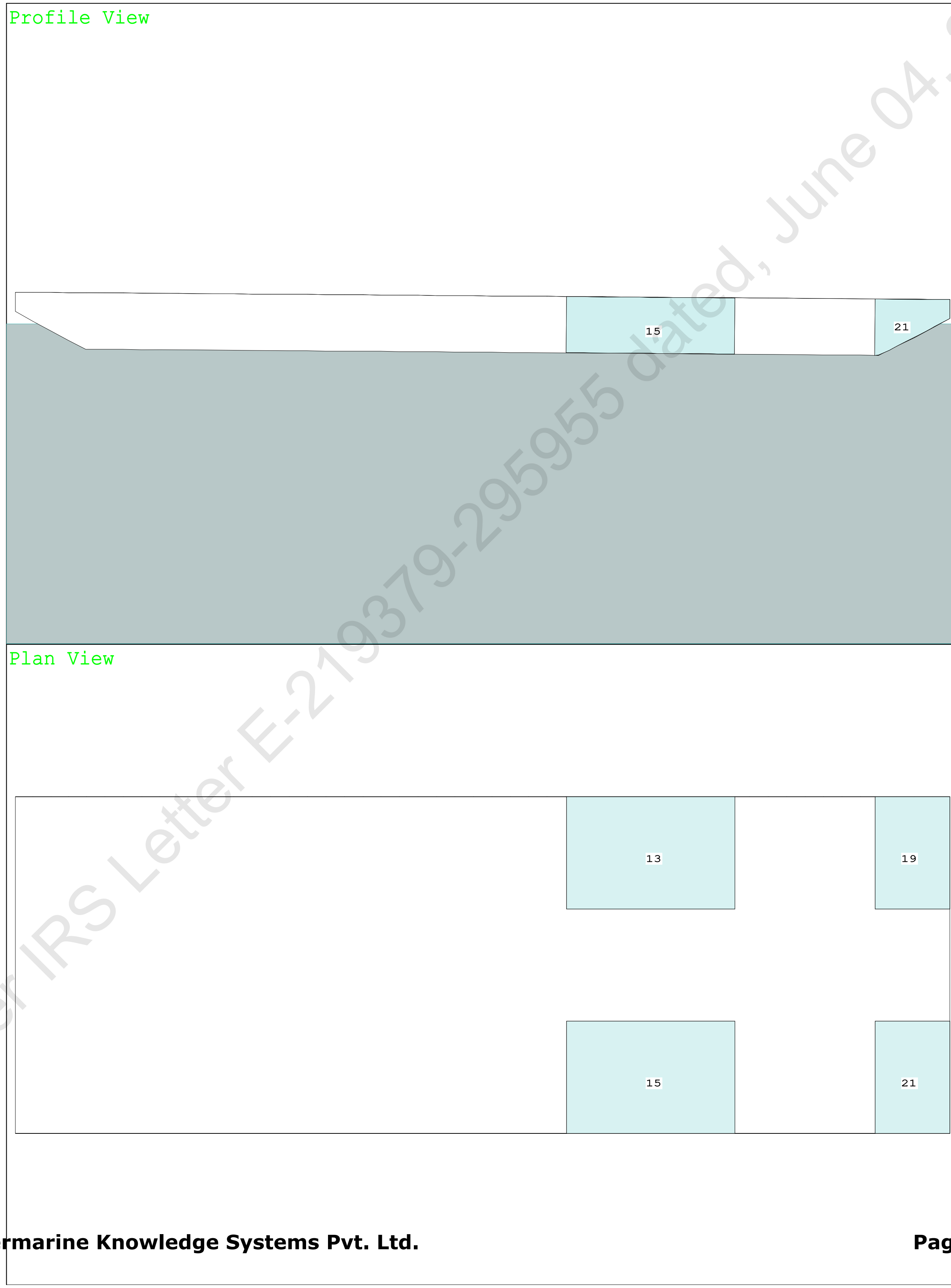
LIM--------------RINA PT E, CH 19, SEC 2: 2.2.3		-------Min/Max--------	Attained
(1) Residual Area Ratio from abs -4 deg to RAzero	>	1.000	59.749 P
(2) Angle from abs 4 deg to RAzero	>	20.00 deg	63.72 P


Degrees of Heel --> Port
Righting Arm
Heeling Arm
M.
Righting Area
Heeling Area
M.-Rad

24-04-10 18:59:33 Cybermarine Technologies PTE. Ltd.
GHS 18.90 **PONTOON**

Condition 09 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Fwd

Condition Graphic - Draft: 1.731 @ 50.000f, 1.341 @ 0.000 Heel: zero



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24-04-10 18:59:33    Cybermarine Technologies PTE. Ltd.
GHS 18.90                           PONTOON
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Condition 09 : Operating Condition of 350T Crane

Crane Lifting 40.0T @ 35m Radius Fwd

```
                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 1.731 @ 50.00f, 1.536 @ 25.00f, 1.341 @ 0.00
                    Trim: Fwd 0.390/50.000,  Heel: zero
Part----------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                               465.25  25.062f  0.000   3.000
Crane                                    350.00  11.500f  0.000   6.325
Crane Load                                40.00  46.500f  0.000  21.478
  Total Fixed--------->                  855.25  20.515f  0.000   5.225
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025         161.60  34.000f  5.980p  1.509   162.73*
05_WBTK.S      1.000    1.025         161.60  34.000f  5.980s  1.509   162.73*
07_WBTK.P      1.000    1.025          49.60  47.660f  5.971p  1.880    72.32*
07_WBTK.S      1.000    1.025          49.60  47.660f  5.971s  1.880    72.32*
  Total Tanks--------->               422.40  37.208f  0.000   1.596   470.09
  Total Weight-------->             1,277.65  26.034f  0.000   4.025
                                    Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025       1,277.56  26.059f  0.000   0.783
------------------------------------------------------------------------------
             Righting Arms:                        0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025           870.2  25.335f  0.000    135.71  18.490*
                                        MT/cm--------m.-MT/cm-------GML-----GMT
                                         8.92           33.85    132.47  15.247
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


                          SUMMARY OF LOADING
       412.1 Cu.M.  (16%) SALT WATER

      350.00 MT  Crane                        40.00 MT  Crane Load

DRAFT AT FP (M) =     1.731
DRAFT AT MS (M) =     1.536
DRAFT AT AP (M) =     1.341
DRAFT AT FWD MARK (M) =    1.700
DRAFT AT AFT MARK (M) =    1.372
                           HYDROSTATIC PROPERTIES
                 Trim: Fwd 0.390/50.000, No Heel, VCG = 4.025

    LCF   Displacement   Buoyancy-Ctr.  Weight/         Moment/
  Draft----Weight(MT)----LCB-----VCB-------cm-----LCF---cm trim----GML-----GMT
  1.539     1,277.56  26.059f  0.783     8.92  25.335f   33.85  132.47  15.247
 Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
                           Trim is per 50.00m.
 Draft is from USK.                                Formal Free Surface included.

      Note:  GMT includes the formal free surface moment 470.1 m.-MT
```

24-04-10 18:59:33 Cybermarine Technologies PTE. Ltd.
GHS 18.90 **PONTOON**

Condition 09 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Fwd

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 1.731 @ 50.00f, 1.536 @ 25.00f, 1.341 @ 0.00
Trim: Fwd 0.390/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	73.7	0.752	1.505	0.05500	6.10
CRANE	57.0	4.092	4.845	0.05500	15.19
Total wind heeling moment to starboard------------->					21.30

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 26.034f TCG = 0.000 VCG = 4.025
Free Surface Adjustment: 0.368
Adjusted CG: LCG = 26.031f TCG = 0.000 VCG = 4.393

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area	50% DeckImm Angle
1.329	0.45f	0.00	1,277.6	0.000	-0.016	0.0000	4.117
1.329	0.45f	0.05s	1,277.7	0.000	0.000	-0.0000	4.064
1.318	0.45f	4.12s	1,277.7	0.000	1.087	0.0385	0.000
1.312	0.45f	5.05s	1,277.7	0.000	1.336	0.0583	-0.936
1.253	0.49f	10.05s	1,277.7	0.000	2.590	0.2302	-5.936
1.092	0.74f	15.05s	1,277.7	0.000	3.031	0.4822	-10.936
1.060	0.78f	15.99s	1,277.7	0.000	3.037	0.5318	-11.873
0.915	0.99f	20.05s	1,277.6	0.000	2.946	0.7459	-15.936
0.726	1.25f	25.05s	1,277.7	0.000	2.689	0.9939	-20.936
0.526	1.52f	30.05s	1,277.6	0.000	2.353	1.2145	-25.936
0.316	1.78f	35.05s	1,277.6	0.000	1.970	1.4034	-30.936
0.099	2.05f	40.05s	1,277.6	0.000	1.558	1.5576	-35.936
-0.122	2.30f	45.05s	1,277.6	0.000	1.124	1.6748	-40.936
-0.344	2.55f	50.05s	1,277.6	0.000	0.676	1.7534	-45.936
-0.565	2.77f	55.05s	1,277.6	0.000	0.220	1.7926	-50.936
-0.669	2.87f	57.45s	1,277.7	0.000	0.000	1.7972	-53.335
-0.780	2.97f	60.05s	1,277.7	0.000	-0.239	1.7918	-55.936

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.30 (constant)

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GHS 18.90 **PONTOON**

Condition 09 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Fwd

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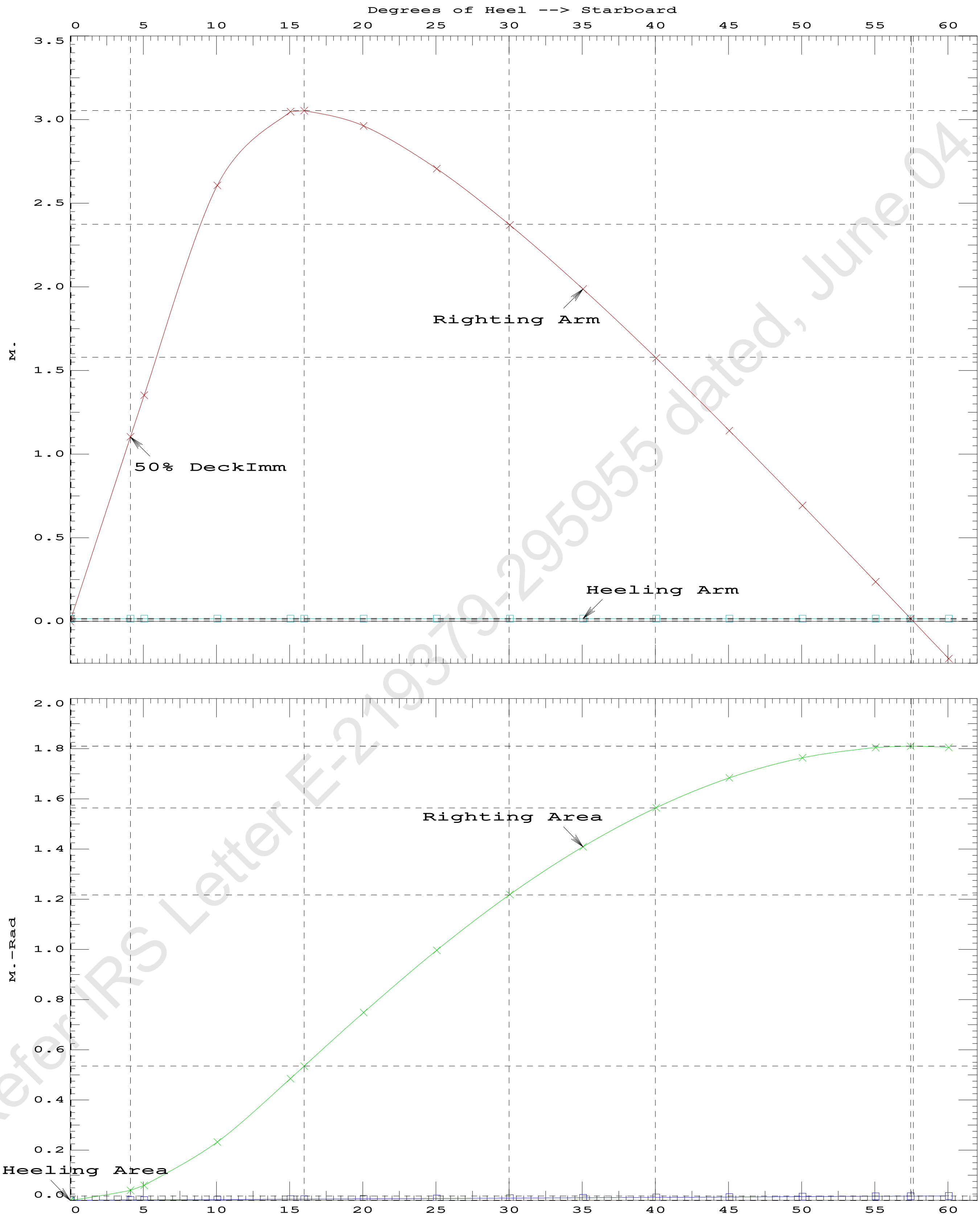
LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max--------Attained
(1) Residual Area from abs 0 deg to MaxRA         >   0.0800 m.-Rad 0.5318 P
(2) Angle from Equilibrium to RAzero or Flood     >    20.00  deg    57.40 P
(3) Angle from Equilibrium to 50% Dk Imm Angle    >     0.00  deg     4.06 P
-----------------------------------------------------------------------------
```

24-04-10 18:59:33 Cybermarine Technologies PTE. Ltd.
GHS 18.90 **PONTOON**

Condition 09 : Operating Condition of 350T Crane

Crane Lifting 40.0T @ 35m Radius Fwd



24-04-10 18:59:33 Cybermarine Technologies PTE. Ltd.
GHS 18.90 **PONTOON**

Condition 09 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 26.034f TCG = 0.000 VCG = 4.025
Free Surface Adjustment: 0.368
Adjusted CG: LCG = 26.031f TCG = 0.000 VCG = 4.393

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.329	0.45f	0.00	1,277.6	0.000	-0.002	0.0000
1.313	0.45f	5.00s	1,277.7	0.000	1.336	0.0582
1.254	0.49f	10.00s	1,277.7	0.000	2.594	0.2303
1.094	0.73f	15.00s	1,277.7	0.000	3.045	0.4832
1.059	0.78f	16.00s	1,277.7	0.000	3.052	0.5363
0.917	0.99f	20.00s	1,277.6	0.000	2.963	0.7482
0.728	1.25f	25.00s	1,277.6	0.000	2.707	0.9977
0.528	1.52f	30.00s	1,277.6	0.000	2.372	1.2199
0.318	1.78f	35.00s	1,277.6	0.000	1.989	1.4105
0.102	2.04f	40.00s	1,277.6	0.000	1.577	1.5664
-0.120	2.30f	45.00s	1,277.6	0.000	1.144	1.6852
-0.342	2.54f	50.00s	1,277.6	0.000	0.696	1.7656
-0.563	2.77f	55.00s	1,277.6	0.000	0.240	1.8065
-0.676	2.88f	57.61s	1,277.7	0.000	0.000	1.8120
-0.777	2.97f	60.00s	1,277.6	0.000	-0.220	1.8074

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 2.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM	STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained	
(1)	Absolute Angle at Equilibrium	<	15.00	deg	0.00	P
(2)	Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	194961	P
(3)	Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		1152.16	P

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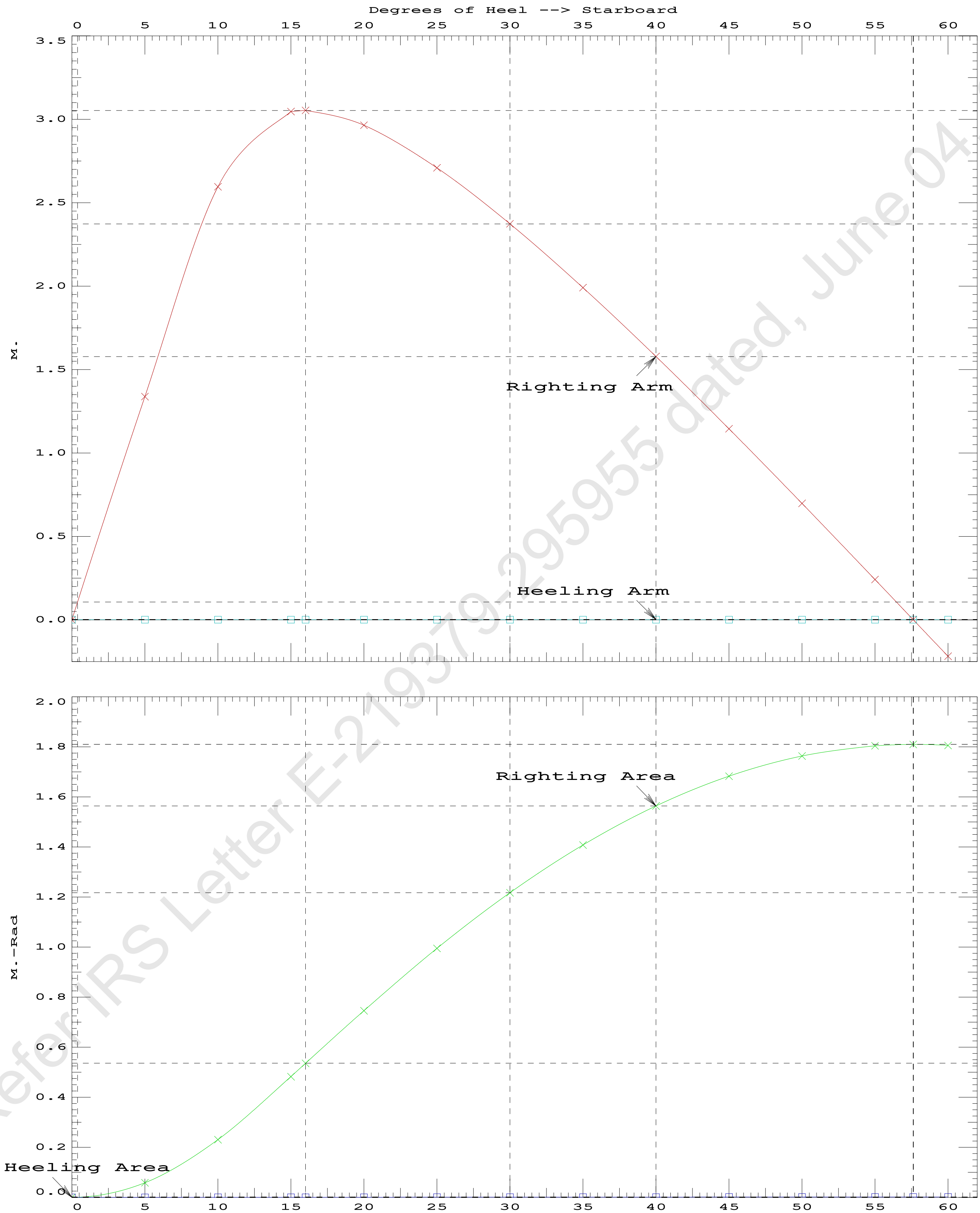
GHS 18.90

PONTOON

Condition 09 : Operating Condition of 350T Crane

Crane Lifting 40.0T @ 35m Radius Fwd



Condition 09 : Operating Condition of 350T Crane

Crane Lifting 40.0T @ 35m Radius Fwd

Crane in operation with load in place
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.731 @ 50.00f, 1.536 @ 25.00f, 1.341 @ 0.00
Trim: Fwd 0.390/50.000, Heel: zero

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			465.25	25.062f	0.000	3.000	
Crane			350.00	11.500f	0.000	6.325	
Crane Load			40.00	46.500f	0.000	21.478	
Total Fixed--------->			855.25	20.515f	0.000	5.225	
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks--------->			422.40	37.208f	0.000	1.596	470.09
Total Weight-------->			1,277.65	26.034f	0.000	4.025	
			Displ(MT)	LCB	TCB	VCB	
HULL		1.025	1,277.64	26.059f	0.000	0.783	

Righting Arms:	0.000	0.000
External Arms:	0.000	0.001
Residual Righting Arms:	0.000	-0.001s

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	870.2	25.335f	0.000	135.70	18.489*
		MT/cm		m.-MT/cm	GML	GMT
		8.92		33.85	132.46	15.246

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

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24-04-10 19:00:57    Cybermarine Technologies PTE. Ltd.
GHS 18.90                        PONTOON

                         After hook load drop
               WEIGHT and DISPLACEMENT and WATERPLANE STATUS
          USK draft: 1.561 @ 50.00f, 1.492 @ 25.00f, 1.424 @ 0.00
                    Trim: Fwd 0.137/50.000,  Heel: zero
Part-------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                            465.25  25.062f  0.000   3.000
Crane                                 350.00  11.500f  0.000   6.325
  Total Fixed--------->               815.25  19.240f  0.000   4.427
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025         161.60  34.000f  5.980p  1.509    162.73*
05_WBTK.S      1.000    1.025         161.60  34.000f  5.980s  1.509    162.73*
07_WBTK.P      1.000    1.025          49.60  47.660f  5.971p  1.880     72.32*
07_WBTK.S      1.000    1.025          49.60  47.660f  5.971s  1.880     72.32*
  Total Tanks--------->               422.40  37.208f  0.000   1.596   470.09
  Total Weight-------->             1,237.65  25.372f  0.000   3.461
                                    Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025       1,237.64  25.379f  0.000   0.756
------------------------------------------------------------------------------
               Righting Arms:                  0.000   0.000
               External Arms:                  0.000   0.001
      Residual Righting Arms:                  0.000  -0.001s
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025         867.5  25.108f  0.000    138.79  19.034*
                                      MT/cm--------m.-MT/cm-------GML-----GMT
                                       8.89            33.68    136.08  16.328
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


limmarg: INFINITY; t3: 30; t: 15.00
limmarg: INFINITY; t3: 15; t: 7.50
limmarg: INFINITY; t3: 8; t: 3.75
limmarg: INFINITY; t3: 4; t: 1.88
limmarg: INFINITY; t3: 2; t: 0.94
limmarg: INFINITY; t3: 1; t: 0.47
limmarg: INFINITY; t3: 1; t: 0.23
limmarg: INFINITY; t3: 1; t: 0.12
limmarg: INFINITY; t3: 1; t: 0.06
limmarg: INFINITY; t3: 1; t: 0.03
limmarg: INFINITY; t3: 1; t: 0.01

               Theta3 is 0.97 degrees, as demonstrated below.
```

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 19.240f TCG = 0.000 VCG = 4.427

Origin Depth	Degrees of Trim	Heel	Displacement Weight (MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.412	0.16f	0.00	1,237.6	0.000	-0.001	0.0000
1.410	0.16f	0.97s	1,237.6	0.000	0.287	0.0024
1.397	0.16f	5.00s	1,237.7	0.000	1.465	0.0640
1.347	0.17f	10.00s	1,237.7	0.000	2.860	0.2533
1.233	0.25f	15.00s	1,237.7	0.000	3.420	0.5333
1.171	0.30f	17.52s	1,237.6	0.000	3.454	0.6853
1.108	0.34f	20.00s	1,237.7	0.000	3.421	0.8339
0.972	0.43f	25.00s	1,237.7	0.000	3.241	1.1268
0.828	0.52f	30.00s	1,237.7	0.000	2.974	1.3986

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 815.25 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM	STABILITY CRITERION		Min/Max	Attained
(1)	Residual Area Ratio from abs 0 deg to abs 1	>	1.000	LARGE

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 19.240f TCG = 0.000 VCG = 4.427

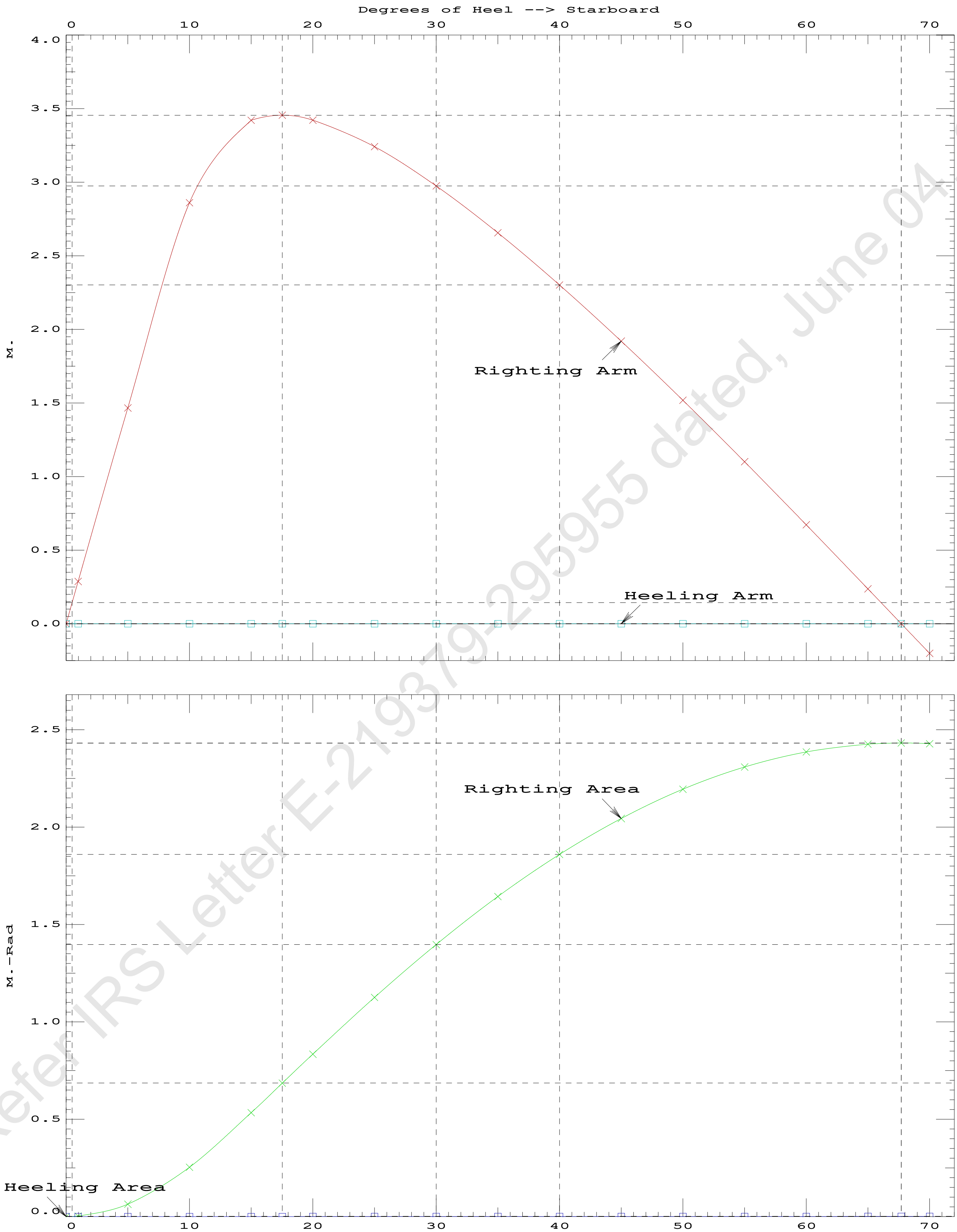
Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.412	0.16f	0.00	1,237.6	0.000	-0.001	0.0000
1.410	0.16f	0.97s	1,237.6	0.000	0.287	0.0024
1.397	0.16f	5.00s	1,237.7	0.000	1.465	0.0640
1.347	0.17f	10.00s	1,237.7	0.000	2.860	0.2533
1.233	0.25f	15.00s	1,237.7	0.000	3.420	0.5333
1.171	0.30f	17.52s	1,237.6	0.000	3.454	0.6853
1.108	0.34f	20.00s	1,237.7	0.000	3.421	0.8339
0.972	0.43f	25.00s	1,237.7	0.000	3.241	1.1268
0.828	0.52f	30.00s	1,237.7	0.000	2.974	1.3986
0.675	0.61f	35.00s	1,237.7	0.000	2.656	1.6446
0.515	0.70f	40.00s	1,237.7	0.000	2.301	1.8612
0.350	0.78f	45.00s	1,237.7	0.000	1.919	2.0455
0.182	0.87f	50.00s	1,237.7	0.000	1.517	2.1956
0.011	0.94f	55.00s	1,237.7	0.000	1.100	2.3099
-0.159	1.02f	60.00s	1,237.7	0.000	0.672	2.3873
-0.328	1.08f	65.00s	1,237.7	0.000	0.236	2.4269
-0.418	1.11f	67.70s	1,237.6	0.000	0.000	2.4325
-0.493	1.13f	70.00s	1,237.7	0.000	-0.202	2.4285

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 815.25 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM--------------RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Residual Area Ratio from abs 0 deg to RAzero	>	1.000		LARGE
(2) Angle from abs 1 deg to RAzero	>	20.00	deg	66.73 P


Degrees of Heel --> Starboard
0
10
20
30
40
50
60
70
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
M.
Righting Arm
Heeling Arm
2.5
2.0
1.5
1.0
0.5
0.0
M.-Rad
Righting Area
Heeling Area

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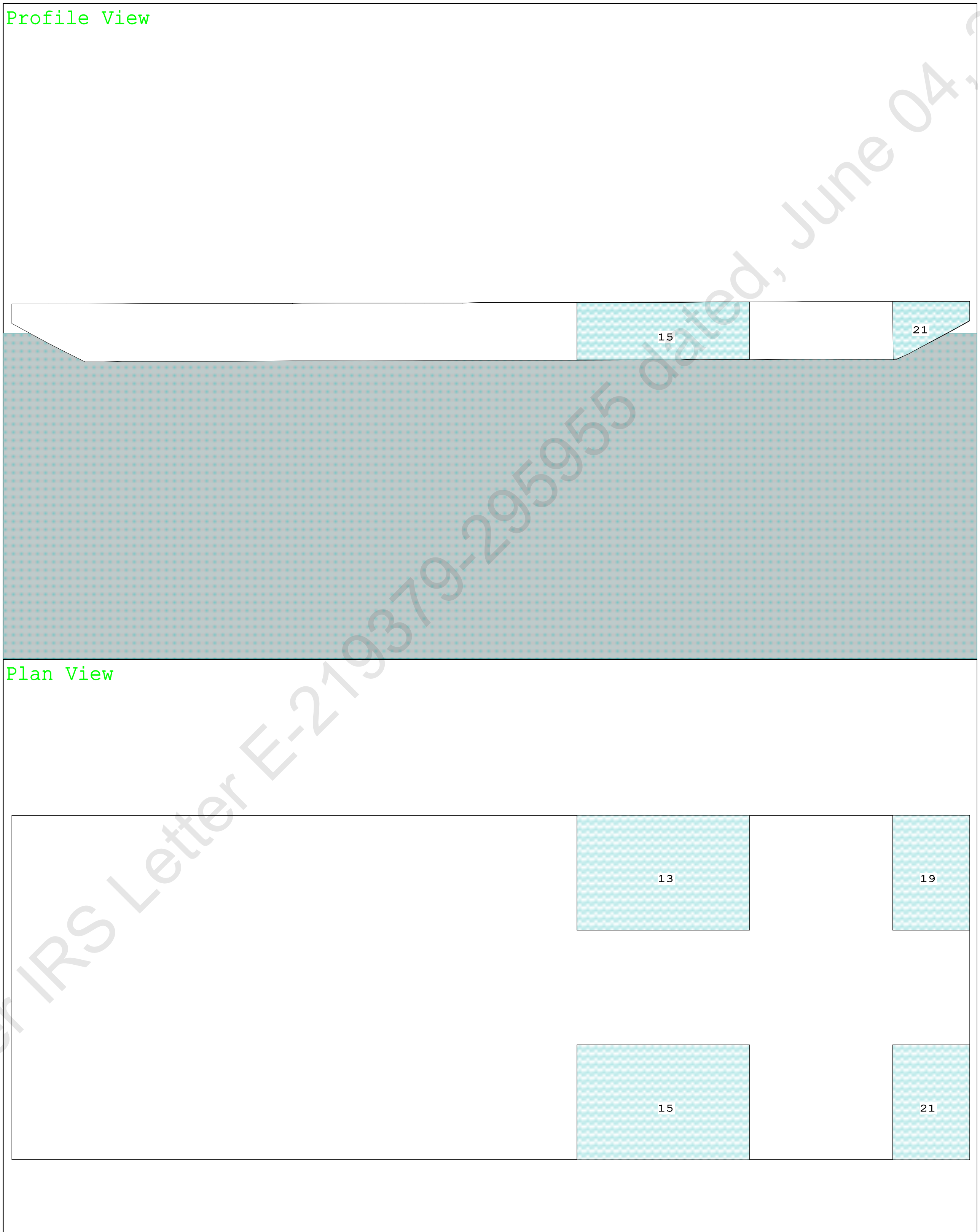
GHS 18.90 **PONTOON**

Condition 10 : Operating Condition of 250T Crane

Crane Lifting 36.6T @ 18m Radius Aft

Condition Graphic - Draft: 1.346 @ 50.000f, 1.495 @ 0.000 Heel: zero



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24-03-19 19:01:31    Cybermarine Technologies PTE. Ltd.
GHS 18.90                         PONTOON
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Condition 10 : Operating Condition of 250T Crane

Crane Lifting 36.6T @ 18m Radius Aft

```
                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 1.346 @ 50.00f, 1.421 @ 25.00f, 1.495 @ 0.00
                    Trim: Aft 0.149/50.000,  Heel: zero
Part----------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                              465.25  25.062f  0.000   3.000
Crane                                   250.00   7.500f  0.000   5.200
Crane Load                               36.60  10.500a  0.000  76.977
  Total Fixed--------->                 751.85  17.491f  0.000   7.333
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025           161.60  34.000f  5.980p  1.509   162.73*
05_WBTK.S      1.000    1.025           161.60  34.000f  5.980s  1.509   162.73*
07_WBTK.P      1.000    1.025            49.60  47.660f  5.971p  1.880    72.32*
07_WBTK.S      1.000    1.025            49.60  47.660f  5.971s  1.880    72.32*
  Total Tanks--------->                 422.40  37.208f  0.000   1.596   470.09
  Total Weight-------->               1,174.25  24.584f  0.000   5.269
                                      Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025         1,174.23  24.570f  0.000   0.718
------------------------------------------------------------------------------
               Righting Arms:                    0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025             862.7  24.872f  0.000    143.84  19.940*
                                        MT/cm---------m.-MT/cm-------GML-----GMT
                                          8.84           32.71    139.28  15.389
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


                           SUMMARY OF LOADING
       412.1 Cu.M.  (16%) SALT WATER

      250.00 MT  Crane                         36.60 MT  Crane Load

DRAFT AT FP (M) =    1.346
DRAFT AT MS (M) =    1.421
DRAFT AT AP (M) =    1.495
DRAFT AT FWD MARK (M) =    1.358
DRAFT AT AFT MARK (M) =    1.483
                          HYDROSTATIC PROPERTIES
                Trim: Aft 0.149/50.000, No Heel, VCG = 5.269

   LCF   Displacement   Buoyancy-Ctr.  Weight/         Moment/
  Draft----Weight(MT)----LCB-----VCB-------cm-----LCF---cm trim----GML-----GMT
  1.421     1,174.23  24.570f  0.718     8.84  24.872f   32.71  139.28  15.389
 Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
                           Trim is per 50.00m.
 Draft is from USK.                           Formal Free Surface included.

      Note:  GMT includes the formal free surface moment 470.1 m.-MT
```

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GHS 18.90 **PONTOON**

Condition 10 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Aft

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 1.346 @ 50.00f, 1.421 @ 25.00f, 1.495 @ 0.00
Trim: Aft 0.149/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	79.4	0.807	1.502	0.05500	6.56
CRANE	57.0	4.047	4.742	0.05500	14.87
Total wind heeling moment to starboard------------->					21.43

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 24.584f TCG = 0.000 VCG = 5.269
Free Surface Adjustment: 0.400
Adjusted CG: LCG = 24.585f TCG = 0.000 VCG = 5.670

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight (MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area	50% DeckImm Angle
1.483	0.17a	0.00	1,174.3	0.000	-0.018	0.0000	4.914
1.483	0.17a	0.06s	1,174.2	0.000	0.000	-0.0000	4.856
1.469	0.17a	4.91s	1,174.3	0.000	1.312	0.0556	0.000
1.469	0.17a	5.06s	1,174.3	0.000	1.351	0.0589	-0.144
1.425	0.18a	10.06s	1,174.2	0.000	2.606	0.2323	-5.144
1.351	0.28a	15.06s	1,174.3	0.000	3.001	0.4843	-10.144
1.270	0.37a	20.06s	1,174.3	0.000	2.825	0.7426	-15.144
1.182	0.47a	25.06s	1,174.3	0.000	2.466	0.9748	-20.144
1.088	0.57a	30.06s	1,174.3	0.000	2.023	1.1713	-25.144
0.987	0.68a	35.06s	1,174.3	0.000	1.534	1.3268	-30.144
0.880	0.78a	40.06s	1,174.3	0.000	1.016	1.4383	-35.144
0.768	0.87a	45.06s	1,174.3	0.000	0.481	1.5037	-40.144
0.665	0.96a	49.47s	1,174.3	0.000	0.000	1.5223	-44.558
0.651	0.97a	50.06s	1,174.3	0.000	-0.064	1.5219	-45.144
0.528	1.05a	55.06s	1,174.3	0.000	-0.613	1.4924	-50.144
0.401	1.13a	60.06s	1,174.3	0.000	-1.160	1.4151	-55.144

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.43 (constant)

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GHS 18.90 **PONTOON**

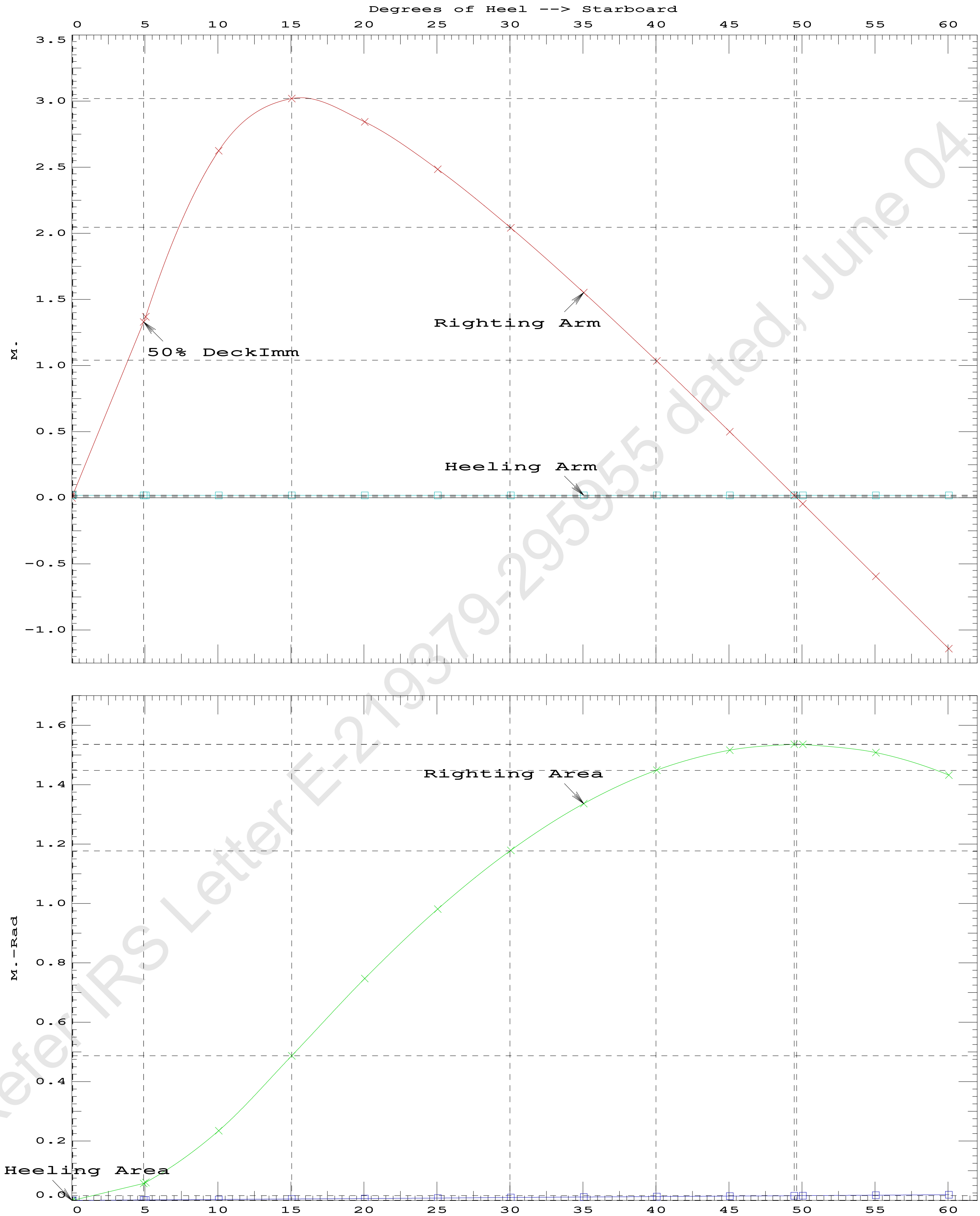
Condition 10 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Aft

```
LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max--------Attained
(1) Residual Area from abs 0 deg to MaxRA         >   0.0800 m.-Rad 0.4843 P
(2) Angle from Equilibrium to RAzero or Flood     >    20.00  deg    49.41 P
(3) Angle from Equilibrium to 50% Dk Imm Angle    >     0.00  deg     4.86 P
-----------------------------------------------------------------------------
```

PONTOON

Condition 10 : Operating Condition of 250T Crane

Crane Lifting 36.6T @ 18m Radius Aft



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GHS 18.90 **PONTOON**

Condition 10 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

```
     Total CG:  LCG = 24.584f  TCG = 0.000   VCG =  5.269
                        Free Surface Adjustment:   0.400
  Adjusted CG:  LCG = 24.585f  TCG = 0.000   VCG =  5.670
```

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.483	0.17a	0.01s	1,174.3	0.000	0.000	0.0000
1.469	0.17a	5.01s	1,174.2	0.000	1.351	0.0590
1.425	0.18a	10.01s	1,174.2	0.000	2.610	0.2325
1.352	0.27a	15.01s	1,174.3	0.000	3.015	0.4853
1.271	0.37a	20.01s	1,174.3	0.000	2.842	0.7451
1.183	0.47a	25.01s	1,174.3	0.000	2.483	0.9788
1.089	0.57a	30.01s	1,174.3	0.000	2.041	1.1768
0.988	0.67a	35.01s	1,174.3	0.000	1.552	1.3339
0.881	0.78a	40.01s	1,174.3	0.000	1.035	1.4470
0.769	0.87a	45.01s	1,174.3	0.000	0.500	1.5141
0.662	0.96a	49.60s	1,174.3	0.000	0.000	1.5342
0.652	0.97a	50.01s	1,174.3	0.000	-0.045	1.5340
0.529	1.05a	55.01s	1,174.3	0.000	-0.594	1.5062
0.402	1.13a	60.01s	1,174.3	0.000	-1.140	1.4305

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM	STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained	
(1)	Absolute Angle at Equilibrium	<	15.00	deg	0.01	P
(2)	Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	70806.3	P
(3)	Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		417.301	P

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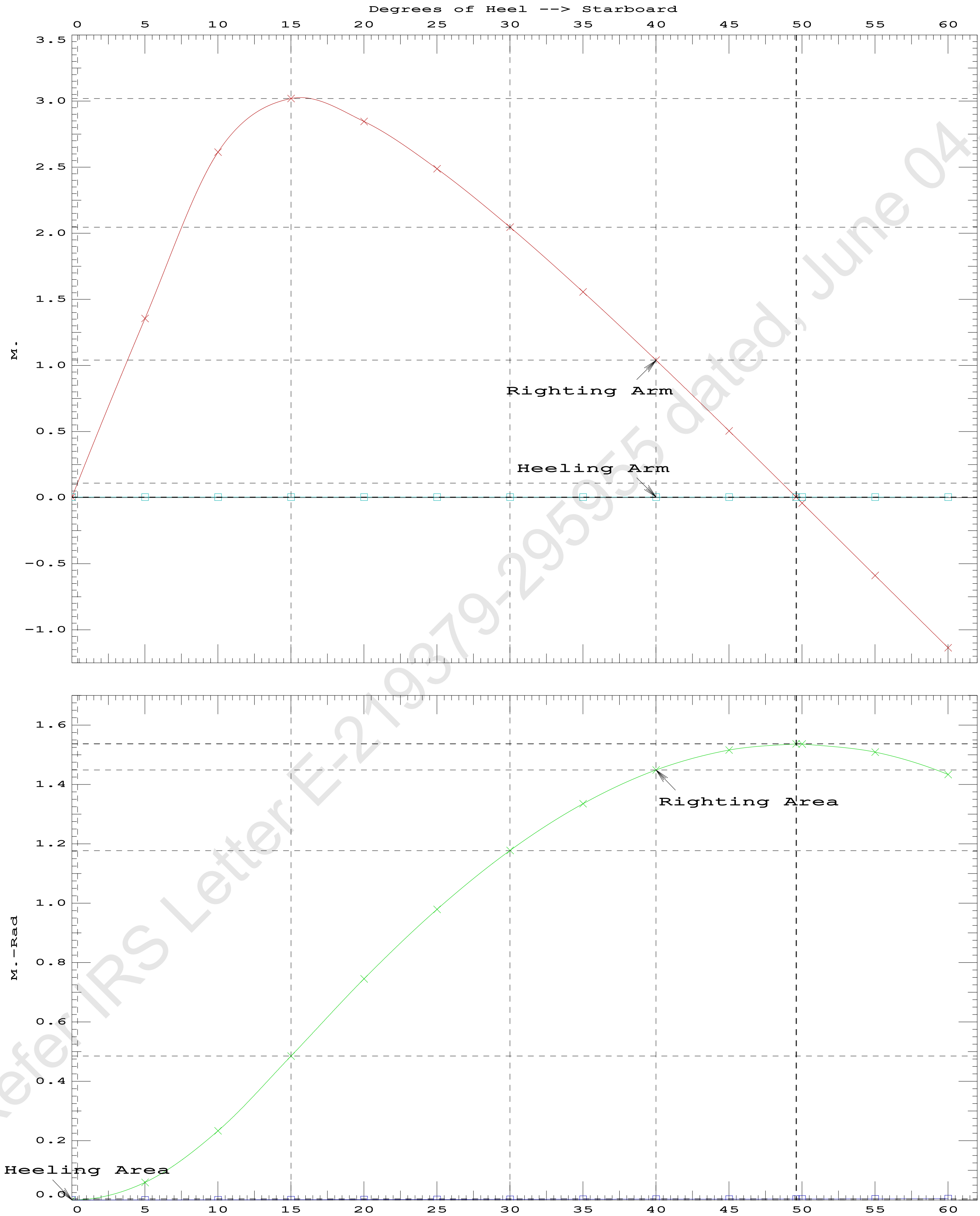
GHS 18.90 **PONTOON**

Condition 10 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Aft



Condition 10 : Operating Condition of 250T Crane

Crane Lifting 36.6T @ 18m Radius Aft

Crane in operation with load in place
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.346 @ 50.00f, 1.421 @ 25.00f, 1.495 @ 0.00
Trim: Aft 0.149/50.000, Heel: Stbd 0.01 deg.

Part	Weight(MT)	LCG	TCG	VCG
LIGHT SHIP	465.25	25.062f	0.000	3.000
Crane	250.00	7.500f	0.000	5.200
Crane Load	36.60	10.500a	0.000	76.977
Total Fixed--------->	751.85	17.491f	0.000	7.333

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks--------->			422.40	37.208f	0.000	1.596	470.09
Total Weight-------->			1,174.25	24.584f	0.000	5.269	

Part	SpGr	Displ(MT)	LCB	TCB	VCB
HULL	1.025	1,174.25	24.570f	0.005s	0.718

Righting Arms:	0.000	0.004s
External Arms:	0.000	0.004s
Residual Righting Arms:	0.000	0.000

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	858.0	24.741f	0.031s	141.51	19.820*
		MT/cm				
		8.79				

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.01

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GHS 18.90                         PONTOON

                           After hook load drop
                  WEIGHT and DISPLACEMENT and WATERPLANE STATUS
             USK draft: 1.502 @ 50.00f, 1.379 @ 25.00f, 1.256 @ 0.00
                    Trim: Fwd 0.246/50.000,  Heel: 0.00 deg.
Part----------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                               465.25  25.062f  0.000   3.000
Crane                                    250.00   7.500f  0.000   5.200
  Total Fixed--------->                  715.25  18.924f  0.000   3.769
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025            161.60  34.000f  5.980p  1.509    162.73*
05_WBTK.S      1.000    1.025            161.60  34.000f  5.980s  1.509    162.73*
07_WBTK.P      1.000    1.025             49.60  47.660f  5.971p  1.880     72.32*
07_WBTK.S      1.000    1.025             49.60  47.660f  5.971s  1.880     72.32*
  Total Tanks--------->                  422.40  37.208f  0.000   1.596   470.09
  Total Weight-------->                1,137.65  25.712f  0.000   2.962
                                       Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025          1,137.65  25.724f  0.001s  0.698
-------------------------------------------------------------------------------
               Righting Arms:                     0.000   0.001s
               External Arms:                     0.000   0.001
      Residual Righting Arms:                     0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025          856.9  25.295f  0.021s   145.50  20.411*
                                      MT/cm---------m.-MT/cm-------GML-----GMT
                                        8.78            32.59   143.24  18.147
Distances in METERS.------------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


limmarg: 681952512.00; t3: 30.0; t: 15.00
limmarg: 255564096.00; t3: 15.0; t: 7.50
limmarg: 69277144.00; t3: 7.5; t: 3.75
limmarg: 17393760.00; t3: 3.8; t: 1.88
limmarg: 4573686.00; t3: 1.9; t: 0.94
limmarg: 1150046.63; t3: 1.0; t: 0.47
limmarg: 353837.94; t3: 0.5; t: 0.23
limmarg: 93409.45; t3: 0.3; t: 0.12
limmarg: 41951.43; t3: 0.2; t: 0.06
limmarg: 25543.31; t3: 0.1; t: 0.03
limmarg: 6544.12; t3: 0.1; t: 0.01

               Theta3 is 0.07 degrees, as demonstrated below.
```

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 18.924f TCG = 0.000 VCG = 3.769

Origin Depth	Degrees of Trim	Heel	Displacement Weight (MT)	Residual Arms in Trim	in Heel	Res. Area
1.244	0.28f	0.01s	1,137.7	0.000	-0.003	0.0000
1.244	0.28f	0.00	1,137.7	0.000	0.000	-0.0000
1.243	0.28f	0.07p	1,137.7	0.000	0.025	0.0000
1.229	0.28f	5.00p	1,137.6	0.000	1.640	0.0716
1.167	0.31f	10.00p	1,137.6	0.000	3.159	0.2819
0.975	0.44f	15.00p	1,137.7	0.000	3.811	0.5923
0.826	0.55f	18.57p	1,137.9	0.000	3.881	0.8334
0.765	0.59f	20.00p	1,137.7	0.000	3.870	0.9303
0.547	0.75f	25.00p	1,137.7	0.000	3.730	1.2643
0.321	0.90f	30.00p	1,137.6	0.000	3.495	1.5803

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 715.25 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM	STABILITY CRITERION	Min/Max	Attained
(1)	Residual Area Ratio from abs 0 deg to abs 0.1 >	1.000	66.441 P

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 18.924f TCG = 0.000 VCG = 3.769

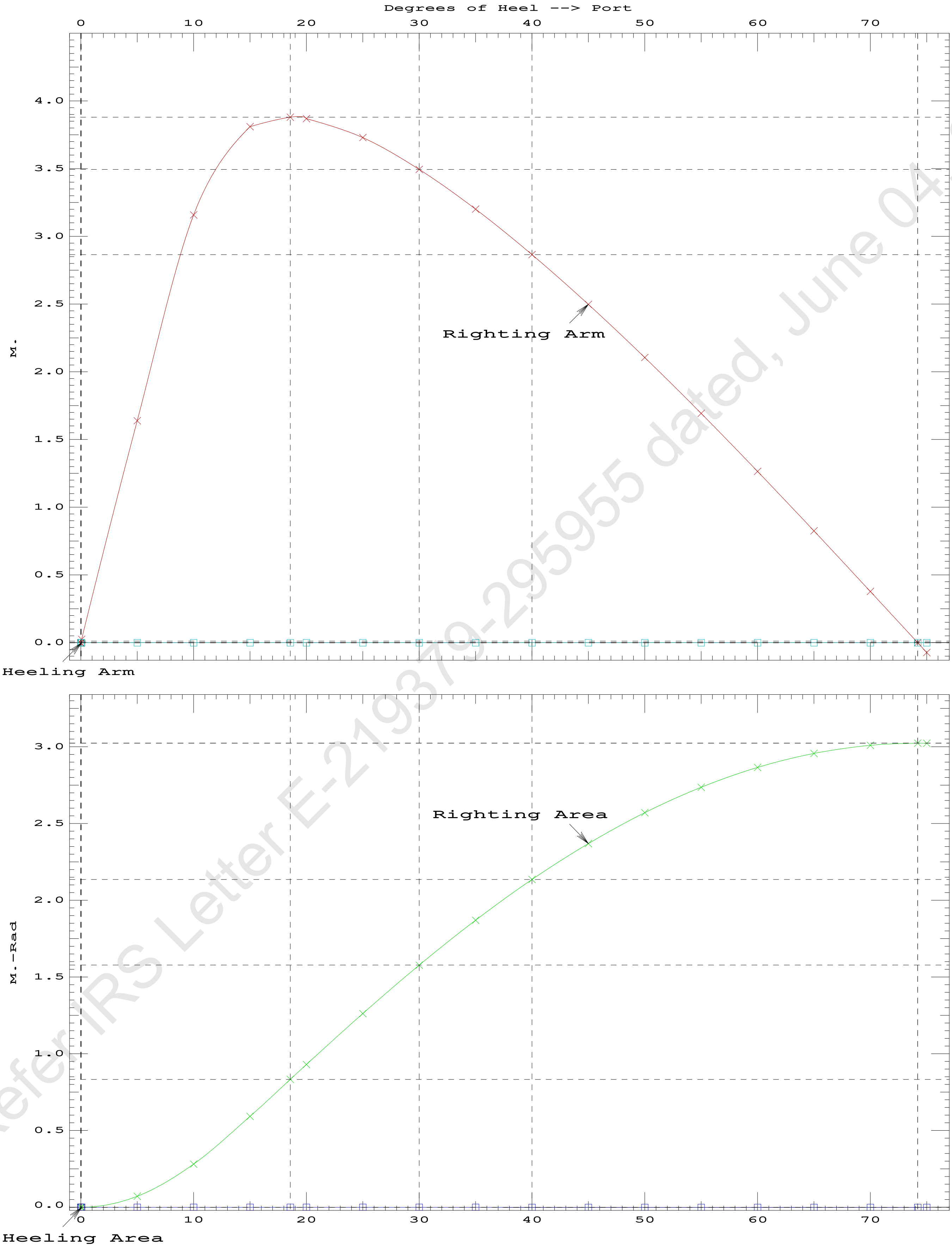
Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.244	0.28f	0.01s	1,137.7	0.000	-0.003	0.0000
1.244	0.28f	0.00	1,137.7	0.000	0.000	-0.0000
1.243	0.28f	0.07p	1,137.7	0.000	0.025	0.0000
1.229	0.28f	5.00p	1,137.6	0.000	1.640	0.0716
1.167	0.31f	10.00p	1,137.6	0.000	3.159	0.2819
0.975	0.44f	15.00p	1,137.7	0.000	3.811	0.5923
0.826	0.55f	18.57p	1,137.9	0.000	3.881	0.8334
0.765	0.59f	20.00p	1,137.7	0.000	3.870	0.9303
0.547	0.75f	25.00p	1,137.7	0.000	3.730	1.2643
0.321	0.90f	30.00p	1,137.6	0.000	3.495	1.5803
0.091	1.05f	35.00p	1,137.6	0.000	3.201	1.8729
-0.143	1.21f	40.00p	1,137.6	0.000	2.866	2.1380
-0.379	1.35f	45.00p	1,137.6	0.000	2.499	2.3723
-0.613	1.50f	50.00p	1,137.7	0.000	2.106	2.5734
-0.845	1.63f	55.00p	1,137.7	0.000	1.694	2.7393
-1.070	1.75f	60.00p	1,137.7	0.000	1.265	2.8685
-1.286	1.86f	65.00p	1,137.7	0.000	0.826	2.9599
-1.491	1.95f	70.00p	1,137.7	0.000	0.379	3.0125
-1.653	2.01f	74.20p	1,137.6	0.000	0.000	3.0264
-1.682	2.02f	75.00p	1,137.7	0.000	-0.072	3.0258

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 715.25 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM--------------RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Residual Area Ratio from abs 0 deg to RAzero	>	1.000		LARGE
(2) Angle from abs 0.1 deg to RAzero	>	20.00	deg	74.13 P


Degrees of Heel --> Port
0
10
20
30
40
50
60
70
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
M.
Righting Arm
Heeling Arm
3.0
2.5
2.0
1.5
1.0
0.5
0.0
M.-Rad
Righting Area
Heeling Area

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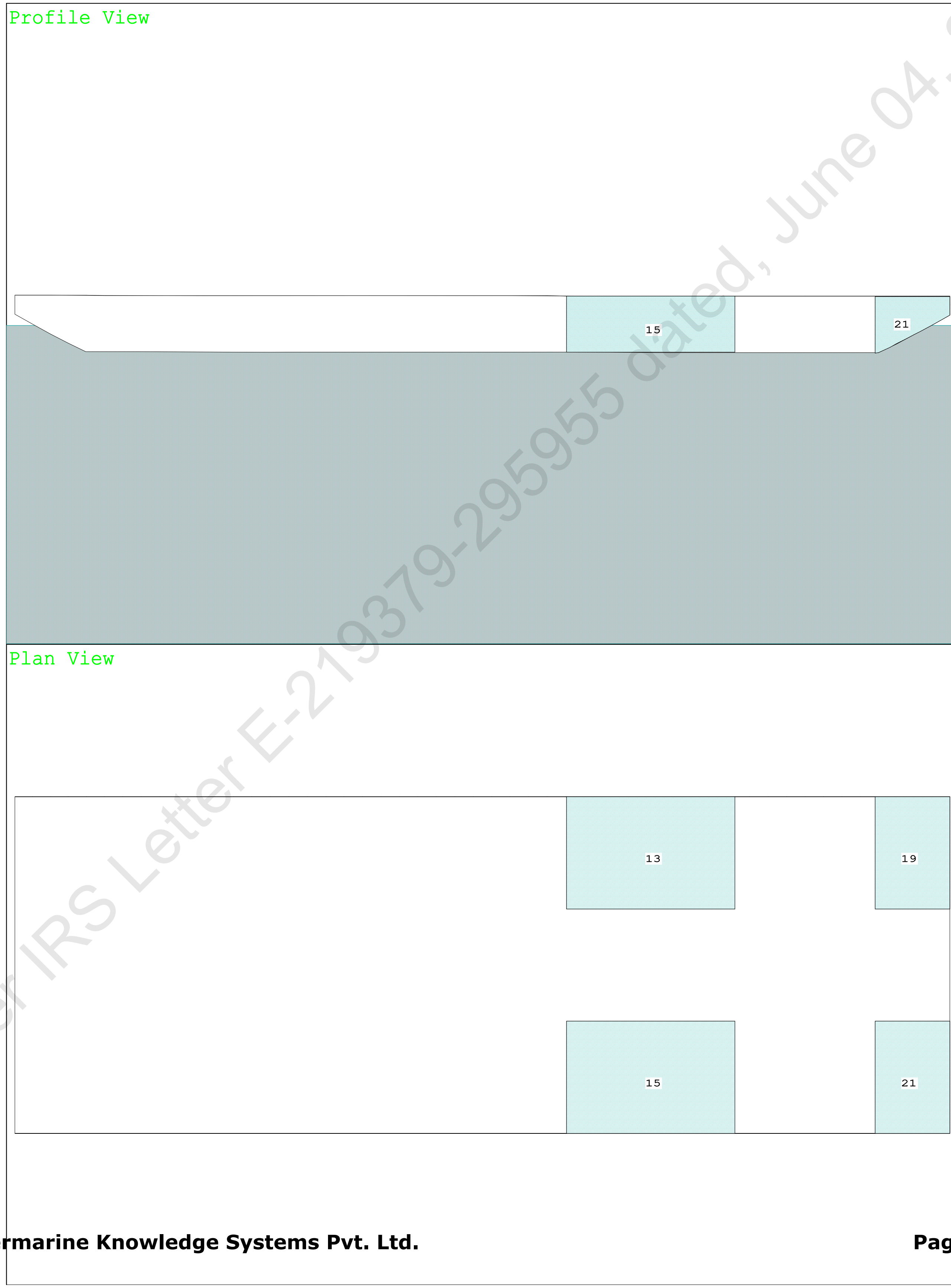
GHS 18.90 **PONTOON**

Condition 11 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Stbd

Condition Graphic - Draft: 1.447 @ 50.000f, 1.395 @ 0.000 Heel: zero



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24-03-19 19:01:33    Cybermarine Technologies PTE. Ltd.
GHS 18.90                             PONTOON
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Condition 11 : Operating Condition of 250T Crane

Crane Lifting 36.6T @ 18m Radius Stbd

```
                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 1.447 @ 50.00f, 1.421 @ 25.00f, 1.395 @ 0.00
                     Trim: Fwd 0.052/50.000,  Heel: zero
Part----------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                              465.25  25.062f  0.000   3.000
Crane                                   250.00   7.500f  0.000   5.200
Crane Load                               36.60   7.500f  0.000  76.977
  Total Fixed--------->                 751.85  18.367f  0.000   7.333
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025           161.60  34.000f  5.980p  1.509   162.73*
05_WBTK.S      1.000    1.025           161.60  34.000f  5.980s  1.509   162.73*
07_WBTK.P      1.000    1.025            49.60  47.660f  5.971p  1.880    72.32*
07_WBTK.S      1.000    1.025            49.60  47.660f  5.971s  1.880    72.32*
  Total Tanks--------->                 422.40  37.208f  0.000   1.596   470.09
  Total Weight-------->               1,174.25  25.145f  0.000   5.269
                                      Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025         1,174.21  25.151f  0.000   0.718
------------------------------------------------------------------------------
               Righting Arms:                    0.001f  0.000
Part---------------------SpGr------------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025              863.2  25.051f  0.000    144.11  19.977*
                                           MT/cm--------m.-MT/cm-------GML-----GMT
                                            8.85            32.77    139.56  15.426
Distances in METERS.----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


                              SUMMARY OF LOADING
       412.1 Cu.M.  (16%) SALT WATER

      250.00 MT  Crane                          36.60 MT  Crane Load

DRAFT AT FP (M) =     1.447
DRAFT AT MS (M) =     1.421
DRAFT AT AP (M) =     1.395
DRAFT AT FWD MARK (M) =    1.443
DRAFT AT AFT MARK (M) =    1.399
                              HYDROSTATIC PROPERTIES
                   Trim: Fwd 0.052/50.000, No Heel, VCG = 5.269

    LCF   Displacement   Buoyancy-Ctr.  Weight/         Moment/
  Draft----Weight(MT)----LCB-----VCB-------cm-----LCF---cm trim----GML-----GMT
  1.421     1,174.21  25.151f  0.718     8.85  25.051f   32.77  139.56  15.426
 Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
                             Trim is per 50.00m.
 Draft is from USK.                                Formal Free Surface included.

      Note:  GMT includes the formal free surface moment 470.1 m.-MT
```

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GHS 18.90 **PONTOON**

Condition 11 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Stbd

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 1.447 @ 50.00f, 1.421 @ 25.00f, 1.395 @ 0.00
Trim: Fwd 0.052/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	79.4	0.806	1.502	0.05500	6.56
CRANE	57.0	4.107	4.802	0.05500	15.06
Total wind heeling moment to starboard------------->					21.61

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 25.145f TCG = 0.000 VCG = 5.269
Free Surface Adjustment: 0.400
Adjusted CG: LCG = 25.144f TCG = 0.000 VCG = 5.670

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. 50% Area	DeckImm Angle
1.383	0.06f	0.00	1,174.2	0.000	-0.018	0.0000	5.091
1.383	0.06f	0.05s	1,174.3	0.000	0.000	-0.0000	5.046
1.368	0.06f	5.05s	1,174.3	0.000	1.347	0.0588	0.046
1.368	0.06f	5.09s	1,174.3	0.000	1.359	0.0599	0.000
1.316	0.06f	10.05s	1,174.3	0.000	2.605	0.2319	-4.954
1.189	0.10f	15.05s	1,174.3	0.000	3.005	0.4841	-9.954
1.051	0.13f	20.05s	1,174.3	0.000	2.829	0.7428	-14.954
0.905	0.16f	25.05s	1,174.3	0.000	2.470	0.9753	-19.954
0.750	0.20f	30.05s	1,174.3	0.000	2.027	1.1722	-24.954
0.589	0.24f	35.05s	1,174.3	0.000	1.538	1.3281	-29.954
0.423	0.27f	40.05s	1,174.3	0.000	1.021	1.4400	-34.954
0.254	0.30f	45.05s	1,174.3	0.000	0.485	1.5058	-39.954
0.101	0.33f	49.50s	1,174.3	0.000	0.000	1.5247	-44.408
0.082	0.34f	50.05s	1,174.3	0.000	-0.060	1.5245	-44.954
-0.090	0.37f	55.05s	1,174.3	0.000	-0.609	1.4953	-49.954
-0.262	0.39f	60.05s	1,174.3	0.000	-1.156	1.4183	-54.954

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.61 (constant)

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GHS 18.90 **PONTOON**

Condition 11 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Stbd

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LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max--------Attained
(1) Residual Area from abs 0 deg to MaxRA         >   0.0800 m.-Rad 0.4841 P
(2) Angle from Equilibrium to RAzero or Flood     >    20.00  deg    49.45 P
(3) Angle from Equilibrium to 50% Dk Imm Angle    >     0.00  deg     5.05 P
-----------------------------------------------------------------------------
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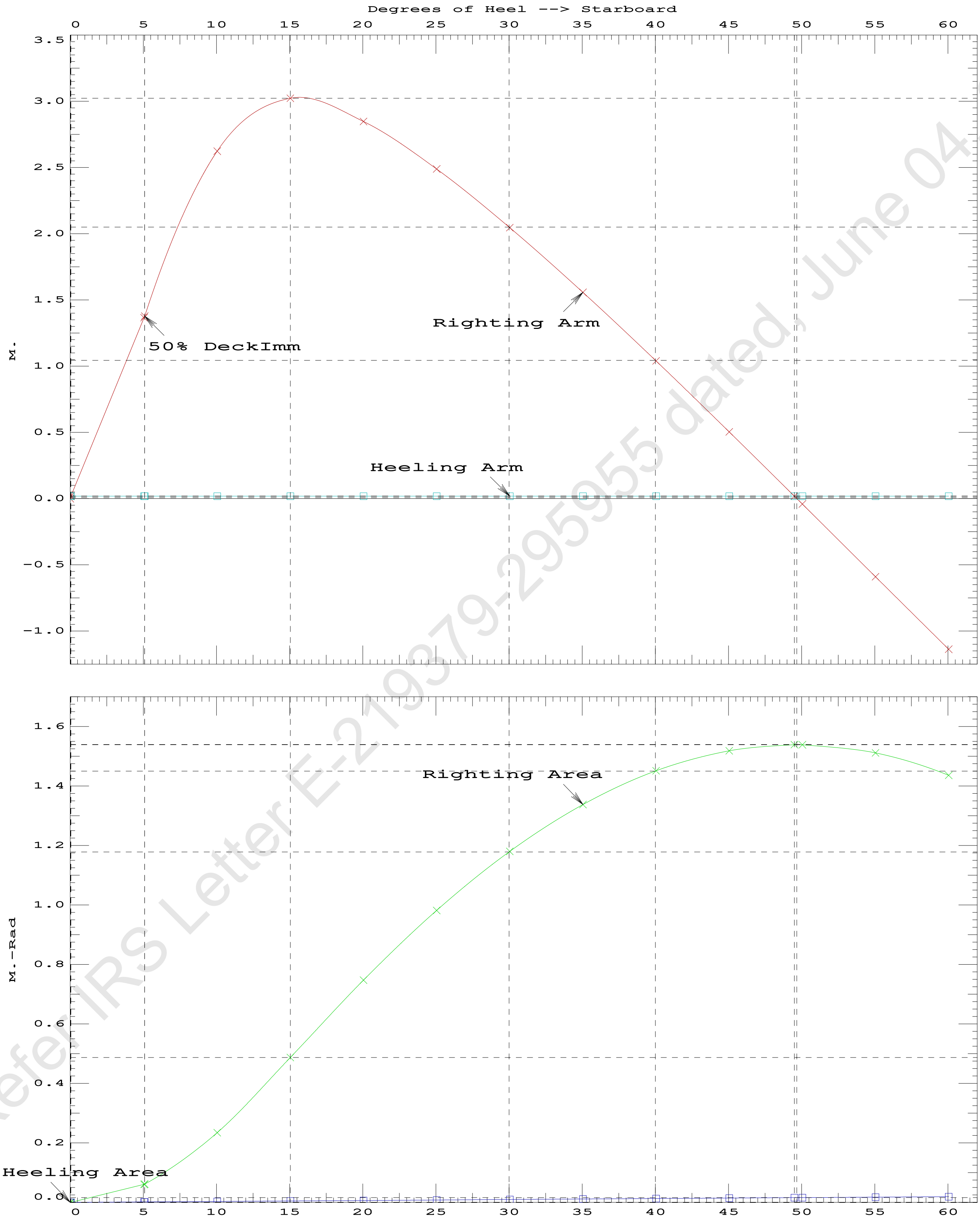
GHS 18.90 **PONTOON**

Condition 11 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Stbd



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GHS 18.90 **PONTOON**

Condition 11 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.145f TCG = 0.000 VCG = 5.269
Free Surface Adjustment: 0.400
Adjusted CG: LCG = 25.144f TCG = 0.014p VCG = 5.669

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.379	0.06f	2.02s	1,174.2	0.000	0.000	0.0000
1.355	0.06f	7.02s	1,174.4	0.000	1.342	0.0585
1.268	0.08f	12.02s	1,174.3	0.000	2.362	0.2225
1.184	0.10f	15.24s	1,174.0	0.000	2.475	0.3606
1.136	0.11f	17.02s	1,174.3	0.000	2.441	0.4368
0.995	0.14f	22.02s	1,174.3	0.000	2.172	0.6420
0.845	0.18f	27.02s	1,174.3	0.000	1.772	0.8150
0.688	0.21f	32.02s	1,174.3	0.000	1.308	0.9499
0.525	0.25f	37.02s	1,174.3	0.000	0.805	1.0424
0.357	0.28f	42.02s	1,174.3	0.000	0.279	1.0899
0.269	0.30f	44.61s	1,174.3	0.000	0.000	1.0962
0.186	0.32f	47.02s	1,174.3	0.000	-0.262	1.0907
0.014	0.35f	52.02s	1,174.3	0.000	-0.810	1.0440
-0.158	0.38f	57.02s	1,174.3	0.000	-1.360	0.9493
-0.329	0.40f	62.02s	1,174.3	0.000	-1.906	0.8068

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 658.80

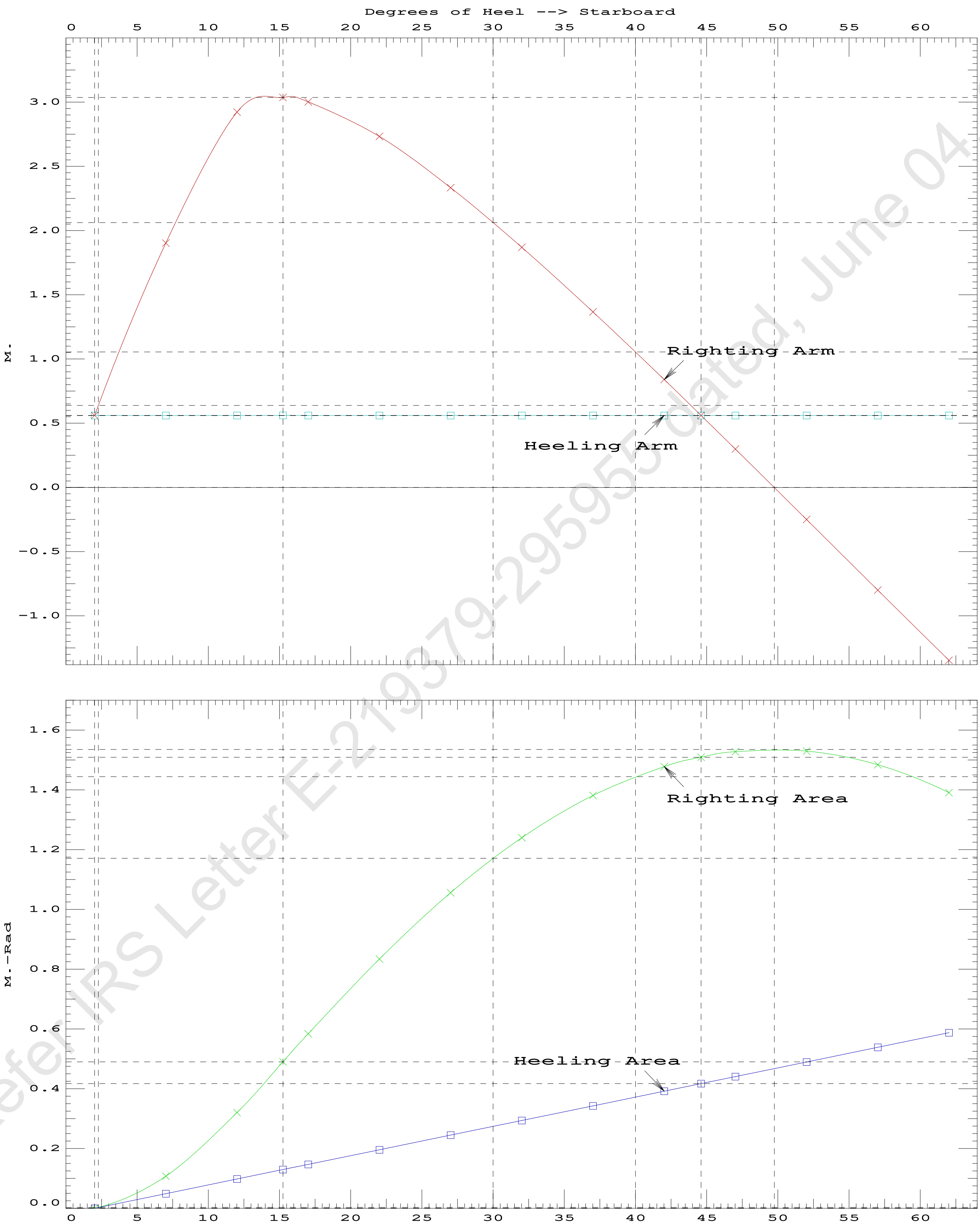
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM	STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained	
(1)	Absolute Angle at Equilibrium	<	15.00	deg	2.02	P
(2)	Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	441.2	P
(3)	Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		3.628	P

PONTOON

Condition 11 : Operating Condition of 250T Crane

Crane Lifting 36.6T @ 18m Radius Stbd



Condition 11 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Stbd
Crane in operation with load in place
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.445 @ 50.00f, 1.418 @ 25.00f, 1.392 @ 0.00
Trim: Fwd 0.052/50.000, Heel: Stbd 2.02 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			465.25	25.062f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Crane Load			36.60	7.500f	0.000	76.977	
Total Fixed--------->			751.85	18.367f	0.000	7.333	
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks--------->			422.40	37.208f	0.000	1.596	470.09
Total Weight-------->			1,174.25	25.145f	0.000	5.269	
			Displ(MT)	LCB	TCB	VCB	
HULL		1.025	1,174.25	25.150f	0.721s	0.731	

Righting Arms:	0.000	0.561s
External Arms:	0.000	0.561s
Residual Righting Arms:	0.000	0.000s

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	860.3	25.054f	0.102s	142.62	19.857*
		MT/cm				
		8.82				

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 2.02

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GHS 18.90                           PONTOON

                        After hook load drop
                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 1.502 @ 50.00f, 1.379 @ 25.00f, 1.256 @ 0.00
                   Trim: Fwd 0.246/50.000,  Heel: 0.00 deg.
Part------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                           465.25  25.062f  0.000   3.000
Crane                                250.00   7.500f  0.000   5.200
  Total Fixed--------->              715.25  18.924f  0.000   3.769
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025        161.60  34.000f  5.980p  1.509    162.73*
05_WBTK.S      1.000    1.025        161.60  34.000f  5.980s  1.509    162.73*
07_WBTK.P      1.000    1.025         49.60  47.660f  5.971p  1.880     72.32*
07_WBTK.S      1.000    1.025         49.60  47.660f  5.971s  1.880     72.32*
  Total Tanks--------->              422.40  37.208f  0.000   1.596   470.09
  Total Weight-------->            1,137.65  25.712f  0.000   2.962
                                   Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025      1,137.66  25.724f  0.001s  0.698
------------------------------------------------------------------------------
               Righting Arms:                 0.000   0.001s
               External Arms:                 0.000   0.001
      Residual Righting Arms:                 0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025          856.9  25.295f  0.021s   145.50  20.411*
                                       MT/cm--------m.-MT/cm-------GML-----GMT
                                        8.78           32.59    143.24  18.147
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


limmarg: 13437.11; t3: 30.00; t: 15.00
limmarg: 4973.24; t3: 15.00; t: 7.35
limmarg: 1330.40; t3: 7.65; t: 3.60
limmarg: 302.59; t3: 4.05; t: 1.76
limmarg: 29.00; t3: 2.29; t: 0.86
limmarg: -49.54; t3: 1.43; t: 0.42
limmarg: -15.73; t3: 1.85; t: 0.21
limmarg: 4.44; t3: 2.06; t: 0.10
limmarg: -5.44; t3: 1.96; t: 0.05
limmarg: -0.56; t3: 2.01; t: 0.02
limmarg: 1.42; t3: 2.03; t: 0.01

               Theta3 is 2.03 degrees, as demonstrated below.
```

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GHS 18.90 **PONTOON**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 18.924f TCG = 0.000 VCG = 3.769

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.240	0.28f	2.02s	1,137.6	0.000	-0.663	0.0000
1.244	0.28f	0.00	1,137.7	0.000	0.000	-0.0117
1.240	0.28f	2.03p	1,137.6	0.000	0.669	0.0002
1.229	0.28f	5.00p	1,137.6	0.000	1.640	0.0600
1.167	0.31f	10.00p	1,137.6	0.000	3.159	0.2701
0.975	0.44f	15.00p	1,137.7	0.000	3.811	0.5805
0.826	0.55f	18.57p	1,137.9	0.000	3.881	0.8216
0.765	0.59f	20.00p	1,137.7	0.000	3.870	0.9186
0.547	0.75f	25.00p	1,137.7	0.000	3.730	1.2526
0.321	0.90f	30.00p	1,137.7	0.000	3.495	1.5685

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 715.25 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM	STABILITY CRITERION		Min/Max	Attained
(1)	Residual Area Ratio from abs -2 deg to abs 2	>	1.000	1.014 P

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 18.924f TCG = 0.000 VCG = 3.769

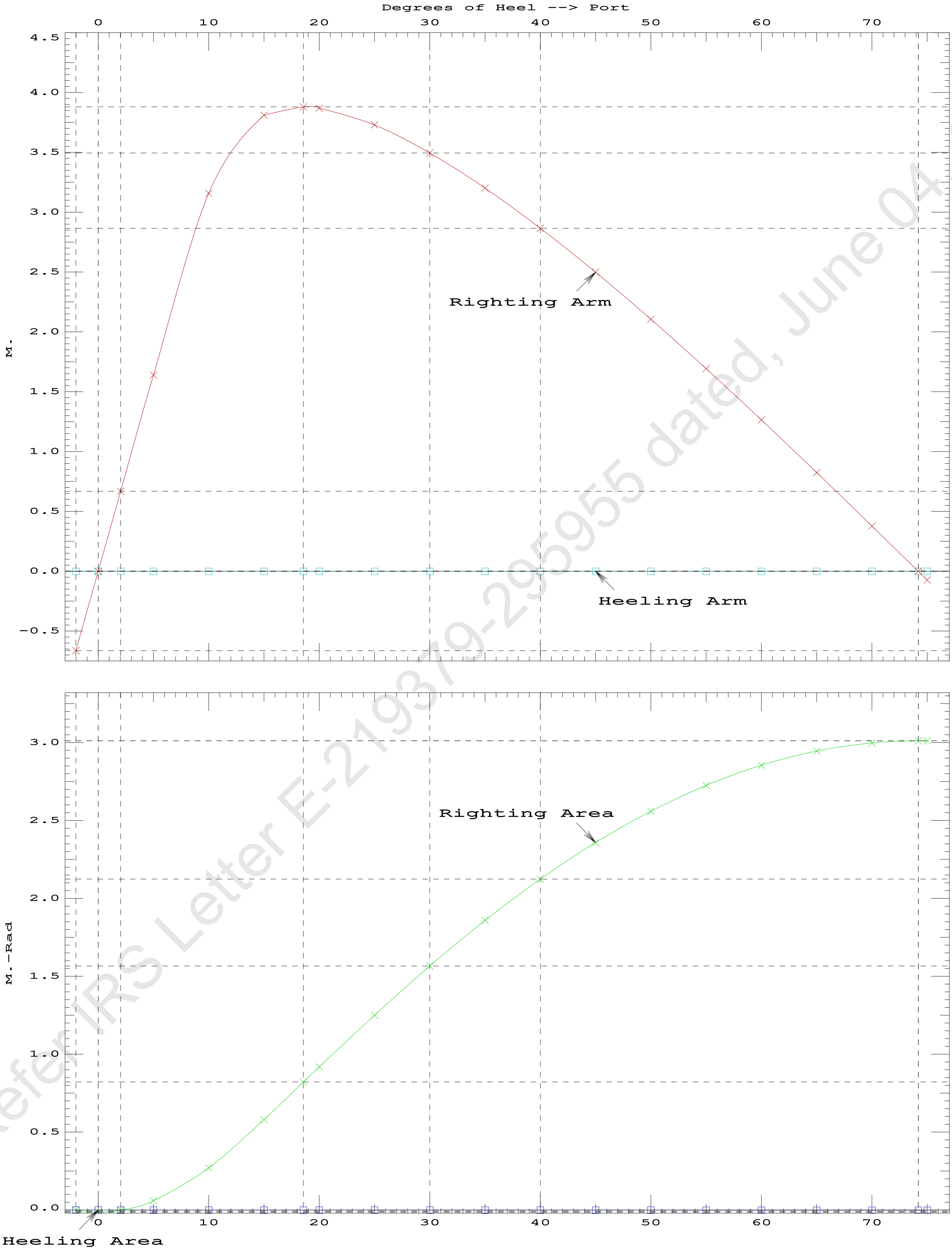
Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.240	0.28f	2.02s	1,137.6	0.000	-0.663	0.0000
1.244	0.28f	0.00	1,137.7	0.000	0.000	-0.0117
1.240	0.28f	2.03p	1,137.6	0.000	0.669	0.0002
1.229	0.28f	5.00p	1,137.6	0.000	1.640	0.0600
1.167	0.31f	10.00p	1,137.6	0.000	3.159	0.2701
0.975	0.44f	15.00p	1,137.7	0.000	3.811	0.5805
0.826	0.55f	18.57p	1,137.9	0.000	3.881	0.8216
0.765	0.59f	20.00p	1,137.7	0.000	3.870	0.9186
0.547	0.75f	25.00p	1,137.7	0.000	3.730	1.2526
0.321	0.90f	30.00p	1,137.7	0.000	3.495	1.5685
0.091	1.05f	35.00p	1,137.6	0.000	3.201	1.8612
-0.143	1.21f	40.00p	1,137.6	0.000	2.866	2.1262
-0.379	1.35f	45.00p	1,137.7	0.000	2.499	2.3605
-0.613	1.50f	50.00p	1,137.6	0.000	2.106	2.5616
-0.845	1.63f	55.00p	1,137.7	0.000	1.694	2.7276
-1.070	1.75f	60.00p	1,137.7	0.000	1.265	2.8568
-1.286	1.86f	65.00p	1,137.7	0.000	0.826	2.9481
-1.491	1.95f	70.00p	1,137.7	0.000	0.379	3.0007
-1.653	2.01f	74.20p	1,137.6	0.000	0.000	3.0146
-1.682	2.02f	75.00p	1,137.7	0.000	-0.072	3.0141

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 715.25 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM--------------RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Residual Area Ratio from abs -2 deg to RAzero	>	1.000		259.244 P
(2) Angle from abs 2 deg to RAzero	>	20.00	deg	72.17 P

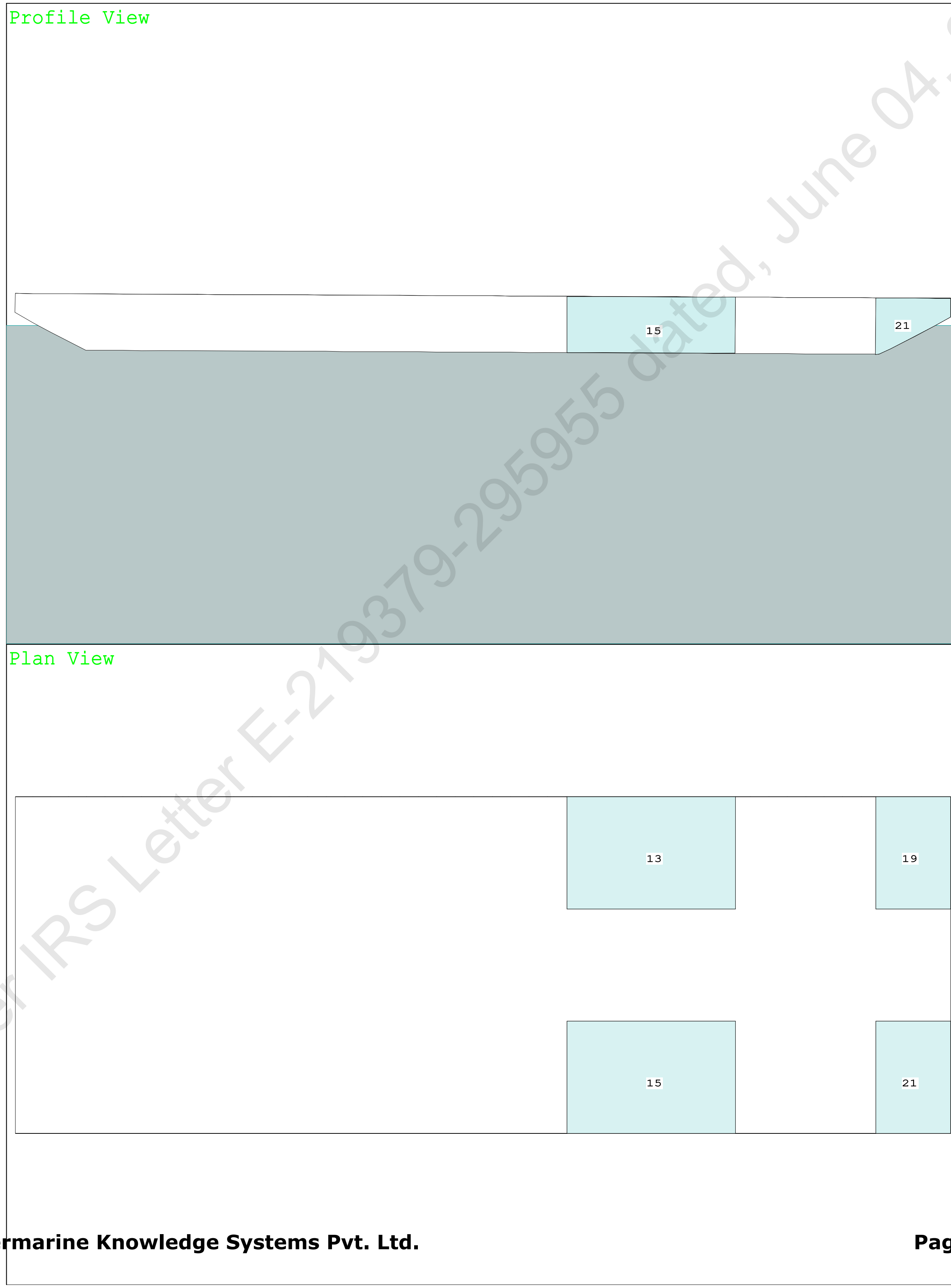

Degrees of Heel --> Port
M.
Righting Arm
Heeling Arm
M.-Rad
Righting Area
Heeling Area

PONTOON

Condition 12 : Operating Condition of 250T Crane

Crane Lifting 36.6T @ 18m Radius Fwd

Condition Graphic - Draft: 1.547 @ 50.000f, 1.294 @ 0.000 Heel: zero



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24-03-19 19:01:37    Cybermarine Technologies PTE. Ltd.
GHS 18.90                          PONTOON
```

Condition 12 : Operating Condition of 250T Crane

Crane Lifting 36.6T @ 18m Radius Fwd

```
                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 1.547 @ 50.00f, 1.420 @ 25.00f, 1.294 @ 0.00
                    Trim: Fwd 0.253/50.000,  Heel: zero
Part---------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                              465.25  25.062f  0.000   3.000
Crane                                   250.00   7.500f  0.000   5.200
Crane Load                               36.60  25.500f  0.000  76.977
  Total Fixed--------->                 751.85  19.244f  0.000   7.333
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025           161.60  34.000f  5.980p  1.509   162.73*
05_WBTK.S      1.000    1.025           161.60  34.000f  5.980s  1.509   162.73*
07_WBTK.P      1.000    1.025            49.60  47.660f  5.971p  1.880    72.32*
07_WBTK.S      1.000    1.025            49.60  47.660f  5.971s  1.880    72.32*
  Total Tanks--------->                 422.40  37.208f  0.000   1.596   470.09
  Total Weight-------->               1,174.25  25.706f  0.000   5.269
                                      Displ(MT)----LCB-----TCB-----VCB
HULL                            1.025 1,174.20  25.729f  0.000   0.720
------------------------------------------------------------------------------
               Righting Arms:                     0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025           862.5  25.220f  0.000    143.76  19.930*
                                        MT/cm--------m.-MT/cm-------GML-----GMT
                                         8.84             32.69    139.21  15.381
Distances in METERS.----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


                         SUMMARY OF LOADING
       412.1 Cu.M.  (16%) SALT WATER

      250.00 MT  Crane                        36.60 MT  Crane Load

DRAFT AT FP (M) =    1.547
DRAFT AT MS (M) =    1.420
DRAFT AT AP (M) =    1.294
DRAFT AT FWD MARK (M) =    1.527
DRAFT AT AFT MARK (M) =    1.314
                        HYDROSTATIC PROPERTIES
                Trim: Fwd 0.253/50.000, No Heel, VCG = 5.269

   LCF   Displacement   Buoyancy-Ctr.  Weight/         Moment/
  Draft----Weight(MT)----LCB-----VCB-------cm-----LCF---cm trim----GML-----GMT
  1.421     1,174.20  25.729f  0.720     8.84  25.220f   32.69  139.21  15.381
 Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
                             Trim is per 50.00m.
 Draft is from USK.                              Formal Free Surface included.

      Note:  GMT includes the formal free surface moment 470.1 m.-MT
```

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GHS 18.90 **PONTOON**

Condition 12 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Fwd

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 1.547 @ 50.00f, 1.420 @ 25.00f, 1.294 @ 0.00
Trim: Fwd 0.253/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	79.4	0.808	1.504	0.05500	6.57
CRANE	57.0	4.167	4.863	0.05500	15.25
Total wind heeling moment to starboard------------->					21.81

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 25.706f TCG = 0.000 VCG = 5.269
Free Surface Adjustment: 0.400
Adjusted CG: LCG = 25.704f TCG = 0.000 VCG = 5.670

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area	50% DeckImm Angle
1.282	0.29f	0.00	1,174.3	0.000	-0.018	0.0000	4.734
1.282	0.29f	0.06s	1,174.3	0.000	0.000	-0.0000	4.672
1.268	0.29f	4.73s	1,174.3	0.000	1.263	0.0515	0.000
1.266	0.29f	5.06s	1,174.2	0.000	1.352	0.0590	-0.328
1.206	0.31f	10.06s	1,174.2	0.000	2.603	0.2322	-5.328
1.026	0.47f	15.06s	1,174.3	0.000	2.993	0.4837	-10.328
0.831	0.63f	20.06s	1,174.2	0.000	2.817	0.7413	-15.328
0.626	0.80f	25.06s	1,174.3	0.000	2.457	0.9728	-20.328
0.412	0.97f	30.06s	1,174.2	0.000	2.014	1.1685	-25.328
0.191	1.15f	35.06s	1,174.2	0.000	1.525	1.3232	-30.328
-0.034	1.32f	40.06s	1,174.2	0.000	1.008	1.4339	-35.328
-0.261	1.48f	45.06s	1,174.2	0.000	0.473	1.4986	-40.328
-0.458	1.62f	49.40s	1,174.3	0.000	0.000	1.5166	-44.671
-0.487	1.64f	50.06s	1,174.3	0.000	-0.072	1.5162	-45.328
-0.709	1.78f	55.06s	1,174.2	0.000	-0.620	1.4860	-50.328
-0.924	1.91f	60.06s	1,174.3	0.000	-1.166	1.4081	-55.328

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.81 (constant)

```
24-03-19 19:01:37    Cybermarine Technologies PTE. Ltd.
GHS 18.90                        PONTOON
```

Condition 12 : Operating Condition of 250T Crane

Crane Lifting 36.6T @ 18m Radius Fwd

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LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max--------Attained
(1) Residual Area from abs 0 deg to MaxRA         >   0.0800 m.-Rad 0.4837 P
(2) Angle from Equilibrium to RAzero or Flood     >    20.00  deg    49.34 P
(3) Angle from Equilibrium to 50% Dk Imm Angle    >     0.00  deg     4.67 P
-----------------------------------------------------------------------------
```

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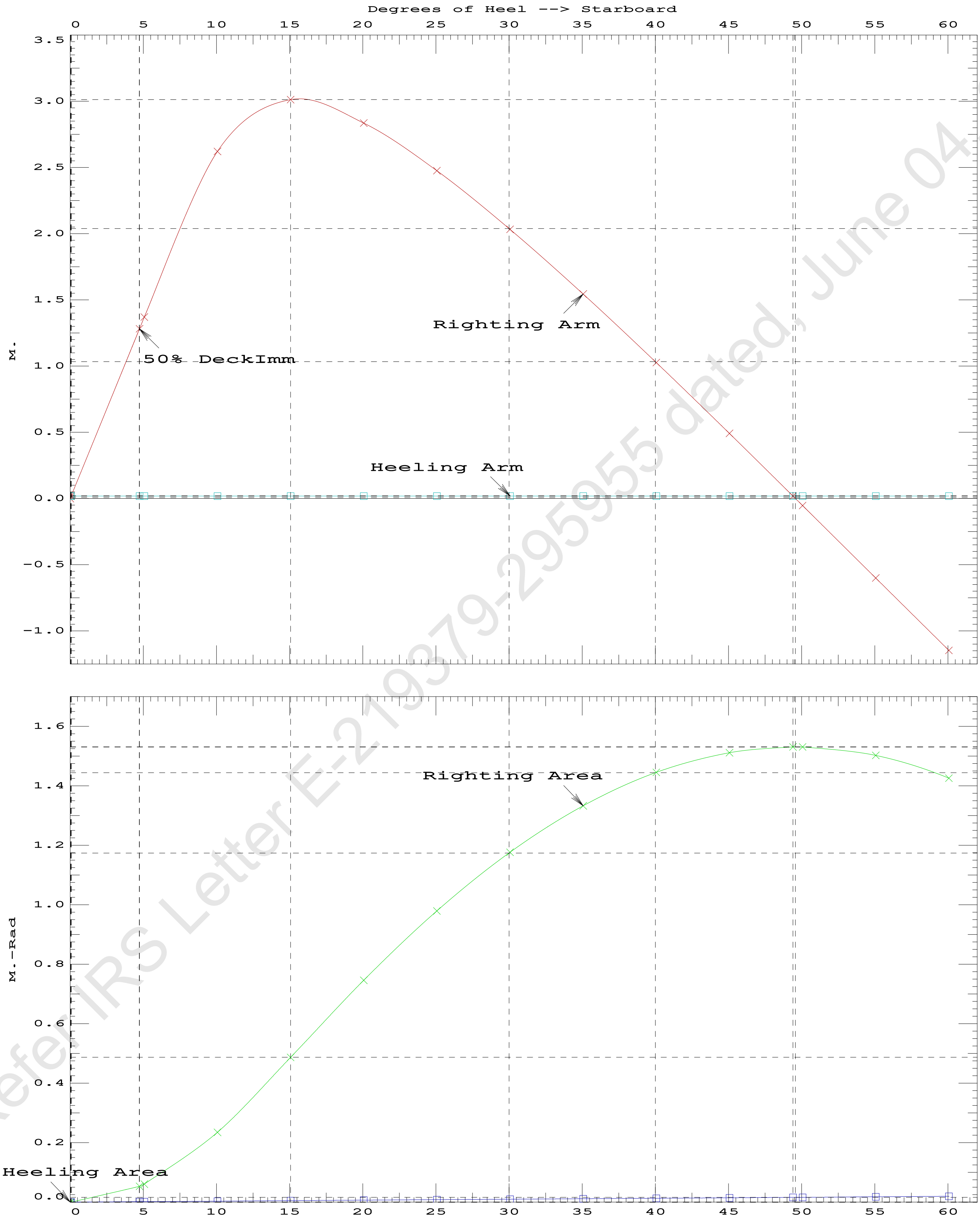
GHS 18.90 **PONTOON**

Condition 12 : Operating Condition of 250T Crane

Crane Lifting 36.6T @ 18m Radius Fwd



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GHS 18.90 **PONTOON**

Condition 12 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.706f TCG = 0.000 VCG = 5.269
Free Surface Adjustment: 0.400
Adjusted CG: LCG = 25.704f TCG = 0.000 VCG = 5.670

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.281	0.29f	0.01s	1,174.3	0.000	0.000	0.0000
1.266	0.29f	5.01s	1,174.2	0.000	1.352	0.0590
1.207	0.31f	10.01s	1,174.2	0.000	2.606	0.2324
1.028	0.47f	15.01s	1,174.3	0.000	3.007	0.4847
0.834	0.63f	20.01s	1,174.2	0.000	2.834	0.7437
0.628	0.80f	25.01s	1,174.3	0.000	2.475	0.9767
0.414	0.97f	30.01s	1,174.2	0.000	2.033	1.1741
0.194	1.14f	35.01s	1,174.2	0.000	1.544	1.3305
-0.032	1.32f	40.01s	1,174.2	0.000	1.027	1.4429
-0.259	1.48f	45.01s	1,174.2	0.000	0.492	1.5093
-0.463	1.62f	49.53s	1,174.3	0.000	0.000	1.5288
-0.485	1.64f	50.01s	1,174.3	0.000	-0.052	1.5286
-0.707	1.78f	55.01s	1,174.2	0.000	-0.600	1.5001
-0.922	1.91f	60.01s	1,174.2	0.000	-1.146	1.4239

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

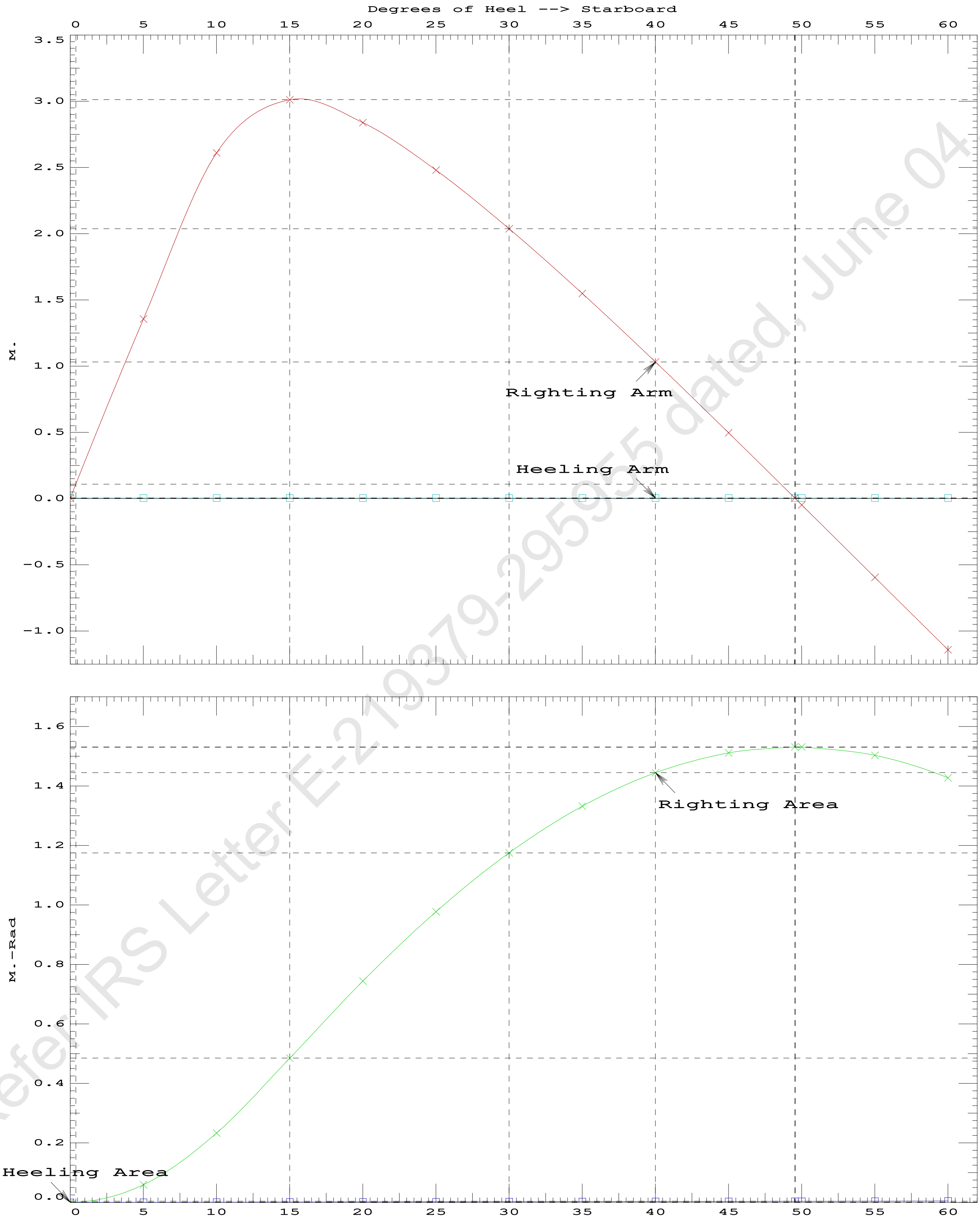
Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM---STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained	
(1) Absolute Angle at Equilibrium	<	15.00	deg	0.01	P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	70630.8	P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		416.384	P

Condition 12 : Operating Condition of 250T Crane

Crane Lifting 36.6T @ 18m Radius Fwd



Condition 12 : Operating Condition of 250T Crane

Crane Lifting 36.6T @ 18m Radius Fwd

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.547 @ 50.00f, 1.420 @ 25.00f, 1.294 @ 0.00

Trim: Fwd 0.253/50.000, Heel: Stbd 0.01 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			465.25	25.062f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Crane Load			36.60	25.500f	0.000	76.977	
Total Fixed--------->			751.85	19.244f	0.000	7.333	
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks--------->			422.40	37.208f	0.000	1.596	470.09
Total Weight-------->			1,174.25	25.706f	0.000	5.269	

	SpGr	Displ(MT)	LCB	TCB	VCB
HULL	1.025	1,174.25	25.729f	0.005s	0.720

Righting Arms:	0.000	0.004s
External Arms:	0.000	0.004s
Residual Righting Arms:	0.000	0.000

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	858.8	25.316f	0.027s	141.92	19.822*
		MT/cm 8.80				

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.01

```
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                          After hook load drop
                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 1.502 @ 50.00f, 1.379 @ 25.00f, 1.256 @ 0.00
                  Trim: Fwd 0.246/50.000,  Heel: 0.00 deg.
Part------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                            465.25  25.062f  0.000   3.000
Crane                                 250.00   7.500f  0.000   5.200
  Total Fixed--------->               715.25  18.924f  0.000   3.769
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025         161.60  34.000f  5.980p  1.509    162.73*
05_WBTK.S      1.000    1.025         161.60  34.000f  5.980s  1.509    162.73*
07_WBTK.P      1.000    1.025          49.60  47.660f  5.971p  1.880     72.32*
07_WBTK.S      1.000    1.025          49.60  47.660f  5.971s  1.880     72.32*
  Total Tanks--------->               422.40  37.208f  0.000   1.596   470.09
  Total Weight-------->             1,137.65  25.712f  0.000   2.962
                                  Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025       1,137.66  25.723f  0.001s  0.698
------------------------------------------------------------------------------
               Righting Arms:                  0.000   0.001s
               External Arms:                  0.000   0.001
      Residual Righting Arms:                  0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025          856.9  25.295f  0.021s   145.50  20.411*
                                       MT/cm--------m.-MT/cm-------GML-----GMT
                                        8.78           32.59    143.24  18.147
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


limmarg: 454640160.00; t3: 30; t: 15.00
limmarg: 170378032.00; t3: 15; t: 7.50
limmarg: 46185264.00; t3: 8; t: 3.75
limmarg: 14882572.00; t3: 4; t: 1.88
limmarg: 3715840.25; t3: 2; t: 0.94
limmarg: 933791.88; t3: 1; t: 0.47
limmarg: 235861.30; t3: 1; t: 0.23
limmarg: 494596.00; t3: 1; t: 0.12
limmarg: 644861.19; t3: 1; t: 0.06
limmarg: 735230.69; t3: 1; t: 0.03
limmarg: 782643.00; t3: 1; t: 0.01

             Theta3 is 0.97 degrees, as demonstrated below.
```

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 18.924f TCG = 0.000 VCG = 3.769

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.244	0.28f	0.01s	1,137.7	0.000	-0.004	0.0000
1.244	0.28f	0.00	1,137.7	0.000	0.000	-0.0000
1.241	0.29f	0.97p	1,137.6	0.000	0.321	0.0027
1.229	0.28f	5.00p	1,137.7	0.000	1.639	0.0717
1.167	0.31f	10.00p	1,137.6	0.000	3.159	0.2819
0.975	0.44f	15.00p	1,137.7	0.000	3.811	0.5923
0.826	0.55f	18.57p	1,137.9	0.000	3.881	0.8334
0.765	0.59f	20.00p	1,137.7	0.000	3.870	0.9304
0.547	0.75f	25.00p	1,137.7	0.000	3.730	1.2644
0.321	0.90f	30.00p	1,137.7	0.000	3.495	1.5804

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 715.25 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM	STABILITY CRITERION		Min/Max	Attained
(1)	Residual Area Ratio from abs 0 deg to abs 1	>	1.000	7827.43 P

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 18.924f TCG = 0.000 VCG = 3.769

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.244	0.28f	0.01s	1,137.7	0.000	-0.004	0.0000
1.244	0.28f	0.00	1,137.7	0.000	0.000	-0.0000
1.241	0.29f	0.97p	1,137.6	0.000	0.321	0.0027
1.229	0.28f	5.00p	1,137.7	0.000	1.639	0.0717
1.167	0.31f	10.00p	1,137.6	0.000	3.159	0.2819
0.975	0.44f	15.00p	1,137.7	0.000	3.811	0.5923
0.826	0.55f	18.57p	1,137.9	0.000	3.881	0.8334
0.765	0.59f	20.00p	1,137.7	0.000	3.870	0.9304
0.547	0.75f	25.00p	1,137.7	0.000	3.730	1.2644
0.321	0.90f	30.00p	1,137.7	0.000	3.495	1.5804
0.091	1.05f	35.00p	1,137.6	0.000	3.201	1.8730
-0.143	1.21f	40.00p	1,137.6	0.000	2.866	2.1380
-0.379	1.35f	45.00p	1,137.7	0.000	2.499	2.3723
-0.613	1.50f	50.00p	1,137.6	0.000	2.106	2.5734
-0.845	1.63f	55.00p	1,137.7	0.000	1.694	2.7394
-1.070	1.75f	60.00p	1,137.7	0.000	1.265	2.8686
-1.286	1.86f	65.00p	1,137.7	0.000	0.826	2.9599
-1.491	1.95f	70.00p	1,137.7	0.000	0.379	3.0125
-1.653	2.01f	74.20p	1,137.6	0.000	0.000	3.0264
-1.682	2.02f	75.00p	1,137.7	0.000	-0.072	3.0259

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 715.25 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM--------------RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Residual Area Ratio from abs 0 deg to RAzero	>	1.000		8707224 P
(2) Angle from abs 1 deg to RAzero	>	20.00	deg	73.23 P

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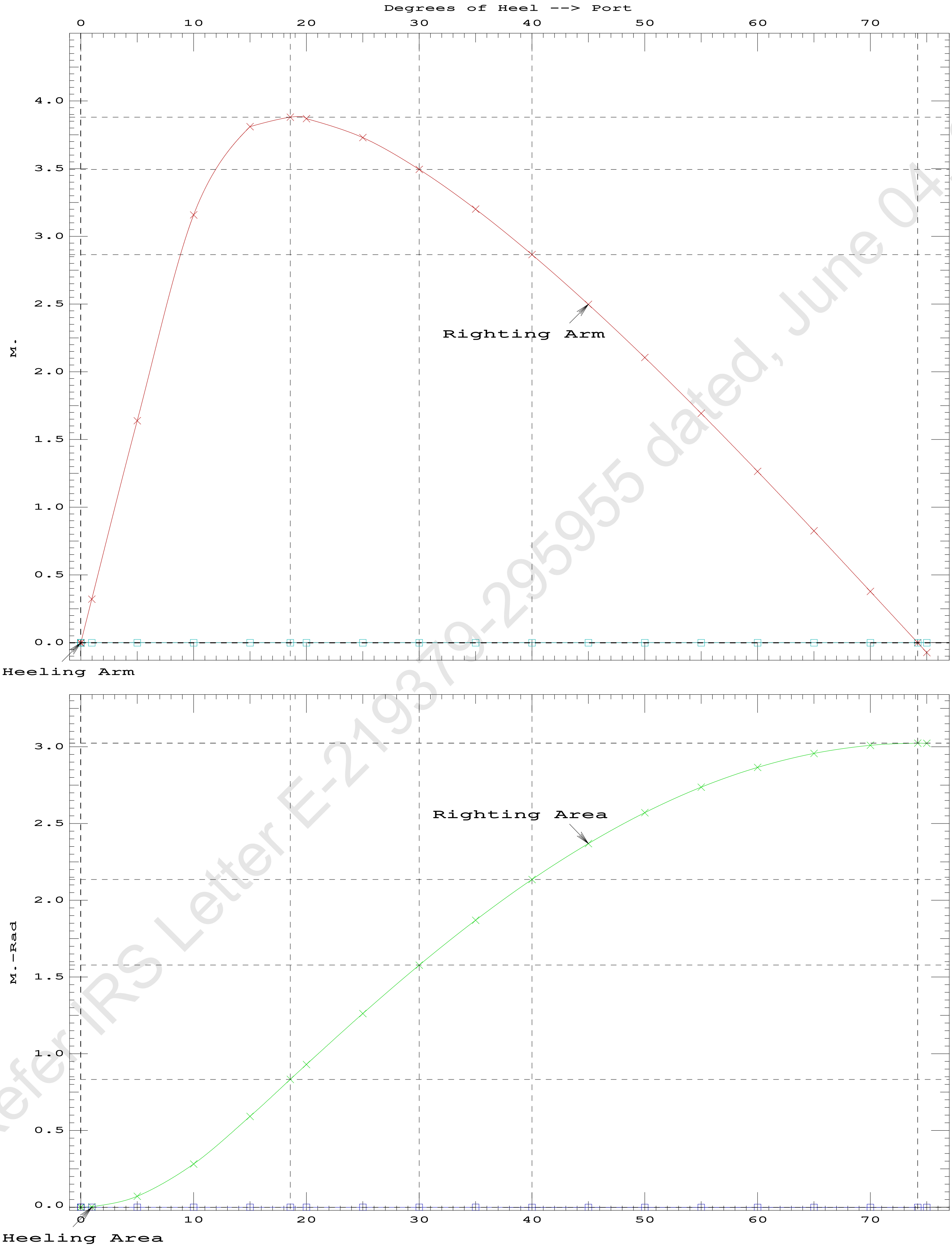
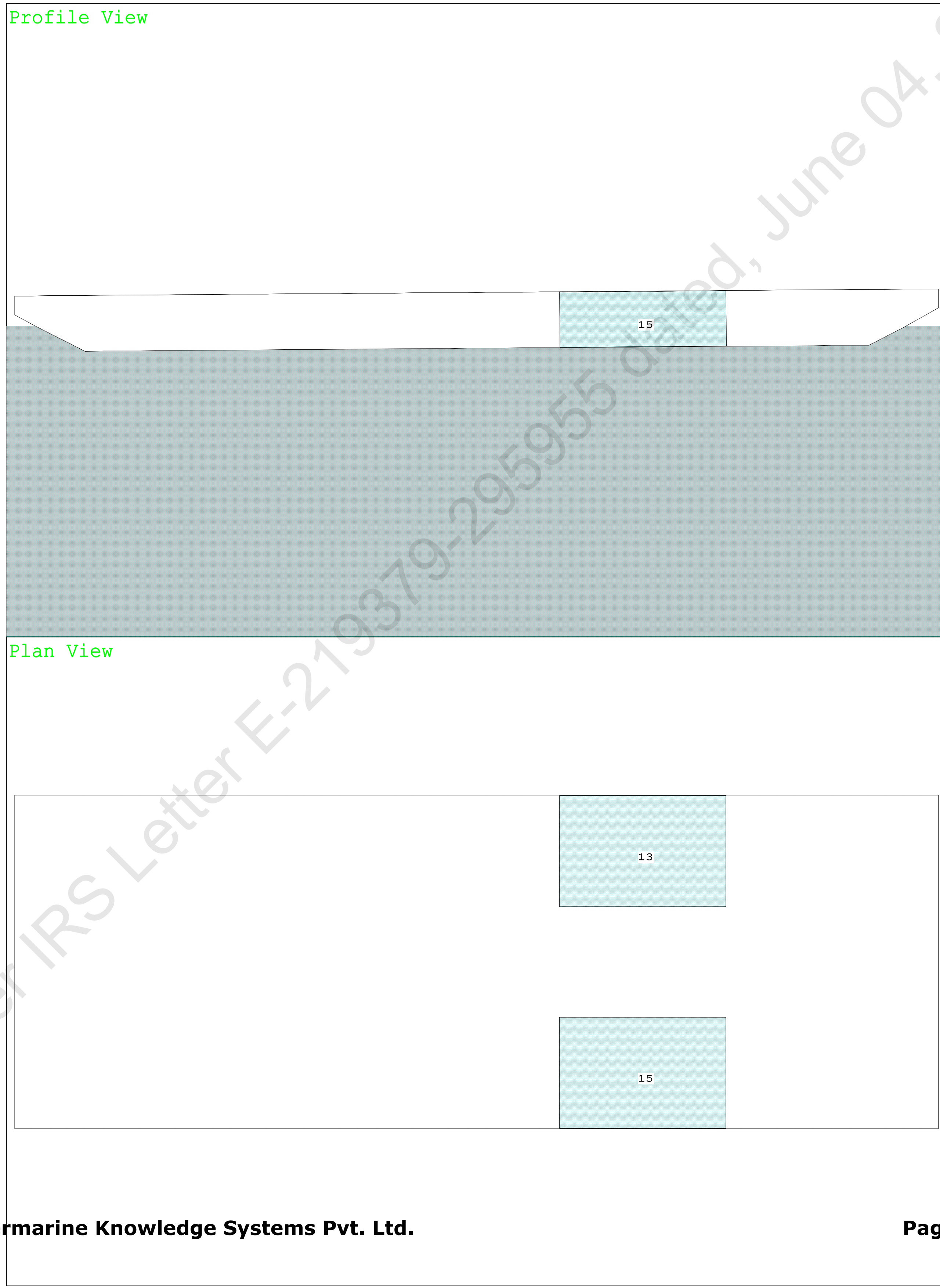
GHS 18.90 **PONTOON**



PONTOON

Condition 13 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Aft

Condition Graphic - Draft: 1.018 @ 50.000f, 1.400 @ 0.000 Heel: zero



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24-03-19 19:08:55    Cybermarine Technologies PTE. Ltd.
GHS 18.90                        PONTOON
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Condition 13 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Aft

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                 WEIGHT and DISPLACEMENT and WATERPLANE STATUS
            USK draft: 1.018 @ 50.00f, 1.209 @ 25.00f, 1.400 @ 0.00
                      Trim: Aft 0.383/50.000,  Heel: zero
Part----------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                              465.25  25.062f  0.000   3.000
Crane                                   150.00   7.500f  0.000   5.200
Crane Load                               51.30   4.500a  0.000  33.218
  Total Fixed--------->                 666.55  18.835f  0.000   5.821
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025       161.60  34.000f  5.980p  1.509    162.73*
05_WBTK.S      1.000    1.025       161.60  34.000f  5.980s  1.509    162.73*
  Total Tanks--------->             323.20  34.000f  0.000   1.509    325.46
  Total Weight-------->             989.75  23.787f  0.000   4.413
                                  Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025       989.65  23.758f  0.000   0.613
------------------------------------------------------------------------------
               Righting Arms:                0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025        848.7  24.656f  0.000    162.51  23.415*
                                     MT/cm--------m.-MT/cm-------GML-----GMT
                                      8.70            31.41    158.71  19.616
Distances in METERS.----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


                            SUMMARY OF LOADING
       315.3 Cu.M.  (13%) SALT WATER

      150.00 MT  Crane                        51.30 MT  Crane Load

DRAFT AT FP (M) =    1.018
DRAFT AT MS (M) =    1.209
DRAFT AT AP (M) =    1.400
DRAFT AT FWD MARK (M) =    1.048
DRAFT AT AFT MARK (M) =    1.370
                           HYDROSTATIC PROPERTIES
                Trim: Aft 0.383/50.000, No Heel, VCG = 4.413

   LCF  Displacement   Buoyancy-Ctr.  Weight/         Moment/
  Draft----Weight(MT)----LCB-----VCB-------cm-----LCF---cm trim----GML-----GMT
  1.212       989.65  23.758f  0.613     8.70  24.656f   31.41  158.71  19.616
 Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
                            Trim is per 50.00m.
 Draft is from USK.                              Formal Free Surface included.

      Note:  GMT includes the formal free surface moment 325.5 m.-MT
```

24-03-19 19:08:55 Cybermarine Technologies PTE. Ltd.
GHS 18.90 **PONTOON**

Condition 13 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Aft

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 1.018 @ 50.00f, 1.209 @ 25.00f, 1.400 @ 0.00
Trim: Aft 0.383/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	89.4	0.919	1.516	0.05500	7.45
CRANE	57.0	4.189	4.786	0.05500	15.00
Total wind heeling moment to starboard------------->					22.46

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 23.787f TCG = 0.000 VCG = 4.413
Free Surface Adjustment: 0.329
Adjusted CG: LCG = 23.789f TCG = 0.000 VCG = 4.742

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area	50% DeckImm Angle
1.388	0.44a	0.00	989.74	0.000	-0.022	0.0000	5.257
1.388	0.44a	0.06s	989.75	0.000	0.000	-0.0000	5.200
1.373	0.44a	5.06s	989.74	0.000	1.716	0.0748	0.200
1.372	0.44a	5.26s	989.69	0.000	1.783	0.0810	0.000
1.326	0.50a	10.06s	989.65	0.000	3.162	0.2898	-4.800
1.196	0.67a	15.06s	989.75	0.000	3.686	0.5948	-9.800
1.157	0.74a	16.64s	989.75	0.000	3.705	0.6969	-11.382
1.069	0.90a	20.06s	989.75	0.000	3.626	0.9168	-14.800
0.939	1.13a	25.06s	989.75	0.000	3.346	1.2234	-19.800
0.806	1.37a	30.06s	989.75	0.000	2.964	1.4995	-24.800
0.671	1.60a	35.06s	989.75	0.000	2.523	1.7393	-29.800
0.535	1.83a	40.06s	989.75	0.000	2.043	1.9388	-34.800
0.398	2.06a	45.06s	989.75	0.000	1.535	2.0951	-39.800
0.260	2.27a	50.06s	989.75	0.000	1.009	2.2063	-44.800
0.121	2.47a	55.06s	989.75	0.000	0.471	2.2710	-49.800
-0.001	2.63a	59.38s	989.75	0.000	0.000	2.2888	-54.127
-0.020	2.65a	60.06s	989.75	0.000	-0.074	2.2884	-54.800

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 325.5 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 22.46 (constant)

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GHS 18.90 **PONTOON**

Condition 13 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Aft

```
LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max--------Attained
(1) Residual Area from abs 0 deg to MaxRA         >   0.0800 m.-Rad 0.6969 P
(2) Angle from Equilibrium to RAzero or Flood     >    20.00  deg    59.33 P
(3) Angle from Equilibrium to 50% Dk Imm Angle    >     0.00  deg     5.20 P
-----------------------------------------------------------------------------
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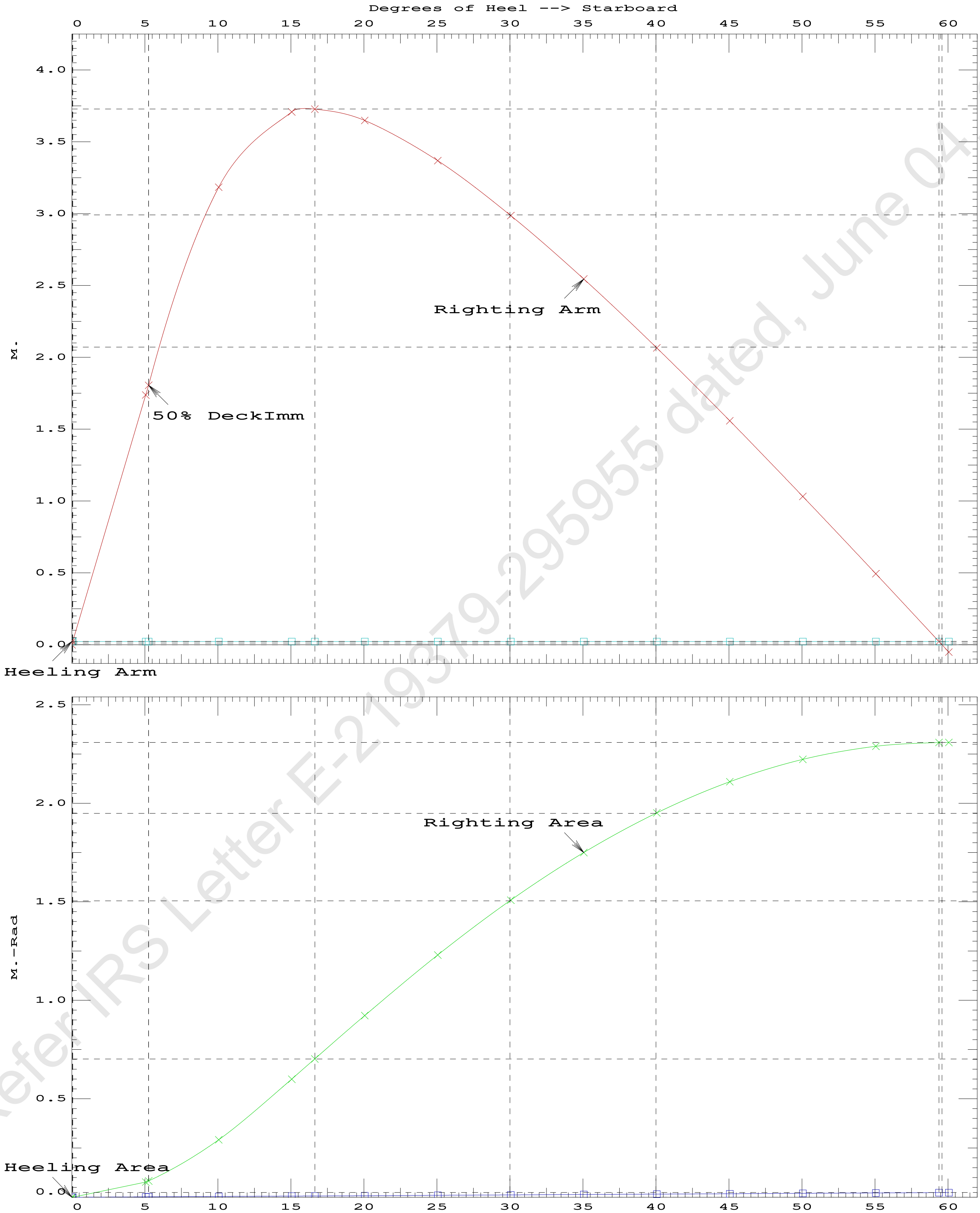
Condition 13 : Operating Condition of 150T Crane

Crane Lifting 51.3T @ 12m Radius Aft



24-03-19 19:08:55 Cybermarine Technologies PTE. Ltd.
GHS 18.90 **PONTOON**

Condition 13 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 23.787f TCG = 0.000 VCG = 4.413
Free Surface Adjustment: 0.329
Adjusted CG: LCG = 23.789f TCG = 0.000 VCG = 4.742

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.388	0.44a	0.01s	989.75	0.000	0.000	0.0000
1.374	0.44a	5.01s	989.75	0.000	1.716	0.0749
1.327	0.50a	10.01s	989.65	0.000	3.169	0.2899
1.198	0.66a	15.01s	989.75	0.000	3.701	0.5960
1.157	0.74a	16.64s	989.75	0.000	3.722	0.7020
1.071	0.89a	20.01s	989.75	0.000	3.645	0.9194
0.940	1.13a	25.01s	989.75	0.000	3.367	1.2279
0.807	1.36a	30.01s	989.75	0.000	2.985	1.5058
0.672	1.60a	35.01s	989.75	0.000	2.545	1.7475
0.536	1.83a	40.01s	989.75	0.000	2.065	1.9490
0.399	2.06a	45.01s	989.75	0.000	1.558	2.1072
0.262	2.27a	50.01s	989.75	0.000	1.032	2.2204
0.122	2.47a	55.01s	989.75	0.000	0.494	2.2871
-0.005	2.63a	59.54s	989.75	0.000	0.000	2.3067
-0.019	2.65a	60.01s	989.75	0.000	-0.051	2.3065

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 325.5 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM	STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained	
(1)	Absolute Angle at Equilibrium	<	15.00	deg	0.01	P
(2)	Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	73682.0	P
(3)	Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		440.409	P

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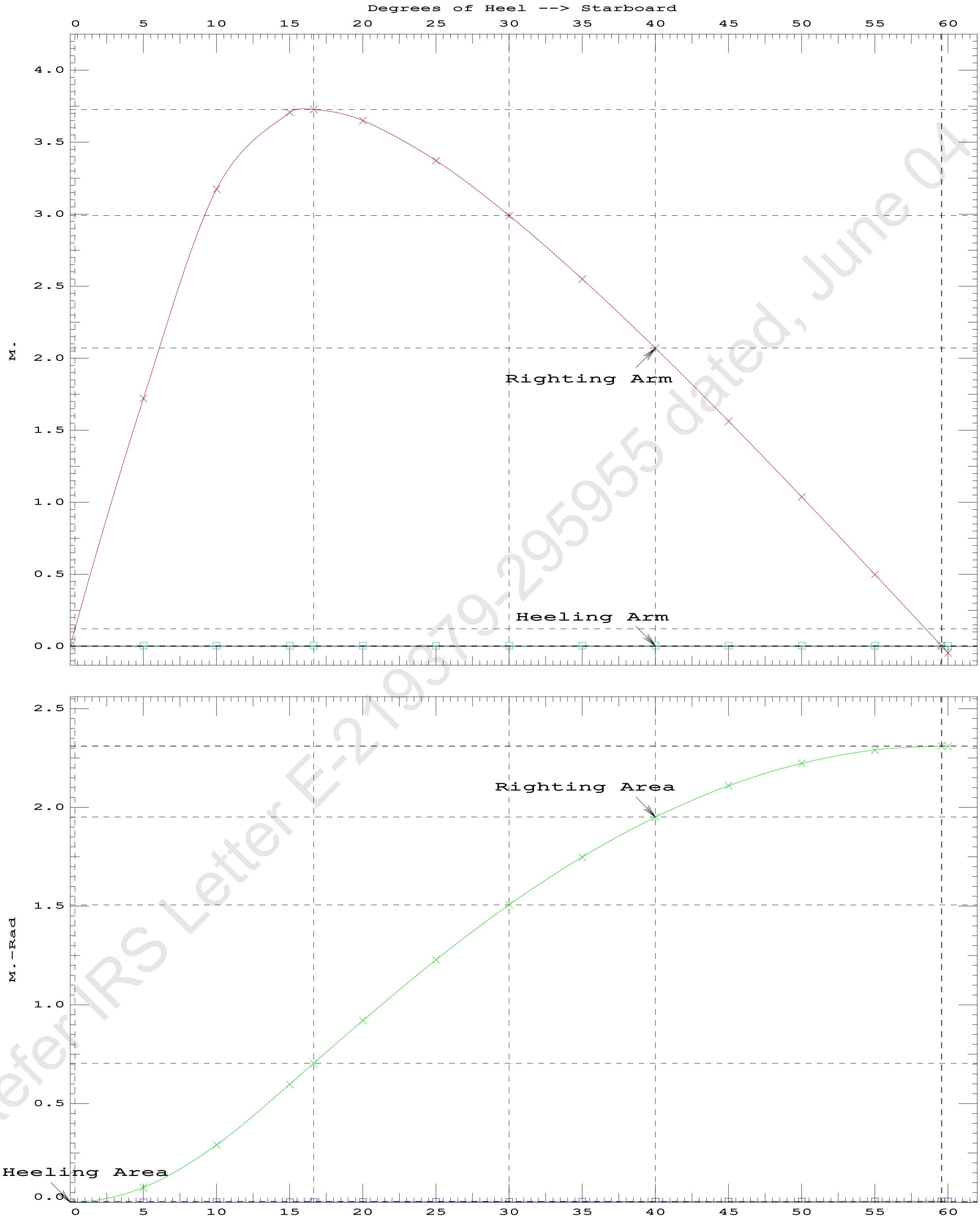
GHS 18.90 **PONTOON**

Condition 13 : Operating Condition of 150T Crane

Crane Lifting 51.3T @ 12m Radius Aft



24-03-20 13:29:00 Cybermarine Technologies PTE. Ltd.
GHS 18.90 **PONTOON**

Condition 13 : Operating Condition of 150T Crane

Crane in operation with load in place
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.018 @ 50.00f, 1.209 @ 25.00f, 1.400 @ 0.00
Trim: Aft 0.383/50.000, Heel: zero

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			465.25	25.062f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Crane Load			51.30	4.500a	0.000	33.218	
Total Fixed--------->			666.55	18.835f	0.000	5.821	
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
Total Tanks--------->			323.20	34.000f	0.000	1.509	325.46
Total Weight-------->			989.75	23.787f	0.000	4.413	

	SpGr	Displ(MT)	LCB	TCB	VCB
HULL	1.025	989.74	23.758f	0.000	0.613

Righting Arms:	0.000	0.000
External Arms:	0.000	0.001s
Residual Righting Arms:	0.000	-0.001s

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	848.7	24.656f	0.000	162.50	23.414*
		MT/cm		m.-MT/cm	GML	GMT
		8.70		31.42	158.70	19.614

Distances in METERS.---Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

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24-03-20 13:29:00    Cybermarine Technologies PTE. Ltd.
GHS 18.90                        PONTOON

                          After hook load drop
                  WEIGHT and DISPLACEMENT and WATERPLANE STATUS
              USK draft: 1.201 @ 50.00f, 1.151 @ 25.00f, 1.101 @ 0.00
                    Trim: Fwd 0.100/50.000,  Heel: 0.00 deg.
Part------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                           465.25  25.062f  0.000   3.000
Crane                                150.00   7.500f  0.000   5.200
  Total Fixed--------->              615.25  20.780f  0.000   3.536
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025        161.60  34.000f  5.980p  1.509    162.73*
05_WBTK.S      1.000    1.025        161.60  34.000f  5.980s  1.509    162.73*
  Total Tanks--------->              323.20  34.000f  0.000   1.509    325.46
  Total Weight-------->              938.45  25.333f  0.000   2.838
                                   Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025        938.45  25.338f  0.001s  0.578
------------------------------------------------------------------------------
               Righting Arms:                 0.000   0.001s
               External Arms:                 0.000   0.001s
      Residual Righting Arms:                 0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025         841.9  25.057f  0.020s   167.30  24.428*
                                      MT/cm--------m.-MT/cm-------GML-----GMT
                                       8.63           30.98    165.04  22.168
Distances in METERS.----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


limmarg: INFINITY; t3: 30.0; t: 15.00
limmarg: INFINITY; t3: 15.0; t: 7.50
limmarg: INFINITY; t3: 7.5; t: 3.75
limmarg: INFINITY; t3: 3.8; t: 1.88
limmarg: INFINITY; t3: 1.9; t: 0.94
limmarg: INFINITY; t3: 1.0; t: 0.47
limmarg: INFINITY; t3: 0.5; t: 0.23
limmarg: INFINITY; t3: 0.3; t: 0.12
limmarg: INFINITY; t3: 0.2; t: 0.06
limmarg: INFINITY; t3: 0.1; t: 0.03
limmarg: INFINITY; t3: 0.1; t: 0.01

                Theta3 is 0.07 degrees, as demonstrated below.

                      RESIDUAL RIGHTING ARMS vs HEEL ANGLE
          Fixed CG:  LCG = 20.780f  TCG = 0.000   VCG =  3.536

      Origin    Degrees of   Displacement    Residual Arms       Res.
       Depth---Trim----Heel----Weight(MT)---in Trim--in Heel---> Area
       1.089   0.12f   0.00       938.44     0.000    0.000    0.0000
       1.089   0.11f   0.07s      938.51     0.000    0.027    0.0000
       1.075   0.11f   5.00s      938.45     0.000    1.984    0.0866
       0.988   0.13f  10.00s      938.44     0.000    3.635    0.3342
       0.744   0.17f  15.00s      938.45     0.000    4.345    0.6892
       0.503   0.22f  19.29s      938.45     0.000    4.466    1.0215
       0.462   0.23f  20.00s      938.45     0.000    4.463    1.0772
```

```
  0.176   0.29f  25.00s      938.45     0.000    4.342    1.4640
 -0.112   0.34f  30.00s      938.45     0.000    4.106    1.8334
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.
```

Note: The Center of Gravity shown above is for the Fixed Weight of 615.25 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

```
LIM--------------------STABILITY CRITERION------------Min/Max--------Attained
(1) Residual Area Ratio from abs 0 deg to abs 0.1  >    1.000          LARGE
-----------------------------------------------------------------------------
```

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 20.780f TCG = 0.000 VCG = 3.536

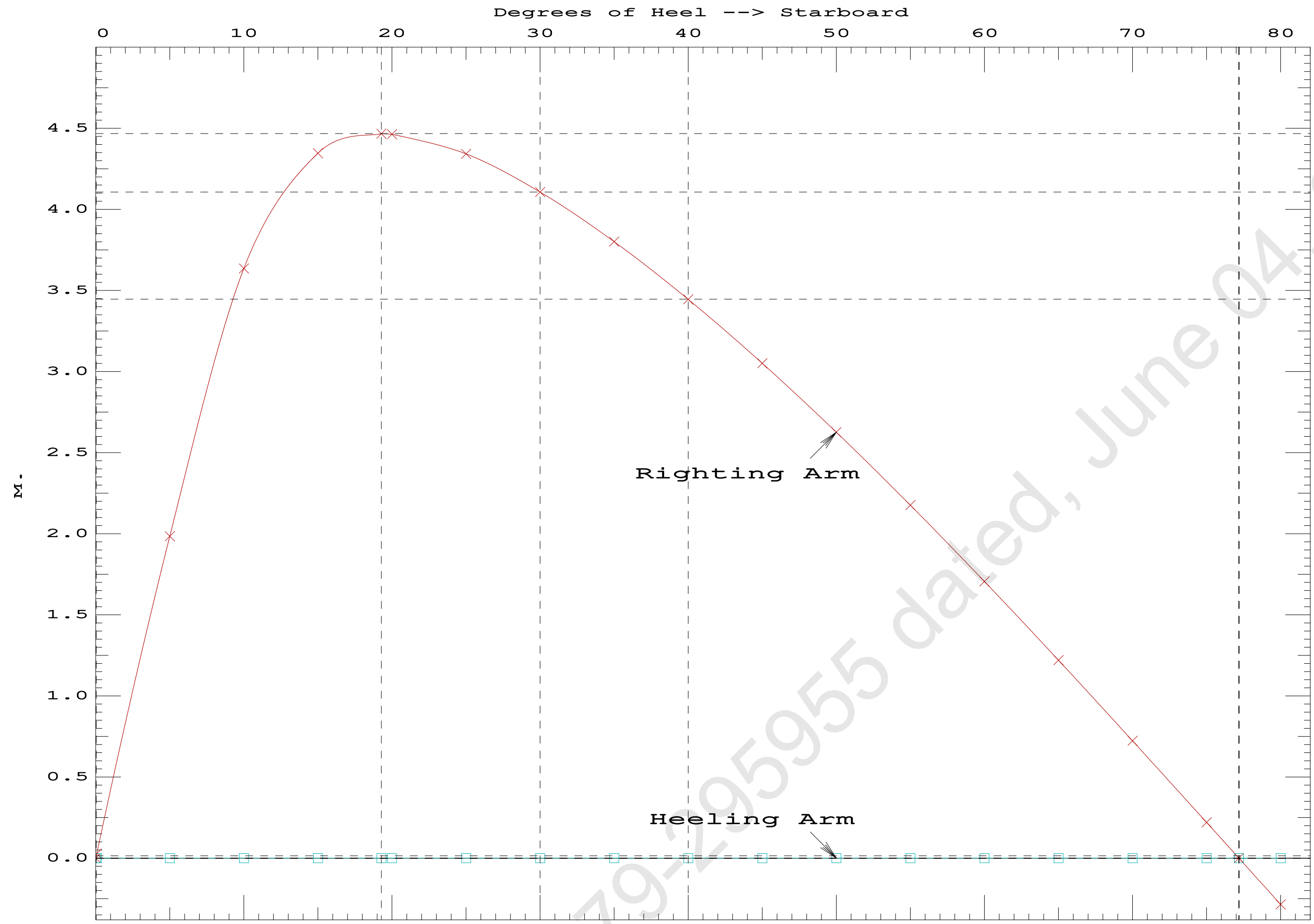
Origin Depth	Degrees of Trim	Heel	Displacement Weight (MT)	Residual Arms in Trim	in Heel	Res. Area
1.089	0.12f	0.00	938.44	0.000	0.000	0.0000
1.089	0.11f	0.07s	938.51	0.000	0.027	0.0000
1.075	0.11f	5.00s	938.45	0.000	1.984	0.0866
0.988	0.13f	10.00s	938.44	0.000	3.635	0.3342
0.744	0.17f	15.00s	938.45	0.000	4.345	0.6892
0.503	0.22f	19.29s	938.45	0.000	4.466	1.0215
0.462	0.23f	20.00s	938.45	0.000	4.463	1.0772
0.176	0.29f	25.00s	938.45	0.000	4.342	1.4640
-0.112	0.34f	30.00s	938.45	0.000	4.106	1.8334
-0.400	0.40f	35.00s	938.45	0.000	3.800	2.1789
-0.687	0.46f	40.00s	938.45	0.000	3.445	2.4954
-0.968	0.51f	45.00s	938.45	0.000	3.050	2.7791
-1.244	0.57f	50.00s	938.45	0.000	2.625	3.0269
-1.510	0.62f	55.00s	938.45	0.000	2.175	3.2366
-1.765	0.66f	60.00s	938.45	0.000	1.705	3.4060
-2.007	0.70f	65.00s	938.45	0.000	1.219	3.5338
-2.232	0.74f	70.00s	938.45	0.000	0.723	3.6186
-2.440	0.77f	75.00s	938.45	0.000	0.220	3.6598
-2.525	0.77f	77.18s	938.45	0.000	0.000	3.6640
-2.629	0.78f	80.00s	938.45	0.000	-0.286	3.6569

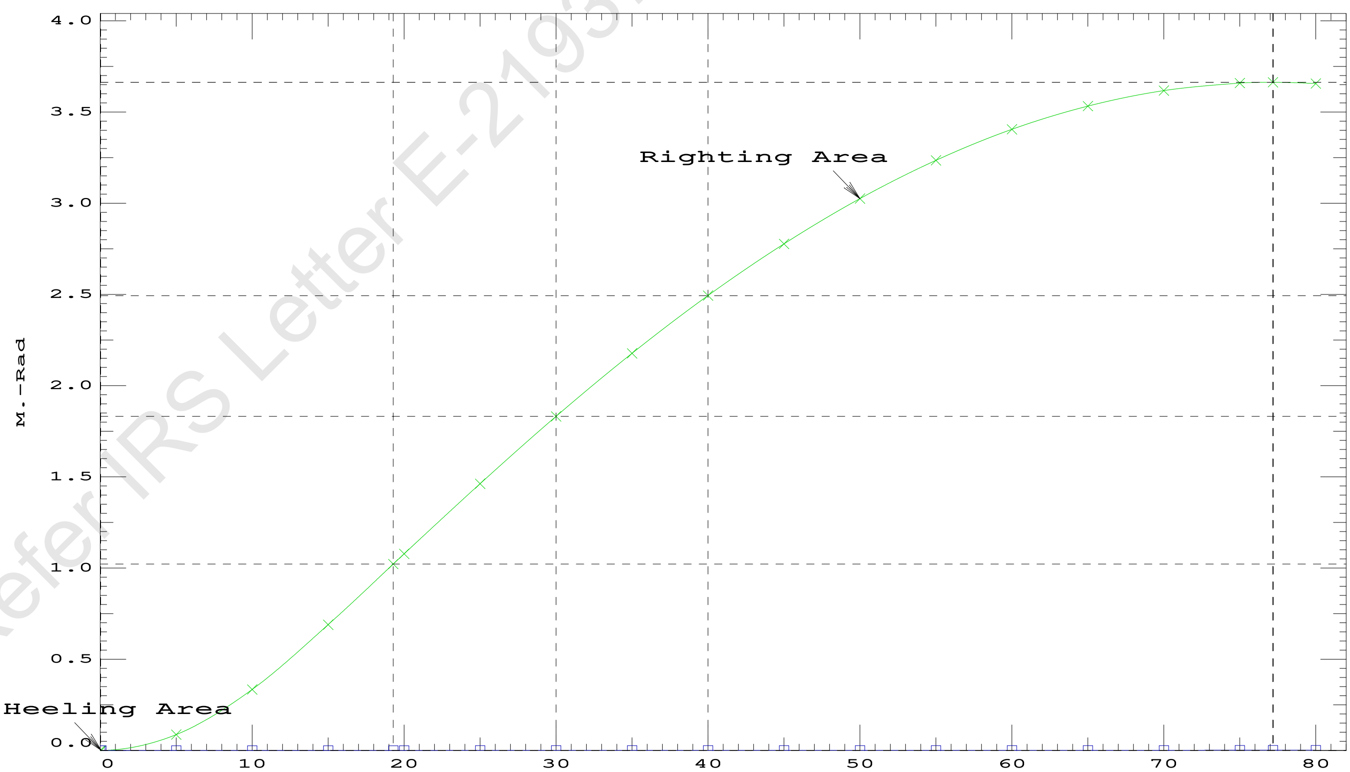
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 615.25 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM--------------RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Residual Area Ratio from abs 0 deg to RAzero	>	1.000		LARGE
(2) Angle from abs 0.1 deg to RAzero	>	20.00	deg	77.11 P

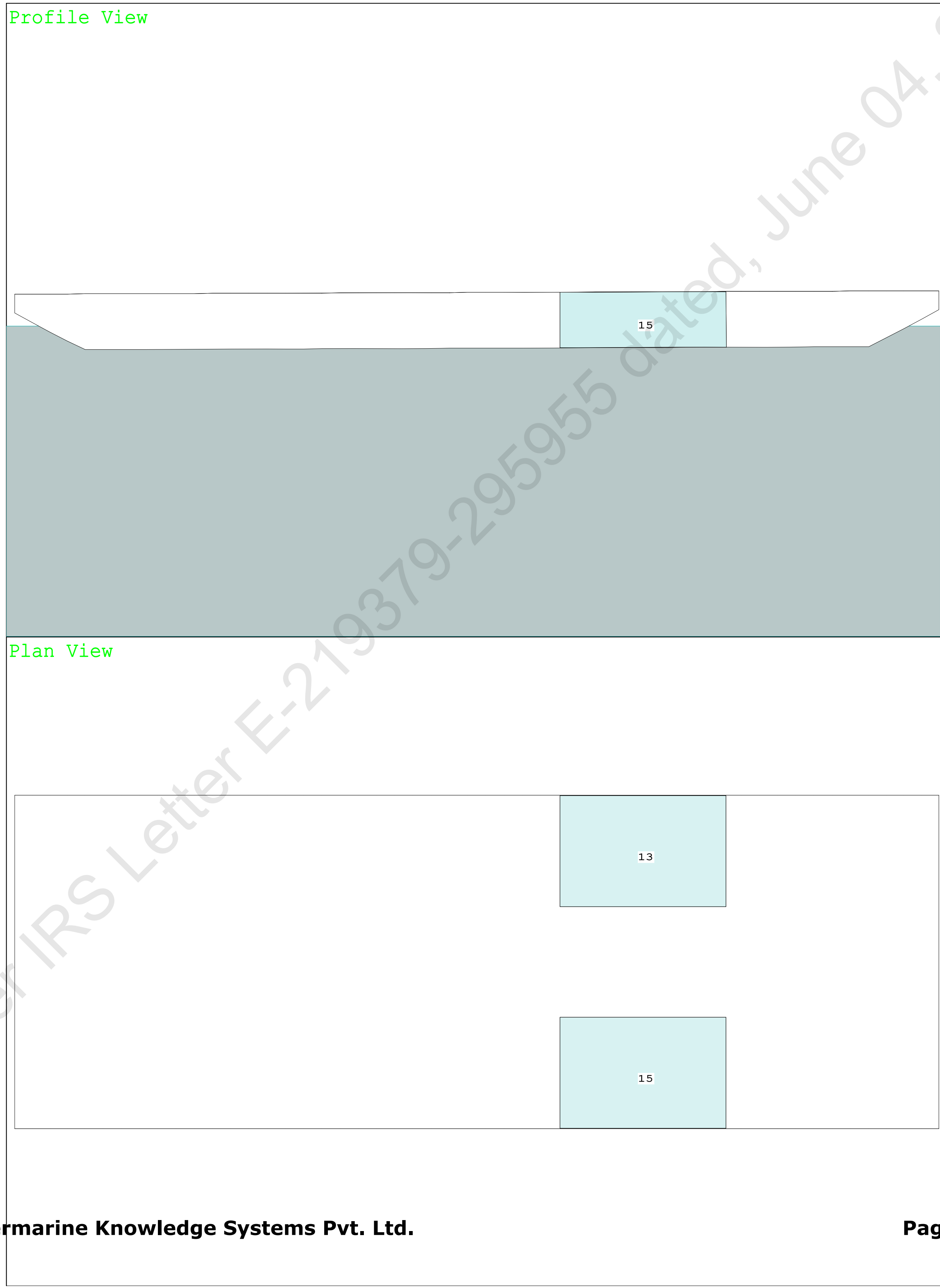

Degrees of Heel --> Starboard
0
10
20
30
40
50
60
70
80
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
M.
Righting Arm
Heeling Arm


4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
M.-Rad
Righting Area
Heeling Area
0
10
20
30
40
50
60
70
80

24-03-19 19:08:57 Cybermarine Technologies PTE. Ltd.
GHS 18.90 **PONTOON**

Condition 14 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Stbd

Condition Graphic - Draft: 1.117 @ 50.000f, 1.303 @ 0.000 Heel: zero



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24-03-19 19:08:57    Cybermarine Technologies PTE. Ltd.
GHS 18.90                        PONTOON
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Condition 14 : Operating Condition of 150T Crane

Crane Lifting 51.3T @ 12m Radius Stbd

```
                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 1.117 @ 50.00f, 1.210 @ 25.00f, 1.303 @ 0.00
                     Trim: Aft 0.186/50.000,  Heel: zero
Part-------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                            465.25  25.062f  0.000   3.000
Crane                                 150.00   7.500f  0.000   5.200
Crane Load                             51.30   7.500f  0.000  33.218
  Total Fixed--------->               666.55  19.758f  0.000   5.821
               Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025         161.60  34.000f  5.980p  1.509    162.73*
05_WBTK.S      1.000    1.025         161.60  34.000f  5.980s  1.509    162.73*
  Total Tanks--------->               323.20  34.000f  0.000   1.509    325.46
  Total Weight-------->               989.75  24.409f  0.000   4.413
                                   Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025         989.71  24.395f  0.000   0.610
------------------------------------------------------------------------------
               Righting Arms:                   0.001   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025          848.4  24.833f  0.000    162.34  23.394*
                                      MT/cm--------m.-MT/cm-------GML-----GMT
                                       8.70            31.38    158.54  19.591
Distances in METERS.----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


                           SUMMARY OF LOADING
       315.3 Cu.M.  (13%) SALT WATER

      150.00 MT  Crane                       51.30 MT  Crane Load

DRAFT AT FP (M) =    1.117
DRAFT AT MS (M) =    1.210
DRAFT AT AP (M) =    1.303
DRAFT AT FWD MARK (M) =    1.132
DRAFT AT AFT MARK (M) =    1.288
                             HYDROSTATIC PROPERTIES
                  Trim: Aft 0.186/50.000, No Heel, VCG = 4.413

   LCF  Displacement   Buoyancy-Ctr.  Weight/          Moment/
  Draft----Weight(MT)----LCB-----VCB-------cm-----LCF---cm trim----GML-----GMT
  1.211       989.71  24.395f  0.610     8.70  24.833f   31.38  158.54  19.591
 Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
                             Trim is per 50.00m.
 Draft is from USK.                               Formal Free Surface included.

      Note:  GMT includes the formal free surface moment 325.5 m.-MT
```

Condition 14 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Stbd

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 1.117 @ 50.00f, 1.210 @ 25.00f, 1.303 @ 0.00
Trim: Aft 0.186/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	89.3	0.916	1.510	0.05500	7.42
CRANE	57.0	4.246	4.840	0.05500	15.17
Total wind heeling moment to starboard------------->					22.59

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 24.409f TCG = 0.000 VCG = 4.413
Free Surface Adjustment: 0.329
Adjusted CG: LCG = 24.410f TCG = 0.000 VCG = 4.742

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area	50% DeckImm Angle
1.291	0.21a	0.00	989.74	0.000	-0.022	0.0000	5.676
1.291	0.21a	0.06s	989.75	0.000	0.000	-0.0000	5.615
1.276	0.21a	5.06s	989.75	0.000	1.719	0.0750	0.615
1.273	0.21a	5.68s	989.72	0.000	1.925	0.0945	0.000
1.217	0.24a	10.06s	989.72	0.000	3.172	0.2910	-4.385
1.047	0.33a	15.06s	989.75	0.000	3.707	0.5962	-9.385
0.991	0.36a	16.64s	989.75	0.000	3.726	0.6988	-10.967
0.869	0.44a	20.06s	989.75	0.000	3.647	0.9200	-14.385
0.686	0.55a	25.06s	989.75	0.000	3.367	1.2285	-19.385
0.500	0.67a	30.06s	989.75	0.000	2.985	1.5065	-24.385
0.313	0.78a	35.06s	989.75	0.000	2.544	1.7481	-29.385
0.125	0.89a	40.06s	989.75	0.000	2.063	1.9494	-34.385
-0.062	1.00a	45.06s	989.75	0.000	1.555	2.1075	-39.385
-0.248	1.11a	50.06s	989.75	0.000	1.028	2.2204	-44.385
-0.432	1.21a	55.06s	989.75	0.000	0.488	2.2866	-49.385
-0.593	1.28a	59.52s	989.75	0.000	0.000	2.3057	-53.848
-0.612	1.29a	60.06s	989.75	0.000	-0.059	2.3054	-54.385

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 325.5 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 22.59 (constant)

24-03-19 19:08:57 Cybermarine Technologies PTE. Ltd.
GHS 18.90 **PONTOON**

Condition 14 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Stbd

```
LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max--------Attained
(1) Residual Area from abs 0 deg to MaxRA         >   0.0800 m.-Rad 0.6988 P
(2) Angle from Equilibrium to RAzero or Flood     >    20.00  deg    59.46 P
(3) Angle from Equilibrium to 50% Dk Imm Angle    >     0.00  deg     5.62 P
-----------------------------------------------------------------------------
```

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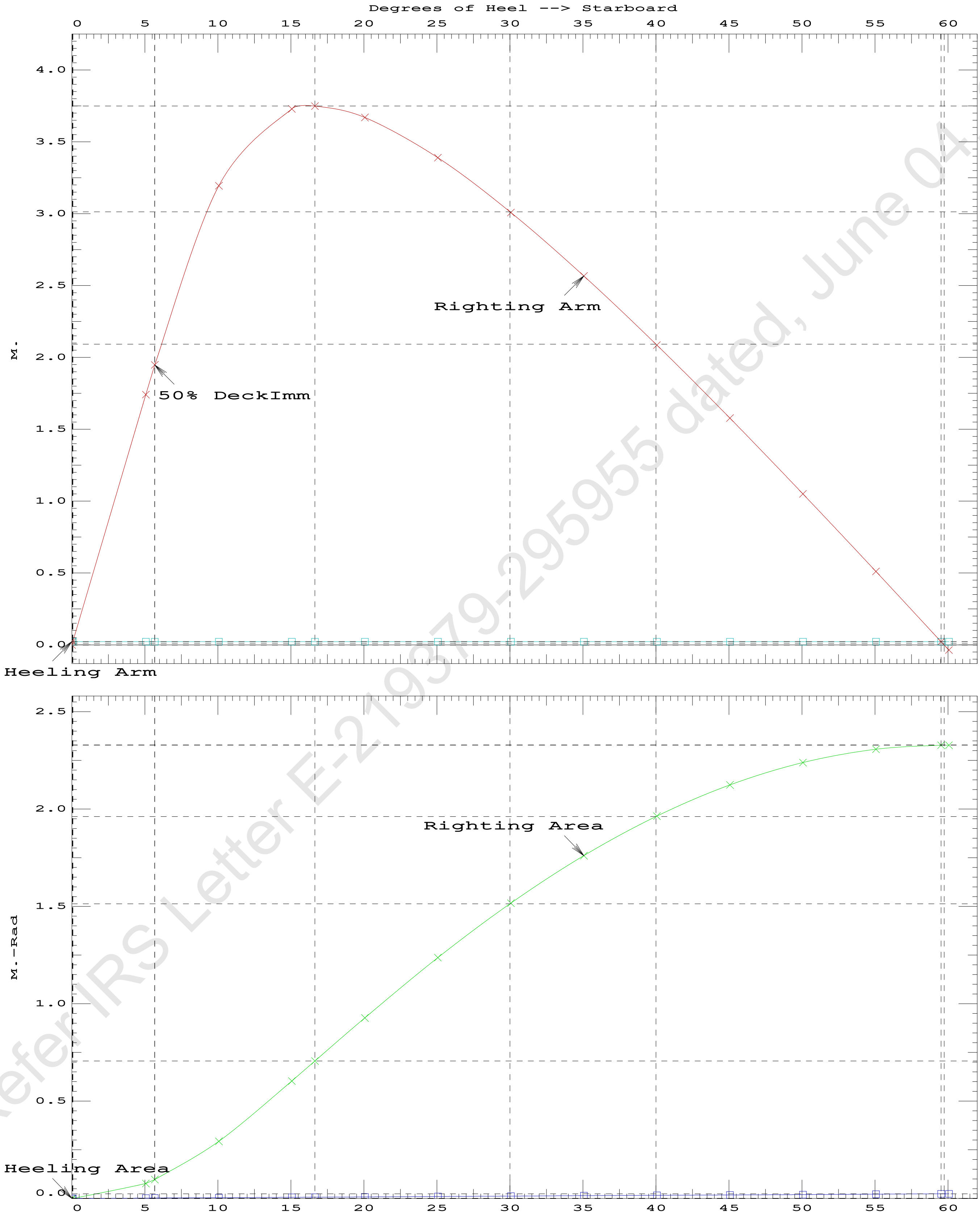
GHS 18.90 **PONTOON**

Condition 14 : Operating Condition of 150T Crane

Crane Lifting 51.3T @ 12m Radius Stbd



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GHS 18.90 **PONTOON**

Condition 14 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

```
     Total CG:  LCG = 24.409f  TCG = 0.000   VCG =  4.413
                        Free Surface Adjustment:   0.329
  Adjusted CG:  LCG = 24.410f  TCG = 0.010p  VCG =  4.742
```

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight (MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.288	0.21a	1.77s	989.75	0.000	0.000	0.0000
1.265	0.22a	6.77s	989.75	0.000	1.691	0.0738
1.163	0.26a	11.77s	989.70	0.000	2.862	0.2762
0.996	0.36a	16.51s	989.75	0.000	3.136	0.5304
0.987	0.36a	16.77s	989.75	0.000	3.136	0.5442
0.807	0.48a	21.77s	989.75	0.000	2.977	0.8141
0.623	0.59a	26.77s	989.75	0.000	2.655	1.0610
0.436	0.71a	31.77s	989.75	0.000	2.249	1.2756
0.249	0.82a	36.77s	989.75	0.000	1.792	1.4523
0.061	0.93a	41.77s	989.75	0.000	1.301	1.5876
-0.126	1.04a	46.77s	989.75	0.000	0.785	1.6787
-0.311	1.14a	51.77s	989.75	0.000	0.252	1.7241
-0.396	1.19a	54.09s	989.75	0.000	0.000	1.7292
-0.493	1.24a	56.77s	989.75	0.000	-0.292	1.7224
-0.673	1.32a	61.77s	989.75	0.000	-0.840	1.6731

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 325.5 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 615.60

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM------------STABILITY CRITERIA FOR CRANE OPERAT		----Min/Max		--------Attained
(1) Absolute Angle at Equilibrium	<	15.00	deg	1.77 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	504.3 P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		4.044 P

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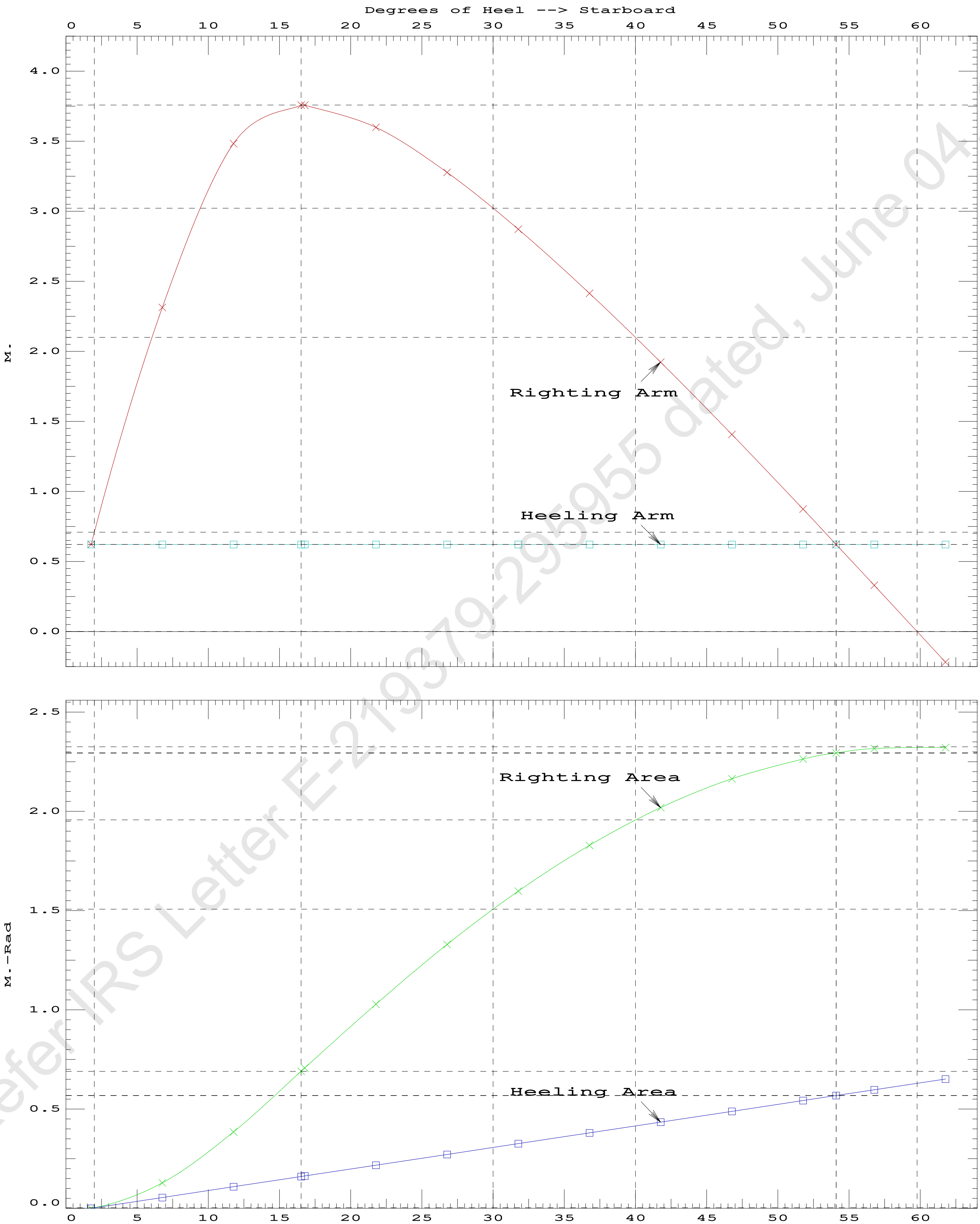
GHS 18.90 **PONTOON**

Condition 14 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Stbd



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24-03-19 17:25:28    Cybermarine Technologies PTE. Ltd.
GHS 18.90                        PONTOON
```

Condition 14 : Operating Condition of 150T Crane

Crane Lifting 51.3T @ 12m Radius Stbd

```
                  Crane in operation with load in place
               WEIGHT and DISPLACEMENT and WATERPLANE STATUS
          USK draft: 1.115 @ 50.00f, 1.208 @ 25.00f, 1.300 @ 0.00
               Trim: Aft 0.185/50.000,  Heel: Stbd 1.77 deg.
Part---------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                              465.25  25.062f  0.000   3.000
Crane                                   150.00   7.500f  0.000   5.200
Crane Load                               51.30   7.500f  0.000  33.218
  Total Fixed--------->                 666.55  19.758f  0.000   5.821
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025           161.60  34.000f  5.980p  1.509    162.73*
05_WBTK.S      1.000    1.025           161.60  34.000f  5.980s  1.509    162.73*
  Total Tanks--------->                 323.20  34.000f  0.000   1.509    325.46
  Total Weight-------->                 989.75  24.409f  0.000   4.413
                                     Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025           989.75  24.394f  0.739s  0.621
------------------------------------------------------------------------------
               Righting Arms:                    0.000   0.622s
               External Arms:                    0.000   0.622s
      Residual Righting Arms:                    0.000   0.000s
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025            847.7  24.865f  0.101s   161.88  23.380*
                                         MT/cm
                                          8.69
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


                            Angle Theta1 = 1.77
```

```
24-03-19 17:25:28    Cybermarine Technologies PTE. Ltd.
GHS 18.90                         PONTOON

                        After hook load drop
                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 1.201 @ 50.00f, 1.151 @ 25.00f, 1.101 @ 0.00
                  Trim: Fwd 0.100/50.000,  Heel: 0.00 deg.
Part-------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                            465.25  25.062f  0.000   3.000
Crane                                 150.00   7.500f  0.000   5.200
  Total Fixed--------->               615.25  20.780f  0.000   3.536
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025         161.60  34.000f  5.980p  1.509    162.73*
05_WBTK.S      1.000    1.025         161.60  34.000f  5.980s  1.509    162.73*
  Total Tanks--------->               323.20  34.000f  0.000   1.509    325.46
  Total Weight-------->               938.45  25.333f  0.000   2.838
                                   Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025         938.47  25.338f  0.001s  0.578
------------------------------------------------------------------------------
               Righting Arms:                  0.000   0.001s
               External Arms:                  0.000   0.001s
      Residual Righting Arms:                  0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025          841.9  25.057f  0.020s   167.30  24.427*
                                      MT/cm--------m.-MT/cm-------GML-----GMT
                                       8.63             30.98    165.04  22.167
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


limmarg: 16750.93; t3: 30.00; t: 15.00
limmarg: 6235.42; t3: 15.00; t: 7.50
limmarg: 1682.59; t3: 7.50; t: 3.75
limmarg: 349.58; t3: 3.75; t: 1.88
limmarg: 12.53; t3: 1.87; t: 0.94
limmarg: -72.10; t3: 0.93; t: 0.47
limmarg: -36.85; t3: 1.40; t: 0.23
limmarg: -14.44; t3: 1.63; t: 0.12
limmarg: -1.40; t3: 1.75; t: 0.06
limmarg: 5.46; t3: 1.81; t: 0.03
limmarg: 2.00; t3: 1.78; t: 0.01

               Theta3 is 1.78 degrees, as demonstrated below.

                  RESIDUAL RIGHTING ARMS vs HEEL ANGLE
            Fixed CG:  LCG = 20.780f  TCG = 0.000   VCG =  3.536

      Origin    Degrees of   Displacement    Residual Arms       Res.
       Depth---Trim----Heel----Weight(MT)---in Trim--in Heel---> Area
       1.087   0.11f   1.77s      938.45      0.000   -0.707    0.0000
       1.089   0.11f   0.00       938.44      0.000    0.000   -0.0109
       1.087   0.11f   1.78p      938.45      0.000    0.715    0.0002
       1.075   0.11f   5.00p      938.46      0.000    1.985    0.0761
       0.988   0.13f  10.00p      938.44      0.000    3.637    0.3235
       0.744   0.17f  15.00p      938.45      0.000    4.347    0.6787
       0.503   0.22f  19.29p      938.45      0.000    4.468    1.0113
```

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GHS 18.90 **PONTOON**

0.462	0.23f	20.00p	938.45	0.000	4.465	1.0669
0.176	0.29f	25.00p	938.45	0.000	4.344	1.4538
-0.112	0.34f	30.00p	938.45	0.000	4.108	1.8235

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 615.25 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

```
LIM--------------------STABILITY CRITERION------------Min/Max--------Attained
(1) Res. Area Ratio from abs -1.8 deg to abs 1.8   >    1.000        1.020 P
-----------------------------------------------------------------------------
```

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 20.780f TCG = 0.000 VCG = 3.536

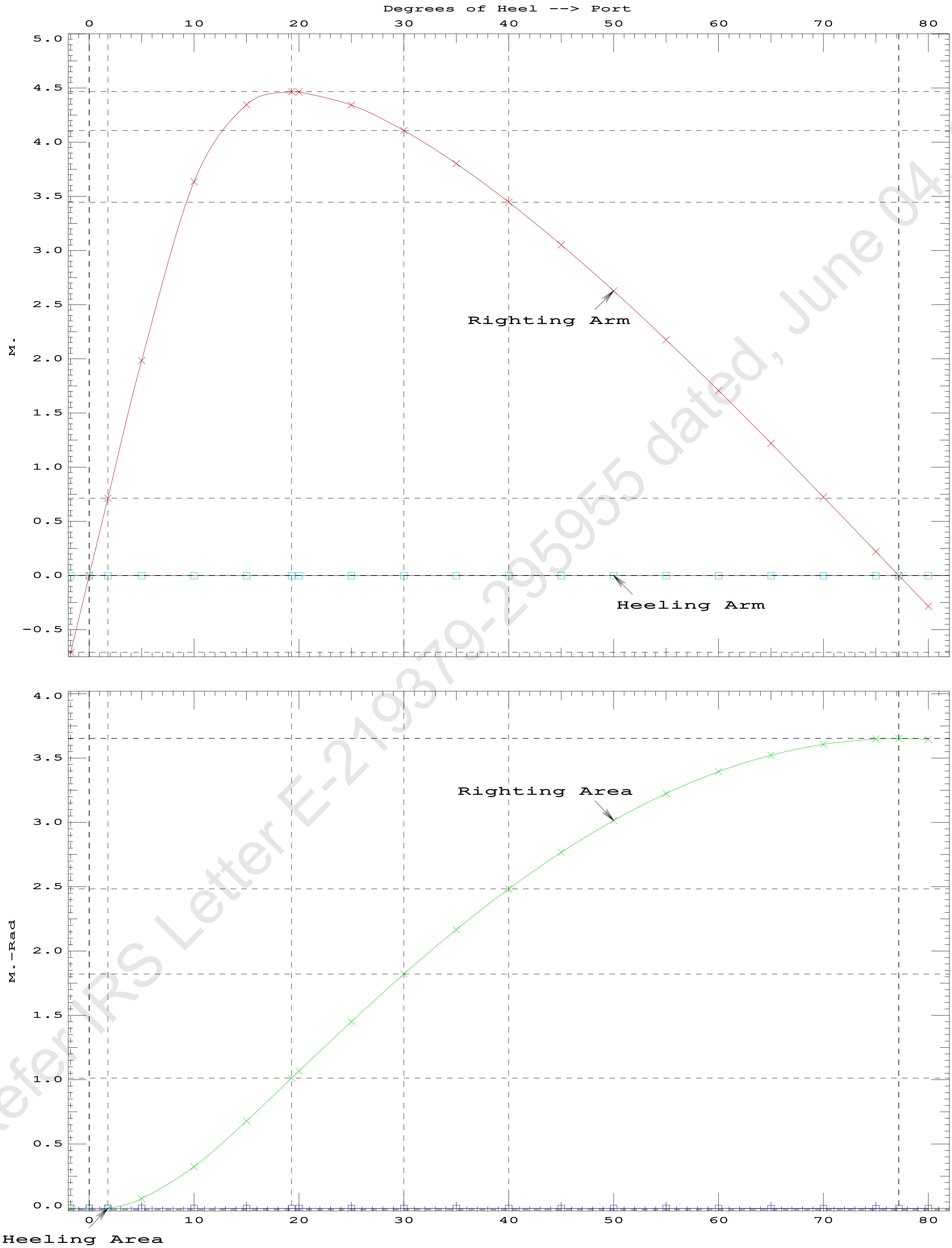
Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.087	0.11f	1.77s	938.45	0.000	-0.707	0.0000
1.089	0.11f	0.00	938.44	0.000	0.000	-0.0109
1.087	0.11f	1.78p	938.45	0.000	0.715	0.0002
1.075	0.11f	5.00p	938.46	0.000	1.985	0.0761
0.988	0.13f	10.00p	938.44	0.000	3.637	0.3235
0.744	0.17f	15.00p	938.45	0.000	4.347	0.6787
0.503	0.22f	19.29p	938.45	0.000	4.468	1.0113
0.462	0.23f	20.00p	938.45	0.000	4.465	1.0669
0.176	0.29f	25.00p	938.45	0.000	4.344	1.4538
-0.112	0.34f	30.00p	938.45	0.000	4.108	1.8235
-0.400	0.40f	35.00p	938.45	0.000	3.803	2.1692
-0.686	0.46f	40.00p	938.45	0.000	3.447	2.4859
-0.968	0.51f	45.00p	938.45	0.000	3.053	2.7697
-1.244	0.57f	50.00p	938.45	0.000	2.628	3.0178
-1.510	0.62f	55.00p	938.45	0.000	2.178	3.2277
-1.765	0.66f	60.00p	938.45	0.000	1.707	3.3973
-2.007	0.70f	65.00p	938.45	0.000	1.222	3.5253
-2.232	0.74f	70.00p	938.45	0.000	0.725	3.6103
-2.440	0.77f	75.00p	938.45	0.000	0.222	3.6517
-2.526	0.77f	77.20p	938.45	0.000	0.000	3.6560
-2.629	0.78f	80.00p	938.45	0.000	-0.283	3.6490

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 615.25 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM--------------RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Res. Area Ratio from abs -1.8 deg to RAzero	>	1.000		336.824 P
(2) Angle from abs 1.8 deg to RAzero	>	20.00	deg	75.42 P


Degrees of Heel --> Port
M.
Righting Arm
Heeling Arm
M.-Rad
Righting Area
Heeling Area

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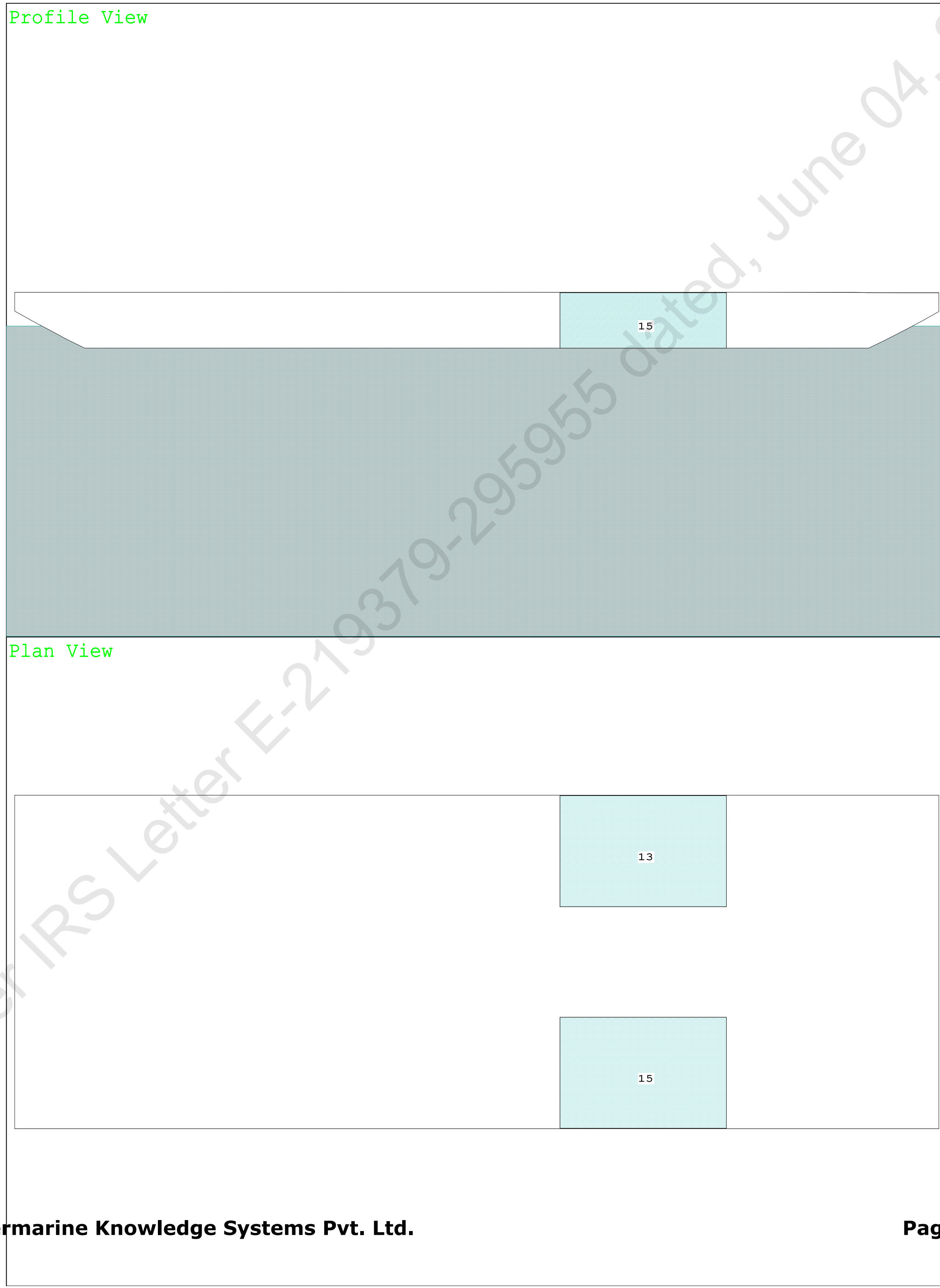
GHS 18.90 **PONTOON**

Condition 15 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Fwd

Condition Graphic - Draft: 1.215 @ 50.000f, 1.206 @ 0.000 Heel: zero



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24-03-19 19:08:57    Cybermarine Technologies PTE. Ltd.
GHS 18.90                        PONTOON
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Condition 15 : Operating Condition of 150T Crane

Crane Lifting 51.3T @ 12m Radius Fwd

```
                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 1.215 @ 50.00f, 1.210 @ 25.00f, 1.206 @ 0.00
                     Trim: Fwd 0.010/50.000,  Heel: zero
Part---------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                              465.25  25.062f  0.000   3.000
Crane                                   150.00   7.500f  0.000   5.200
Crane Load                               51.30  19.500f  0.000  33.218
  Total Fixed--------->                 666.55  20.682f  0.000   5.821
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025           161.60  34.000f  5.980p  1.509   162.73*
05_WBTK.S      1.000    1.025           161.60  34.000f  5.980s  1.509   162.73*
  Total Tanks--------->                 323.20  34.000f  0.000   1.509   325.46
  Total Weight-------->                 989.75  25.031f  0.000   4.413
                                       Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025           989.74  25.031f  0.000   0.609
------------------------------------------------------------------------------
               Righting Arms:                    0.000   0.000
Part---------------------SpGr------------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025            848.3  25.009f  0.000    162.28  23.387*
                                         MT/cm---------m.-MT/cm-------GML-----GMT
                                          8.70           31.37    158.48  19.583
Distances in METERS.----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


                            SUMMARY OF LOADING
       315.3 Cu.M.  (13%) SALT WATER

      150.00 MT  Crane                          51.30 MT  Crane Load

DRAFT AT FP (M) =     1.215
DRAFT AT MS (M) =     1.210
DRAFT AT AP (M) =     1.206
DRAFT AT FWD MARK (M) =    1.215
DRAFT AT AFT MARK (M) =    1.206
                              HYDROSTATIC PROPERTIES
                Trim: Fwd 0.010/50.000, No Heel, VCG = 4.413

   LCF  Displacement   Buoyancy-Ctr.  Weight/          Moment/
  Draft----Weight(MT)----LCB-----VCB-------cm-----LCF---cm trim----GML-----GMT
  1.210       989.74  25.031f  0.609     8.70  25.009f   31.37  158.48  19.583
 Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
                              Trim is per 50.00m.
 Draft is from USK.                             Formal Free Surface included.

      Note:  GMT includes the formal free surface moment 325.5 m.-MT
```

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GHS 18.90 **PONTOON**

Condition 15 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Fwd

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 1.215 @ 50.00f, 1.210 @ 25.00f, 1.206 @ 0.00
Trim: Fwd 0.010/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	89.3	0.915	1.508	0.05500	7.41
CRANE	57.0	4.304	4.897	0.05500	15.35
Total wind heeling moment to starboard------------->					22.76

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 25.031f TCG = 0.000 VCG = 4.413
Free Surface Adjustment: 0.329
Adjusted CG: LCG = 25.031f TCG = 0.000 VCG = 4.742

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. 50% Area	DeckImm Angle
1.193	0.01f	0.00	989.67	0.000	-0.023	0.0000	6.099
1.194	0.01f	0.06s	989.75	0.000	0.000	-0.0000	6.036
1.179	0.01f	5.06s	989.75	0.000	1.720	0.0750	1.036
1.172	0.01f	6.10s	989.72	0.000	2.065	0.1093	0.000
1.107	0.01f	10.06s	989.81	0.000	3.175	0.2916	-3.964
0.898	0.02f	15.06s	989.91	0.000	3.713	0.5972	-8.964
0.822	0.02f	16.71s	989.75	0.000	3.732	0.7045	-10.615
0.668	0.02f	20.06s	989.75	0.000	3.653	0.9215	-13.964
0.433	0.03f	25.06s	989.75	0.000	3.374	1.2306	-18.964
0.194	0.04f	30.06s	989.75	0.000	2.992	1.5091	-23.964
-0.046	0.04f	35.06s	989.75	0.000	2.550	1.7514	-28.964
-0.286	0.05f	40.06s	989.75	0.000	2.069	1.9532	-33.964
-0.523	0.05f	45.06s	989.75	0.000	1.561	2.1118	-38.964
-0.757	0.06f	50.06s	989.75	0.000	1.034	2.2252	-43.964
-0.985	0.06f	55.06s	989.75	0.000	0.493	2.2919	-48.964
-1.185	0.07f	59.57s	989.75	0.000	0.000	2.3113	-53.468
-1.206	0.07f	60.06s	989.74	0.000	-0.054	2.3111	-53.964

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 325.5 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 22.76 (constant)

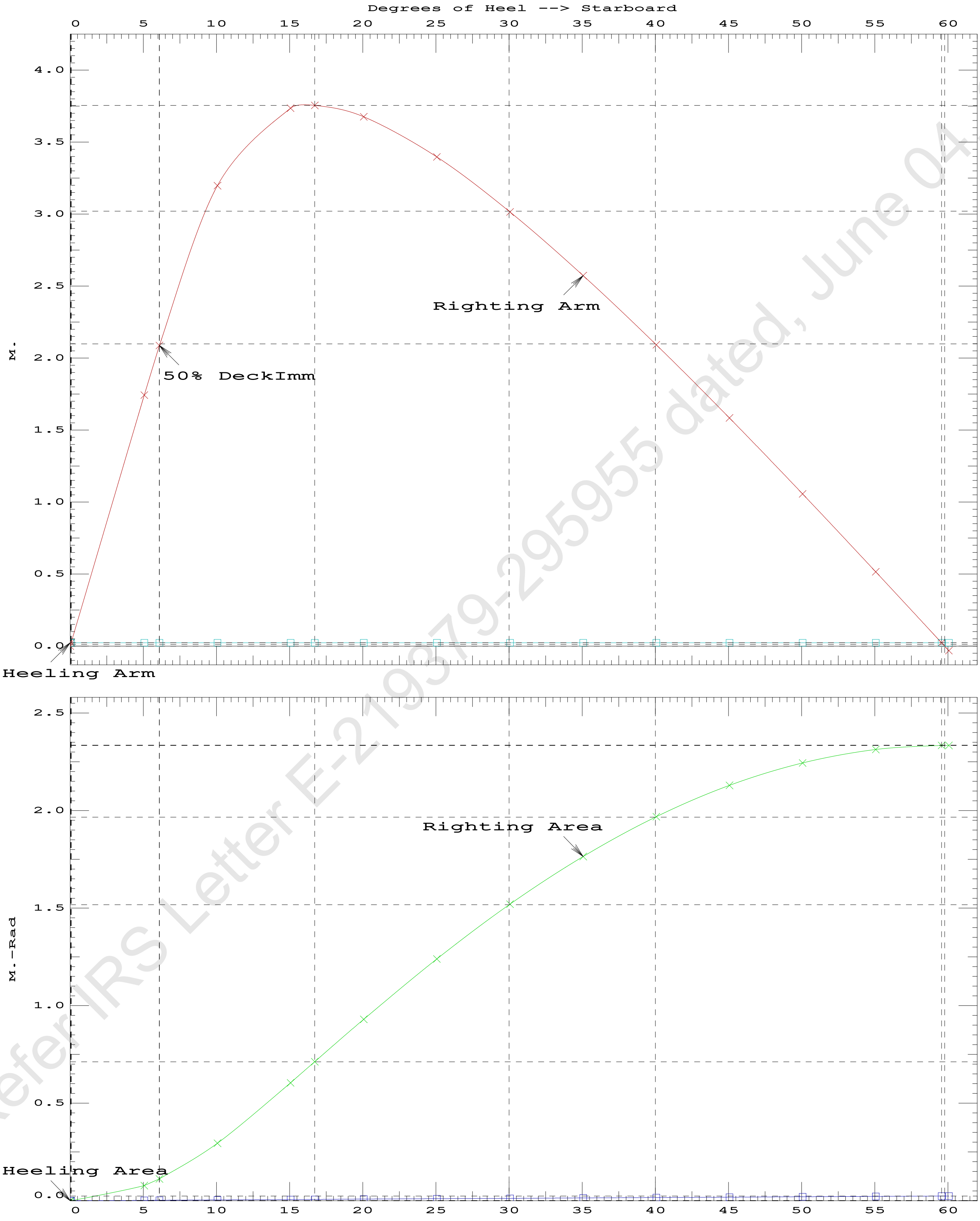
24-03-19 19:08:57 Cybermarine Technologies PTE. Ltd.
GHS 18.90 **PONTOON**

Condition 15 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Fwd

```
LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max--------Attained
(1) Residual Area from abs 0 deg to MaxRA         >   0.0800 m.-Rad 0.7045 P
(2) Angle from Equilibrium to RAzero or Flood     >    20.00  deg    59.50 P
(3) Angle from Equilibrium to 50% Dk Imm Angle    >     0.00  deg     6.04 P
-----------------------------------------------------------------------------
```

24-03-19 19:08:57 Cybermarine Technologies PTE. Ltd.
GHS 18.90 **PONTOON**

Condition 15 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Fwd



24-03-19 19:08:57 Cybermarine Technologies PTE. Ltd.
GHS 18.90 **PONTOON**

Condition 15 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.031f TCG = 0.000 VCG = 4.413
Free Surface Adjustment: 0.329
Adjusted CG: LCG = 25.031f TCG = 0.000 VCG = 4.742

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.193	0.01f	0.00	989.75	0.000	0.000	0.0000
1.179	0.01f	5.00s	989.66	0.000	1.721	0.0751
1.108	0.01f	10.00s	989.75	0.000	3.185	0.2910
0.901	0.02f	15.00s	989.91	0.000	3.733	0.5995
0.822	0.02f	16.72s	989.75	0.000	3.754	0.7121
0.671	0.02f	20.00s	989.75	0.000	3.677	0.9259
0.436	0.03f	25.00s	989.75	0.000	3.399	1.2372
0.197	0.04f	30.00s	989.75	0.000	3.018	1.5180
-0.043	0.04f	35.00s	989.75	0.000	2.578	1.7626
-0.283	0.05f	40.00s	989.75	0.000	2.097	1.9669
-0.521	0.05f	45.00s	989.75	0.000	1.589	2.1279
-0.754	0.06f	50.00s	989.75	0.000	1.062	2.2438
-0.983	0.06f	55.00s	989.75	0.000	0.522	2.3129
-1.193	0.07f	59.77s	989.74	0.000	0.000	2.3347
-1.203	0.07f	60.00s	989.74	0.000	-0.026	2.3346

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 325.5 m.-MT was applied to artificially modify the CG.

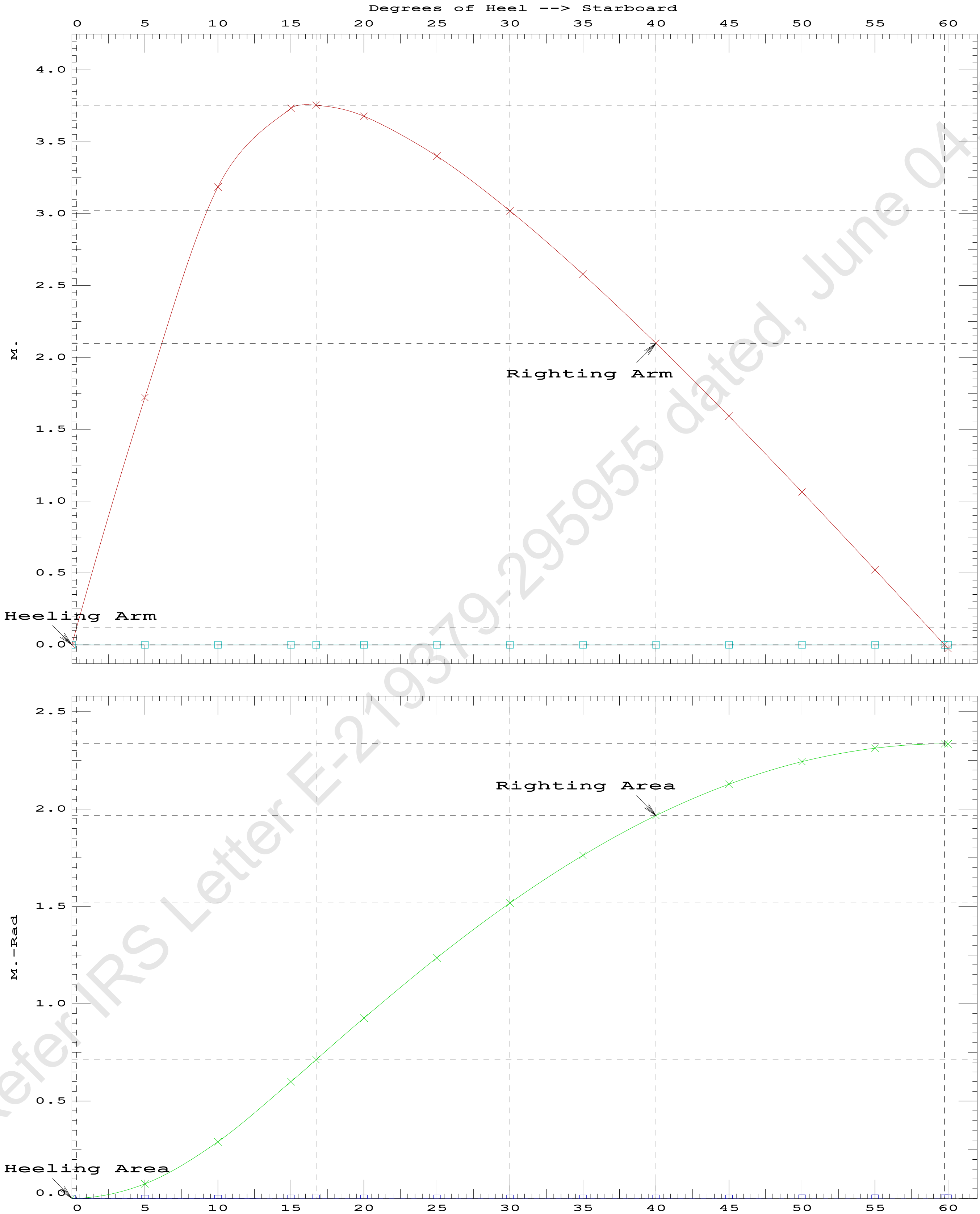
Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM---STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained
(1) Absolute Angle at Equilibrium	<	15.00	deg	0.00 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	371536 P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		2216.28 P

Condition 15 : Operating Condition of 150T Crane

Crane Lifting 51.3T @ 12m Radius Fwd



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24-03-19 17:25:31    Cybermarine Technologies PTE. Ltd.
GHS 18.90                          PONTOON

             Condition 15 : Operating Condition of 150T Crane
                   Crane Lifting 51.3T @ 12m Radius Fwd
                   Crane in operation with load in place
                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 1.215 @ 50.00f, 1.210 @ 25.00f, 1.206 @ 0.00
                 Trim: Fwd 0.010/50.000,  Heel: 0.00 deg.
Part----------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                               465.25  25.062f  0.000   3.000
Crane                                    150.00   7.500f  0.000   5.200
Crane Load                                51.30  19.500f  0.000  33.218
  Total Fixed--------->                  666.55  20.682f  0.000   5.821
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025            161.60  34.000f  5.980p  1.509   162.73*
05_WBTK.S      1.000    1.025            161.60  34.000f  5.980s  1.509   162.73*
  Total Tanks--------->                  323.20  34.000f  0.000   1.509   325.46
  Total Weight-------->                  989.75  25.031f  0.000   4.413
                                       Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025            989.75  25.031f  0.001s  0.609
------------------------------------------------------------------------------
               Righting Arms:                      0.000   0.001s
               External Arms:                      0.000   0.001s
      Residual Righting Arms:                      0.000   0.000s
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025             844.4  25.005f  0.031s   160.05  23.224*
                                          MT/cm---------m.-MT/cm-------GML-----GMT
                                           8.66           30.93    156.25  19.420
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


                            Angle Theta1 = 0.00
```

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24-03-19 17:25:31    Cybermarine Technologies PTE. Ltd.
GHS 18.90                        PONTOON

                         After hook load drop
                 WEIGHT and DISPLACEMENT and WATERPLANE STATUS
            USK draft: 1.201 @ 50.00f, 1.151 @ 25.00f, 1.101 @ 0.00
                  Trim: Fwd 0.100/50.000,  Heel: 0.00 deg.
Part------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                           465.25  25.062f  0.000   3.000
Crane                                150.00   7.500f  0.000   5.200
  Total Fixed--------->              615.25  20.780f  0.000   3.536
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025        161.60  34.000f  5.980p  1.509    162.73*
05_WBTK.S      1.000    1.025        161.60  34.000f  5.980s  1.509    162.73*
  Total Tanks--------->              323.20  34.000f  0.000   1.509    325.46
  Total Weight-------->              938.45  25.333f  0.000   2.838
                                  Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025        938.45  25.337f  0.001s  0.578
------------------------------------------------------------------------------
               Righting Arms:                 0.000   0.001s
               External Arms:                 0.000   0.001s
      Residual Righting Arms:                 0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025         841.9  25.057f  0.020s   167.30  24.428*
                                      MT/cm---------m.-MT/cm-------GML-----GMT
                                       8.63             30.98    165.04  22.168
Distances in METERS.----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


limmarg: INFINITY; t3: 30; t: 15.00
limmarg: INFINITY; t3: 15; t: 7.50
limmarg: INFINITY; t3: 8; t: 3.75
limmarg: INFINITY; t3: 4; t: 1.88
limmarg: INFINITY; t3: 2; t: 0.94
limmarg: INFINITY; t3: 1; t: 0.47
limmarg: INFINITY; t3: 1; t: 0.23
limmarg: INFINITY; t3: 1; t: 0.12
limmarg: INFINITY; t3: 1; t: 0.06
limmarg: INFINITY; t3: 1; t: 0.03
limmarg: INFINITY; t3: 1; t: 0.01

                 Theta3 is 0.97 degrees, as demonstrated below.

                       RESIDUAL RIGHTING ARMS vs HEEL ANGLE
            Fixed CG:  LCG = 20.780f  TCG = 0.000   VCG =  3.536

      Origin    Degrees of   Displacement    Residual Arms       Res.
       Depth---Trim----Heel----Weight(MT)---in Trim--in Heel---> Area
       1.089   0.12f   0.00       938.45      0.000    0.000    0.0000
       1.089   0.11f   0.97s      938.45      0.000    0.388    0.0033
       1.075   0.11f   5.00s      938.45      0.000    1.984    0.0867
       0.988   0.13f  10.00s      938.44      0.000    3.635    0.3343
       0.744   0.17f  15.00s      938.45      0.000    4.345    0.6893
       0.503   0.22f  19.29s      938.45      0.000    4.466    1.0216
       0.462   0.23f  20.00s      938.45      0.000    4.463    1.0773
```

```
  0.176   0.29f  25.00s      938.45     0.000    4.342    1.4641
 -0.112   0.34f  30.00s      938.45     0.000    4.106    1.8335
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.
```

Note: The Center of Gravity shown above is for the Fixed Weight of 615.25 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

```
LIM--------------------STABILITY CRITERION------------Min/Max--------Attained
(1) Residual Area Ratio from abs 0 deg to abs 1    >    1.000          LARGE
-----------------------------------------------------------------------------
```

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 20.780f TCG = 0.000 VCG = 3.536

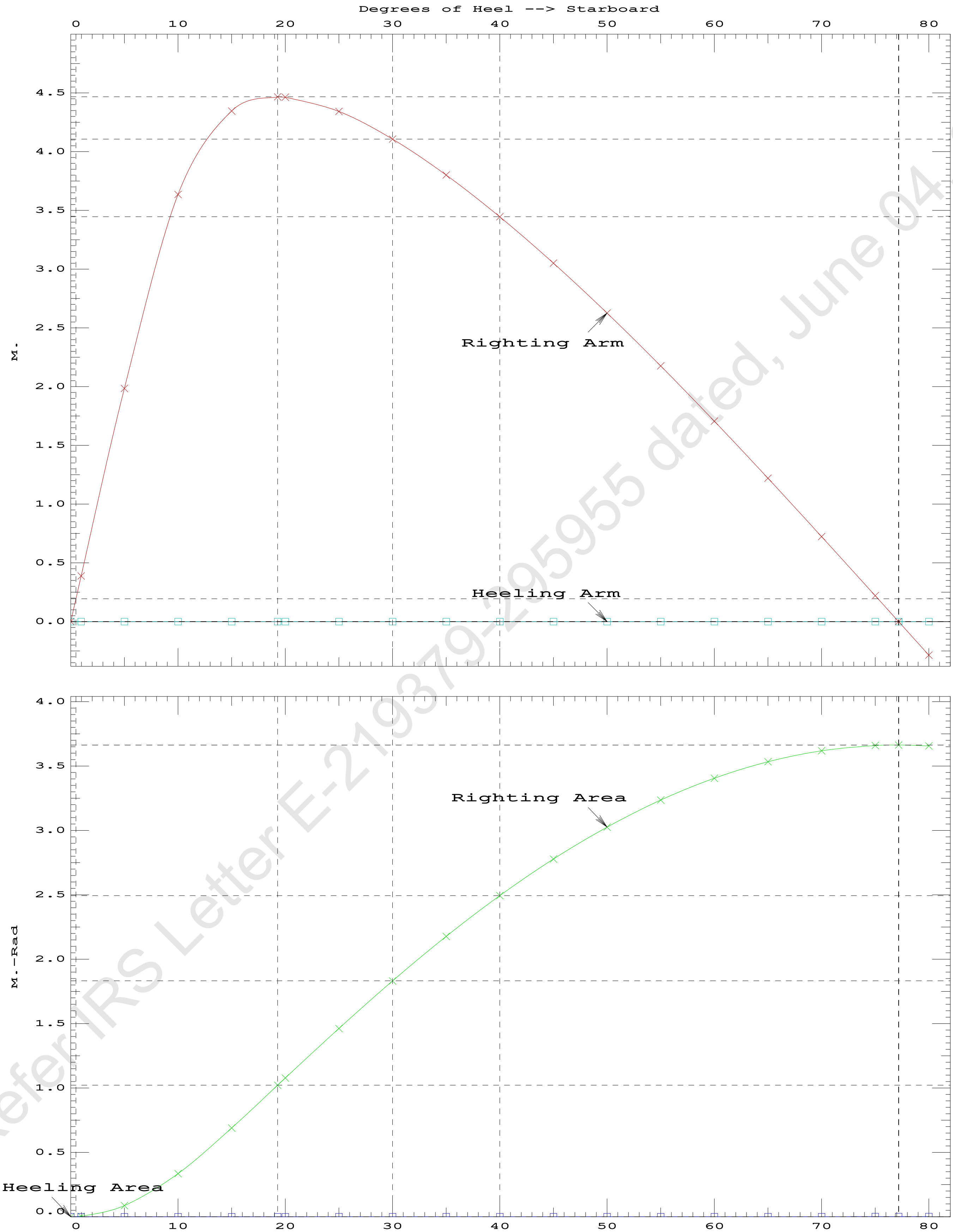
Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.089	0.12f	0.00	938.45	0.000	0.000	0.0000
1.089	0.11f	0.97s	938.45	0.000	0.388	0.0033
1.075	0.11f	5.00s	938.45	0.000	1.984	0.0867
0.988	0.13f	10.00s	938.44	0.000	3.635	0.3343
0.744	0.17f	15.00s	938.45	0.000	4.345	0.6893
0.503	0.22f	19.29s	938.45	0.000	4.466	1.0216
0.462	0.23f	20.00s	938.45	0.000	4.463	1.0773
0.176	0.29f	25.00s	938.45	0.000	4.342	1.4641
-0.112	0.34f	30.00s	938.45	0.000	4.106	1.8335
-0.400	0.40f	35.00s	938.45	0.000	3.800	2.1790
-0.687	0.46f	40.00s	938.45	0.000	3.445	2.4955
-0.968	0.51f	45.00s	938.45	0.000	3.050	2.7792
-1.244	0.57f	50.00s	938.45	0.000	2.625	3.0271
-1.510	0.62f	55.00s	938.45	0.000	2.175	3.2367
-1.765	0.66f	60.00s	938.45	0.000	1.705	3.4062
-2.007	0.70f	65.00s	938.45	0.000	1.219	3.5339
-2.232	0.74f	70.00s	938.45	0.000	0.723	3.6187
-2.440	0.77f	75.00s	938.45	0.000	0.220	3.6599
-2.525	0.77f	77.18s	938.45	0.000	0.000	3.6641
-2.629	0.78f	80.00s	938.45	0.000	-0.286	3.6570

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 615.25 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM---RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Residual Area Ratio from abs 0 deg to RAzero	>	1.000		LARGE
(2) Angle from abs 1 deg to RAzero	>	20.00	deg	76.21 P


Degrees of Heel --> Starboard
M.
Righting Arm
Heeling Arm
M.-Rad
Righting Area
Heeling Area

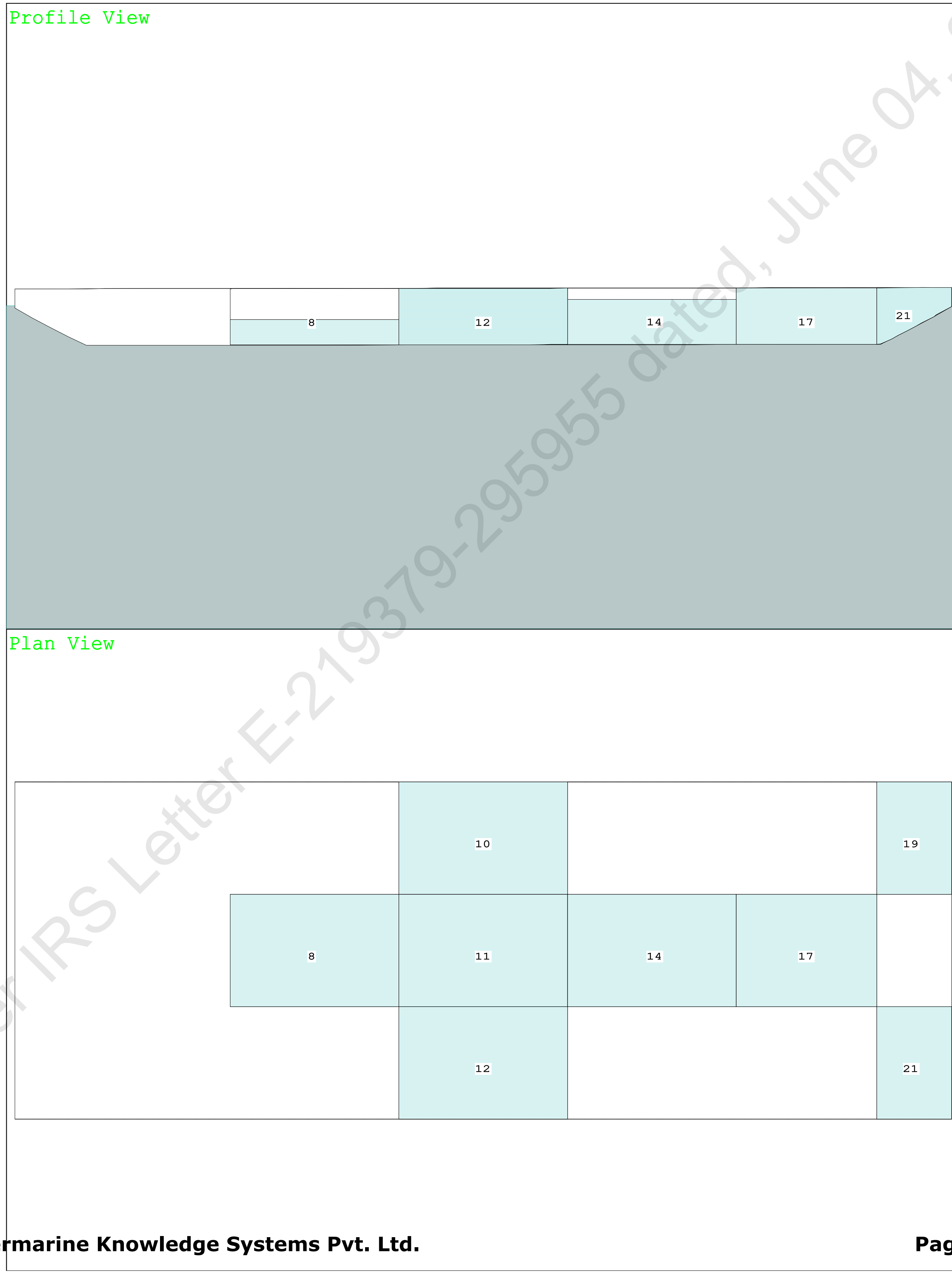
24-03-19 17:27:37 Cybermarine Technologies PTE. Ltd.
GHS 18.90

PONTOON

Condition 16 : Lightship + Crane + Ballast

TOWING CONDITION

Condition Graphic - Draft: 2.064 @ 50.000f, 2.136 @ 0.000 Heel: zero



24-03-19 17:27:37 Cybermarine Technologies PTE. Ltd.
GHS 18.90 **PONTOON**

Condition 16 : Lightship + Crane + Ballast

TOWING CONDITION

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 2.064 @ 50.00f, 2.100 @ 25.00f, 2.136 @ 0.00
Trim: Aft 0.072/50.000, Heel: zero

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			465.25	25.062f	0.000	3.000	
Crane			400.00	11.500f	0.000	6.325	
Total Fixed--------->			865.25	18.792f	0.000	4.537	
03_WBTK.C	0.450	1.025	73.23	15.993f	0.000	0.675	162.73
04_WBTK.P	1.000	1.025	161.60	25.000f	5.980p	1.509	162.73*
04_WBTK.C	1.000	1.025	162.73	25.000f	0.000	1.500	162.73*
04_WBTK.S	1.000	1.025	161.60	25.000f	5.980s	1.509	162.73*
05_WBTK.C	0.800	1.025	130.18	33.996f	0.000	1.200	162.73
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000	1.499	135.61*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks--------->			924.21	30.518f	0.000	1.436	1093.89
Total Weight-------->			1,789.46	24.848f	0.000	2.936	
			Displ(MT)	LCB	TCB	VCB	
HULL		1.025	1,789.46	24.846f	0.000	1.073	
Righting Arms:				0.000	0.000		

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	900.2	24.998f	0.000	106.84	13.319*
		MT/cm		m.-MT/cm	GML	GMT
		9.23		37.57	104.98	11.457

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

901.7 Cu.M. (36%) SALT WATER

400.00 MT Crane

DRAFT AT FP (M) = 2.064
DRAFT AT MS (M) = 2.100
DRAFT AT AP (M) = 2.136
DRAFT AT FWD MARK (M) = 2.070
DRAFT AT AFT MARK (M) = 2.130

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GHS 18.90 **PONTOON**

Condition 16 : Lightship + Crane + Ballast

TOWING CONDITION

HYDROSTATIC PROPERTIES
Trim: Aft 0.072/50.000, No Heel, VCG = 2.936

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
2.100	1,789.46	24.846f	1.073	9.23	24.998f	37.57	104.98	11.457

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 1093.9 m.-MT

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 2.064 @ 50.00f, 2.100 @ 25.00f, 2.136 @ 0.00
Trim: Aft 0.072/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	46.1	0.461	1.481	0.05500	3.76
CRANE	57.0	3.391	4.410	0.05500	13.83
Total wind heeling moment to starboard------------->					17.58

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

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GHS 18.90 **PONTOON**

Condition 16 : Lightship + Crane + Ballast

TOWING CONDITION

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.848f TCG = 0.000 VCG = 2.936
Free Surface Adjustment: 0.611
Adjusted CG: LCG = 24.849f TCG = 0.000 VCG = 3.547

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area	50% DeckImm Angle
2.124	0.08a	0.00	1,789.5	0.000	-0.009	0.0000	2.818
2.124	0.08a	0.05s	1,789.5	0.000	0.000	-0.0000	2.772
2.121	0.08a	2.82s	1,789.5	0.000	0.557	0.0135	0.000
2.113	0.09a	5.05s	1,789.5	0.000	0.999	0.0437	-2.228
2.182	0.12a	10.05s	1,789.5	0.000	1.643	0.1616	-7.228
2.385	0.15a	15.05s	1,789.5	0.000	1.797	0.3153	-12.228
2.434	0.16a	16.07s	1,789.4	0.000	1.801	0.3474	-13.252
2.628	0.20a	20.05s	1,789.5	0.000	1.746	0.4712	-17.228
2.856	0.26a	25.05s	1,789.5	0.000	1.580	0.6174	-22.228
3.063	0.31a	30.05s	1,789.5	0.000	1.359	0.7460	-27.228
3.248	0.36a	35.05s	1,789.5	0.000	1.107	0.8539	-32.228
3.409	0.41a	40.05s	1,789.5	0.000	0.836	0.9388	-37.228
3.545	0.46a	45.05s	1,789.5	0.000	0.553	0.9995	-42.228
3.654	0.51a	50.05s	1,789.5	0.000	0.260	1.0351	-47.228
3.727	0.55a	54.44s	1,789.5	0.000	0.000	1.0451	-51.619
3.735	0.56a	55.05s	1,789.5	0.000	-0.036	1.0449	-52.228
3.788	0.59a	60.05s	1,789.5	0.000	-0.335	1.0287	-57.228

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 1093.9 m.-MT was applied to artificially modify the CG.

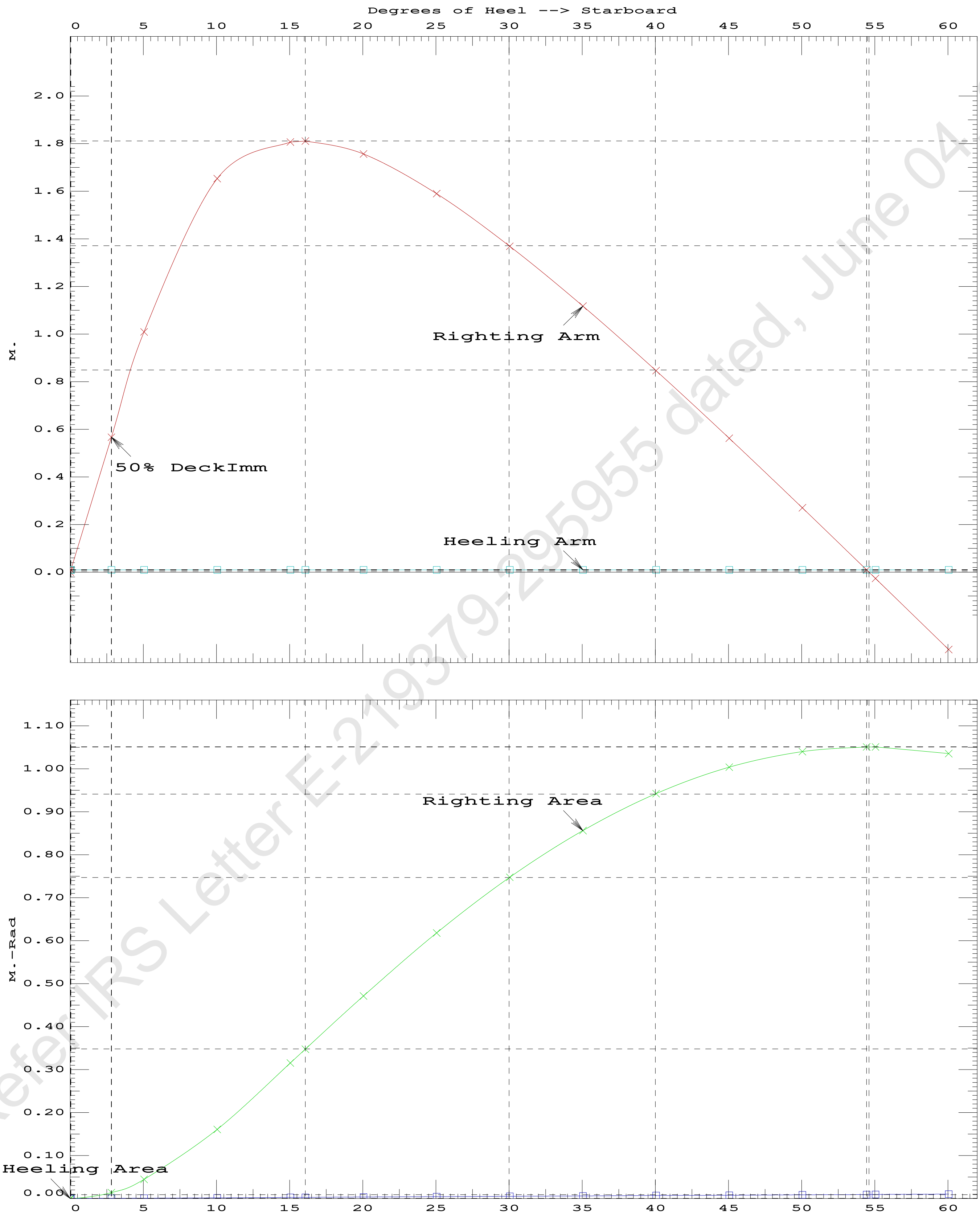
Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 17.58 (constant)

LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2		Min/Max		Attained	
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	0.3474	P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00	deg	54.39	P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	2.77	P

PONTOON

Condition 16 : Lightship + Crane + Ballast

TOWING CONDITION



LIGHTSHIP PARTICULARS

Lightship Weight	=	465 25	T
VCG	=	3.000	M above Baseline
LCG	=	25.062	M fwd of AP (Fr.0)
TCG	=	0.000	M off centerline

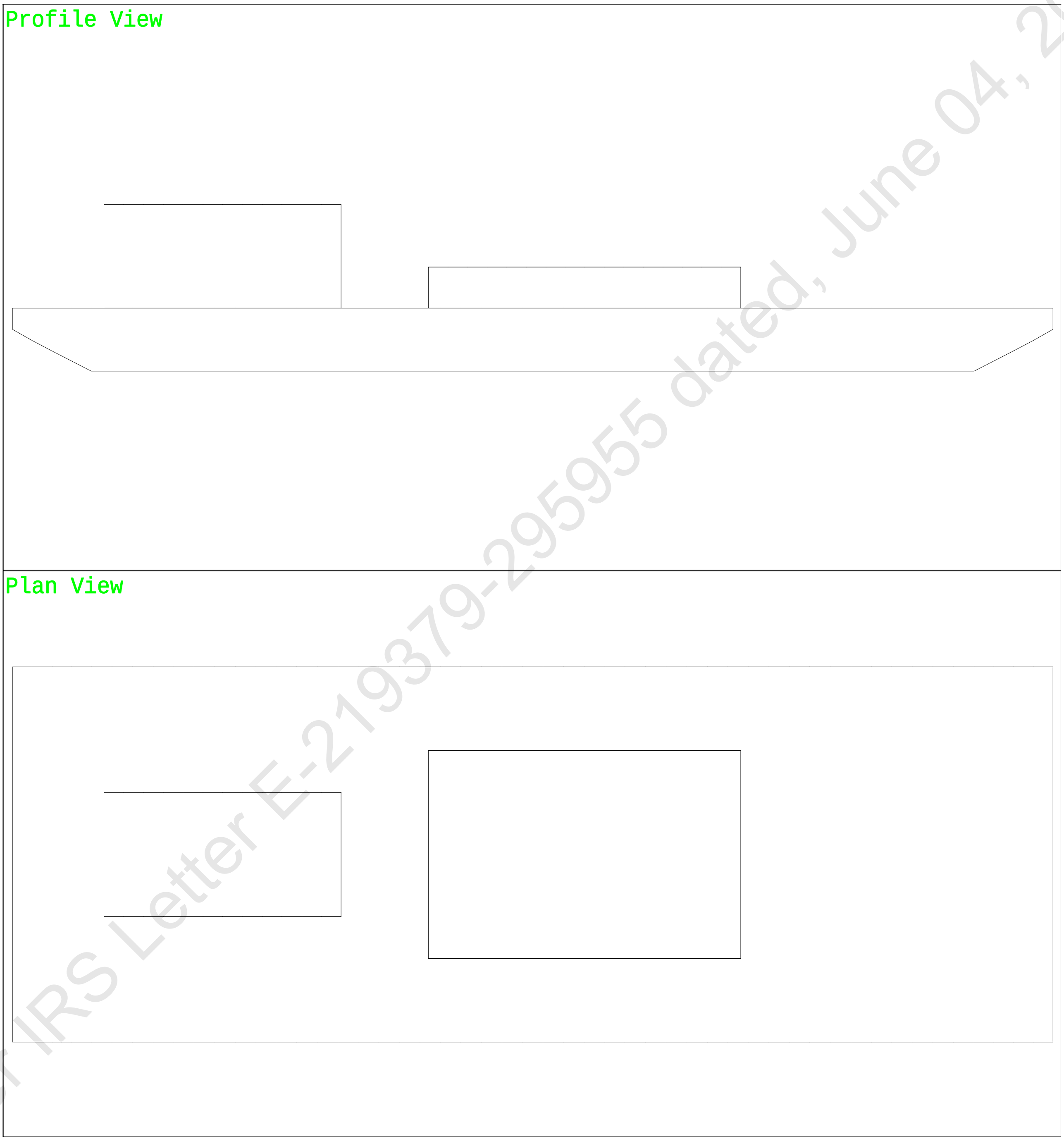
(Reference: Lightship particulars are taken from approved draft survey report.)

01-08-20 21:20:15   Cybermarine Technologies PTE. Ltd.
GHS 17.22   **PONTOON**

USK draft refers to the line:
0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000

WINDAGE AREA CALCULATION AT DIFFERENT DRAFTS

Condition Graphic



LATERAL PLANE STATUS
USK draft: 0.512 @ 50.00f, 0.512 @ 25.00f, 0.512 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	22.3	25.000f	-0.253	121.4	25.000f	1.281
CRANE				57.0	10.100f	5.000
DECKCARGO.C				29.4	27.500f	3.500
Total Lateral Plane->	22.3	25.000f	-0.253	207.8	21.267f	2.615

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 0.612 @ 50.00f, 0.612 @ 25.00f, 0.612 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	26.8	25.000f	-0.302	116.9	25.000f	1.229
CRANE				57.0	10.100f	4.900
DECKCARGO.C				29.4	27.500f	3.400
Total Lateral Plane->	26.8	25.000f	-0.302	203.3	21.184f	2.572

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 0.712 @ 50.00f, 0.712 @ 25.00f, 0.712 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	31.3	25.000f	-0.351	112.4	25.000f	1.176
CRANE				57.0	10.100f	4.800
DECKCARGO.C				29.4	27.500f	3.300
Total Lateral Plane->	31.3	25.000f	-0.351	198.8	21.097f	2.529

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 0.812 @ 50.00f, 0.812 @ 25.00f, 0.812 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	35.9	25.000f	-0.400	107.8	25.000f	1.123
CRANE				57.0	10.100f	4.700
DECKCARGO.C				29.4	27.500f	3.200
Total Lateral Plane->	35.9	25.000f	-0.400	194.2	21.006f	2.487

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 0.912 @ 50.00f, 0.912 @ 25.00f, 0.912 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	40.4	25.000f	-0.450	103.3	25.000f	1.070
CRANE				57.0	10.100f	4.600
DECKCARGO.C				29.4	27.500f	3.100
Total Lateral Plane->	40.4	25.000f	-0.450	189.7	20.911f	2.445

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.012 @ 50.00f, 1.012 @ 25.00f, 1.012 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	45.0	25.000f	-0.497	98.7	25.000f	1.018
CRANE				57.0	10.100f	4.500
DECKCARGO.C				29.4	27.500f	3.000
Total Lateral Plane->	45.0	25.000f	-0.497	185.1	20.809f	2.405

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.112 @ 50.00f, 1.112 @ 25.00f, 1.112 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	49.7	25.000f	-0.545	94.0	25.000f	0.966
CRANE				57.0	10.100f	4.400
DECKCARGO.C				29.4	27.500f	2.900
Total Lateral Plane->	49.7	25.000f	-0.545	180.4	20.699f	2.367

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.212 @ 50.00f, 1.212 @ 25.00f, 1.212 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	54.4	25.000f	-0.594	89.3	25.000f	0.914
CRANE				57.0	10.100f	4.300
DECKCARGO.C				29.4	27.500f	2.800
Total Lateral Plane->	54.4	25.000f	-0.594	175.7	20.584f	2.329

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.312 @ 50.00f, 1.312 @ 25.00f, 1.312 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	59.1	25.000f	-0.642	84.5	25.000f	0.863
CRANE				57.0	10.100f	4.200
DECKCARGO.C				29.4	27.500f	2.700
Total Lateral Plane->	59.1	25.000f	-0.642	170.9	20.462f	2.291

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.412 @ 50.00f, 1.412 @ 25.00f, 1.412 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	63.9	25.000f	-0.691	79.8	25.000f	0.810
CRANE				57.0	10.100f	4.100
DECKCARGO.C				29.4	27.500f	2.600
Total Lateral Plane->	63.9	25.000f	-0.691	166.2	20.333f	2.255

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.512 @ 50.00f, 1.512 @ 25.00f, 1.512 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	68.7	25.000f	-0.739	75.0	25.000f	0.759
CRANE				57.0	10.100f	4.000
DECKCARGO.C				29.4	27.500f	2.500
Total Lateral Plane->	68.7	25.000f	-0.739	161.4	20.194f	2.220

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.612 @ 50.00f, 1.612 @ 25.00f, 1.612 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	73.6	25.000f	-0.786	70.1	25.000f	0.708
CRANE				57.0	10.100f	3.900
DECKCARGO.C				29.4	27.500f	2.400
Total Lateral Plane->	73.6	25.000f	-0.786	156.5	20.044f	2.188

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.712 @ 50.00f, 1.712 @ 25.00f, 1.712 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	78.5	25.000f	-0.834	65.2	25.000f	0.657
CRANE				57.0	10.100f	3.800
DECKCARGO.C				29.4	27.500f	2.300
Total Lateral Plane->	78.5	25.000f	-0.834	151.6	19.883f	2.157

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.812 @ 50.00f, 1.812 @ 25.00f, 1.812 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	83.4	25.000f	-0.882	60.3	25.000f	0.607
CRANE				57.0	10.100f	3.700
DECKCARGO.C				29.4	27.500f	2.200
Total Lateral Plane->	83.4	25.000f	-0.882	146.7	19.712f	2.128

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.912 @ 50.00f, 1.912 @ 25.00f, 1.912 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	88.3	25.000f	-0.930	55.4	25.000f	0.556
CRANE				57.0	10.100f	3.600
DECKCARGO.C				29.4	27.500f	2.100
Total Lateral Plane->	88.3	25.000f	-0.930	141.8	19.529f	2.100

Distances in METERS.

01-08-20 21:20:15 Cybermarine Technologies PTE. Ltd.
GHS 17.22 **PONTOON**

LATERAL PLANE STATUS
USK draft: 2.012 @ 50.00f, 2.012 @ 25.00f, 2.012 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	93.2	25.000f	-0.978	50.5	25.000f	0.505
CRANE				57.0	10.100f	3.500
DECKCARGO.C				29.4	27.500f	2.000
Total Lateral Plane->	93.2	25.000f	-0.978	136.9	19.333f	2.073

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 2.112 @ 50.00f, 2.112 @ 25.00f, 2.112 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	98.2	25.000f	-1.025	45.5	25.000f	0.455
CRANE				57.0	10.100f	3.400
DECKCARGO.C				29.4	27.500f	1.900
Total Lateral Plane->	98.2	25.000f	-1.025	131.9	19.118f	2.050

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 2.212 @ 50.00f, 2.212 @ 25.00f, 2.212 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	103.2	25.000f	-1.073	40.5	25.000f	0.405
CRANE				57.0	10.100f	3.300
DECKCARGO.C				29.4	27.500f	1.800
Total Lateral Plane->	103.2	25.000f	-1.073	126.9	18.887f	2.029

Distances in METERS.

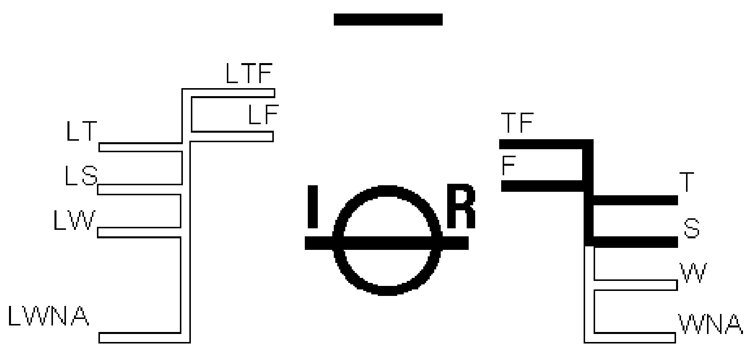
MEMORANDUM OF FREEBOARDS
15 March 2024

Chisti Marine Engineering Works.	Type B freeboard			
Yard No. CHISTI-037	Freeboard Length	:	48.000 m	
	Summer WL Extreme Draught - load line amidships	:	2.212 m	

The upper edge of the deckline from which these freeboards are measured is 0 **mm below top of * ~~Wood~~/ * steel Main Deck * at side / *~~continued to side~~. (*Delete words not applicable).**

Freeboards from Deck Line			**Seasonal, Fresh Water, or Timber allowances**	
Tropical Fresh Water	713	**mm.**	97	**mm. above summer**
Fresh Water	759	**mm.**	51	**mm. above summer**
Tropical	764	**mm.**	46	**mm. above summer**
Summer	810	**mm.**		**Upper edge of line through centre**
Winter	Not	**mm.**	Assigned	**mm. below summer**
Winter North Atlantic	Not	**mm.**	Assigned	**mm. below summer**
Timber Tropical fresh water	—	**mm.**	—	**mm. above timber summer**
Timber fresh water	—	**mm.**	—	**mm. above timber summer**
Timber Tropical	—	**mm.**	—	**mm. above timber summer**
Timber Summer	—	**mm.**	—	**mm. above centre of ring**
Timber Winter	—	**mm.**	—	**mm. below timber summer**
Timber Winter North Atlantic	—	**mm.**	—	**mm. below timber summer**

Valid only for unmanned operation while under tow along the Indian Coast, during the course of which the vessel does not go more than 20 nautical miles from the nearest land, and may cross gulfs or similar features recognised by the Administration.

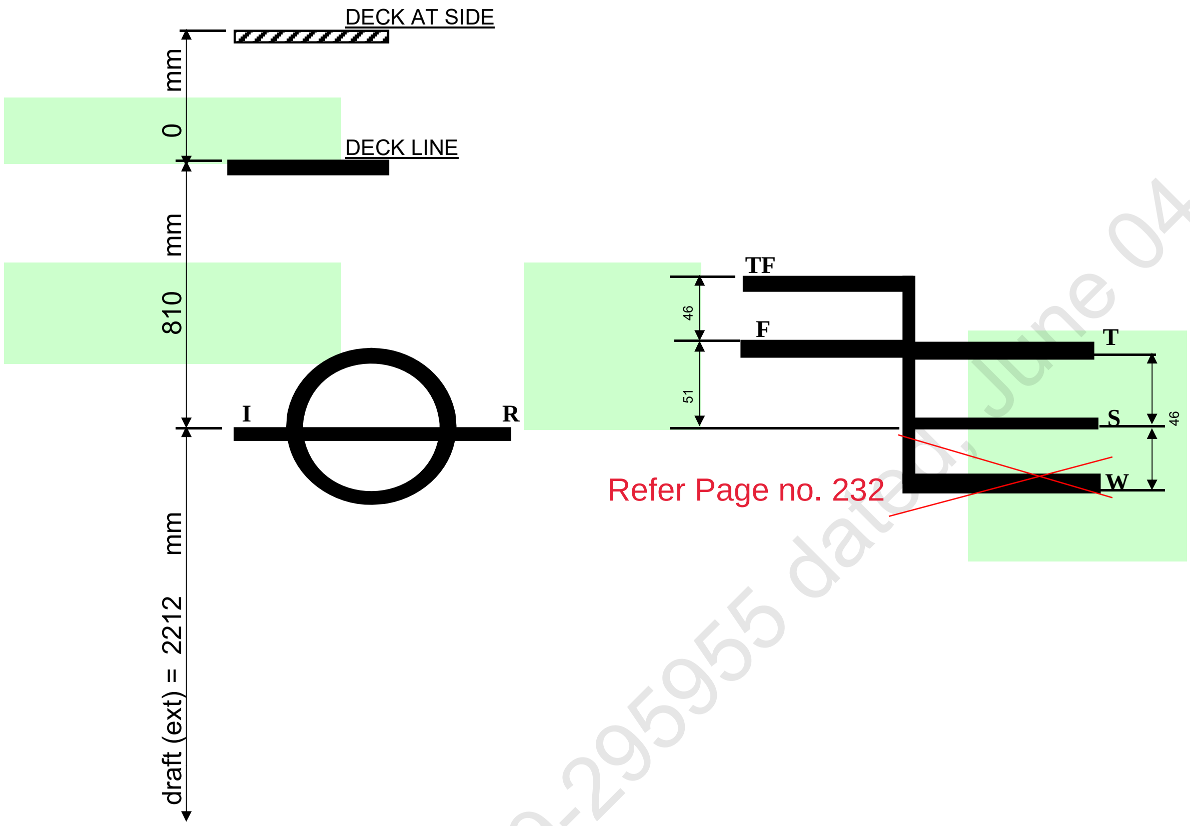


At Fr. 52

Form No. 4111 (07-83)

TABLE OF FREEBOARDS

Depth mld	=	3000	mm
Lightship	=	465	T



LOADLINE	FREEBOARD (mm)	DRAFT (ext) BL (m)	DISPLACEMENT (ext) T	DEADWEIGHT T
SUMMER	810	2.212	1893.06	1427.81
FRESH WATER	759	2.263	1892.81	1427.56
TROPICAL	764	2.258	1935.5	1470.3
TROPICAL FRESH WATER	713	2.309	1934.2	1469.0

Deck Plate Thickness =	10	mm
Keel Plate Thickness =	12	mm

IRCLASS
Indian Register of Shipping
CIN: U61100MH1975NPL018244
52 A, Adi Shankaracharya Marg, Opp. Powai Lake,
Powai, Mumbai - 400 072, India
Phone : +91 22 30519400 / +91 22 7119 9400
Fax : +91 22 2570 3611
E-mail : ho@irclass.org ★ Website : www.irclass.org

Our ref: E-218594-290106

15-March-2024

Company : Chisti Marine Engineering Works

Dear Sir/Madam,

Your document (detailed below) has been reviewed.

Project Details	
Project number	NC003020
Project name **Specific Yard No(s):**	Chisti Marine Engineering Works - CHISTI-036 to CHISTI-038 CHISTI-037
Project Manager	Varun Yalvatkar
Project In-Charge	Amit Waje

Document Details	
Plan No.	-
Plan Title	Memorandum of Freeboard – CHISTI-037
Rev.No.	1
Upload.No.	1
Submitted Date	13-March-2024
Review Status	Reviewed
Parent Plan Title	

Notes

1. The review status Reviewed indicates:

   Under the terms of International Convention on Load Lines 1966 and Protocol of 1988, as amended, the committee has presently assigned freeboards to the subject vessel as shown on the enclosed Memorandum.

   You may have these freeboards marked on the sides of the vessel in accordance with the diagram on the Memorandum. On completion, these markings should be verified by the Society's surveyor with a view to issue a Load Line Certificate.

   Please note that the validity of any such certificate will be subject to the stability of the vessel being approved.

   These freeboards are also subject to satisfactory compliance with other requirements of the International Convention on Load Lines 1966 and Protocol of1988, as amended.

Thanking you,
Yours Sincerely,
For INDIAN REGISTER OF SHIPPING

Vimal R
Surveyor

This is an electronically published document and does not require signature.

DRAFT SURVEY REPORT
OF
ARCADIA PARSHVA
(OFFICIAL NO. JMR-0025)

LIGHTSHIP PARTICULARS:

Lightship Displacement - 465.25 T
Lightship L.C.G - ~~25.000~~ m from AP(Fr.0) 25.062 m
Lightship V.C.G - 3.00 m above BL (VCG taken at the level of main deck).
Lightship T.C.G - 0.00 m.

Rev.	Description	Date	Prepared	Review

Issued for : **Approval**

Owner: **ARVIND AND COMPANY SHIPPING AGENCIES LTD.** Builder: **CHISTI MARINE ENGINEERING WORKS**	Project: **PONTOON** **(YARD NO. CHISTI-037)**

Doc #				Document Title:	Prepared	**YVG**
CM23-VI-110-0207B				**DRAFT SURVEY REPORT**	Checked	**NM**
Scale	Size	Sheet	Date		Review 1	
NTS	A4	1 of 15	15-03-2024		Review 2	

Cybermarine
www.cybermarine.net

10 Bukit Batok Crescent
#12-01 The Spire
Singapore 658 079

501 Technocity IT Premises
Plot No X-5/3 Mahape
Navi Mumbai 400 710 India

GENERAL PARTICULARS

Name of the Vessel	ARCADIA PARSHVA
Client	SMT. PARUL A. SHAH
Builder	CHISTI MARINE ENGINEERING WORKS
Yard No.	CHISTI-037
Type of Vessel	PONTOON
Officila No.	JMR-0025
Class Notation	卐SU,PONTOON, INDIAN COASTAL SERVICE FOR CRANE OPERATION DECK STRENGTHENED FOR 10T/M^2

Main Particulars

LOA	50.000 M
LBP	50.000 M
Breadth (Mld.)	18.000 M
Depth (Mld.)	3.000 M
Draft (Mld.)	2.200 M

References

AP	at Fr.0
FP	at Fr.100
Midship	at Fr.50

Frame spacing

Throughout	500 mm

DRAFT SURVEY REPORT : ARCADIA PARSHVA (YARD: CHISTI-037)

Location of Test	Sikka, Jamnagar	
Date	15-Mar-24	
Weather	CALM	
Wind	LIGHT	
Specific Gravity	1.025	
Restraining Line	Loose	
Witnessed by	Mr. Ravi Karaka	(IRS Surveyor)

Drafts :

(w.r.t keel line)

Location		Draft Port (m)	Draft Stbd (m)	Mean Draft (m)
Fwd	(Fr.92)	0.590	0.600	0.595
Mid	(Fr.50)	0.585	0.600	0.593
Aft	(Fr.8)	0.580	0.590	0.585

Hydrostatics

Correction for Drafts

To correct for observed drafts at marks and obtain Drafts at perps.

LBP	-	50.000	m

Observed Drafts (wrt keel line)

Fwd	-	0.595	m
Aft	-	0.585	m

Correction to Drafts (wrt Keel line)

Location	At Fwd (Fr.92)	At Aft (Fr.8)	
Draft at Marks	0.595	0.585	m
Correction w.r.t. baseline	0.000	0.000	
corrected Drafts w.r.t. baseline	0.595	0.585	
Trim at Marks		-0.010	m
Distance of Marks from PP	4.000	4.000	m
Distance between Marks		42.000	m
Draft correction factor		0.000	
Correction at PP	0.001	-0.001	
Corrected Draft at PP	0.596	0.584	
Trim		-0.012	m
Mean Draft		0.590	m
From Hydrostatics against above Mean Draft and Trim			
Displacement at density(T/M^3)=1.025 T/m^3		465.25	T
LCB		25.062	m
VCB		0.289	m
KMt		48.197	m

NOTE

1. Hydrostatic Particulars computed by Computer Program at Sp.Gr. 1.025
2. LCB, LCG are defined from AP, Fwd of AP is +ve and Aft of AP is -ve
3. Trim by Fwd is -ve and Trim by Aft is +ve.
4. KB is measured from Base Line.
5. Draft is measured from Keel Line

HYDROSTATIC PROPERTIES
Trim: Fwd 0.010/50.000, No Heel, Fixed VCG = 0.000

```
 Draft@  Displacement   Buoyancy-Ctr.  Weight/          Moment/
 25.000f---Weight(MT)----LCB-----VCB-------cm-----LCF---cm trim----KML-----KMT
  0.590       465.25  25.062f  0.289     8.26  25.009f   27.55  296.09  48.197
 Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
                            Trim is per 50.00m.
 Draft is from USK.
```

Weights to go off Board

No	Item	WEIGHT T	LCG m	L-MOM T-m	VCG m	V-MOM T-m	TCG m	T-MOM T-m
-	-	0.00	0.000	0.00	0.000	0.00	0.000	0.00
	TOTAL	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Weights to go on Board

No	Item	WEIGHT T	LCG m	L-MOM T-m	VCG m	V-MOM T-m	TCG m	T-MOM T-m
-	-	0.000	0.000	0.00	0.000	0.00	0.000	0.00
	TOTAL	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Lightship Particulars

Item	WEIGHT T	LCG m	L-MOM T-m	VCG m	V-MOM T-m	TCG m	T-MOM T-m
Particulars as Draft Survey	465.25	25.062	11660.096	0.000	0.000	0.000	0.000
Weights to go Off board	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Weights to go on board	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Lightship Particulars	465.25	25.062	11660.096	0.000	0.000	0.000	0.000

Lightship Particulars

Displacement	**465.25**	**T**
LCG	**25.062**	**m From AP**

NOTE

1. Hydrostatic Particulars computed by Computer Program at Sp.Gr. 1.025
2. LCB, LCG are defined from AP, Fwd of AP is +ve and Aft of AP is -ve
3. Trim by Fwd is -ve and Trim by Aft is +ve.
4. KB is measured from Base Line.
5. Draft is measured from Keel Line

सत्यमेव जयते

रजिस्ट्री का प्रमाणपत्र/ CERTIFICATE OF REGISTRY

कोसटींग वैसील्स कानून सन 1838 ई की XIX वी धार के अन्तगर्त/ Issued under the Coasting Vessels Act (Act XIX of 1838)

रजिस्टर नंबर Registered No.	JMR-0025

प्रमाणित किया जाता है कि This is to Certify that	ARVIND AND COMPANY SHIPPING AGENCIES LTD						
जिनका स्थायी निवास स्थान / व्यापार का प्रमुख स्थान है ~~residinent permanently~~ at / having principal place of business at	5th floor,701-702, City point, NearTown Hall, Jamnagar-361001						
ओर जिन्होंने ये घोषित किया है की वह / वे भारत के/में नागरिक/ रेजिस्टर्ड कंपनी है ओर एक मात्र मालिक है and having declared that ~~he~~/it is the citizen/s/ Company registerd ~~of~~/in India and sole owner/s of the			PONTOON BARGE		जिसका नाम called the	ARCADIA PARSHVA	
ओर जिसका नेट टनभार है which is the burthen of - NT	206	ओर यह भी की and that the said vessel	ARCADIA PARSHVA		मे बना था । was built at	SIKKA	
वर्ष in the year	2024	उक्त वेसल सन 1838 के अधिनियम XIX अनुसार कर लिया गया है। The said vessel has been duly registered at the port of Bedi under Act XIX of 1938					
यह निम्नलिखित हस्ताक्षर दवारा तारीख को प्रमाणित किया गया। certified under my hand this the	13TH	day of	MARCH	2024			

खुदा हुवा ओर लिखा हुवा रजिस्टर्ड नंबर Branded and painted Registerd Number	JMR-0025

	मिटर / Meters
पूरी लबाइ /Extreme Length	50.00 M
लंबाइ / Length(L.L)	48.00 M
चौड़ाइ / Breadth	18.00 M
गहराई / Depth	03.00 M

सकल टनभार / GROSS TONAGE	685
नेट टनभार / NET TONNAGE	206

समुद्री वाणिज्य विभाग, जामनगर • MERCANTILE MARINE DEPT., JAMNAGAR • सत्यमेव जयते

(M. BEHL)

भारतीय पोत रजिस्ट्रार/ Registar of indian Ships
समुद्री वाणिज्य विभाग /Mercantile Marine Department
जामनगर /Jamnagar District.
Issued under the authority of PO,MMD, Mumbai

सूचना -1. यह जहाज केवल भारतीय लाल एनसाइन झंडा लगायेगा कोई दूसरी तरह का झंडा लगाने पर यह प्रमाण पत्र वापिस लिया जायेगा ।

2. यदि यह जहाज विलीन हो जाए या टूट जाए तो इस घटना की सूचना इस प्रमाण पत्र सहित भारतीय पोत रजिस्ट्रार ऑफ शिपिंग, समुद्री वाणिज्य विभाग,जामनगर को शिघ्र भेज दी जाय ।

Notes : (1) The Vessel Shall hoist the Indian Red Ensign Only. If any other kind of flag is hoisted this Certificate Will be Withdrawn.

(2) Should the vessel be lost or broken up, notice thereof, together With the Certificate of Registry if in existence, Should immediately given to the Registrar of India Ship, Bedi.

No: 77
13/3/24

LIGHT-SHIP SURVEY PROCEDURE AND REPORTING FORMAT
Part I

Pre requisites for Conducting Light-Ship Survey
(To be provided /confirmed by Builder/Owner/Operator/Consultant prior commencement of Light-Ship Survey)

Name Of Builder/owner rep conducting trial	CHISTI MARINE ENGG. WORKS
Hull Number/Vessel name	CHISTI-037 | ARCADIA PARSHVA
Type of Vessel (Cargo/passenger)	PONTOON
Location of Vessel	SIKKA
Date of Proposed Light-Ship Survey	15.03.2024, 1500 HRS.
Draft of Vessel and trim	0.6 , and 0.01
Depth of Water	5.5 M
Anticipated tide on date of survey	NORMAL
Weather prediction for proposed date	NORMAL
Availability of following reports from builder/Owner/Owners rep	
Copy of Approved Inclining experiment report	☐ YES ☐ NO NA
Details of Periodical Draft Survey report/any Draft Survey report conducted after the approval of Final TSB.	☑ YES ☐ NO
Declaration of weights to go on	☐ YES ☐ NO NA
Declaration of weights to go off -tank contents - Loose items	☐ YES ☐ NO NA

Facilities/Arrangement for conduct of Light-Ship Survey	
Availability of boat to take draft reading	☑ YES ☐ NO
Availability of draft mark plan/approved document like Prelim TSB/Final TSB for confirming location of draft marks	☑ YES ☐ NO
Safety Requirements- lighting, gas freeing, accessibility of areas requiring inspection as part of Survey	☑ YES ☐ NO
Availability of sounding tapes, hydrometer etc	☑ YES ☐ NO
Closing/isolating cross connected tanks	☐ YES ☐ NO NA
Removal of gangway during draft reading/experiment	☑ YES ☐ NO
Permanent ballast used ? ; if any and its details	☐ YES ☑ NO
If vessel has any list, reason for the list is established.	☐ YES ☑ NO
A trial is conducted to see whether the vessel's draft corresponding to the loading condition is matching with the approved hydrostatics.	☐ YES ☑ NO

NOTE : The responsibility of the IRS Surveyor is to verify that the survey is conducted according to accepted procedures and that all basic measurements and data are correctly taken and recorded. The responsibility for making preparation, conducting lightship survey and preparing report rest with owner/operator.

Confirm Readiness for test

Name & Signature of Shipyard /Owners rep

Sheet for recording Weights to Go-on, Go-off, relocated and Tank Soundings

Weights To Go-on						
Sl No.	Weight Description	Weight (t)	L.C.G (m)	V.C.G (m)	T.C.G (m)	Surveyor Verification
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐

LCG: Longitudinal Center of Gravity from AP/Transom/Midship/Frame -----

Weights To Go-off

Sl No.	Weight Description	Weight (t)	L.C.G (m)	V.C.G (m)	T.C.G (m)	Surveyor Verification
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
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						☐
						☐
						☐
						☐

LCG: Longitudinal Center of Gravity from AP/Transom/Midship/Frame -----

Tank Soundings				
Sl No.	Tank Name	Sounding (cm /m)	Density	Surveyor Endorsement

Note: Tank name should be mentioned as per the approved TSB / Capacity Plan.

Weights To Be Relocated									
Sl No.	Weight Description	Weight (t)	L.C.G (m)		V.C.G (m)		T.C.G (m)		Surveyor Verification
			Initial	Final	Initial	Final	Initial	Final	
									☐
									☐
									☐
									☐
									☐
									☐
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									☐
									☐
									☐

LCG: Longitudinal Center of Gravity from AP/Transom/Midship/Frame -----

LIGHTSHIP SURVEY PROCEDURE AND REPORTING FORMAT
Part II

REPORTING FORM FOR LIGHTSHIP SURVEY

Attending Surveyor should witness Light-Ship Survey only if the Part-I submitted by Owner/Builder/Operator/Consultant is found satisfactory

General Particulars

Name of Vessel	ARCADIA PARSHVA
Builder/ Owner	CHISTI MARINE ENGG WORKS/K.B SHIPPING
Yard Number/ IR Number	CHISTI 037
Type of Vessel Passenger / Non Passenger	PONTOON
Date(dd/mm/yyyy);Local Time of inspection/test	15/03/24, 1500HRS.
Wind / Weather condition	CALM
Location of Test	SIKKA.
Depth of water where vessel is moored, m	5.5M
Method of reading Draft(By Boat-Y/N)	YES

Location of Draft Marks	Frame Number	Draft Reading (Port side),m	Draft Reading (STBD Side),m
Aft	8	0.58	0.59
Midship	50	0.585	0.60
Forward	92	0.59	0.60
Note : *Recommended to attach Photographs of draft; as record*			

Density of Water

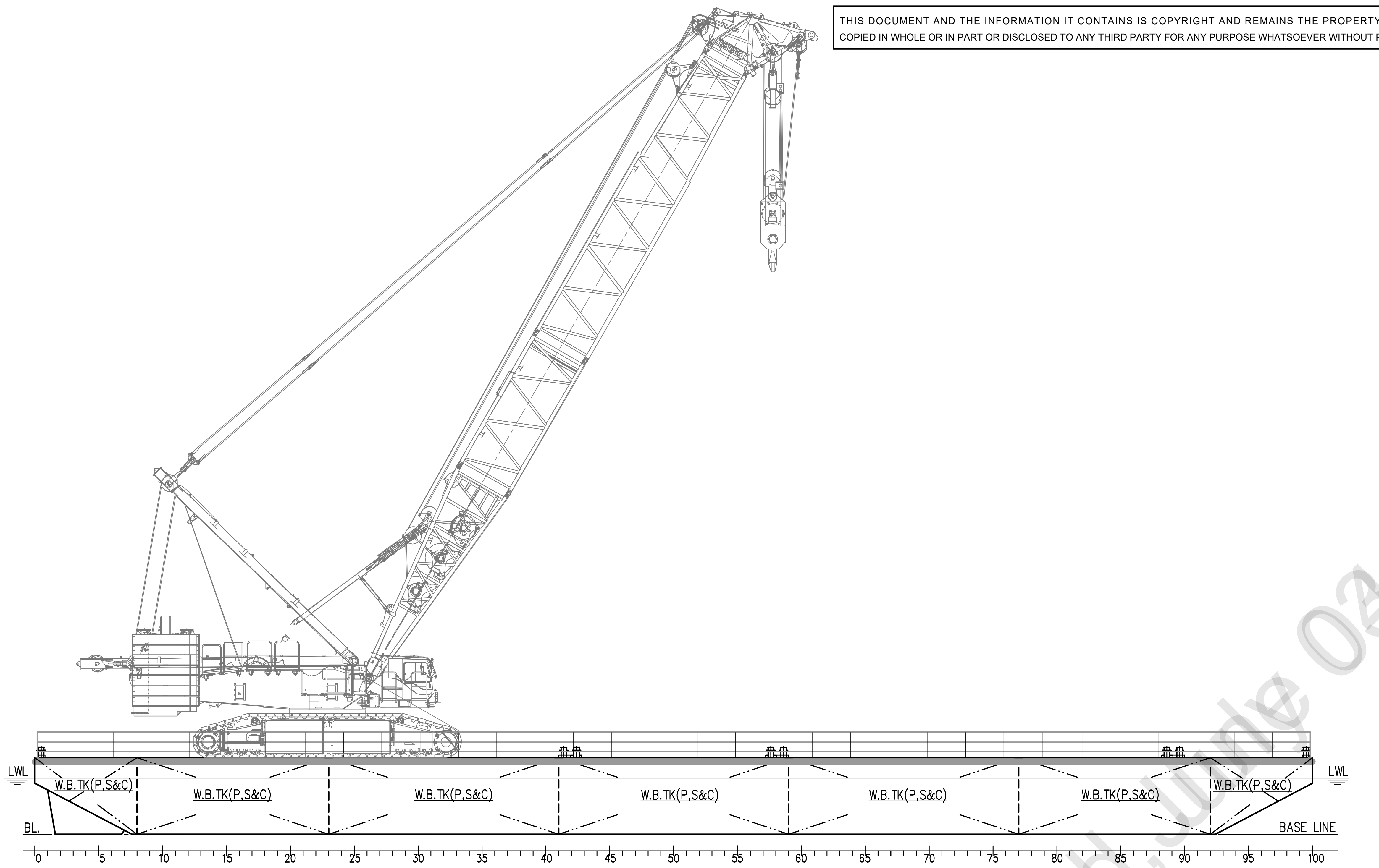
Results of 3 samples from Forward, Aft and Mid of the vessel		
Sample 1: (ford)	1.024	**t/m3**
Sample 2: (mid)	1.025	**t/m3**
Sample 3: (aft)	1.025	**t/m3**

Verification of weights to go on / go off /re-location	*SEE PART 1*
Note: *See Part 1 (the sheet for recording the above data) , verify and endorse.*	

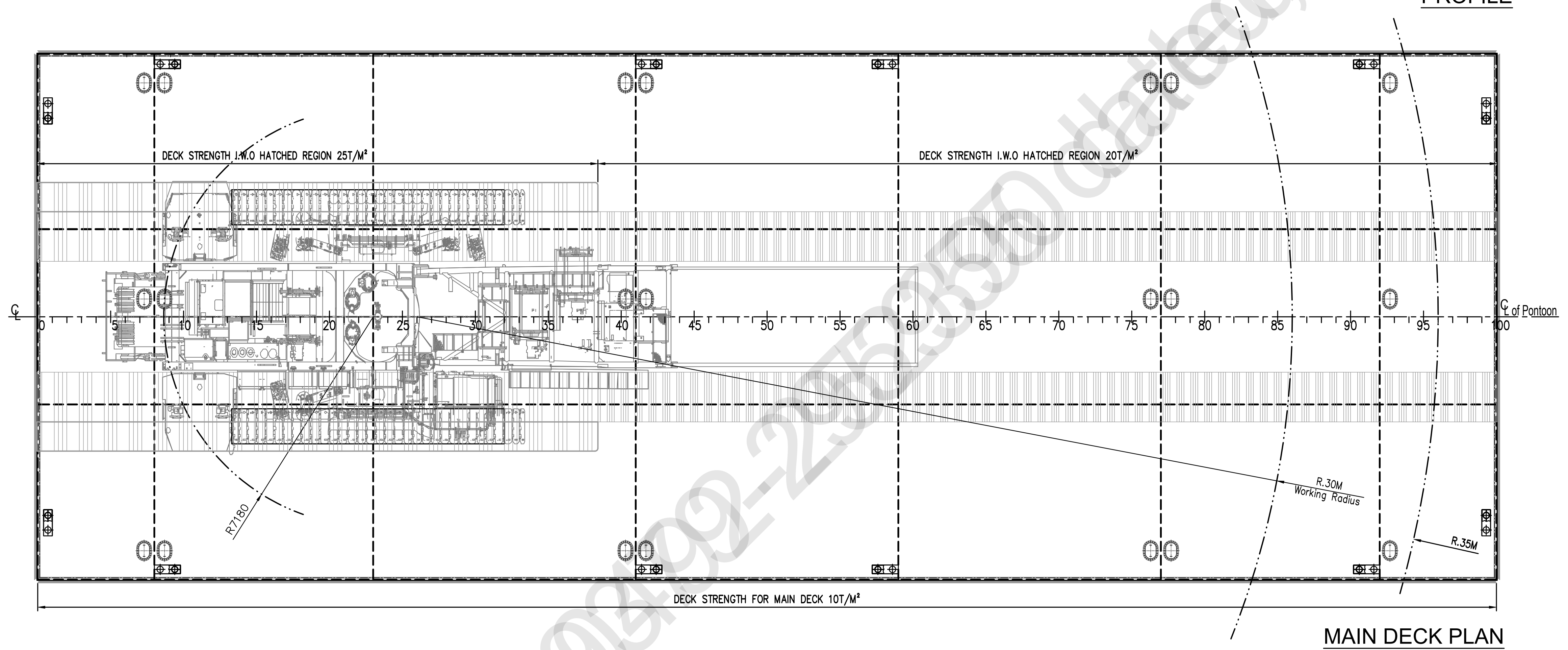
Special Comments on Light Ship Survey, if any.
Draft SURVEY CARRIED OUT AS PER APPROVED Draft SURVEY procedure CM23- VI-110-0206, Approved on 29 JUNE 2023, READINGS ARE RECORDED

Attending surveyor to forward both forms Part 1 and Part 2 together with completed attachments 1 and 2 to HO.

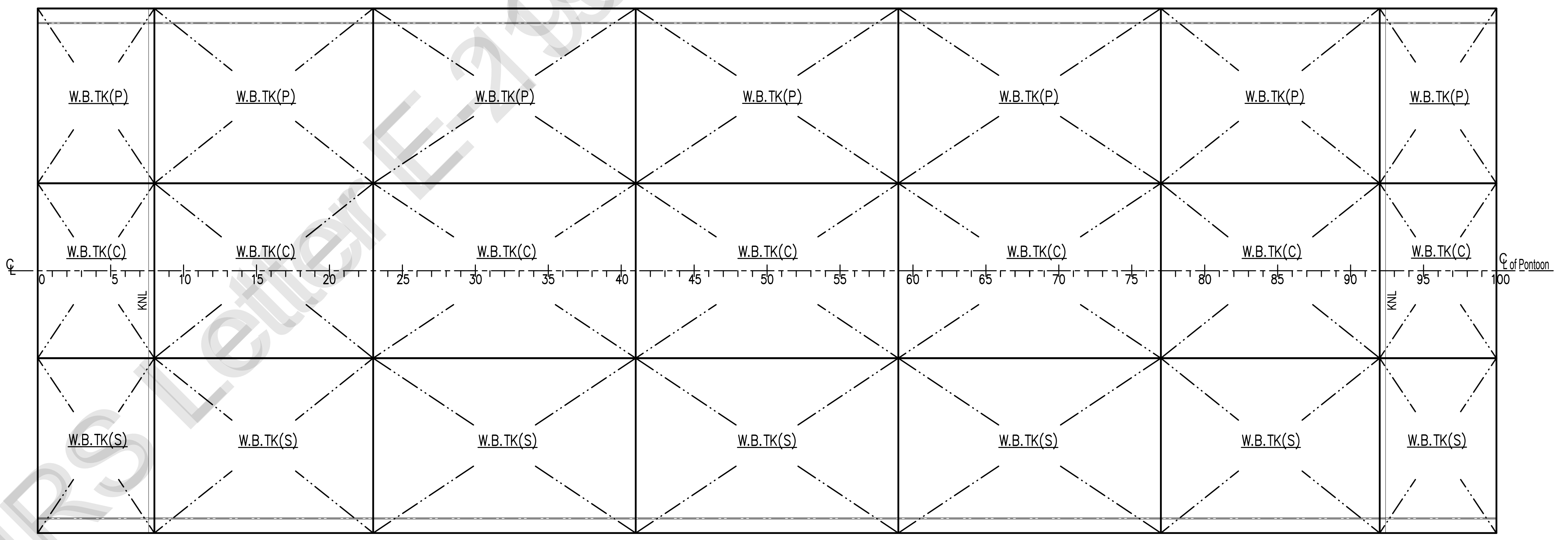
Name & designation of Attending Surveyor: Ravi Karaka



PROFILE



MAIN DECK PLAN



BELOW DECK PLAN

PRINCIPAL PARTICULARS:-

LENGTH O.A	:	50.00 M
BREADTH MLD.	:	18.00 M
DEPTH MLD.	:	3.00 M
DESIGN DRAFT	:	2.20 M
SCANTLING DRAFT	:	2.60 M
FRAME SPG.	:	500 mm

CRANE SPECIFICATION:-

CRAINE WEIGHT	=	400T
CRANE CAPACITY	=	40 T
WORKING RADIUS	=	30 M

SEE LETTER E-190492-252390

INDIAN REGISTER OF SHIPPING
MUMBAI
DATE: 03-Jul-2023
NOTED FOR INFORMATION

CLASS NOTATION:-

卐 SU, PONTOON, "INDIAN COASTAL SERVICE"
"DECK STRENGTHENED FOR 10 T/M^2 ",
"FOR CRANE OPERATION"

NOTE:

1) ALL DIMENSION ARE IN MM.
2) CRANE WILL OPERATE ONLY IN THE HATCHED REGION.

Rev.#	Description	Date	Drawn	Review

Issued for : Approval

Client: K.B. SHIPPING & CO.

Builder: CHISTI MARINE ENGINEERING WORKS

Project: PONTOON 036-038
(YARD NO. CHISTI - ~~036 & 037~~)

Drg.#	Rev.#	Drawing Title	Drawn	ISHWAR
CM23-VI-110-0001	00	GENERAL ARRANGEMENT	Checked	JASIM
Scale / Size / Date	Sheet		Review-1	MOHIT
1:175 / A2 / 29.05.2023	1 OF 1		Review-2	SUDHEESH

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